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## Cartographie de la biodiversité mondiale, profilage des espèces envahissantes et élaboration d'un indice global du changement de la biodiversité basé sur les facettes taxonomiques, fonctionnelles et phylogénétiques de la biodiversité des poissons d'eau douce

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PhD Thesis

# Mapping global biodiversity, profiling invasive species and developing a global index of biodiversity change based on taxonomic, functional and phylogenetic facets of freshwater fish biodiversity

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路漫漫其修远兮，吾将上下而求索.

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# ABSTRACT

Biodiversity is a multifaceted concept that includes three main components, namely taxonomic, phylogenetic and functional diversity. Biogeographical studies have paid more attention to the first two facets while the patterns and drivers of functional diversity and their changes because of global change remains largely unknown at the global scale. These knowledge gaps are especially large for freshwater fishes, because they not only account for a quarter of vertebrates and support the functioning and stability of ecosystems, but are also one the most threatened vertebrates groups in the world. Thus this thesis aims to improve the understanding of the functional diversity of global freshwater fishes and bridge the gap between taxonomic, functional and phylogenetic facets to evaluate the impacts of human activities on the multifaceted biodiversity of global fishes.

Towards this aim, we first built a trait database describing the morphology of 10 600 species occurring in over 2 400 river basins all over the six terrestrial realms.

First, we assessed the distributions of the morphological traits within the fauna of each realm. We revealed that fish morphological traits are different between realms and that morphologically extreme species are distributed in all realms. Second, using a multi-traits approach at the basin scale we found that the historical functional diversities have been shaped by habitat, energy and history-related variables. Third, we demonstrated that morphology differs between species that have never been introduced species and those that were introduced and those that were even established. Last, using a novel cumulative index combining changes in six facets of biodiversity we found that human activities have markedly affected fish biodiversity in more than half of the world river (52.8%, 1 297 rivers). Those biodiversity changes were primarily due to alterations of water connectivity and introductions of non-native species.

This work underlined the potential of morphological features in the study of global freshwater fish functional diversity, and the combination of functional phylogenetic and taxonomic features in a novel multifaceted biodiversity change index will constitute a useful tool for biological conservation.

# RESUME

La biodiversité est un concept à multiples facettes qui comprend trois composantes principales, à savoir la diversité taxonomique, phylogénétique et fonctionnelle. Les études biogéographiques ont jusqu'à présent accordé plus d'attention aux deux premières facettes tandis que les déterminants de la diversité fonctionnelle et ses changements sous l'effet de l'anthropisation restent largement inconnus à l'échelle du globe. Ces lacunes sont particulièrement importantes pour les poissons d'eau douce, car ils représentent non seulement un quart des vertébrés et soutiennent le fonctionnement et la stabilité des écosystèmes, mais sont également l'un des groupes de vertébrés les plus menacés au monde. Ainsi, l'objectif de cette thèse est d'améliorer la compréhension de la diversité fonctionnelle des poissons d'eau douce du globe et de combler le fossé entre les facettes taxonomiques, fonctionnelles et phylogénétiques pour évaluer les impacts des activités humaines sur la biodiversité des poissons d'eau douce.

Dans ce but, nous avons d'abord construit une base de données de traits décrivant la morphologie de 10 600 espèces présentes dans plus de 2 400 bassins fluviaux dans les six royaumes terrestres.

Nous avons d'abord testé les distributions des traits morphologiques des poissons dans chaque zone biogéographique Nous avons montré que les traits morphologiques des poissons sont i) très variés à travers le globe et ii) que la distribution de ces traits est diffère entre zones biogéographiques. Cependant, les espèces morphologiquement extrêmes occupent la plus grande partie de l'espace morphologique dans toutes les zones biogéographiques. De plus, en utilisant une approche multi-traits à l'échelle du bassin versant, nous avons constaté que la diversité fonctionnelle historique a été façonnée par des variables liées à l'habitat, à l'énergie et à l'histoire. Nous avons également démontré que la morphologie diffère entre les espèces qui n'ont jamais été introduites et celles qui ont été introduites et celles qui se sont même établies. Enfin, nous avons réalisé une approche holistique en proposant un nouvel indice cumulatif combinant les changements dans les six principales facettes de la biodiversité pour évaluer les impacts des activités humaines sur la biodiversité des poissons. Nous avons constaté que les activités humaines ont considérablement affecté la biodiversité des poissons dans plus de la moitié des fleuves mondiaux (52,8%, 1 297 bassins versants). Parmi ces cours d'eau, les zones tempérées sont les plus affectées par les changements de biodiversité, et les changements de biodiversité sont principalement dus à des altérations de la connectivité de l'eau et à l'introduction d'espèces non indigènes.

Ce travail a souligné le potentiel des caractéristiques morphologiques dans l'étude de la diversité fonctionnelle mondiale des poissons d'eau douce, et le indice de changement de la biodiversité à multiples facettes donnera ici ouvre de nouvelles implications pour la conservation de la biodiversité.

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# Chapter I: General introduction

## I.1.1 Biodiversity and biogeography

Taxonomic experts estimated that the number of species on Earth is as high as 100 million, including some 8.7 million eukaryotic species (May 2010, Mora *et al.* 2011a). Together with the abiotic materials, this tremendous number of species support the Earth's biodiversity and ecosystems and their contribution to human societies. The term 'biological diversity' was firstly used in 1916 by Arthur Harris then commonly used from the beginning of the 1980s (Soule 1980) to describe the variety of life. The term biodiversity was proposed as an abbreviation of biological diversity in the mid-1980s in a scientific Forum (Wilson 1988, Sarkar 2002, Stork 2009). In 1992 the definition of biological diversity was adapted by the Convention of Biological Diversity: *"variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems"* (Convention on Biological Diversity 1992). Biodiversity supports ecosystem functioning and sustainability, species evolutionary processes, and stores and cycles nutrients essential for life, such as carbon, nitrogen, and oxygen (Stork 2009). Biodiversity is the foundation of ecosystem services, which is now called "nature contribution to people" (Diaz *et al.* 2018), closely related to human well-being. It is crucial to human's survival because it serves people food, construction material, raw material for industry, and medicine, as well as the basis for all improvements to domesticated plants and animals. These ecosystem services not only deliver the basic materials for survival, but also contain other aspects of a good life, including health, security, good social relations and freedom of choice (CBD 2006, Fig. 1).

Meanwhile, the study of biogeography that began in the 19th century has been increasing dramatically since the 1990s (Kent 2005, Cox *et al.* 2016). Biogeography aims to identify the distributions of species at temporal and spatial scales and to explore the underlying biotic and abiotic factors, mechanisms, and processes (Brown & Lomolino 1998, Violle *et al.* 2014). Biogeography studies at large scale commonly only use the species composition data without

considering their abundance. It can be studied with a focus on ecological factors that shape the distribution of species, or with a focus on the historical factors that have shaped the current distributions. Historically, biogeography is built on species concept, but recently a branch of biogeography based on the species functional traits and analyzing the patterns, causes, and consequences of the geographic distribution has been rapidly developed, namely functional biogeography (Violle *et al.* 2014). Functional biogeography builds a bridge between species-based biogeography and earth sciences to provide ideas and tools to explain patterns of multifaceted diversity (including species, functional and phylogenetic diversity), and predict global ecosystem functioning and services (Violle *et al.* 2014).

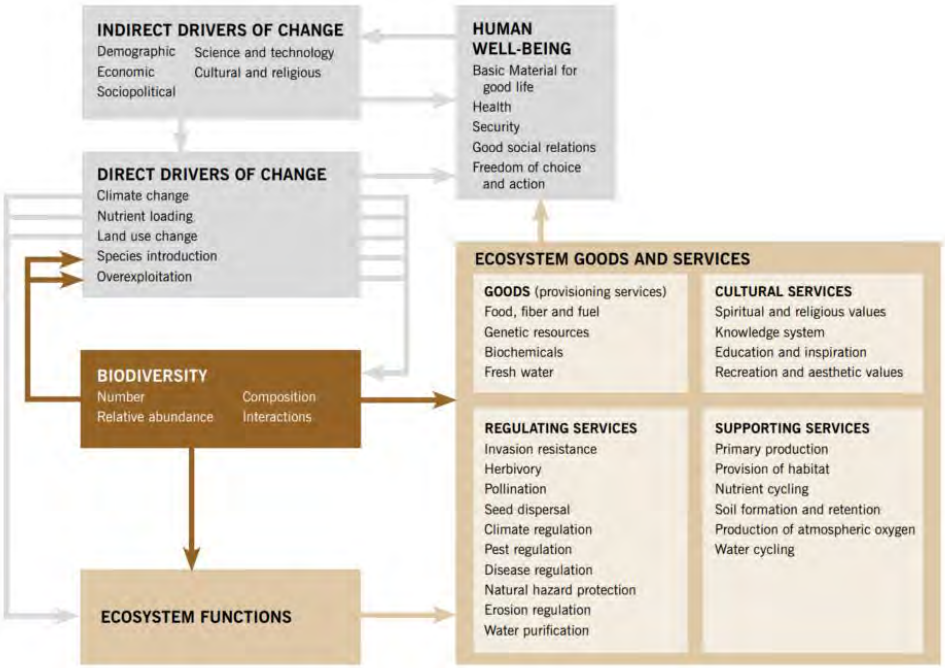



**Fig. 1**. Biodiversity, ecosystem functioning, ecosystem services, and drivers of change (adapted from CBD 2006).

### I.1.2 The multiple dimensions and scales of biodiversity

Biodiversity is a multi-dimensional concept that embraces many aspects of biological variation. Much of our current understanding of biodiversity patterns initials from the analyses of species richness and abundance, which is called taxonomic diversity (TD). TD accounts for the number of species and is the most commonly considered component of diversity. However, TD treats all species as functionally equivalent and evolutionarily independent, thereby

ignoring the fact that species belonging to different phylogenetic trees or with distinct functional traits play different roles in the ecosystem process (e.g. umbrella species, prey or predator). To address these limitations, new biodiversity estimates have been proposed for the last 2 decades to describe the functional (Petchey and Gaston 2006) and phylogenetic (Webb *et al.* 2002, Faith 2008) variations of the species assemblages. Functional diversity (FD), reflects the diversity of morphological, physiological and/or ecological traits within biological assemblages (Petchey & Gaston 2006); and Phylogenetic diversity (PD) accounts for phylogenetic relationships between the organisms (Webb *et al.* 2002, Devictor *et al.* 2010). Furthermore, besides evaluation of the three complementary components of diversity within a given community (alpha diversity), the term beta diversity was also applied to compares the composition of different communities in space or through time (Lande 1996, Dornelas *et al.* 2014).

## I.2 Global biodiversity under anthropogenic pressures

We are currently living in an age of extinction due to multiple human direct and indirect activities on the ecosystems (Barnosky *et al.* 2011, Naeem *et al.* 2012). Diverse indirect and direct drivers of change affect ecosystem functions by altering biodiversity, thereby jeopardizing the welfare of human well-being (Fig. 1). Climate change, habitat degradation, overexploitation, pollution and invasive alien species have been recognized as the most important and widespread direct anthropogenic drivers of biodiversity change (Bowler *et al.* 2020, Diaz *et al.* 2019, Pereira *et al.* 2012). These drivers interactively affect biodiversity change and attribute unequally among different taxa across the world. For instance, Wilcove *et al* (1998) quantifies the threats by major human activities to biodiversity loss for many different species taxa (fishes, birds, mammals, invertebrates, plants, etc.) across the United States. They ranked the threats and found that habitat loss and alien species were the two top-ranked threats, which affected 85% and 49% of the species they analyzed. They also found that alien species affected a higher proportion of imperiled plants (57%) than animals (39%), while for many aquatic animal groups (amphibians, fish, dragonflies and damselflies, freshwater mussels, and

crayfish), pollution being the second important threat after habitat loss.

## I.3 Freshwater ecosystems

Freshwater ecosystems cover less than 1% of the Earth's surface, while support huge and unique biodiversity, including more than 13 000 fish species and speciose amphibians, aquatic reptiles (crocodiles, turtles) and mammals (otters, river dolphins, platypus), which in total represent one-third of global vertebrates (Allen & Pavelsky 2018, Dudgeon *et al.* 2006). Freshwater ecosystems also host numerous invertebrates and microbes, especially in tropical regions that support most of the world's species (Dudgeon *et al.* 2006). Thus, freshwater ecosystems provide a broad variety of valuable goods and services for human beings (Olden *et al.* 2020), some of which are irreplaceable. For instance, fish can be served as food and source of protein (Tidwell & Allan 2001), and also can be used as a biological control agent and leisure pets. However, the uncontrolled use of natural resources from freshwater ecosystems by humans also makes them among the most threatened ecosystems comparing to their terrestrial or marine counterparts (Dudgeon *et al.* 2006). Living Planet Index (LPI) is a measure of the state of the world's biological diversity based on population trends of vertebrate species from terrestrial, freshwater and marine habitats biomes. Living Planet Report in 2018 from WWF showed that the living planet index for freshwater species is declining the most among all the ecosystems (Grooten & Almond 2018). Both fish and freshwater are resources humans need and they have been heavily impacted by human usage and regulation. The increasing threats from dams, flow regulation, pollution, invasive species, land-use change, and overexploitation make freshwater ecosystems more vulnerable than the terrestrial and marine ecosystems (Abell *et al.* 2008, Dudgeon *et al.* 2019). Freshwater fishes also become one of the most threatened groups of vertebrates due to the various threats of human activities (Arthington *et al.* 2016). Sala et al (2000) developed the scenarios of changes in biodiversity for the year 2100 based on the current knowledge about the impact of human activities on biodiversity change. They expected that freshwater ecosystems would be substantially affected by land use, biotic exchange, and climate change in the future. They also emphasized that biotic exchange would

be relatively more important for aquatic (especially lakes) than for terrestrial ecosystems because of both extensive intentional (e.g. fish stocking) and unintentional (e.g. ballast water releases) releases of organisms.

## I.4 Natural and anthropic drivers of fish taxonomic diversity

Fishes are the most diverse group of vertebrates, with more than 33 000 species described to date (Froese & Pauly 2018). 40% of the fish species are permanently inhabiting freshwater systems, which occupies only 0.8% of the Earth's surface (Tedesco *et al.* 2017). Such a striking richness is due to the geographic isolation between freshwater habitats and the low dispersal abilities of fish across landmasses (Hugueny, 1989), leading to distinct speciation directions and markedly different fish faunas between the different geographic realms and river basins (Leveque *et al.* 2008, Tedesco *et al.* 2012, Villeger *et al.* 2011, 2015). Indeed, nearly 40% of the fishes are endemic species that inhabits only one drainage river basin (Tedesco *et al.* 2012, 2017). The over 13 000 freshwater fish species belong to 170 families and most species occur in relatively few orders: Characiformes, Cypriniformes, Siluriformes, Gymnotiformes, Perciformes, and Cyprinodontiformes (Nelson 2006, Leveque *et al.* 2008). The number of fish species is unequal among the six biogeographic realms (i.e. the Afrotropical, Australian, Nearctic, Neotropical, Oriental, and Palearctic realms). Indeed, the two tropical realms (i.e. Neotropical and Afrotropical realms) host the most species, 4 035 and 2 938, respectively, followed by the Oriental (2 345). In contrast, the two arctic realms (i.e. Palearctic and Nearctic realms) host fewer species, 1 844 and 1 411, respectively, even less for the Australian with only 261 species (Leveque *et al.* 2008).

Oberdorff *et al.* (2011) reviewed the patterns and determinants of riverine fish species richness at the global, regional and local scales by dividing the six biogeographic continents into 926 river basins (Fig. 2). A few large river basins in tropical areas have a very high species richness, with Amazon, Congo and Mekong hosting roughly one-third of the world's freshwater fish species (Winemiller *et al.* 2016).

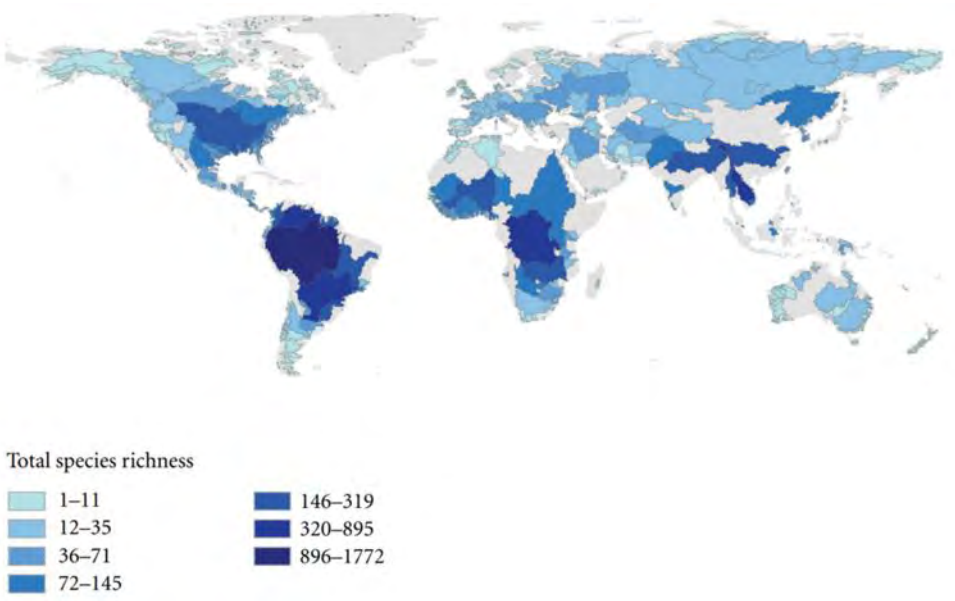



**Fig. 2.** Global freshwater fish species richness patterns at the river basin scale (adapted from Oberdorff *et al.* 2011).

Species richness at the river basin scale is directly influenced by evolutionary and geographical processes operating at regional or even continental scales (Gaston 2000, Ricklefs 1987). Various processes that shape the species richness patterns make a single mechanism not adequate to explain the variability of riverine fish species richness throughout the world. Current patterns of species richness are the result of the interactions between speciation, extinction, colonization and ecological processes (e.g. long-term environmental oscillations) (Jansson & Davies 2008). Thus, three major hypotheses were proposed to explain the global patterns of the fish species richness (Oberdorff *et al.* 2011).

- The species-area hypothesis (MacArthur & Wilson 1963, 2001) predicts that the number of species in a given area is positively related to the size of the area. This hypothesis is mainly based on the three mechanisms which are the size-dependent extinction rate, the size-dependent extinction rate, and the diversity of the habitat.

- The species-energy hypothesis (Wright 1983) predicts that the number of species in a given area is positively related to the energy available within the system. Energy can influence species richness through two very different processes. Some consider energy to be a productivity factor that determines resources available for a given biological community (Wright 1983). In this process, variables such as net primary productivity would be expected as

important determinants of species richness (Hawkins *et al.* 2003). Others consider energy to be a factor that determines the physiological limits of the species (Turner *et al.* 1987, Currie 1991). In this process, variables such as temperature would be expected more important for species richness (Hawkins *et al.* 2003).

- The historical hypothesis (Whittaker 1977), which combines the historical environmental conditions and geographic contingencies (e.g. emergence of geographical barriers), tries to explain the relationship between the species richness and the climate oscillations since the last major climate change (i.e. Last Glacial Maximum, circa 21 thousand years ago) (Dynesius & Jansson 2000).

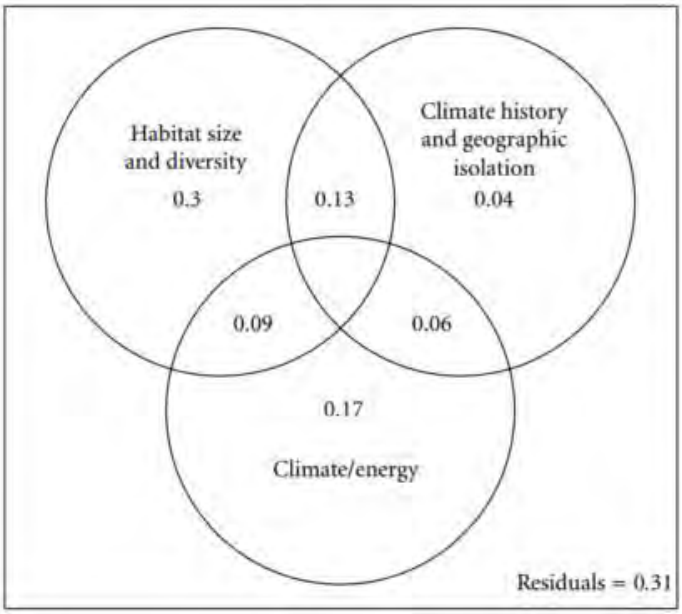



**Fig. 3.** Relative influence of the three hypotheses on global freshwater fish species richness patterns (adapted from Oberdorff *et al.* 2011).

Results from Oberdorff *et al.* (2011) showed that the three hypotheses well explained (77% explanation by a spatial autoregressive model, SAR) the patterns of fish species richness with interaction effects between them (Fig. 3). The area and energy hypotheses make a major contribution to richness patterns whereas the climate hypothesis makes a minor contribution, but the past climatic changes and geographic isolation of drainage basins still have a significant impact on the species richness patterns at the global scale. In contrast, Leprieur *et al.* (2011) found that past climate changes play an important role in shaping the present global patterns of

freshwater fish beta diversity. By decomposing the taxonomic beta diversity into species turnover (i.e. species replacement between communities) and species nestedness-resultant (i.e. that reflect the difference in the number of species among communities), they found the Quaternary climate changes have contrastingly influence on the two components, resulting in their different geographic contributions to beta diversity. For instance, the nestedness component has greater contributions to the beta diversity of the fish faunas in the previously glaciated drainage basins of the Northern Hemisphere, which experienced a larger amplitude of Quaternary climate oscillations. Besides, these northern previously glaciated basins are also dominated by the widespread species with large geographic range sizes (Griffiths 2006). These patterns can attribute to two nonexclusive mechanisms. On one hand, species with small range size distributions were selectively extinct due to the successive glaciations during the Quaternary period (Hocutt & Wiley 1986, Griffiths 2006). On the other hand, species with strong dispersal abilities from southern refuge areas (e.g. the Mississippi and Danube drainage basins in North America and Europe respectively) selectively colonized to the northern basins during intermittent connections among them (Hocutt & Wiley 1986, Leprieur *et al.* 2011).

All the above studies focused on the historical patterns of the global freshwater fishes. However, the patterns have been marked changed due to the various human activities. Freshwater ecosystems among the most highly invaded ecosystems in the world (Moyle & Marchetti 2006). Base on the most recent fish occurrence database by Tedesco *et al.* (2017), over 600 non-native fish species have been introduced and established in other habitats. Under this circumstance, patterns and drivers of change in fish taxonomic diversity were also well explored during the past two decades. Change in fish biodiversity is always drove by a variety of human disturbances (e.g. exploitation, introduction, fishery), which lead to two main consequences, species extinctions and species introductions (the initial stage of the invasive process). Compared to the large number of introduced non-native fish species among the world river basins, the species extinctions were less frequent among the world's fish faunas (Villeger *et al.* 2011), therefore, the main driver of fish biodiversity change is species introduction. Leprieur *et al.* (2008) reported the global patterns of fish invasion in 1 055 river basins and

identified the major human drivers of the fish invasive processes. The process of species invasion consists of three main stages: introduction (initial dispersal), establishment (build self-sustaining populations), and expansion (spread into the recipient habitat) (Puth & Post 2005). Once the invasion passes through the second stage (establishment), it is difficult to reverse (Vitule *et al.* 2009). Three major hypotheses that are human activity (Taylor & Irwin 2004), and biotic acceptance (Levin 2000), biotic resistance (Maron & Vila 2001), have been put forward to disentangle the invasion patterns. The human activity hypothesis argues that human activities facilitate the establishment of nonnative species by disturbing natural habitats and increasing the number of individuals released and the frequency of introductions in a given habitat. The biotic acceptance predicts that if the biotic factors are appropriate for the nonnative species, the species is prone to successfully established in the new area. In contrast, the biotic resistance predicts that strong biotic interactions with native species will reduce the establishment of the nonnative species (Maron & Vila 2001). By testing these three hypotheses on the global freshwater fish invasion, Leprieur *et al.* (2008) found that human activity indicators account for most of the global variation in non-native fish species richness, which is highly consistent with the human activity hypothesis. Their results indicated that the spatial patterns of fish invasions matched the spatial patterns of human impact at the global scale, which means that human activities are the major drivers of fish invasions in the world's river systems. In contrast, there were no obvious clues to support either the biotic acceptance or the biotic resistance at the river basin scale. They also inferred that due to the growing economy in developing countries, the next fish invasion wave could occur in these countries (Leprieur *et al.* 2008).

As a consequence, non-native species introduction and native species extinction will lead to taxonomic homogenization, which is a biodiversity change that increases the similarity among species assemblages. This process has been reported in many taxa (Gossner *et al.* 2016, Winter *et al.* 2009) and was also observed in freshwater fish faunas, which mainly results from the introduction of non-native fish species throughout the world (Villeger *et al.* 2011, Toussaint *et al.* 2016a). In contrast, the native fish species extinction does not play a strong role in this homogenization process at the river basin scale. For instance, Villeger *et al.* (2011) revealed

slight taxonomic homogenization patterns of the world freshwater fish faunas by using a global database on historical and current freshwater fish occurrences over 1 054 river basins (same database as Leprieur *et al.* 2008). The results showed that the average homogenization degree of the freshwater fish faunas was still low at the world scale (0.5%) but reached very high levels (up to 10%) in some highly invaded river basins from the Nearctic and Palearctic realms. It thus indicated that the non-native species introductions rather than native species extinctions play a dominant role in the taxonomic homogenization process in the global freshwater fish faunas (Villeger *et al.* 2011). Moreover, Toussaint *et al.* (2016a) highlighted that the global taxonomic homogenization pattern was driven by a few widespread non-native species, whereas most introduced species slightly contribute to the global change in dissimilarity. They also found that the effect of species on change in dissimilarity was influenced by whether introduced species were translocated (i.e. introduced to other basins within their biogeographic realm) or exotic (i.e. introduced from other biogeographic realms). They concluded that taxonomic homogenization in freshwater fish faunas was dominated by the species translocated within a realm and only by few widespread exotic species whereas the majority of exotics contribute to a differentiation effect.

## I.5 Fish functional diversity

Functional traits are defined as biological attributes measurable at the individual level which impact organism performance in the ecosystems (McGill *et al.* 2006, Violle *et al.* 2007), thus functional traits can reflect the contributions of species to ecosystem functioning. During the last two decades, functional traits have been widely used in plants (Diaz *et al.* 2016, Butler *et al.* 2017), mammals (Safi *et al.* 2011), birds (Ricklefs 2012, Barnagaud *et al.* 2019), insects (Poff *et al.* 2006) and fish (Mouillot *et al.* 2014 to assess the functional diversity of species assemblages and the ecosystems functioning they support. Functional diversity can help us better understand the response of communities to disturbance and the implications for ecosystem functioning (Villeger *et al.* 2017). However, the studies of the functional diversity for animals are much less developed than for plants, mostly because plant traits are easy to

encode functions and have fixed meanings while the same trait for animals can encode for several distinct functions. For example, it has been proved in plants that specific leaf area, leaf N content, and leaf area ratio affect primary productivity while phenology and litter quality affect nutrient cycling rates (Funk *et al.* 2008, Drenovsky *et al.* 2012). In contrast, functional traits for animals can encode several distinct functions. Nevertheless, there are still some studies focusing on the functional diversity of freshwater fish faunas at the regional and global scale (Azzurro *et al.* 2014, Cilleros *et al.* 2016, Toussaint *et al.* 2016b, 2018, Villeger *et al.* 2013, 2014). Indeed, freshwater fish display a remarkable range and variance in functional traits, such as morphological, behavioral, physiological, trophic and life-history traits (Mims *et al.* 2010). For instance, freshwater fish exhibit an unequaled range of morphologies, spanning from tiny killifish and mosquitofish (Rivulus, Gambusia) to giant species (e.g. sturgeons, wells catfish), from deep-bodied Cichlids to elongated snake eel, from blind cavefish to big-eye Poeciliidae, from suckermouth catfish to long jaw needlefish (Su *et al.* 2019). Fish play a key role in the nutritional dynamics of freshwater ecosystems by participating in energy flow and material circulation. Thus, understanding how fish control other organisms through predation was one of the main issues for fish ecologists during the last decades. Although the relationship between morphology and the role a species play in the ecosystem is still unclear, many morphological traits have been proved to relate to fish certain functions (Villeger *et al.* 2017). This benefits from a long history, since the seminal work of Gatz (1979), improved by Winemiller (1991) and Douglas & Matthews (1992) leading to propose many morphological traits to describe fish functions. For instance, body size, which has been used in most studies to assess the functional diversity of fish assemblages, is suggested to indicate food acquisition, trophic level and more generally to all other functions (Romanuk *et al.* 2011, Villeger *et al.* 2017). More precisely, relative eye diameter has been often used as a proxy of visual acuity and/or light sensitivity to assess the prey detection capacity before the food acquisition starts (Winemiller 1991, Bellwood *et al.* 2014). And the eye vertical position often indicates fish position in the water column relative to relative to that of its prey (Villeger *et al.* 2017). Locomotion (i.e. swimming performance) is another key function for fish surviving in the aquatic habitats. Morphological

traits have been also used as proxies to assess the fish swimming capacity. For instance, body size is a basic trait related to the swimming capacity, larger fishes can swim faster and have greater endurance than small ones, while small fishes have better manoeuvrability and can thus move in topographically complex environments such as aquatic plants, tree branches, or roots in rivers (Villeger *et al.* 2017). Other traits, including body shape (elongation), fins size and position have also often been measured to indicate the fish's locomotion function (Blake 1983, Ward *et al.* 2010). Furthermore, morphological traits are more accessible than other functional traits (e.g. life-history, physiology), especially for studies on large number of species, the benefits are obvious. Thus, studies on fish functional diversity always focused on the fish morphological traits.

Based on ten morphological traits of global fishes, Toussaint *et al.* (2016b) compared the functional diversity and taxonomic diversity patterns of the global fish faunas at the realm scale (Fig. 4). They found that the global functional diversity of freshwater fish was concentrated in the Neotropical realm while the functional vulnerability of freshwater fish existed in every biogeographic realm. They concluded that conservation of the Neotropical fish diversity was a key target to maintain world fish functional diversity. However, one of their conclusions that Neotropical freshwater fishes had the highest functional diversity, but low vulnerability given high levels of functional redundancy drew controversy (Vitule *et al.* 2017). Vitule *et al.* (2017) argued that this conclusion underestimates the threat to biodiversity in the tropics freshwater fish faunas and will mislead policymakers in developing countries in tropical areas. More recently, Toussaint *et al.* (2018) assessed how non-native fish species changed the functional diversity of global freshwater fish faunas by using the same ten morphological traits. They found that the non-native species not only increased the local functional richness on average but also caused shifts in functional identity toward higher body sized and less elongated species for most of the assemblages, which would potentially modify ecosystem functioning throughout the world (Toussaint *et al.* 2018). In addition, based on morphological and ecological traits, Villeger *et al.* (2014) found functional homogenization exceeds taxonomic homogenization among 137 European fish assemblages. Their results showed that translocated species (i.e. non-

native species originating from Europe) played a stronger role than exotic species (i.e. those coming from outside Europe) in this homogenization process, while extirpation did not play a significant role. They also proved that change in taxonomic diversity could not be used to predict changes in functional diversity and the latter should be a better indicator of ecosystem functioning and stability (Villeger *et al.* 2014).

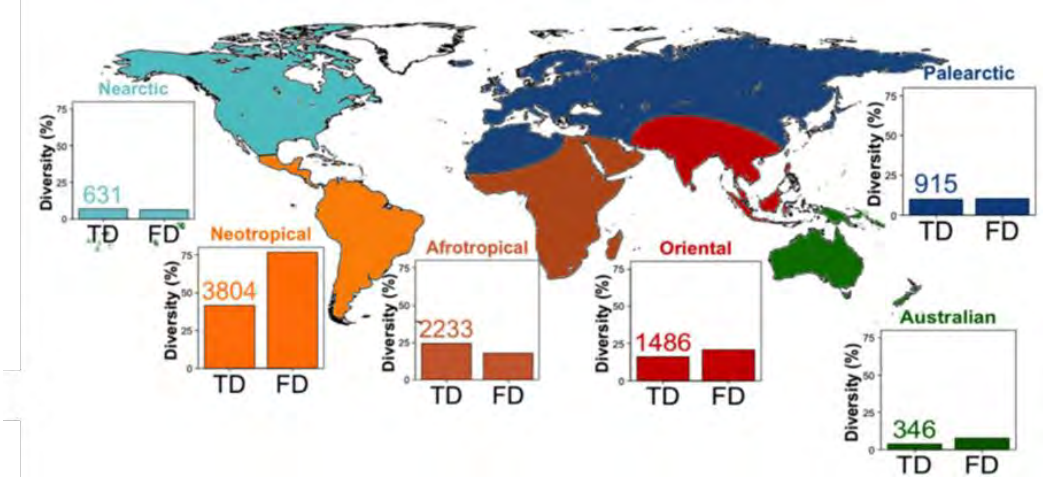



**Fig. 4.** Fish taxonomic and functional diversity in the 6 biogeographic realms (adapted from Toussaint *et al.* 2016b).

## I.6 Fish phylogenetic diversity

Phylogenetic diversity is another key component of the biodiversity, which measures the evolutionary breadth of assemblages. Many previous studies assessed the fish phylogenetic diversity at the regional scale (e.g. Strecker *et al.* 2010, Sternberg & Kennard 2014, Hasnain *et al.* 2013, Kuczynski *et al.* 2018). Nevertheless, due to the insufficiency of the complete phylogenetic information, few studies were exploring the patterns, changes and determinants of the phylogenetic diversity of the global freshwater fishes. However, recently Betancur-R *et al.* (2017) built and updated the first time the phylogenetic tree for nearly 2 000 bony fishes based on phylogenies inferred using molecular and genomic data, which covered 72 orders (410 families). Then Rabosky *et al.* (2018) assembled a time-calibrated phylogeny of all ray-finned fishes (31 526 species). Their work on the global fish phylogenies makes it possible to conduct the phylogenetic analysis for freshwater fish on a global scale.

## I.7 Thesis conduct and structure

From the introduction in the first chapter, we know that questions about the taxonomic diversity of the global freshwater fishes have been thorough studied during the last decades, in contrast, the study related to functional diversity and other facets has just started and remains lots of questions to be answered. Thus, based on the above four main databases, we aim to fill the knowledge blank in the study of global freshwater fish functional diversity. We investigate the biodiversity of the global freshwater fish faunas, from the patterns of functional diversity towards a holistic map of multifaceted biodiversity change. Specifically, surrounding the biodiversity of global freshwater fish, my Ph.D. project consists of four main points, the former two parts focused on the historical patterns of the native species, while the latter two parts focused on the dynamic change patterns from historical to the current period (Fig. 5).

***1) Do the morphological traits differ between realm fish faunas?***

In this chapter, I first compared the distribution of morphological traits between realms and evaluated the possibility that makes the morphological differences. Then I identified the morphologically extreme species and analyzed their contribution to the morphological space in the world and realms. (Chapter III)

***2) How do the historical and environmental variables shape the patterns of freshwater fish functional diversity?***

In this chapter, I showed the patterns for three complementary functional diversity indices (i.e. functional identity, functional richness, functional dispersal), which represent the mean, range and variance of functional diversity. Then I tested the area/energy/historical hypotheses on the patterns of each of the functional diversity by using the relevant environmental variables. (Chapter IV)

***3) Do the introduced fish species have morphological preferences?***

In this chapter, I divided the introduced species into several groups by the invasive stage they reached and the region they were introduced. Then I compared the morphological traits between

them and the non-introduced species. (Chapter V)

***4) How do various human activities affect the multi-faceted biodiversity of the global freshwater fishes?***

In this chapter, I developed a synthetic index to evaluate the cumulative change in the multifaceted biodiversity of the global freshwater fish fauna. The index encompasses the changes in alpha and beta diversity from taxonomic, functional and phylogenetic facets, thus considering six independent diversity values. Then I assessed the impact of human activities on the multifaceted biodiversity change. (Chapter VI)

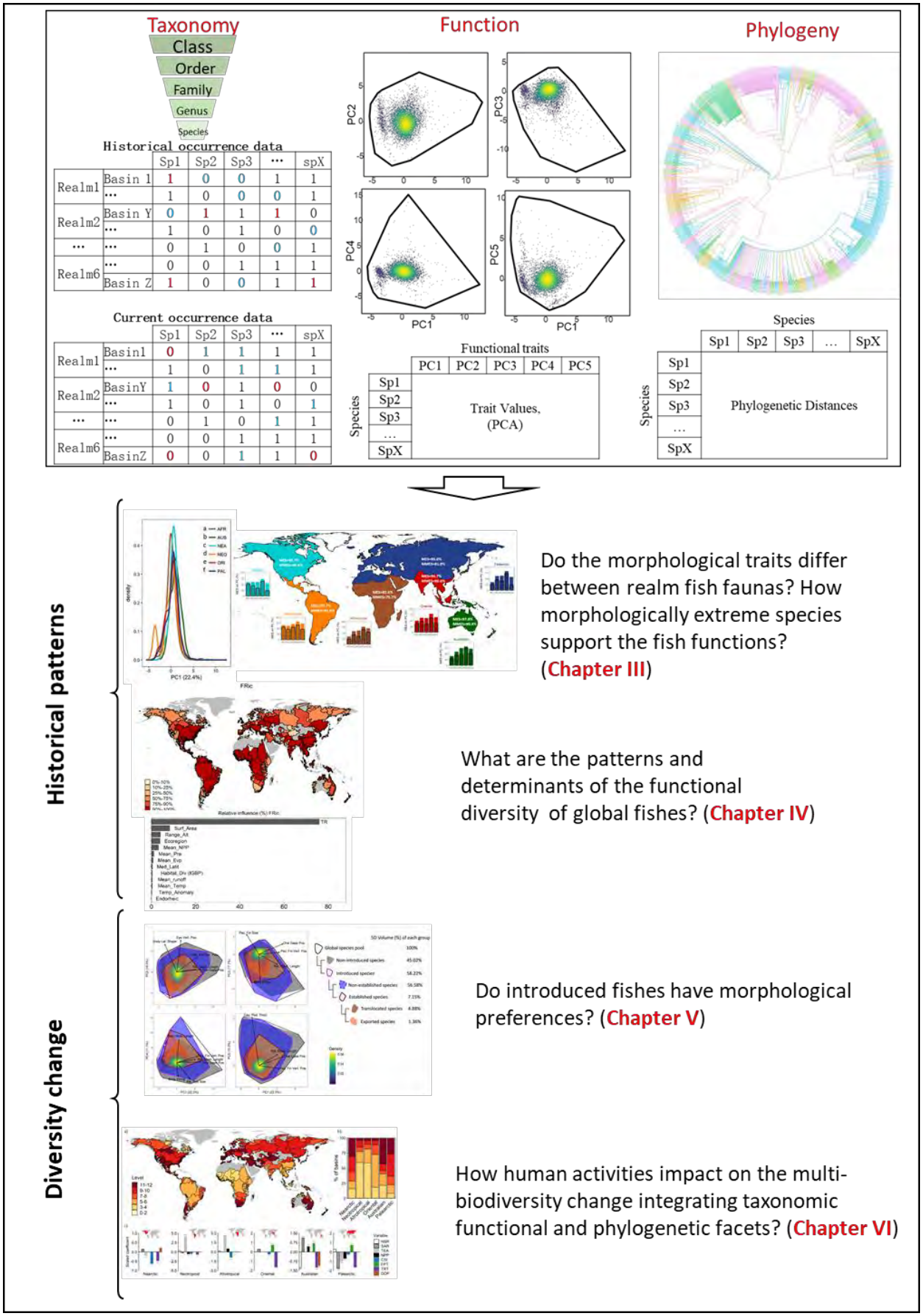



**Fig. 5.** Illustration of three facets of diversity, structure and objective of the thesis.

# Chapter II: Databases and methodology

Previous studies on global freshwater fish have thoroughly explored the mechanisms of natural processes shaping the historical species richness, the species richness and dissimilarity changes under species invasions and expirations led by various environmental drivers and anthropogenic pressures. Nevertheless, our knowledge about the functional and phylogenetic diversity of freshwater fish faunas remains scarce. Thus to complement the research on biodiversity of global freshwater fish, I combined information on 4 main databases: fish occurrence data in basins (historical and current), fish functional (morphological) and phylogenetic data, drivers including anthropogenic and environmental variables for each river basin.

## II.1 Taxonomic data: fish occurrence status in the world river basins

To access the fish species spatial distribution and taxonomic diversity across the world, we used the fish occurrence database from Tedesco *et al.* (2017), which contains the occurrence status (i.e. whether a fish present or absent in a basin) of 14 953 species (more than 90% of the freshwater fish species) in 3 119 river basins, covering more than 80% of the earth surface (Tedesco *et al.* 2017). Compared with the database used in the papers of Toussaint *et al.* (2016b, 2018), this updated database contains many more river basins, as well as fish species. And these river basins span six biogeographic realms (except Antarctica) and 143 countries, and cover a huge diversity of attributes in terms of altitude (0 – 4 200m), latitude (55°S – 83°N), size (1.7 $km^2$– 5.9M $km^2$), *etc*.

Each occurrence is also paired with a status, either native or non-native established if the species was not historically present in the river basin. Each river basin is assigned to one of the six terrestrial biogeographic realms (i.e. Afrotropical, Australian, Nearctic, Neotropical, Oriental and Palearctic) according to Lévêque *et al.* (2008) and Brosse *et al.* (2013). Historical fish assemblages composition in the river basins refers to only native species, and thus roughly corresponds to the preindustrial period (i.e. before the 18th century), from when industrialization began and fish introductions for aquaculture, fishing, and ornamental purposes

sharply increased (Blanchet *et al.* 2010, Leprieur *et al.* 2008, Villeger *et al.* 2011). Current fish assemblage's composition refers to the non-native species (established) and excludes the local extirpated or extinct native ones. To do this, we combined the extirpated/extinct fish species information from Brosse *et al.* (2013) and IUCN Red lists (IUCN 2018) with the occurrence database. In addition, we collected the fish species of interest to humans (e.g. aquaculture, pest control, game fishing) from FishBase (Froese & Pauly, 2018) and Blanchet *et al.* (2010) and assigned this information to the species list in the occurrence database, which would be used in chapter V as an alternative of the species that have been introduced but failed to establish.

To be noted, since the species extinction and introduction is a continuous process and even more severe now, the current number of these species should exceed our estimation.

## II.2 Functional data: fish morphological measures and functional traits

To access the functional facet of the fish faunas, we measured 11 fish morphological traits related to food acquisition and locomotion. 11 morphological measures were assessed on side view pictures (Fig. 6; Table 1a) collected during an extensive literature review from more than 200 scientific literature sources including peer-reviewed articles, books and scientific websites. We collected at least one picture (validated photograph or scientific drawing) per species. Only good quality pictures and scientific side view drawings of entire adult animals were kept. For species with a marked sexual dimorphism, we considered male morphology, as female pictures are scarce for most species (especially for Perciformes and Cyprinodontiformes). Morphological measures were achieved using ImageJ software (http://rsb.info.nih.gov/ij/index.html). These 11 morphological measures were used to compute 9 unitless traits describing the size, shape and/or position of the fish head (including mouth and eye), body, pectoral and caudal fins (Table 1b). For species with unusual morphologies (species without tails, flatfishes) that prevent from measuring some morphological traits, we defined rules for these few exceptions as in Villéger *et al.* (2010) and Toussaint *et al.* (2016b). For instance, for species with no visible caudal fin (e.g. Sternopygidae, Anguillidae), caudal peduncle throttling (CFd/CPd) was set to 0. Then the maximum body length extracted from

FishBase (http://www.fishbase.org/) were added to this morphological trait database. Since some of the values in FishBase were irrelevant, we checked this database and corrected the irrelevant measures based on other appropriate data sources. In addition, because intraspecific variation is unknown for most of the species, as a check, we measured 10 individuals per species for a subset with 55 fish species covering most of the orders and found the intraspecific variation is much lower than the interspecific variation.

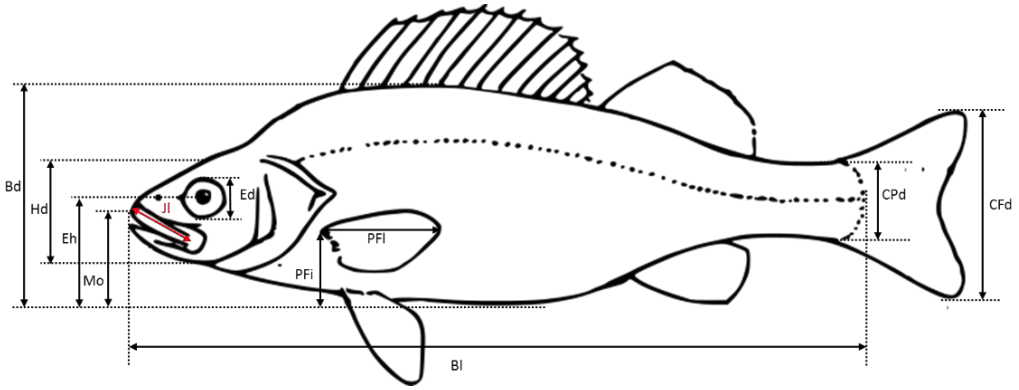



**Fig. 6.** Morphological measurements for normal fish species.

**Table 1** The morphological measurements on each fish species (a) and the functional traits calculated by the measurements (b)

**a. Morphological measurements**

| Code | Name | Protocol for measurement |
|---|---|---|
| Blmax | Maximum Body length | Maximum adult length (obtained from FishBase (Froese & Pauly, 2018)) |
| Bl | Body length | Standard length (snout to caudal fin basis) |
| Bd | Body depth | Maximum body depth |
| Hd | Head depth | Head depth at the vertical of eye |
| CPd | Caudal peduncle depth | Minimum depth of the caudal peduncle |
| CFd | Caudal fin depth | Maximum depth of the caudal fin |
| Ed | Eye diameter | Vertical diameter of the eye |
| Eh | Eye position | Vertical distance between the centre of the eye to the bottom of the body |
| Mo | Oral gape position | Vertical distance from the top of the mouth to the bottom of the body |
| Jl | Maxillary jaw length | Length from snout to the corner of the mouth |
| PFl | Pectoral fin length | Length of the longest ray of the pectoral fin |

| | | |
|---|---|---|
| PFi | Pectoral fin position | Vertical distance between the upper insertion of the pectoral fin to the bottom of the body |

All measurements were made on pictures except Blmax values, which were downloaded from Fishbase.org.

b. Functional traits

| Functional traits | Formula | Potential link with fish functions | References |
|---|---|---|---|
| Maximum body length | BLmax | Size is linked to metabolism, trophic impacts, locomotion ability, nutrient cycling | Toussaint *et al.* 2016b |
| Body elongation | $\frac{Bl}{Bd}$ | Hydrodynamism | Reecht *et al.* 2013 |
| Eye vertical position | $\frac{Eh}{Bd}$ | Position of fish and/or of its prey in the water column | Winemiller 1991 |
| Relative eye size | $\frac{Ed}{Hd}$ | Visual acuity | Boyle & Horn, 2006 |
| Oral gape position | $\frac{Mo}{Bd}$ | Feeding position in the water column | Dumay *et al.* 2004, Lefcheck *et al.* 2014 |
| Relative maxillary length | $\frac{Jl}{Hd}$ | Size of mouth and strength of jaw | Toussaint *et al.* 2016b |
| Body lateral shape | $\frac{Hd}{Bd}$ | Hydrodynamism and head size | Toussaint *et al.* 2016b |
| Pectoral fin vertical position | $\frac{PFi}{Bd}$ | Pectoral fin use for swimming | Dumay *et al.* 2004 |
| Pectoral fin size | $\frac{PFl}{Bl}$ | Pectoral fin use for swimming | Fulton *et al.* 2001 |
| Caudal peduncle throttling | $\frac{CFd}{CPd}$ | Caudal propulsion efficiency through reduction of drag | Webb 1984 |

I updated the database built by Nicolas Charpin and Aurèle Toussaint, which includes 10 morphological traits for 9 170 species. After checking and correcting the database (e.g. merging species with the synonymous names, excluding wrong recording), I got the revised trait database for 9 150 species in more than 1 000 river basins. This database was applied to the analyses in chapter III and chapter V. To further extend the coverage of the trait database I

collected and measured nearly 2 000 more species that were not included in the initial database. After updating and matching with the occurrence database, our fish functional trait database has been expanded to 10 682 species, which was matched with the fish occurrence database in 3 094 river basins. This updated database was applied to the analyses in chapter IV and chapter VI.

### II.3 Phylogenetic data: build the whole phylogenetic tree of fish by genetic data

Phylogenetic data was extracted from Rabosky *et al.* (2018), which is currently the most comprehensive and complete phylogenetic data for the global freshwater fish. This database includes 31 526 ray-finned fishes, which is based on 11 638 species whose position was estimated from genetic data. The remaining 19 888 species were placed in the tree using stochastic polytomy resolution (Rabosky *et al.* 2018).

I matched this dataset to the taxonomic and functional datasets and eventually got the phylogenetic relatedness for the 10 682 species with taxonomic and functional information. These triple matched datasets were applied to the analyses in chapter V to access the multifaceted change in global fish biodiversity.

### II.4 Drivers data: anthropogenic factors and environmental variables selection

Drivers data was used to explored the determinants of two kinds of patterns: the patterns of functional diversity of the global native fish faunas (chapter IV) and the patterns of multifaceted change in global fish biodiversity (integrated taxonomic, functional and phylogenetic facets) (chapter VI). The diversity patterns of native species were mainly caused by the natural processes thus the environmental variables were selected in this part to test the three main ecological hypotheses (i.e. area, energy, historical). To test these hypotheses and propose more specific ones, we collected the related variables from various data sources, which included net primary productivity, temperature, runoff, evapotranspiration, surface area, basin's altitude range, basin's median latitude, habitat diversity, ecoregion, endorheic, temperature anomaly of since the last glacial maximum, etc. However, biodiversity changes were mainly

caused by human activities, thus we collected several human-related variables to explore the determinants of the patterns of multifaceted change in global fish biodiversity. The variables include the river connectivity status index, human footprint, travel to the nearest city and gross domestic product (see the methods in chapter VI for details).

**Table 2** The structure of datasets used in the four main chapters.

| | chapter III | chapter IV | chapter V | chapter VI |
|---|---|---|---|---|
| Spatial scale | realm | basin | realm | basin |
| Number of realms | 6 | 6 | 6 | 6 |
| Number of basins | >1000 | 2453 | >1000 | 2456 |
| Number of species | 9150 | 10682 | 9150 | 10682 |
| Number of morphological traits measured | 10 | 10 | 10 | 10 |
| Number of species with phylogenetic information | --- | --- | --- | 10682 |
| Number of natural variables | --- | 13 | --- | 4 |
| Number of human-related variables | --- | --- | --- | 4 |

# Chapter III

## Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread

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RESEARCH PAPER

WILEY Global Ecology and Biogeography

# Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread

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## Abstract

**Aim**: The aim was to analyse the morphological diversity of the world freshwater fish fauna. We tested to which extent the distributions of morphological traits are supported by extreme morphologies and how those extreme morphologies are distributed among realms and affect the functional vulnerability. We also analysed the contribution of between- and within-order morphological variability to the morphological differences between realms.

**Major taxa studied**: Freshwater fish.

**Location**: Global.

**Time period**: 1960s–2010s.

**Methods**: We used a global database of freshwater fishes from the six realms. Ten morphological traits were measured on 9,150 species, that is, 75% of the *ca.* 13,000 freshwater fish species. A principal components analysis was conducted to combine the 10 traits into a multidimensional space. We used Kolmogorov–Smirnov tests to compare the difference in morphological diversity between the six realms and between and within the major fish orders. We then identified the morphologically extreme species and quantified their contributions to the morphological range to assess the functional vulnerability and redundancy of fish faunas in the six biogeographical realms for freshwater ecosystems.

**Results**: We report a strong morphological variability among freshwater fishes of the world, with significant morphological differences among realm fish faunas, caused by an interplay between taxonomic composition of the realm faunas and morphological differences within orders among the realms. Morphologically extreme species accounted for a large percentage of the filling of the global morphological space and are distributed throughout the world.

**Main conclusions**: Fish morphological diversity is largely supported by a few species with extreme trait combinations, indicating that functional vulnerability exists throughout the world. Our results suggest that more attention should be paid to these morphologically extreme species and that they should be protected to ensure the sustainability of functions they support.

KEYWORDS

biogeographical realms, freshwater fish, functional redundancy, functional vulnerability, morphologically extreme species, morphological traits, rivers

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## 1 | INTRODUCTION

Biodiversity is as a multifaceted concept, encompassing several dimensions, including the number and identity of species (taxonomic facet), their evolutionary breadth (phylogenetic facet) and their variability in functional traits (functional facet) (Garnier, Navas, & Grigulis, 2016; Gaston & Spicer, 2004). Most global and regional scale studies have focused on the taxonomic facet, but the recent development of functional databases encompassing large numbers of species for plants (e.g., Díaz et al., 2016; Kattge et al., 2011) and animals (e.g., Mouillot et al., 2014; Ricklefs, 2012; Toussaint, Charpin, Brosse, & Villéger, 2016) allows the consideration of the functional dimensions of biodiversity at macroecological scales. Most of the previous studies have revealed a mismatch between taxonomic and functional dimensions of diversity (Kuczynski et al., 2018; Parravicini et al., 2014; Toussaint et al., 2016) because of a difference in trait distributions between realm faunas. For instance, Toussaint et al. (2016) reported that the Nearctic realm hosts a functional diversity similar to that of the Australian realm, although it hosts almost twice the number of species.

Freshwater fishes account for more than one-quarter of all vertebrates species, with *ca.* 13,000 species described to date (Nelson, Grande, & Wilson, 2016; Tedesco et al., 2017). In addition, because of their weak ability to disperse both between and within continents, fish faunas from the six biogeographical realms share only a few species (Levêque, Oberdorff, Paugy, Stiassny, & Tedesco, 2008; Villéger, Blanchet, Beauchard, Oberdorff, & Brosse, 2011). Freshwater fishes show a huge diversity of ecological strategies (e.g., diet, mobility) and contribute to key ecosystem processes, such as the regulation of food webs and biogeochemical cycles (Villéger, Brosse, Mouchet, Mouillot, & Vanni, 2017). Fish morphology, although not accounting for the entire range of fish functional roles, is a proxy of some key functions performed by fish and is more informative of fish roles than a taxonomy-based approach (Leitão et al., 2018; Villéger, Miranda, Hernandez, & Mouillot, 2010). For instance, morphological traits provide information on locomotion (Watson & Balon, 1984; Winemiller, 1991) and food acquisition (Albouy et al., 2011; Hugueny & Pouilly, 1999), which are biological features related to the contribution of the fish to the regulation of food webs and nutrient cycles (Villéger et al., 2017). However, the distribution of key fish morphological traits within and among realms and orders remains largely unknown except for body size (Blanchet et al., 2010; Olden, Hogan, & Zanden, 2007; Villéger et al., 2017). Freshwater fish size ranges from <3 cm [e.g., tiny killifish (genus *Rivulus*) and mosquitofish (*Gambusia affinis*)] to >2 m [e.g., sturgeon (*Acipenser sturio*), wels catfish (*Silurus glanis*)] (Blanchet et al., 2010; Stone, 2007). The variability of other morphological traits has been investigated far less, although they can provide complementary information, because fishes of similar body size can have contrasting locomotion and nutrition characteristics (Cucherousset, Blanchet, & Olden, 2012). Using multivariate approaches is a way to consider those traits altogether and to deal with correlations among traits (e.g., Schleuter, Daufresne, Massol, & Argillier, 2010; Villéger et al., 2010). This has been proved an efficient method to reveal overall variations of fish morphology across the globe, and Toussaint et al. (2016) reported large differences in freshwater fish morphological richness among realms. Indeed, fish faunas have different percentages of their species richness in the main fish orders (Levêque et al., 2008), and orders have marked morphological differences (Toussaint et al., 2016). Moreover, differences in environmental constraints and/or evolutionary legacies among realms should also promote differences in trait values between species from a given order, but the magnitude of this variability is still unknown.

Furthermore, trait distribution can also discriminate ecological strategies. For instance, Díaz et al. (2016) demonstrated that trait distribution is bimodal in plants, corresponding to distinct strategies between trees (long life, late maturity, large size) and herbs (short life, early maturity, small size). In contrast, trait distribution is often unimodal in animals, with a core of generalist species and distribution tails indicating specialist species [see Ricklefs (2012) for birds, Mouillot et al. (2014) for coral reef fish and Toussaint et al. (2016) for freshwater fish]. Nevertheless, the identity of morphologically extreme species (i.e., species with values in tails of trait distribution) and their spatial and taxonomic distributions remains to be quantified, because those extreme species probably play unique functions in ecosystems (Leitão et al., 2018; Mouillot et al., 2014). Therefore, the percentage of the morphological space filled by the extreme species can be considered as a measure of the functional vulnerability of each realm. For instance, Mouillot et al. (2014) reported that coral reef fish across the world pack disproportionately into a few particular functions, while leaving many functions highly vulnerable.

Here, we assess the distribution of 10 morphological traits for 9,150 freshwater fish species. We compare the morphological differences between the six biogeographical realms for the entire fish faunas and for the five most species-rich orders to test whether the high distinctiveness of taxonomic diversity of fishes translates into morphological distinctiveness or convergence across realms. We then quantify the proportion of the morphological space filled by the morphologically extreme species and assess the fish functional vulnerability in each realm.

## 2 | MATERIALS AND METHODS

### 2.1 | Datasets

We considered the six terrestrial biogeographical realms [Afrotropical, Australian (including Oceania), Nearctic, Neotropical, Oriental and Palaearctic] commonly used for freshwater fishes (Levêque et al., 2008; Nelson et al., 2016). The identity of the fish species occurring in each realm was obtained using the occurrences database from Brosse et al. (2013) and Tedesco et al. (2017), in >3,000 river basins. Nine thousand one hundred and fifty of these species (i.e., 75% of the species described; Supporting Information Table S1) were described morphologically using 10 traits describing size and shape of body parts involved in food acquisition and locomotion (Supporting Information Figure S1b). More precisely, fish size was described using the maximum body length (Max. Body Length) taken from FishBase (Froese & Pauly, 2018). Those maximum body lengths were carefully

reviewed, and irrelevant measures have been corrected according to the appropriate literature. In addition to size, 11 morphological measures were assessed on side-view pictures (Supporting Information Figure S1a) collected during an extensive literature review from >200 scientific literature sources, including peer-reviewed articles, books and scientific Websites. We collected at least one picture (validated photograph or scientific drawing) per species. Only good-quality pictures and scientific side-view drawings of entire adult animals were kept. For species with marked sexual dimorphism, we considered male morphology, because female pictures are scarce for most species (especially for Perciformes and Cyprinodontiformes). Morphological measures were achieved using ImageJ software (https://rsb.info.nih.gov/ij/index.html). As a check, we compared those measures with a set of measures taken from 10 individuals per species of 55 preserved fish specimens belonging to the Palaearctic and Neotropical realms (Cilleros, Allard, Grenouillet, & Brosse, 2016), and all our measures fell within the trait range measured on preserved specimens. The 11 morphological measures were used to compute nine unitless traits describing the morphology of the fish head (including mouth and eye), body, pectoral and caudal fins (Supporting Information Figure S1b). For 35% of the species, one or more morphological trait remains unmeasured because the fish position or picture quality did not permit us to achieve one or several relevant measures. Traits were therefore measured on different numbers of species, ranging from 6,891 species for relative maxillary size to 8,531 species for maximum body length (Supporting Information Table S2).

Some species have unusual morphologies (species without tails, flatfishes) that prevented us from measuring some morphological traits. We thus defined rules for these few exceptions as in Villéger et al. (2010) and Toussaint et al. (2016): (a) for species with no visible caudal fin (e.g., Sternopygidae, Anguillidae), caudal peduncle throttling (CFd/CPd) was set to zero (Supporting Information Figure S1b); (b) for species with the mouth positioned under the body (e.g., Loricaridae, some Balitoridae, such as *Gastromyzon*) mouth vertical position (Mo) was set to zero; (c) for the species having a specialized sucker mouth for grazing algae (e.g., Loricariidae, *Chiloglanis*, *Gastromyzon*), relative maxillary length (Jl/Hd) was set to zero; (d) for the species without pectoral fins (e.g., Synbranchiformes and some Anguiliformes), pectoral fin vertical position (Pec. Fin Vert. Pos., PFi/Bd) was set to zero; and (e) for flatfishes, body depth (Bd) was the body width as the fish lies on one side of its body. We thus assumed that Pleuronectiformes are morphologically closer to dorsoventrally flattened fishes (e.g., *Gastromyzon*) than to laterally compressed fishes (e.g., *Symphysodon*). All these conventions are relevant from a biological point of view and contribute to the quality of the data because, for instance, a fish without a caudal fin has nil propulsion effect of its caudal fin, and therefore setting caudal peduncle throttling to zero accounts for the absence of caudal propulsion in this species. Likewise, setting mouth position to zero in the Loricariids accounts for the ventral position of the mouth. All traits apart from size are ratios of morphological measurements (length or area); therefore, all values are positive and unitless. Three traits, namely relative maxillary length (Rel. Maxil. Length), body elongation (Body Elong.) and caudal peduncle throttling (Cau. Ped. Throt.), could potentially reach high values and are hereafter called non-rightbounded traits. Six traits, namely relative eye size (Rel. Eye Size), oral gape position (Oral Gape Pos.), eye vertical position (Eye Vert. Pos.), body lateral shape (Body Lat. Shape), pectoral fin vertical position (Pec. Fin Vert. Pos.) and pectoral fin size (Pec. Fin size), range from zero to one and are hereafter called bounded traits.

Owing to the deficiency of phylogenetic information on the global freshwater fishes, phylogenetic independent contrasts (PICs) analyses were applied to only a subset of 493 species from phylogeny by Betancur-R et al. (2017) matching our morphological traits dataset. We first computed the value of Pagel's $\lambda$ (Pagel, 1999) to test the phylogenetic signal of each trait and then compared the correlation between traits before and after accounting for phylogeny (Felsenstein, 1985).

## 2.2 | Statistical analyses

Missing values among the morphological traits dataset were filled using a random forest algorithm, which takes the taxonomic information into account (Penone et al., 2014). Then, after standardizing and centralizing values for the 10 traits, all species from the global pool were ordered in a multidimensional morphological space using a principal components analysis (PCA). The first five principal component (PC) axes, which account for 78.6% of the total variance (each selected axis had an eigenvalue ≥1; Supporting Information Figure S2a) were selected to conduct the following analyses. We assessed the robustness of our findings given traits accounted for, using a sensitivity procedure. We tested the effect of trait identity on the morphological distance between species by rerunning PCA based on all combinations of nine morphological traits out of 10. The distance between species was hardly affected by morphological traits accounted for (Mantel tests, $r > 0.900$, $p < 0.001$).

We compared the distribution of species coordinates on the first five PC axes among realms using the Kolmogorov–Smirnov test (hereafter K–S test). We then tested, for each of the five orders represented by >15 species in at least two realms, whether distribution of species in the morphological space differed between realms using the K–S test.

In order to determine the contribution of extreme morphologies to the filling of the morphological space, we identified the 2.5% of species with the lowest values and the 2.5% of species with the highest values on each of the five PC axes as morphologically extreme species (MES). We then considered the 0.5% of species with the lowest values and the 0.5% of species with the highest values on each of the five PC axes as the most morphologically extreme species (MMES). The remaining species with trait values among the 95% closest to that of the average among all species from the world were considered to be part of the morphological core species (MCS). Furthermore, in order to assess the distribution of morphologically extreme species among biogeographical realms, we subsequently computed the number of morphologically extreme species in each realm and their contribution to the species richness of the realm. We then computed the contributions of these extreme species to

the value ranges of the five PC axes and to the five-dimensional space volume in the global and realm fish faunas. In addition, we calculated the functional originality (FOri) and specialization (FSpe) values for the entire fish faunas and for only the extreme species assemblages according to Mouillot, Graham, Villéger, Mason, and Bellwood (2013). The FOri measures the distance between each species and its nearest neighbour in a multidimensional space and indicates the functional redundancy between species, with a lower FOri value indicating higher functional redundancy. The FSpe measures the dominance of specialist species (species having extreme trait combinations) in a given assemblage, with a higher FSpe value indicating a higher proportion of specialist species (Mouillot et al., 2013; Rodrigues-Filho, Leitão, Zuanon, Sánchez-Botero & Baccaro, 2018).

We used scripts from Mouillot et al (2013) to compute FSpe and FOri metrics. We used the "convhulln" function from the "geometry" R package to compute the five-dimensional space volume and the "ks.test" function from the "stats" R package to conduct the K–S tests. The PICs analyses were done using the "phylosig" function from the "phytools" R package and the "pic" function from the "ape" R package. All statistical analyses were performed with R software version 3.3 (R Core Team, 2017).

## 3 | RESULTS

The 10 morphological traits all showed unimodal continuous distributions (Supporting Information Figure S3). Among the six bounded traits, none filled the entire range of possible values (i.e., zero to one). The proportion of potential range filled varied from 51% (pectoral fin size) to 94% (oral gape position). The four remaining non-right-bounded traits exhibited a skewed to the right distribution (Supporting Information Figure S3), with skewness values higher than three (Supporting Information Table S2). See Supporting Information Figure S4 for an illustration of the morphological diversity in freshwater fishes. The 10 traits were significantly although weakly correlated between each other among the 9,150 species considered (Spearman test, |rho| ≤ 0.46). Analysis on the subset of 493 species present in the phylogeny from Betancur-R et al. (2017) showed that all the traits had strong phylogenetic signals (Pagel's $\lambda > 0.5$, $p < 0.001$; Supporting Information Table S3). Nevertheless, the effect of phylogeny on the correlation between traits remained weak (Supporting Information Figure S5).

### 3.1 | Differences in morphological diversity between realm fish faunas

The distribution of fish species along the five axes of the morphological space was significantly different between most realm pairs (K–S tests, $p < 0.05$; Figure 1). Distributions on PC1 significantly differed between all the six realm faunas. Only one or two pairwise comparisons of the distributions on PC2–PC5 showed no significant differences, which were pairs of Nearctic versus Neotropical on PC2, Australian versus Oriental on PC3, Australian versus Palaearctic and Nearctic versus Oriental on PC4, Australian versus Nearctic and Neotropical versus Palaearctic on PC5. Those differences paired with differences in the distribution of the 10 morphological traits between biogeographical realms (K–S tests, $p < 0.05$; Supporting Information Figure S6).

Comparison of species position along morphological axes between the five most species-rich orders (Siluriformes, Cypriniformes, Perciformes, Characiformes and Cyprinodontiformes) revealed significant differences between all the orders (K–S tests, $p < 0.05$; Supporting Information Figure S7). Within those five orders, the distribution of the species on the morphological axes between realms also differed significantly (K–S tests, $p < 0.05$; Table 1; Figure 2). For all the 220 pairwise comparisons within the five orders between realms, 70% (154 pairs) were significantly different (Table 1). Differences among realms were the most marked for Siluriformes and Perciformes, which are the most species-rich orders while being present in all the six realms (Table 1).

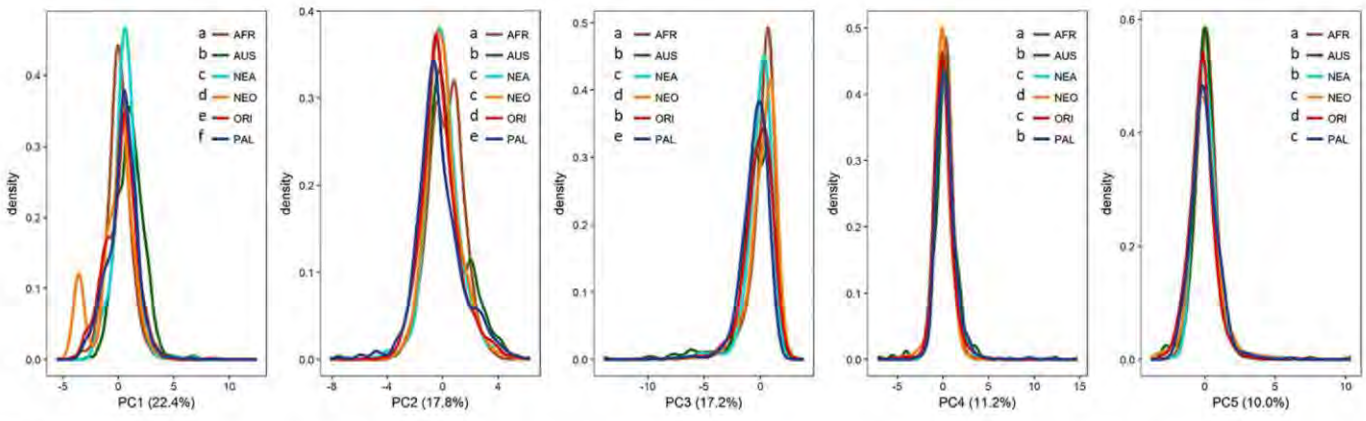



**FIGURE 1** Kernel density estimation for the distributions of fish species from the six realms along five morphological axes (PCs). Letters in front of keys indicate the difference in trait distributions between realms; different letters indicate a significant difference between realms (Kolmogorov–Smirnov tests, $p < 0.05$). AFR = Afrotropical; AUS = Australian; NEA = Nearctic; NEO = Neotropical; ORI = Oriental; PAL = Palaearctic [Colour figure can be viewed at wileyonlinelibrary.com]

**TABLE 1** Difference between realms in distribution of fish species from the five most species-rich fish orders along five morphological axes (PCs)

| | Realm number | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|---|
| Siluriformes | 6 | 13/15 | 13/15 | 12/15 | 10/15 | 6/15 |
| Perciformes | 6 | 14/15 | 13/15 | 14/15 | 10/15 | 8/15 |
| Cypriniformes | 5 | 6/10 | 5/10 | 9/10 | 4/10 | 8/10 |
| Cyprinodontiformes | 3 | 1/3 | 0/3 | 0/3 | 2/3 | 3/3 |
| Characiformes | 2 | 0/1 | 1/1 | 0/1 | 1/1 | 1/1 |

*Note*. Values are the number of significantly different pairs (Kolmogorov–Smirnov tests, $p < 0.05$) out of all realm pairs considered. Only realms with >15 species in each taxonomic order were considered. PC = principal component.

### 3.2 | Morphologically extreme species among the global and realm fish faunas

Of the 9,150 species considered in this study, 1,663 species (18.2% of the total species number) identified as morphologically extreme species filled >50% of the range of each morphological axis (Figure 3), and even up to 80% for the fourth PC axis, which is driven by maximum body length, body elongation and relative eye size (Supporting Information Figure S2b). Overall, the extreme species filled a large percentage of the five-dimensional morphological volume (97.8%), whereas the other species were packed into the remaining 2.2% of the space (Figure 3). Restricting the most morphologically extreme species to the 0.5% tails of the distribution on each morphological axis limited their contribution to 356 species, which filled 89.7% of the morphological volume (Figure 3).

Even though the number of morphologically extreme species varied markedly among biogeographical realms, these species were widespread. Indeed, fish fauna in the Neotropical realm accounted for most of the species with extreme trait combinations (791 species, 21.6% of the Neotropical fauna), followed by the Oriental (338 species, 22.5%), the Afrotropical (267 species, 11.5%) and the Palaearctic (176 species, 20.0%) realm. In contrast, Australian and Nearctic fish faunas host only 106 (28.2%) and 93 (13.8%) species with extreme trait combinations, respectively (Table 2).

The contributions of extreme species to the range filled by each realm fauna for each morphological axis exceeded 40% (Figure 4). Likewise, extreme species filled >90% of the morphological volume filled by species from each realm (Figure 4). The values of FSpe and FOri for the entire faunas of the globe and in the six realms remained low (*ca*. 10%), but the extreme species had higher FSpe and FOri levels than the core species, with values *ca*. 20% for MES and *ca*. 30% for MMES (Table 2).

## 4 | DISCUSSION

The morphological diversity of freshwater fish across the globe is packed around one core combination of trait values, with a few extreme species remaining outside this core (Figure 3). More precisely, the distribution of trait values among the global freshwater fish fauna was continuous (i.e., no marked gaps) and unimodal, with distributions of trait values peaking around median values for all traits (Supporting Information Figure S3). Hence, most of the world freshwater fishes have a similar morphology characterized by average trait values (e.g., maximum body length *ca*. 15 cm, body elongation ratio *ca*. 4, terminal and medium-sized mouth), which fits with the most commonly represented morphology in freshwater fishes, such as roaches and chubs (e.g., *Gila brevicauda* in the Nearctic realm or *Rutilus rutilus* in the Palaearctic realm). Trait distributions of freshwater fishes thus differ markedly from that of world plant flora (Díaz et al., 2016), which is clustered between woody and non-woody plants, with few intermediate species, illustrating two distinct ecological strategies (long-lived and tall ligneous species with a late sexual maturity versus short-lived and small non-ligneous species with an early sexual maturity). In contrast to plants, we did not detect any clustering between opposite strategies in freshwater fish, but a continuum between extreme strategies, as in coral reef fishes (Mouillot et al., 2014) or passerine birds (Ricklefs, 2012).

Nevertheless, morphological characteristics of the fish varied significantly among realms, and these differences were driven partly by the differences in the taxonomic order membership of the fish inhabiting each realm (Supporting Information Table S4). For instance, a substantial proportion of the Siluriformes has a ventral mouth (oral gape position = 0) and a pectoral fin at the bottom of the body (pectoral fin position = 0). Given that Siluriformes are the most species-rich order in Neotropical fish fauna (Levêque et al., 2008; Nelson et al., 2016), they contribute disproportionally to the morphological distinctiveness of the Neotropical compared with other realm fish faunas (Toussaint et al., 2016). Therefore, the evolutionary processes that have promoted order composition distinctiveness (Supporting Information Table S4) between realms partly explain the current morphological differences between realms. Thus, the difference of trait distributions is attributable, at least in part, to distinct evolutionary histories between realm fish faunas, as pointed out by Lamouroux, Poff, and Angermeier (2002) and Heino, Schmera, and Erős (2013) for European fish faunas.

In addition, differences in the distribution of morphologies between realm faunas were also driven by morphological differences among species belonging to the same order. Given that freshwater

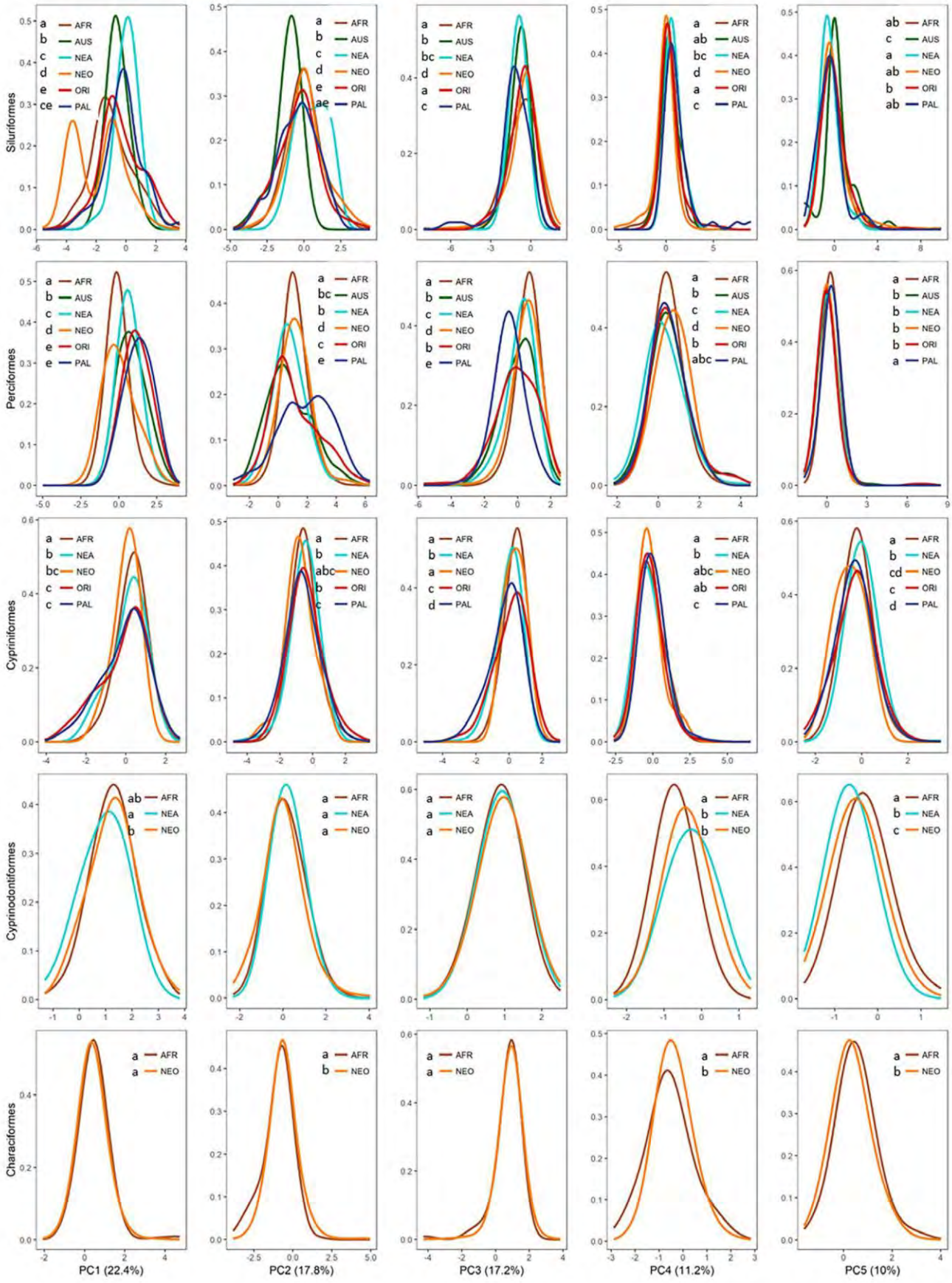



**FIGURE 2** Kernel density estimation for the distributions of fish species between realms from the five most species-rich fish orders along five morphological axes (PCs). Letters in front of keys depict the difference in trait distributions between realms; different letters indicate a significant difference between realms (Kolmogorov–Smirnov tests, $p < 0.05$). AFR = Afrotropical; AUS = Australian; NEA = Nearctic; NEO = Neotropical; ORI = Oriental; PAL = Palaearctic [Colour figure can be viewed at wileyonlinelibrary.com]

fish cannot easily disperse between realms (Hugueny, 1989), realm faunas have evolved independently for millions of years, leading to distinct fish species inhabiting each realm (Leprieur et al., 2011; Villéger et al., 2011). Fish species from the same order occurring in distinct realms have thus evolved in independent and probably distinct environments (e.g., tropical rivers have a different flow regime and diversity of prey from arctic ones (Dudgeon, 2008)), which could ultimately have affected their morphological diversity. This was the case for oral gape position, relative maxillary length and pectoral fin position (represented by PC1) of the Siluriformes, which differed significantly between the Neotropical species and those belonging to the five other realms (Figure 2). Such distinctiveness of the Siluriformes in the Neotropical realm can be attributed, in part, to the Loricariidae family, which has a modified suckermouth adapted to graze on algae and biofilms, with small jaws in the ventral position and with ventral pectoral fins (Covain & Fisch-Muller, 2007). Our results therefore confirm the prediction that the fauna having the same deep evolutionary history (i.e., fish belonging to the same taxonomic order) evolved under distinct environmental constraints, leading to different morphological trait distributions among realms. This also means that fish from a taxonomic order exhibit a distinct range of ecological strategies among realms.

Considering the distribution of traits, we show that the 1,663 morphologically extreme species, which account for 18.2% of the world freshwater fish fauna, fill 97.8% of the morphological space. The limited functional specialization and originality measured for the world fish fauna (Table 2) hence translate into a core of 81.8% of the species with trait values close to the average (i.e., centre of functional space defined by the world species pool). This contrasts with the remaining 18.2% of species with extreme morphologies. In addition, there is a diversity of morphologies among these extreme species, with no marked gap in the filling of morphological axes (Figure 3). This pattern is even more marked when considering the most extreme species (0.5% tails of the distribution), which account for only 3.9% of the global fish fauna but support 89.7% of the morphological space. These results parallel those of Mouillot et al. (2014), who found, using ecological–behavioural traits, that a substantial proportion of functional diversity is supported by a few coral reef fish species. In the same way, we extend the findings of Rodrigues-Filho et al. (2018) demonstrating that extreme species at

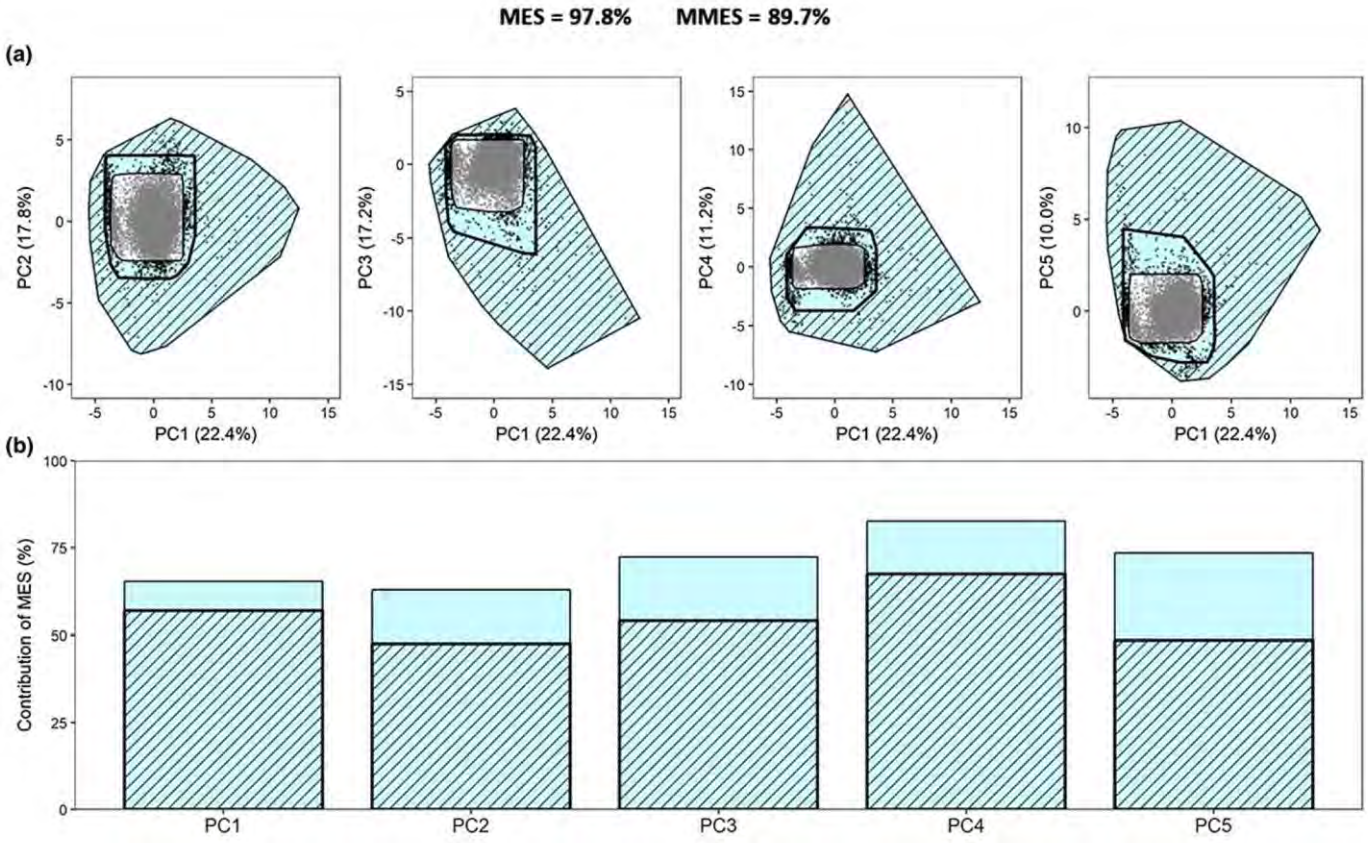



**FIGURE 3** Contribution of morphologically extreme species to morphological diversity of the world freshwater fish fauna. Values of MES and MMES above the convex hulls represent the contributions of the morphologically extreme species [MES; i.e., the 2.5% of species with the lowest or highest values on each of the five principal component (PC) axes] and the most morphologically extreme species (MMES; i.e., the 0.5% of species with the lowest or highest values on each of the five PC axes) to the five-dimensional space volume. (a) The global convex hull, including 9,150 species, is delimited by the external black line. Blue-filled areas show the morphological volume filled by MES, and hatched parts represent MMES. The white-filled areas show the morphological volume filled by the morphological core species (MCS; i.e., species among the 95% closest to the average position of all species). Black dots represent the 9,150 species. (b) Bars represent the contributions of MES (blue) and of MMES (hatched) to PC1–PC5 [Colour figure can be viewed at wileyonlinelibrary.com]

**TABLE 2** Functional specialization (FSpe) and functional originality (FOri) of fish, and number and percentage of morphologically extreme species [MES; i.e., the 2.5% of species with the lowest or highest values on each of the five principal component (PC) axes] and most morphologically extreme species (MMES; i.e., the 0.5% of species with the lowest or highest values on each of the five PC axes) in the six realms and in the global freshwater fish faunas

| | | | MES (2.5%) | | | MMES (0.5%) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | FSpe (%) | FOri (%) | Number of species | Percentage of fauna | FSpe (%) | FOri (%) | Number of species | Percentage of fauna | FSpe (%) | FOri (%) |
| Afrotropical | 10.2 | 8.6 | 267 | 11.5 | 19.7 | 18.0 | 27 | 1.2 | 33.7 | 29.9 |
| Australian | 13.9 | 11.8 | 106 | 28.2 | 23.4 | 19.5 | 28 | 7.4 | 36.7 | 35.2 |
| Nearctic | 10.7 | 9.5 | 93 | 13.8 | 23.0 | 21.5 | 23 | 3.4 | 37.2 | 36.5 |
| Neotropical | 12.6 | 9.1 | 791 | 21.6 | 23.2 | 16.0 | 191 | 5.2 | 34.9 | 24.2 |
| Oriental | 12.3 | 10.4 | 338 | 22.5 | 21.4 | 18.9 | 74 | 4.9 | 32.4 | 30.7 |
| Palaearctic | 12.7 | 10.7 | 176 | 20.0 | 23.2 | 20.3 | 46 | 5.2 | 37.2 | 31.4 |
| Global | 11.7 | 9.3 | 1,663 | 18.2 | 22.2 | 17.5 | 356 | 3.9 | 34.6 | 27.3 |

the margins of morphological space tend to be more original than the other species of Neotropical stream fishes, indicating that extinction of a single extreme species could have marked consequences for the functional diversity of the fish fauna at local and regional scales. Given that functionally extreme species are also often threatened, as shown for fish body size by Olden et al. (2007) and, more recently, by Ripple et al. (2017) for large vertebrates, it is likely that other extreme morphological traits indicate particular habitat or life strategies that make the extreme species more vulnerable to anthropogenic disturbances. For instance, the species having extremely elongated bodies and large mouths represent predators specialized in the capture of small fishes (e.g., some Beloniformes and some Cyprinodontiformes) with a strict range of prey sizes (Nelson et al., 2016), or eyeless fishes that live only in caves (Borowsky, 2018).

Our assessment of the number of morphologically extreme species per realm reveals strong variations that are in large part attributable to the species richness in each realm, with the exception of the Australian fauna, which contains the fewest species but hosts the highest proportion of morphologically extreme species among the six realm faunas (Table 2). Regardless of how the extreme species were distributed among realms, functional specialization and originality remained low in all the realms. Nevertheless, these two metrics doubled for the extreme species assemblages (Table 2), which indicates a consistent pattern of high functional specialization and low functional redundancy among extreme species. Considering the huge proportion of functional space occupied by these extreme species (Figure 4), a remarkable functional vulnerability is supported by MES and MMES among all the realms, therefore paralleling the results of Mouillot et al. (2014) on ecological–behavioural traits of coral reef fishes. This finding contrasts with the discrepancy of functional vulnerability among realms revealed by Toussaint et al. (2016). This last measure was based on the morphological diversity supported by endangered species, but as underlined by Vitule et al. (2017), this measure only accounts for the species currently known to be endangered. Indeed, only 7% of the MES (and 10% of the MMES) are currently listed as threatened (IUCN, 2018; see Supporting Information Table S5), meaning that most of the MES and MMES species are currently not considered to be endangered. Nevertheless, given that climate change is affecting most of the species and ecosystems (Pörtner & Knust, 2007; Thomas et al., 2004), and thousands of dams are planned on most of the world's rivers, although they are known potentially to be detrimental to aquatic biodiversity (Anderson et al., 2018; Winemiller et al., 2016), a substantial proportion of those species will be pushed to risk of extinction in the next decades. Given that the most morphologically extreme species account for almost 90% of the morphological diversity of the world's freshwater fish, we suggest that their vulnerability to increasing anthropogenic pressures should be monitored accurately to ensure the sustainability of the unique functions they support in all the six realms (see Supporting Information Table S5). We therefore suggest that the current functional vulnerability of fish faunas (as in Toussaint et al., 2016) and the potential vulnerability supported by the morphologically

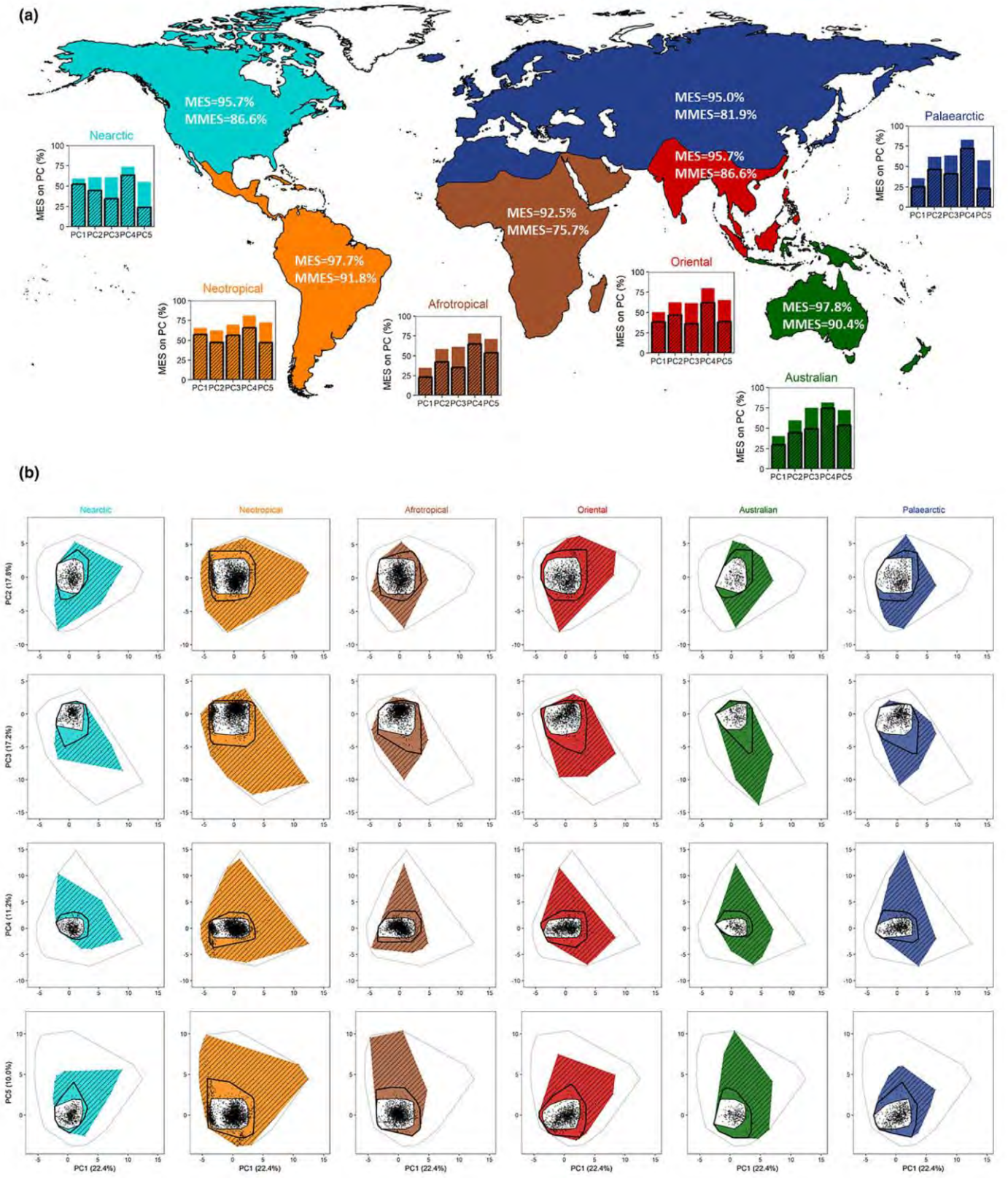



**FIGURE 4** Contribution of morphologically extreme species to morphological diversity of the six realm fish faunas. (a) Values of MES and MMES on the map represent the contributions of the morphologically extreme species [MES; i.e., the 2.5% of species with the lowest or highest values on each of the five principal component (PC) axes] and the most morphologically extreme species (MMES; i.e., the 0.5% of species with the lowest or highest values on each of the five PC axes) to the five-dimensional space volume in each realm. Bars represent the contributions of MES to PC1–PC5. The hatched part of the bars represents MMES. (b) The global convex hull, including 9,150 species, is in grey. Colour-filled areas show the morphological volume filled by MES (hatched parts represent MMES) in each realm fauna. The white-filled areas show the morphological volume filled by the morphological core species (MCS; i.e., species among the 95% closest to the average position of all species). Black dots represent species belonging to each realm fauna [Colour figure can be viewed at wileyonlinelibrary.com]

extreme species in each realm should be considered together to develop biodiversity conservation guidelines.

In conclusion, our findings suggest that the difference in morphological distributions between the fish faunas of the six biogeographic realms is driven not only by the evolutionary divergence between realm fish faunas (Heino et al., 2013; Hugueny, 1989; Tedesco et al., 2017), but also by the distinct environmental constraints acting in each realm. Moreover, such morphological differences among realms translates into a shared strong contribution of morphologically extreme species among realms. Therefore, in all the six realms, >90% of the morphological diversity is supported by a few morphologically extreme species, accounting for <20% of the fish fauna (i.e., MES) that deserve to be conserved. Among those species, the <5% MMES are of paramount importance to maintain the morphological diversity of the fish faunas, because they support *ca*. 90% of the morphological diversity of the realms and therefore also support a wide range of associated functions.

## ACKNOWLEDGMENTS

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## DATA ACCESSIBILITY

*Fish occurrence data can be retrieved freely from https://doi.org/10.6084/m9.figshare.c.3739145, and metadata are available in Tedesco et al. (2017). Information about the 9,150 species with combined trait values (PC1–PC5) used in this study is provided as a supplementary (Table S5), which also indicates the morphologically extreme and most morphologically extreme species and the IUCN Red List status for each species.*

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## REFERENCES

Albouy, C., Guilhaumon, F., Villéger, S., Mouchet, M., Mercier, L., Culioli, J., … Mouillot, D. (2011). Predicting trophic guild and diet overlap from functional traits: Statistics, opportunities and limitations for marine ecology. *Marine Ecology Progress Series*, *436*, 17–28. https://doi.org/10.3354/meps09240

Anderson, E. P., Jenkins, C. N., Heilpern, S., Maldonado-Ocampo, J. A., Carvajal-Vallejos, F. M., Encalada, A. C., … Ortega, H. (2018). Fragmentation of Andes-to-Amazon connectivity by hydropower dams. *Science Advances*, *4*, eaao1642. https://doi.org/10.1126/sciadv.aao1642

Betancur-R, R., Wiley, E. O., Arratia, G., Acero, A., Bailly, N., Miya, M., … Orti, G. (2017). Phylogenetic classification of bony fishes. *BMC Evolutionary Biology*, *17*, 162. https://doi.org/10.1186/s12862-017-0958-3

Blanchet, S., Grenouillet, G., Beauchard, O., Tedesco, P. A., Leprieur, F., Dürr, H. H., … Brosse, S. (2010). Non-native species disrupt the worldwide patterns of freshwater fish body size: Implications for Bergmann's rule. *Ecology Letters*, *13*, 421–431. https://doi.org/10.1111/j.1461-0248.2009.01432.x

Borowsky, R. (2018). Cavefishes. *Current Biology*, *28*, R60–R64. https://doi.org/10.1016/j.cub.2017.12.011

Brosse, S., Beauchard, O., Blanchet, S., Dürr, H. H., Grenouillet, G., Hugueny, B., … Villéger, S. (2013). Fish-SPRICH: a database of freshwater fish species richness throughout the World. *Hydrobiologia*, *700*, 343–349. https://doi.org/10.1007/s10750-012-1242-6

Cilleros, K., Allard, L., Grenouillet, G., & Brosse, S. (2016). Taxonomic and functional diversity patterns reveal different processes shaping European and Amazonian stream fish assemblages. *Journal of Biogeography*, *43*, 1832–1843. https://doi.org/10.1111/jbi.12839

R Core Team. (2017). *R: A language and environment for statistical computing*. Retrieved from https://www.r-project.org/

Covain, R., & Fisch-Muller, S. (2007). The genera of the neotropical armored catfish subfamily Loricariinae (Siluriformes: Loricariidae): A practical key and synopsis. *Zootaxa*, *1462*, 1–40.

Cucherousset, J., Blanchet, S., & Olden, J. D. (2012). Non-native species promote trophic dispersion of food webs. *Frontiers in Ecology and the Environment*, *10*, 406–408. https://doi.org/10.1890/12.WB.018

Díaz, S., Kattge, J., Cornelissen, J. H., Wright, I. J., Lavorel, S., Dray, S., … Prentice, I. C. (2016). The global spectrum of plant form and function. *Nature*, *529*(7585), 167–171.

Dudgeon, D. (2008). *Tropical stream ecology*. London: Academic Press.

Felsenstein, J. (1985). Phylogenies and the comparative method. *The American Naturalist*, *125*, 1–15. https://doi.org/10.1086/284325

Froese, R., & Pauly, D. (2018). FishBase. Retrieved from www.fishbase.org

Garnier, E., Navas, M.-L., & Grigulis, K. (2016). *Plant functional diversity: Organism traits, community structure, and ecosystem properties*. Oxford: Oxford University Press.

Gaston, K. J., & Spicer, J. I. (2004). *Biodiversity: An introduction*. Oxford: Blackwell Publishing.

Heino, J., Schmera, D., & Erős, T. (2013). A macroecological perspective of trait patterns in stream communities. *Freshwater Biology*, *58*, 1539–1555. https://doi.org/10.1111/fwb.12164

Hugueny, B. (1989). West African rivers as biogeographic islands: Species richness of fish communities. *Oecologia*, *79*, 236–243. https://doi.org/10.1007/BF00388483

Hugueny, B., & Pouilly, M. (1999). Morphological correlates of diet in an assemblage of West African freshwater fishes. *Journal of Fish Biology*, *54*, 1310–1325. https://doi.org/10.1111/j.1095-8649.1999.tb02057.x

IUCN. (2018). *The IUCN red list of threatened species, Version 2017-3*. Retrieved from https://www.iucnredlist.org

Kattge, J., Diaz, S., Lavorel, S., Prentice, I., Leadley, P., Bönisch, G., … Wright, I. (2011). TRY—a global database of plant traits. *Global Change Biology*, *17*, 2905–2935. https://doi.org/10.1111/j.1365-2486.2011.02451.x

Kuczynski, L., Côte, J., Toussaint, A., Brosse, S., Buisson, L., & Grenouillet, G. (2018). Spatial mismatch in morphological, ecological and phylogenetic diversity, in historical and contemporary European freshwater fish faunas. *Ecography*, *41*, 1665–1674.

Lamouroux, N., Poff, N. L., & Angermeier, P. L. (2002). Intercontinental convergence of stream fish community traits along geomorphic and hydraulic gradients. *Ecology*, *83*, 1792–1807. https://doi.org/10.1890/0012-9658(2002)083[1792:ICOSFC]2.0.CO;2

Leitão, R. P., Zuanon, J., Mouillot, D., Leal, C. G., Hughes, R. M., Kaufmann, P. R., … de Paula, F. R. (2018). Disentangling the pathways of land use impacts on the functional structure of fish assemblages in Amazon streams. *Ecography*, *41*, 219–232. https://doi.org/10.1111/ecog.02845

Leprieur, F., Tedesco, P. A., Hugueny, B., Beauchard, O., Dürr, H. H., Brosse, S., & Oberdorff, T. (2011). Partitioning global patterns of freshwater fish beta diversity reveals contrasting signatures of

past climate changes. *Ecology Letters*, *14*, 325–334. https://doi.org/10.1111/j.1461-0248.2011.01589.x

Levêque, C., Oberdorff, T., Paugy, D., Stiassny, M., & Tedesco, P. (2008). Global diversity of fish (Pisces) in freshwater. *Hydrobiologia*, *595*, 545–567. https://doi.org/10.1007/s10750-007-9034-0

Mouillot, D., Graham, N. A., Villéger, S., Mason, N. W., & Bellwood, D. R. (2013). A functional approach reveals community responses to disturbances. *Trends in Ecology and Evolution*, *28*, 167–177. https://doi.org/10.1016/j.tree.2012.10.004

Mouillot, D., Villéger, S., Parravicini, V., Kulbicki, M., Arias-González, J. E., Bender, M., … Vigliola, L. (2014). Functional over-redundancy and high functional vulnerability in global fish faunas on tropical reefs. *Proceedings of the National Academy of Sciences of the USA*, *111*, 13757–13762. https://doi.org/10.1073/pnas.1317625111

Nelson, J. S., Grande, T. C., & Wilson, M. V. (2016). *Fishes of the world*. Hoboken: John Wiley & Sons.

Olden, J. D., Hogan, Z. S., & Zanden, M. J. V. (2007). Small fish, big fish, red fish, blue fish: Size-biased extinction risk of the world's freshwater and marine fishes. *Global Ecology and Biogeography*, *16*, 694–701. https://doi.org/10.1111/j.1466-8238.2007.00337.x

Pagel, M. (1999). Inferring the historical patterns of biological evolution. *Nature*, *401*(6756), 877. https://doi.org/10.1038/44766

Parravicini, V., Villéger, S., McClanahan, T. R., Arias-González, J. E., Bellwood, D. R., Belmaker, J., … Guilhaumon, F. (2014). Global mismatch between species richness and vulnerability of reef fish assemblages. *Ecology Letters*, *17*, 1101–1110. https://doi.org/10.1111/ele.12316

Penone, C., Davidson, A. D., Shoemaker, K. T., Di Marco, M., Rondinini, C., Brooks, T. M., … Costa, G. C. (2014). Imputation of missing data in life-history trait datasets: Which approach performs the best? *Methods in Ecology and Evolution*, *5*, 961–970. https://doi.org/10.1111/2041-210X.12232

Pörtner, H. O., & Knust, R. (2007). Climate change affects marine fishes through the oxygen limitation of thermal tolerance. *Science*, *315*(5808), 95–97.

Ricklefs, R. E. (2012). Species richness and morphological diversity of passerine birds. *Proceedings of the National Academy of Sciences of the USA*, *109*, 14482–14487. https://doi.org/10.1073/pnas.1212079109

Ripple, W. J., Wolf, C., Newsome, T. M., Hoffmann, M., Wirsing, A. J., & McCauley, D. J. (2017). Extinction risk is most acute for the world's largest and smallest vertebrates. *Proceedings of the National Academy of Sciences of the USA*, *114*, 10678–10683. https://doi.org/10.1073/pnas.1702078114

Rodrigues-Filho, C. A., Leitão, R. P., Zuanon, J., Sánchez-Botero, J. I., & Baccaro, F. B. (2018). Historical stability promoted higher functional specialization and originality in Neotropical stream fish assemblages. *Journal of Biogeography*, *45*, 1345–1354. https://doi.org/10.1111/jbi.13205

Schleuter, D., Daufresne, M., Massol, F., & Argillier, C. (2010). A user's guide to functional diversity indices. *Ecological Monographs*, *80*, 469–484. https://doi.org/10.1890/08-2225.1

Stone, R. (2007). The last of the Leviathans. *Science*, *316*(5832), 1684–1688.

Tedesco, P. A., Beauchard, O., Bigorne, R., Blanchet, S., Buisson, L., Conti, L., … Hugueny, B. (2017). A global database on freshwater fish species occurrence in drainage basins. *Scientific Data*, *4*, 170141. https://doi.org/10.1038/sdata.2017.141

Thomas, C. D., Cameron, A., Green, R. E., Bakkenes, M., Beaumont, L. J., Collingham, Y. C., … Hannah, L. (2004). Extinction risk from climate change. *Nature*, *427*(6970), 145–148. https://doi.org/10.1038/nature02121

Toussaint, A., Charpin, N., Brosse, S., & Villéger, S. (2016). Global functional diversity of freshwater fish is concentrated in the Neotropics while functional vulnerability is widespread. *Scientific Reports*, *6*, 22125. https://doi.org/10.1038/srep22125

Villéger, S., Blanchet, S., Beauchard, O., Oberdorff, T., & Brosse, S. (2011). Homogenization patterns of the world's freshwater fish faunas. *Proceedings of the National Academy of Sciences of the USA*, *108*, 18003–18008. https://doi.org/10.1073/pnas.1107614108

Villéger, S., Brosse, S., Mouchet, M., Mouillot, D., & Vanni, M. J. (2017). Functional ecology of fish: Current approaches and future challenges. *Aquatic Sciences*, *79*, 783–801.

Villéger, S., Miranda, J. R., Hernandez, D. F., & Mouillot, D. (2010). Contrasting changes in taxonomic vs. functional diversity of tropical fish communities after habitat degradation. *Ecological Applications*, *20*, 1512–1522.

Vitule, J. R., Agostinho, A. A., Azevedo-Santos, V. M., Daga, V. S., Darwall, W. R., Fitzgerald, D. B., … Magalhães, A. L. (2017). We need better understanding about functional diversity and vulnerability of tropical freshwater fishes. *Biodiversity and Conservation*, *26*, 757–762.

Watson, D., & Balon, E. (1984). Ecomorphological analysis of fish taxocenes in rainforest streams of northern Borneo. *Journal of Fish Biology*, *25*, 371–384. https://doi.org/10.1111/j.1095-8649.1984.tb04885.x

Winemiller, K. O. (1991). Ecomorphological diversification in lowland freshwater fish assemblages from five biotic regions. *Ecological Monographs*, *61*, 343–365. https://doi.org/10.2307/2937046

Winemiller, K. O., McIntyre, P., Castello, L., Fluet-Chouinard, E., Giarrizzo, T., Nam, S., … Harrison, I. (2016). Balancing hydropower and biodiversity in the Amazon, Congo, and Mekong. *Science*, *351*(6269), 128–129. https://doi.org/10.1126/science.aac7082

## BIOSKETCH

**Guohuan Su** is a PhD student in the laboratory "Evolution et Diversité Biologique" (University of Toulouse, France). He is interested in the macroecology and diversity of freshwater fish. He is currently investigating determinants of global freshwater fish morphology and the influence of global changes on global morphological patterns.

**Sébastien Villéger** is a researcher at the CNRS in Montpellier. He is interested in the functional ecology of fishes and, in particular, is studying the effect of global change on the functional diversity of coastal fish assemblages.

**Sébastien Brosse** is a professor of Ecology at the University of Toulouse. He is interested in freshwater fish biodiversity patterns and processes across the globe and in the influence of global changes on fish distribution at both macroecological and local scales.

## SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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Supporting information

# Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread

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**Table S1** Number of known freshwater fish species for the main fish orders and corresponding number and proportion of species considered in this study.

| Order | Total number of species | Number of species considered | Percentage of species considered (%) |
|---|---|---|---|
| Siluriformes | 2740 | 2117 | 77.26 |
| Cypriniformes | 3268 | 1969 | 60.25 |
| Perciformes | 2040 | 1965 | 96.32 |
| Characiformes | 1674 | 1467 | 87.63 |
| Cyprinodontiformes | 996 | 589 | 59.14 |
| Others | 1234 | 980 | 79.42 |
| Total | 11952 | 9150 | 76.56 |

**Table S2** Number of species considered per morphological trait and statistics about the distribution of trait values within the global freshwater fish fauna. Maximum Body Length is in centimetres; the nine other traits are unitless ratios, see Fig. S1 for details on traits.

| | Species No. | Min | 2.50% | Median | 97.5% | Max | Mean | SD | Kurtosis | Skewness |
|---|---|---|---|---|---|---|---|---|---|---|
| Maximum Body Length | 8531 | 1.00 | 3.00 | 12.00 | 100.00 | 800.00 | 21.04 | 34.95 | 136.52 | 8.91 |
| Relative Eye Size | 7648 | 0.00 | 0.16 | 0.40 | 0.66 | 0.87 | 0.40 | 0.13 | 2.58 | 0.10 |
| Oral Gape Position | 7648 | 0.00 | 0.00 | 0.41 | 0.71 | 0.94 | 0.38 | 0.20 | 2.66 | -0.45 |
| Relative Maxillary Length | 6891 | 0.00 | 0.00 | 0.37 | 0.99 | 7.06 | 0.41 | 0.32 | 115.15 | 7.36 |
| Eye Vertical Position | 7643 | 0.16 | 0.36 | 0.54 | 0.75 | 1.00 | 0.55 | 0.10 | 4.00 | 0.28 |
| Body Elongation | 7677 | 1.12 | 2.08 | 4.00 | 9.95 | 43.73 | 4.43 | 2.34 | 44.73 | 4.64 |
| Body Lateral Shape | 7671 | 0.19 | 0.35 | 0.57 | 0.83 | 0.99 | 0.57 | 0.12 | 3.01 | 0.27 |
| Pectoral Fin Vertical Position | 7613 | 0.00 | 0.00 | 0.27 | 0.65 | 0.86 | 0.28 | 0.16 | 3.45 | 0.40 |
| Pectoral Fin Size | 7082 | 0.00 | 0.08 | 0.18 | 0.33 | 0.51 | 0.19 | 0.06 | 4.54 | 0.52 |
| Caudal Peduncle Throttling | 7249 | 0.00 | 1.23 | 2.45 | 4.75 | 15.45 | 2.59 | 1.04 | 27.60 | 3.07 |

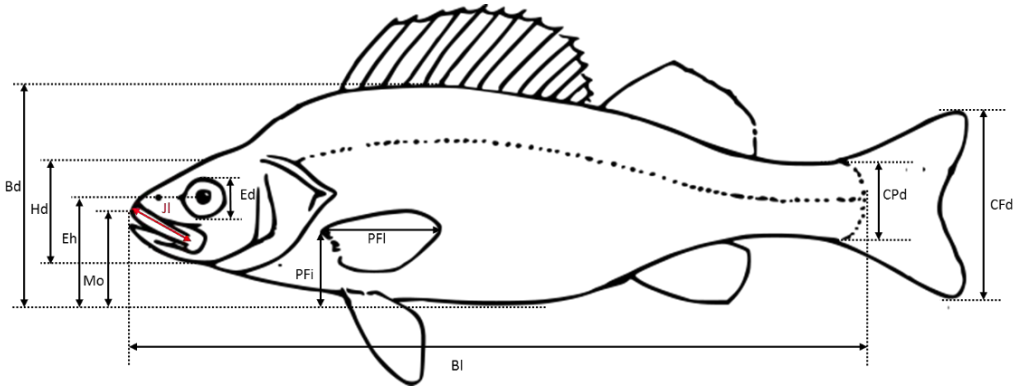
**Table S3** Phylogenetic signals for morphological traits based on the phylogenetic tree of 493 freshwater fish species. Traits with λ values significantly different from zero are in bold (*** $P < 0.001$).

| | Pagel's λ | P |
|---|---|---|
| Maximum body length | **0.935** | *** |
| Relative Eye size | **0.900** | *** |
| Oral gape position | **0.912** | *** |
| Relative maxillary length | **0.944** | *** |
| Eye vertical position | **0.536** | *** |
| Body elongation | **0.999** | *** |
| Body lateral shape | **0.789** | *** |
| Pectoral fin vertical position | **0.935** | *** |
| Pectoral fin size | **0.962** | *** |
| Caudal peduncle throttling | **0.937** | *** |

**Table S4** Number (and proportion) of species per order in each realm considered in this study.

| Order | Afrotropical | Australian | Nearctic | Neotropical | Oriental | Palearctic |
|---|---|---|---|---|---|---|
| Siluriformes | 379 (16.29%) | 44 (11.67%) | 31 (4.59%) | 1323 (36.09%) | 310 (20.6%) | 49 (5.57%) |
| Cypriniformes | 377 (16.2%) | 3 (0.8%) | 268 (39.7%) | 21 (0.57%) | 839 (55.75%) | 527 (59.95%) |
| Perciformes | 952 (40.91%) | 190 (50.4%) | 163 (24.15%) | 447 (12.19%) | 196 (13.02%) | 105 (11.95%) |
| Characiformes | 178 (7.65%) | 0 (0%) | 1 (0.15%) | 1289 (35.16%) | 0 (0%) | 0 (0%) |
| Cyprinodontiformes | 154 (6.62%) | 0 (0%) | 74 (10.96%) | 359 (9.79%) | 5 (0.33%) | 10 (1.14%) |
| Others | 287 (12.33%) | 140 (37.13%) | 138 (20.45%) | 227 (6.20%) | 155 (10.30%) | 188 (21.39%) |
| Total | 2327 (100%) | 377 (100%) | 675 (100%) | 3666 (100%) | 1505 (100%) | 879 (100%) |

**Table S5** is too large, so it is provided as a separate file.




**a. Morphological measurements**

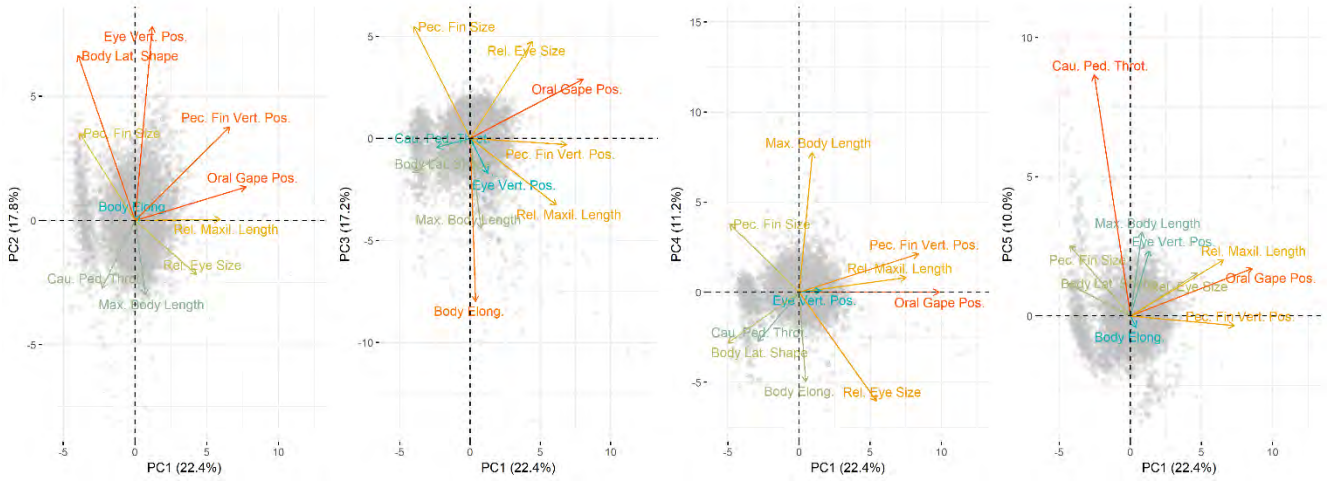
| Code | Name | Protocol for measurement |
|---|---|---|
| Blmax | Maximum Body length | Maximum adult length (obtained from FishBase (Froese & Pauly, 2018)) |
| Bl | Body length | Standard length (snout to caudal fin basis) |
| Bd | Body depth | Maximum body depth |
| Hd | Head depth | Head depth at the vertical of eye |
| CPd | Caudal peduncle depth | Minimum depth of the caudal peduncle |
| CFd | Caudal fin depth | Maximum depth of the caudal fin |
| Ed | Eye diameter | Vertical diameter of the eye |
| Eh | Eye position | Vertical distance between the centre of the eye to the bottom of the body |
| Mo | Oral gape position | Vertical distance from the top of the mouth to the bottom of the body |
| Jl | Maxillary jaw length | Length from snout to the corner of the mouth |
| PFl | Pectoral fin length | Length of the longest ray of the pectoral fin |
| PFi | Pectoral fin position | Vertical distance between the upper insertion of the pectoral fin to the bottom of the body |

All measurements were made on pictures except Blmax values, which were downloaded from Fishbase.org

**b. Morphological traits**

| Morphological traits | Formula | Potential link with fish functions | References |
|---|---|---|---|
| Maximum body length | $BLmax$ | Size is linked to metabolism, trophic impacts, locomotion ability, nutrient cycling | Toussaint *et al.* 2016b |
| Body elongation | $\frac{Bl}{Bd}$ | Hydrodynamism | Reecht *et al.* 2013 |
| Eye vertical position | $\frac{Eh}{Bd}$ | Position of fish and/or of its prey in the water column | Winemiller 1991 |
| Relative eye size | $\frac{Ed}{Hd}$ | Visual acuity | Boyle & Horn, 2006 |
| Oral gape position | $\frac{Mo}{Bd}$ | Feeding position in the water column | Dumay *et al.* 2004, Lefcheck *et al.* 2014 |
| Relative maxillary length | $\frac{Jl}{Hd}$ | Size of mouth and strength of jaw | Toussaint *et al.* 2016b |
| Body lateral shape | $\frac{Hd}{Bd}$ | Hydrodynamism and head size | Toussaint *et al.* 2016b |
| Pectoral fin vertical position | $\frac{PFi}{Bd}$ | Pectoral fin use for swimming | Dumay *et al.* 2004 |
| Pectoral fin size | $\frac{PFl}{Bl}$ | Pectoral fin use for swimming | Fulton *et al.* 2001 |
| Caudal peduncle throttling | $\frac{CFd}{CPd}$ | Caudal propulsion efficiency through reduction of drag | Webb 1984 |

**Fig. S1.** Morphological measurements (a) and morphological traits (b) measured on each fish species.




(a).

| | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|
| Eigenvalues | 2.24 | 1.78 | 1.72 | 1.12 | 1.00 |
| Proportion of Variance | 22.4 | 17.8 | 17.2 | 11.2 | 10.0 |
| Cumulative Proportion | 22.4 | 40.2 | 57.4 | 68.6 | 78.6 |

(b).

| | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|
| Relative Eye size | 9.4 | 3.1 | 14.1 | **23.3** | 2.2 |
| Oral gape position | **30.8** | 1.2 | 5.3 | <1 | 2.7 |
| Relative maxillary length | **18.0** | <1 | 6.6 | <1 | 3.8 |
| Eye vertical position | <1 | **39.4** | 1.8 | <1 | 5.1 |
| Body elongation | <1 | <1 | **39.6** | **15.7** | <1 |
| Body lateral shape | 8.0 | **28.5** | 1.8 | 5.1 | 1.3 |
| Pectoral fin vertical position | **22.4** | 9.2 | <1 | 2.9 | <1 |
| Pectoral fin size | 7.7 | 8.0 | **18.7** | 9.3 | 6.0 |
| Caudal peduncle throttling | 2.7 | 4.8 | <1 | 4.8 | **70.1** |
| Maximum body length | <1 | 5.8 | 12.1 | **38.5** | 8.5 |

The contribution of each morphological trait to the first 5 axes is expressed as percentages and values higher than 15% are in bold font.

**Fig. S2. Results of the Principal Components Analysis on morphological traits.** (a). Eigenvalues and percentages of variance explained by each axis. (b). Percentage of contribution of each morphological trait to each axis.

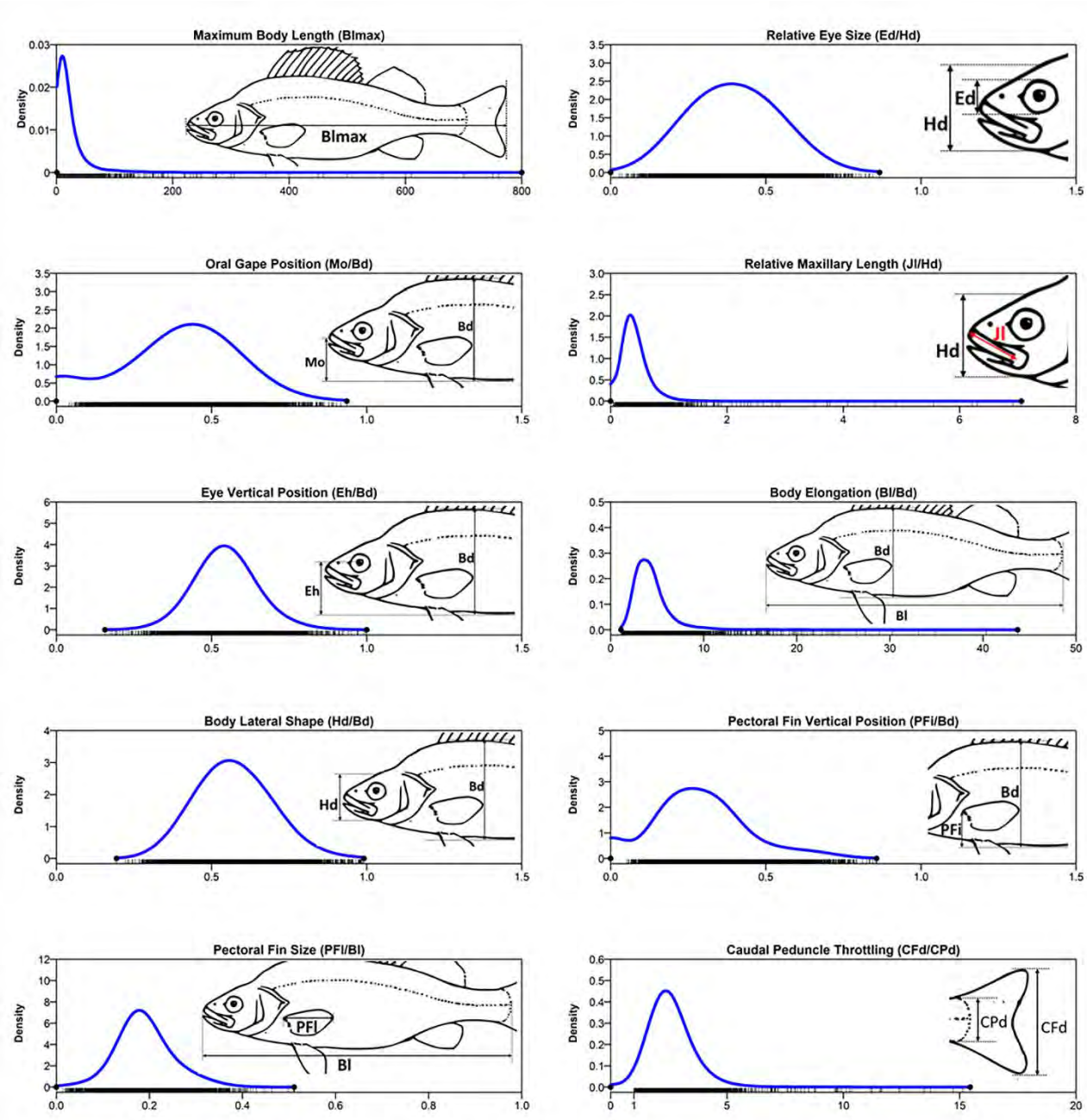



**Fig. S3.** Kernel density estimation for the distributions of the global freshwater fish species along the ten morphological traits. Black points illustrate minimum and maximum values. Black marks on the X-axis indicate species values.

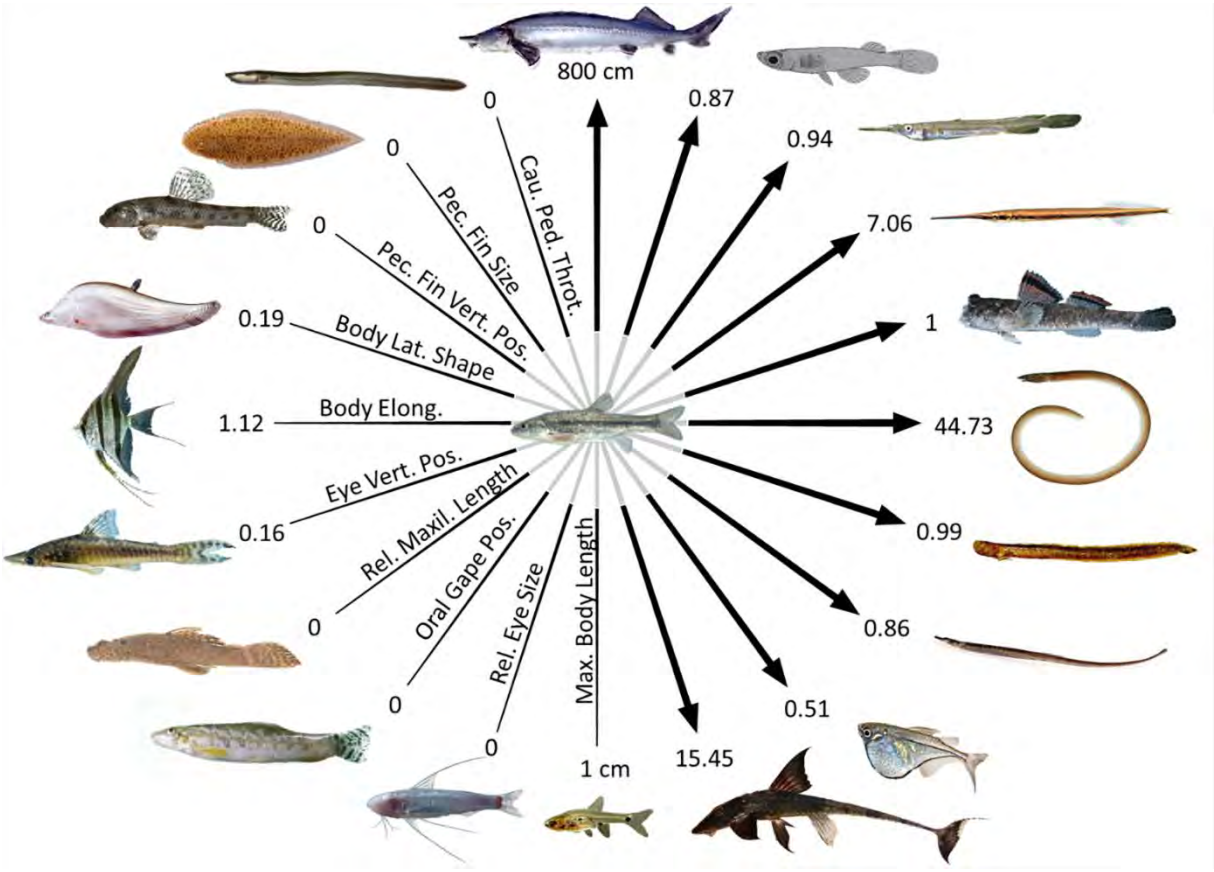



**Fig. S4.** Illustration of the morphological diversity in freshwater fishes. For each trait, minimum and maximum values are given, and pictures illustrate an example of species with minimum or maximum value of each trait. In the centre of the figure, a species (*Gila brevicauda*) with median values of all traits is illustrated. The fish illustrated for highest traits values are: *Huso huso* (maximum body length); *Fluviphylax simplex* (relative eye size); *Dermogenys pusilla* (oral gape position); *Potamorrhaphis guianensis* (relative maxillary length); *Periophthalmus argentilineatus* (eye vertical position); *Lamnostoma orientalis* (body elongation); *Taenioides cirratus* (body lateral shape); *Microphis leiaspis* (pectoral fin vertical position); *Carnegiella schereri* (pectoral fin size); *Loricaria cataphracta* (caudal peduncle throttling). The fish illustrated for lowest traits values are: *Boraras micros* (maximum body length); *Cetopsis oliveirai* (relative eye size); *Acanthocobitis botia* (oral gape position); *Ancistrus brevifilis* (relative maxillary length); *Hypoptopoma gulare* (eye vertical position); *Pterophyllum altum* (body elongation); *Chitala blanci* (body lateral shape); *Abbottina obtusirostris* (pectoral fin vertical position); *Cynoglossus feldmanni* (pectoral fin size); *Anguilla anguilla* (caudal peduncle throttling).

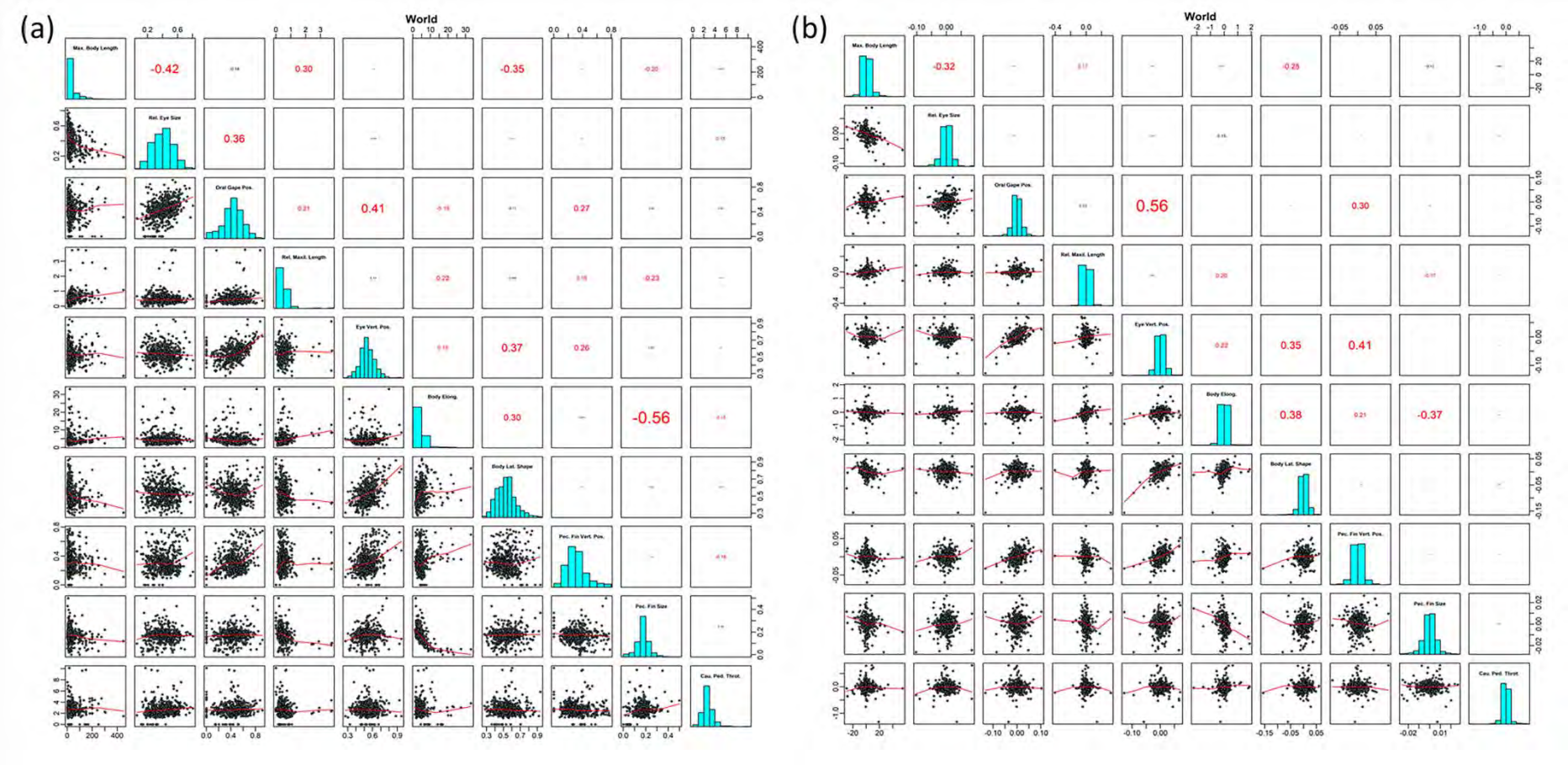



**Fig. S5. Spearman correlation between the ten morphological traits of 493 fish species** (a) using original trait values (b) after phylogenetic correction.

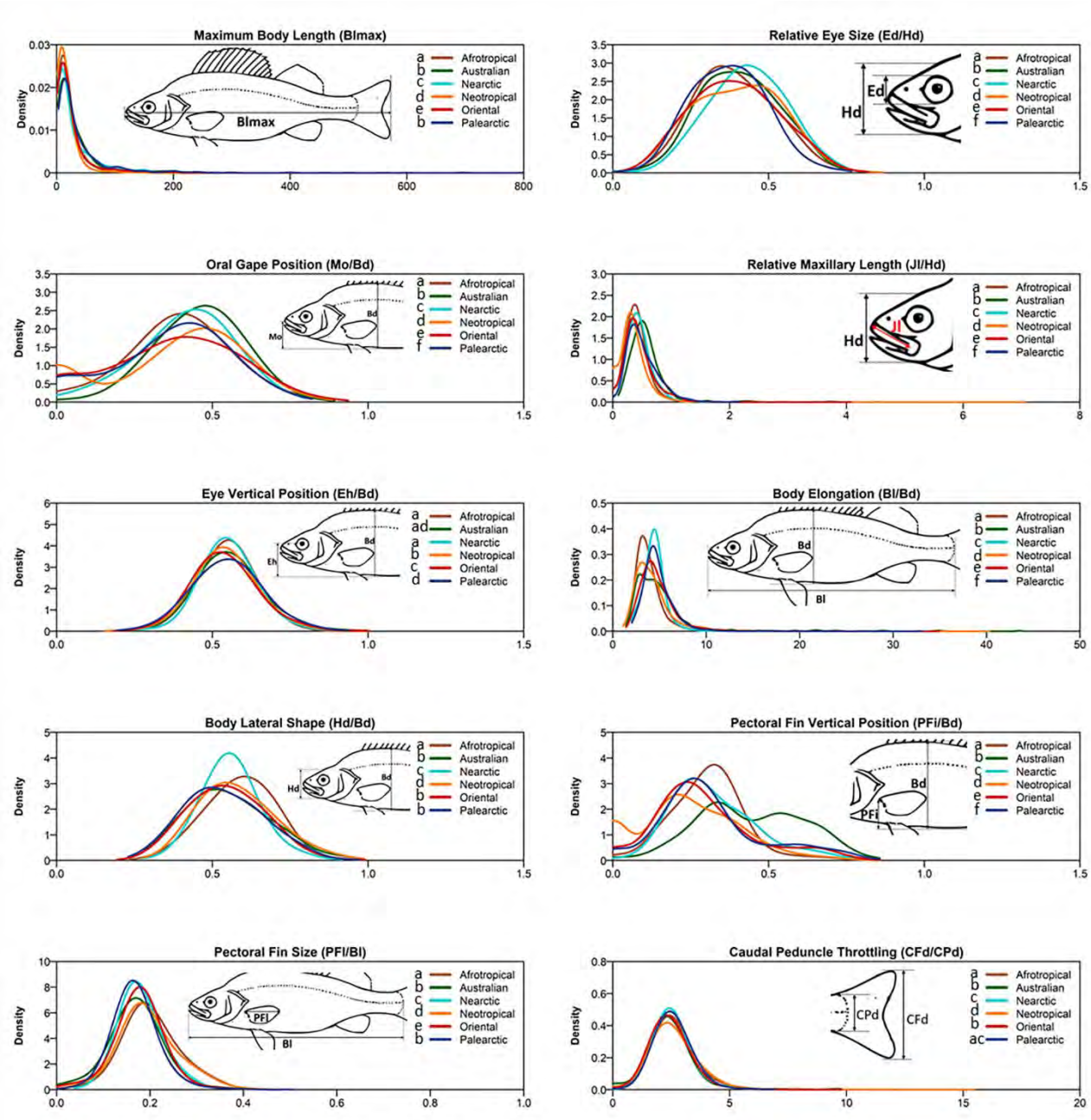



**Fig. S6.** Kernel density estimation of the distribution of ten morphological traits among species from the six biogeographic realms. The letters in front of realm captions depict the difference in trait distributions among realm faunas. Different letters indicate a significant difference between realms (K-S Tests, $P<0.05$).

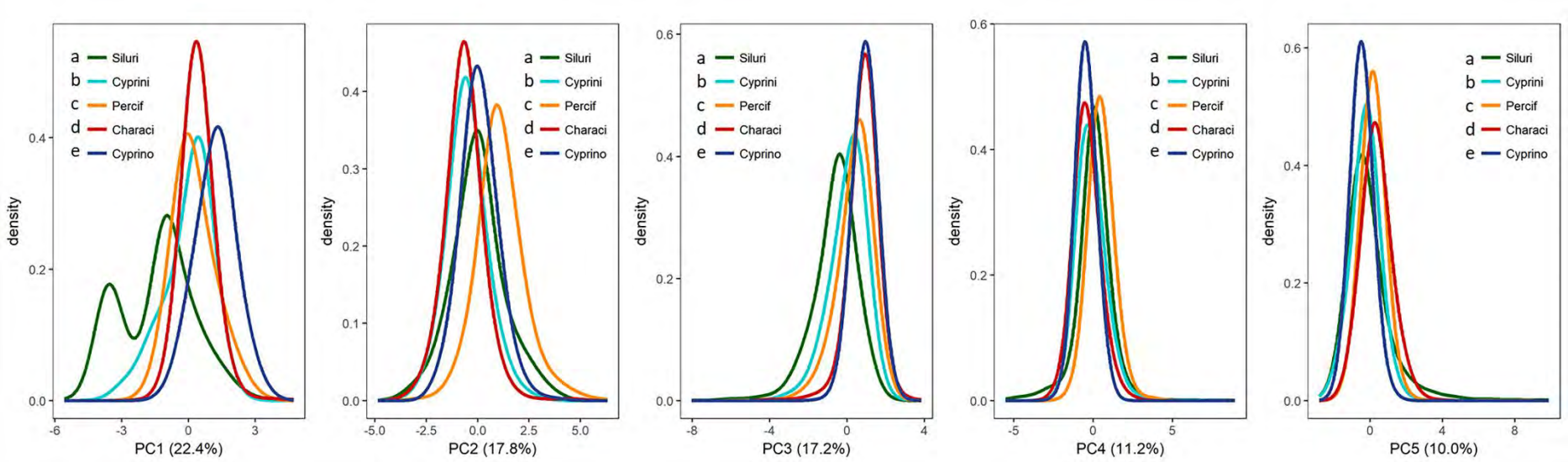



**Fig. S7.** Kernel density estimation for the distributions of fish species from the five most species-rich orders along five morphological axes (PC). Letters in front of captions depict the difference in trait distributions between orders. Different letters indicate a significant difference between orders (K-S tests, $P < 0.05$).

# Chapter IV

## Global patterns and determinants of freshwater fish functional diversity

*Global Ecology and Biogeography, resubmission*

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Abstract

**Aim:** Regional taxonomic diversity (species richness) is strongly influenced by a joint effect of the current processes (habitat and energy availability) and historical legacies, but it is still unclear how those environmental drivers have shaped the functional diversity of species assemblages.

**Major taxa studied:** Freshwater fish

**Location:** Global

**Time period:** 1960s-2000s

**Methods:** We combined the spatial occurrences over 2 400 river basins worldwide and the functional traits measured on 10 682 freshwater fish species to quantify the relative role of the habitat, climate and historical processes on the current global fish functional diversity. To avoid any correlation between taxonomic diversity and functional diversity, we controlled for differences in the number of species (species richness) between rivers. Functional diversity was considered through three complementary facets, functional richness (FRic), functional identity (FIde) and functional dispersion (FDis).

**Results**: The habitat size/diversity hypothesis contributes the most to the FRic due to its strong dependency on species richness, whereas the history/geography legacy markedly imprinted the

FDis and FIde patterns, leading to a balanced influence of the current and historical processes. Indeed, the distribution of morphological traits related to fish dispersal was largely explained by the glaciations events during the Quaternary, leading to strong latitudinal gradients.

**Main conclusions**: This study provides insights into the role of historical and environmental determinants on the functional structure of fish assemblages and strengthen that the independence of facets of functional diversity from the species richness makes them essential biodiversity variables to understand the structure of communities and their responses to global change.

**Keywords:** ecological hypothesis, functional dispersion, functional richness, functional identity, history/geography hypothesis, morphological traits.

## Introduction

Understanding the mechanisms that drove the global distribution of species and the subsequent species richness patterns is a central goal of biogeography and ecology (Gaston 2000). Those mechanisms can be classified in different categories, such as habitat, energy and history. Among ecological, evolutionary and historical processes proposed to explain taxonomic biodiversity patterns, habitat diversity and energy availability (e.g. temperature and precipitation) explained a large part of variations in species richness of freshwater fish over the globe (e.g. Guegan *et al.* 1998, Leprieur *et al*. 2011, Oberdorff *et al*. 2011), while historical legacy (e.g. realm membership, temperature anomaly since the last glaciation event) had a secondary role in explaining those species richness patterns (Guegan *et al*. 1998, Oberdorff *et al*. 2011).
Besides taxonomic diversity, the functional diversity account for the diversity of ecological features of organisms (Villeger *et al.* 2017) and is often summarized as the distribution of functional traits (be they morphological, ecological, behavioral or physiological), in a multidimensional functional space (Villeger *et al.* 2008, Su *et al.* 2019). The distribution of functional diversity is also unequal across the world, but the determinants are still largely unknown (Barnagaud *et al.* 2017, Bennie *et al.* 2014, Collen *et al.* 2014, Parravicini *et al.* 2013). For instance, Toussaint *et al*. (2016) reported the patterns of fish functional diversity at the realm scale but did not elaborate on the underlying mechanisms that shape those patterns.

The functional diversity can be characterized using a series of complementary indices (Villeger *et al.* 2008, Mouillot *et al.* 2013). Functional richness (FRic) measures the minimal volume occupied by the species from an assemblage (i.e. convex hull), and is the most frequently used index to describe the functional diversity of assemblages (Kuczynski *et al.* 2018, Stewart *et al.* 2020, Su et al. 2021). FRic allows going further than species richness since two assemblages with the same number of species can host very different species with distinct functional traits, or two assemblages with contrasted species richness but hosting species with similar extreme trait values can have the same FRic. In such case, considering the environmental determinants of functional diversity might reveal the influence of some variables hidden for the species richness facet. However, FRic is often positively correlated with species richness

(Villeger *et al.* 2008, Mouchet *et al.* 2010), and does not inform on the species position/distribution within the functional space, as it only consider the species with the most extreme traits (Villeger *et al.* 2008). Hence, functional metrics complementing FRic are needed to provide a more comprehensive description of the functional diversity of species assemblages. The distribution of species within assemblages can be assessed by the functional dispersion, which measures the mean distance of species to the centroid of the functional space (FDis, Laliberté & Legendre 2010). In addition, the position of the species assemblage in the functional space can be assessed by the functional identity, which measures the average position of species on each functional axis (FIde, Mouillot *et al.* 2013). Those three above cited indices are independent and, as show on Figure 1, FDis and FIde can take contrasted values for a given FRic. For instance, two species assemblages with the same FRic due to the similar extreme species, can have low FDis when most of the non-extreme species are located around the centroid (Figure 1a, Assemblage 1 & Assemblage 3), or have high FDis when most of the non-extreme species have almost extreme trait values and disperse to the margins of the functional space (Figure 1a, Assemblage 2 & Assemblage 4). Combining those three indices, rather than focusing only on FRic, allows a better understanding of how current assemblages fill the functional space, and should provide new insights on how environmental and historical factors shape functional diversity. As a consequence, assessing the role of habitat, energy and history related hypotheses on these three facets of functional diversity can help understanding how the related mechanisms have promoted ecosystem processes and how these processes may be affected by current and future global and regional anthropogenic changes.

Using a functional trait database providing information on food acquisition and locomotion for more than 10 000 freshwater fish species (*Su et al.* 2021), we evaluated the contributions of three hypotheses (i.e. climate/energy, habitat area/diversity, and history/geography) on three complementary facets of functional diversity (FRic, FDis and FIde) in 2 453 river basins covering almost the entire continental areas of the globe. Freshwater fishes, which count *ca.* 18 000 species, have colonized all freshwater habitats of the globe except the poles. They account for more than one-quarter of all the vertebrates on earth (Tedesco *et al*. 2017). Unlike terrestrial

animals that can disperse freely across the land, freshwater fishes can only live and disperse through the river networks set by the drainage basins, which are separated from one another by natural barriers (i.e. oceans or land) and thus constitute island-like systems (Oberdorff *et al*. 2011, Tedesco *et al*. 2012). Previous studies reported that patterns in species richness are driven by combinations of energy (temperature) and habitat (drainage area) conditions, and to a lesser extent historical legacy (climate fluctuations) (Oberdorff *et al*. 1997, 2011). So far, no study has explored how these main hypotheses explaining species diversity across the globe apply to the different facets of functional diversity for freshwater fishes.

We here intend to disentangle the interplay between species richness, history/geography, habitat size/diversity and climate/energy related variables on the global patterns of freshwater fish functional diversity. and predict i) that FRic patterns are congruent with species richness patterns and hence related to the same drivers, ii) that habitat area negatively affects FDis and FIde, because large river basins present a larger diversity of habitats, promoting low functional dispersion and identity through very diversified morphologies, and iii) that energy availability should also promote low functional dispersion and identity by providing diverse food resources, while lower energy regions should select for particular morphologies leading to increased FDis and FIde. These hypotheses might nevertheless also depend on the historical and geographic context, with rivers subjected to ancient isolation and recent glaciations having their fauna filtered for particular traits and promoting therefore higher FDis and FIde in such rivers.

## Materials and methods

**Occurrence data:** We used the most comprehensive database of freshwater fish species distributions across the world (Tedesco *et al*. 2017; available at http://data.freshwaterbiodiversity.eu). The fish occurrence database gathers the occurrence of 14 953 species (more than 90% of the freshwater fish species) in 3 119 drainage basins, covering more than 80% of the earth surface (Tedesco *et al*. 2017). We only considered the historical occurrence, i.e. records for native species.

**Functional traits:** The functional database was obtained from Su *et al* (2021), which

encompasses ten morphological traits for 10 682 freshwater fish species out of the 14 953 species present in the occurrence database. It hence covers almost 60% of the documented world freshwater fish fauna. The ten traits describe the size and shape of body parts involved in food acquisition and locomotion (Villéger *et al*. 2017). Among them, body size is a key trait related to all functions driven by metabolism (Blanchet *et al*. 2010) and was estimated as the maximum body length registered on FishBase (www.fishbase.org, Froese & Pauly 2018). In addition to size, nine unit-less traits describing the morphology of the fish head (including mouth and eye), body, pectoral and caudal fins were computed as ratios between the 11 morphological measures done using ImageJ software (http://rsb.info.nih.gov/ij/index.html). Please see Figure S4 in Su *et al* (2021) for the detailed measurements. The ten morphological traits (nine unit-less ratios and body size) selected are commonly used in assessments of fish functional diversity (e.g. Bellwood *et al*. 2014; Su *et al*. 2019, 2020, 2021; Toussaint *et al*. 2016; 2018, Villéger *et al.* 2010) and are linked to the feeding and locomotion functions of fish that themselves determine their contribution to key ecosystem processes such as controlling food webs and nutrient cycles (Villéger *et al*. 2017). Overall, 24.1% of the values were missing in the original dataset (from 6.9% for maximum body length to 31.4% for relative maxillary length). We filled these missing values by using a phylogenetic generalized linear model, which includes phylogenetic information to improve the imputation performance (Bruggeman *et al*. 2009, Penone *et al*. 2014). Phylogenetic information was obtained from R package 'fishtree' (Rabosky *et al.* 2018, Chang *et al.* 2019) and the model was performed using the 'phylopars' function from the R package 'Rphylopars' (Bruggeman *et al*. 2009). The efficiency of this model in filling missing

values was tested on a random set of 1 000 species with complete values for all ten traits. We randomly set 25% of the values for the 1 000 species as missing values, and then filled them with simulated values. We then compared the simulated values to the actual values. This procedure was repeated 999 times. Spearman's rho between actual and simulated data was used to measure the efficiency of the procedure. Spearman's rho varied from 0.86 to 0.96, demonstrating the efficiency of the method. As a comparison with a classical imputation method, filling the gaps using the average trait value of the 75% species with data, the Spearman's rho varied from 0.80 to 0.91, showing that the 'phylopars' procedure outperforms the classical imputation method. We then computed a principal component analysis (PCA) using the 10 morphological traits for all the species. We selected the first five PCA axes, which explained 78.4% of the total variance among the world's fish functions, to compute the functional diversity indices (Figure S1).

**Functional diversity indices：** We calculated three complementary functional diversity indices — which are functional richness (FRic), functional dispersion (FDis) and functional identity (FIde) — to assess the functional structure of fish assemblages in each river basin.

Functional richness (FRic, Villeger *et al*. 2008), measures the volume of the minimum convex hull that includes all the species in the five-dimensional functional space. The higher the FRic, the higher the range of functional trait values in the species assemblage considered. Thus, FRic accounts only for the species with the most extreme trait values and thus tends to be positively correlated with species richness through a sampling effect (Villeger *et al*. 2008).

Functional dispersion (FDis, Laliberté & Legendre 2010) measures the mean distance of species in the five-dimensional functional space to the centroid of all species in the assemblage. Here we scaled it between 0 and 1 by dividing its raw value by half of the maximum distance among all the species present in the global species pool, which is the maximum value possible

given species trait values (Mouillot *et al*. 2013). FDis is a priori not related to species richness (Laliberté & Legendre 2010).

Functional identity (FIde) measures how the species distribute in the functional space and is calculated as the average position of the species from an assemblage on each axis of the global functional space (Toussaint *et al*. 2018). Here we consider FIde on the first five PC axes to evaluate the species distributions in the functional space.

We used R scripts from Mouillot *et al* (2013) to compute the three functional diversity indices.

**Predictor variables:** Overall, 13 predictor variables were selected to evaluate the effect of the three hypotheses, i.e. habitat size/diversity, climate/energy and history/geography, on the variation in freshwater fish diversity.

Climate/energy was assessed by five predictors: mean annual temperature (TEM), mean annual precipitation (PRE), mean annual evapotranspiration (EVP), mean annual runoff (ROF) and mean annual net primary productivity (NPP).

Habitat was assessed using four predictors: river basin area (RBA), altitudinal range (ALR), mean slope (MSP) and river basin diversity (RBD). RBD was estimated by applying Shannon's diversity index to proportions of biomes (i.e. vegetation types associated with regional variations in climate) within river basins (Oberdorff *et al*. 2011).

History/geography was assessed by four predictors: temperature anomaly during the quaternary period (TEA), basin median latitude (BML), Ecoregion (ECO, i.e. the six biogeographic realms: Afrotropical, Australian, Nearctic, Neotropical, Oriental and Palearctic; Lévêque *et al*. 2008 and Brosse *et al*. 2013), and the Endorheic/ Exorheic status (EOC) of the river that relates to the level of isolation of the river basin. TEA was measured as the change in the mean annual temperature between the present and Last Glacial Maximum conditions (the average values based on two Global Circulation Models, Table S1, Dias *et al*. 2014).

**Statistical analysis:** Combining the occurrences, functional traits and environmental databases described above permitted to describe the functional diversity of 2 453 freshwater fish assemblages across the world and analyse the relationship between the functional diversity and

the environmental variables. We applied boosted regression trees (BRT) to assess the relative importance of each considered variable in shaping the patterns of FRic, FDis and FIde. In addition to the 13 predictor variables, we also added the total number of native species (i.e. species richness, SR) present in each river basin as a predictor in all the BRT models (see Table S1 for a list of variables, corresponding references and units).

Although BRT are considered as relatively robust to collinearity among predictor variables, it has been recently shown that removing predictor variables showing a Pearson correlation coefficient higher than |0.7| is a desirable practice with techniques based on boosting (Parravicini *et al.* 2013). We hence removed precipitation from the BRT models, which was highly correlated with evapotranspiration and runoff (Figure S2).

BRT were fitted using the 'gbm.step' function in 'dismo' package in R (Elith *et al*. 2008). This technique allows for the specification of four main parameters: bag fraction (*bf*), learning rate (*lr*), tree complexity (*tc*) and the number of trees (*n*t). *bf* is the proportion of samples used at each step, *lr* is the contribution of each fitted tree to the final model, *tc* is the number of nodes of each fitted tree determining the extent to which statistical interactions were fitted, and *nt* represents the number of trees corresponding to the number of boosting iterations.

Optimal setting of the parameters was chosen using 10-fold cross validation (CV). The procedure provides a parsimonious estimate, CV – $D^2$ (i.e. the cross validated proportion of the deviance explained), representing the expected performance of the model when fitted to new data (Elith *et al*. 2008). Using CV, we explored different combinations of the parameters to be set and retained the model showing the highest CV – $D^2$ (Table S2).

The relative influence of predictor variables in the final optimal BRT models was evaluated with the contribution to model fit attributable to each predictor, averaged across all the trees fitted (Friedman 2001). In order to account for uncertainty around these estimates, the relative influence of each predictor variable was calculated on 1000 bootstrap replicates of the original dataset in order to compute 95% confidence intervals by using the 'gbm.fixed' function in 'dismo' package and 'boot' function in 'boot' package. We also estimated the interactions between each pair of predictors by using the 'gbm.interactions' function in the 'dismo' package (Elith *et*

*al*. 2008). Spatial autocorrelation has become an important issue in geographical ecology and macroecology over the last decades (Rangel *et al*. 2010, Leprieur *et al*. 2011). As BRT do not account for spatial autocorrelation in both the dependent and predictor variables, we also performed autoregressive error ($SAR_{error}$) models and compared the respective results of the two models. To fit the $SAR_{error}$ models, we transformed the categorical variables (i.e. Ecoregion and Endorheic) to numeric types. Then we scaled all the 13 predictor variables to a zero mean and unit variance to ensure equal weighting in the models. The spatial autocorrelation analysis was performed using the 'spatialreg' and 'spdep' packages (Bivand & Piras 2015).

All statistical analyses were performed with R software version 3.5.1 (R Core Team, 2019).

**Results**

We first mapped FRic, FDis (Figure 2) and FIde (measured on PC1 to PC5; Figure 3) of freshwater fish assemblages at the global scale. Those three facets of functional diversity were mostly independent of each other (Pearson's $r < |0.51|$, Figure S3). Besides, they highly varied among river basins between and within the six realms. The highest values of FRic were disproportionately concentrated in river basins located in the Neotropical realm (e.g. Amazon river reached the highest FRic value, 44% of the world FRic) and some large basins from other realms, whereas low values were mostly located in the two northern realms (i.e. Nearctic and Palearctic) and Australian realm (Figure 2a). The patterns of FDis and FRic were weakly correlated to each other (Pearson's $r = 0.11$). FDis values ranged from 0.06 to 0.38, with the highest values concentrated in river basins from Europe (Figure 2b). Overall, the patterns of FIde on the five PC axes showed latitudinal gradients for some axes. Specifically, FIde tends to increase from the equator to the poles on PC1, tends to decrease with latitude from south to north on PC2 and PC3, and tends to increase with latitude from south to north on PC4 and PC5 (Figure 3, Figure 4).

The final BRT model for FRic explained 80.2% of the total variation (cross validation procedure; Table S2). The relationship between observed and predicted FRic was remarkably high ($R^2 = 0.987$ for a regression of slope 1 and 0 intercept, Figure S4). As expected, species

richness was found to be the best predictor of FRic, with a relative contribution to the BRT model reaching 72.2%. Other variables had a marginal contribution and only Ecoregion (7.2%) contributed more than 5% to this pattern, whereas all the remaining variables related to the three hypotheses did not play strong roles (i.e. relative contribution < 5%) in shaping the pattern of FRic, independently from species richness (Figure 2a).

For FDis, the BRT model explained 46.9% of the total variation (cross validation procedure; Table S2). The relationship between observed and predicted FDis was also high ($R^2 = 0.994$ for a regression of slope 1 and 0 intercept, Figure S4). Together the climate/ energy related variables contributed the most to the FDis pattern (37.4%), followed by the history/geography related variables (30.2%) and habitat related variables (23.9%). However, median latitude was the most influential variable (15.2%), followed by the net primary productivity (13.4%). All the other variables had a contribution lower than 10% (Figure 2b).

The BRT models explained about 65% of the total deviance for FIde on PC1 to PC5, respectively (Table S2). The relationships between observed and predicted FIde on the five PC axes remained high, with R-squared values higher than 0.9 (Figure S4). However, FIde on the five PC axes displayed contrasting relationships with variables related to the different hypotheses. Indeed, results of FIde on PC1 showed that history/geography related variables (47.0%), especially the Ecoregion (22.6%) and median latitude (15.0%), contributed most to the pattern (Figure 3a). FIde on PC2 and PC3 was mainly influenced by climate/energy related variables (43.5% and 50.5% for PC2 and PC3, respectively), with a secondary influence of habitat size/diversity related variables (29.4%) on PC2 and of history/geography related variables (27.7%) on PC3, (Figures 3b and 3c). In contrast, FIde on PC4 and PC5 was mainly influenced by history/geography related variables (44.9% and 47.5%, for PC4 and PC5, respectively), among which, median latitude alone contributed more than 30% to FIde on both axes (31.2% and 35.2%, respectively). Climate/energy related variables were also influential and contributed to 32.1% and 29.1% to FIde on PC4 and PC5, respectively (Figures 3 e,f).

BRTs considering interactions between predictors performed better than simpler BRTs (Table S2). Although interactions between variables were weak in the models for FRic and FDis

(Figure 5 a,b), models taking interactions into account still had better performances (Table S2). In contrast, some variable interactions were very strong in the models for FIde on the five PC axes (Figure 5 c-g). Basin median latitude, in particular, was the most 'interacting' variable and had relatively strong interactions in most of the models for FIde, especially with ecoregion, temperature, and evapotranspiration, thus highlighting that these factors have different effects in basins located in different latitudes (Figure 5 c-g).

Finally, BRT and $SAR_{error}$ models showed broadly similar results (Table S3). Predictor variables having strong influence on the functional diversity in BRT also showed significant and relatively high coefficients in the $SAR_{error}$ models (Figures 2, 3, Table S4). In addition, the $SAR_{error}$ models reduced the residual spatial autocorrelation to non-significant levels for all the functional diversity indices.

**Discussion**

Considering the functional diversity through three complementary indices (Figure 1), allowed to quantify the role of climate/energy, habitat size/diversity, history/geography processes on the current functional diversity of the world freshwater fish faunas. We also show that the relative role of those processes differs from those known for species richness (Guegan *et al*. 1998, Oberdorff *et al*. 2011). Functional richness was nevertheless strongly influenced by the richness in species, which is itself ruled primarily by habitat diversity and energy availability (Guegan *et al*. 1998, Oberdorff *et al*. 2011). The tight linkage between species and functional richness could be partly explained by the expected increase of the range of traits with increasing number of species in an assemblage (Villeger *et al*. 2008). Nevertheless this relationship is not constrained to linearity, and it might saturate once most of the range of traits are represented. Therefore, above a certain number of species, the relationship between species and functional richness should saturate, indicating a functional redundancy among species. Such saturation was only observed here in the richest river basins (above 400 species, without strong biogeographic realm effect (see Figure 4a)), while at the realms scale, contrasted ratios between species and functional richness have been observed (Toussaint *et al*. 2016).

In contrast, species richness had a limited influence (< 9%) on the patterns of functional dispersion and identity, attesting the multidimensionality of the functional diversity. Instead, variables related to history/geography and climate/energy hypotheses primarily explained the patterns of FDis and FIde. We here reveal a latitudinal gradient in FDis (Figure 4b), where values increased from South to North, and reach maximal values in the high latitude basins of the northern hemisphere. Those basins experienced larger amplitude of Quaternary climatic oscillations (Jansson 2003, Lepieur *et al.* 2011), explaining the overabundance of locomotion-related extreme trait values in the fish assemblages of the temperate and cold areas of the Northern hemisphere. Indeed, those river basins experienced glaciations events during the Quaternary, which caused the extinction of most small sized species with low dispersal ability (Griffiths 2006). Following glaciations, glaciers retreat opened available habitats, which were preferentially recolonized from southern refuge areas (e.g. the Mississippi and Danube drainage basins in Nearctic and Palearctic realms, respectively) through intermittent inter-basin connections during the post-glacial period (Reyjol *et al*. 2007, Leprieur *et al*. 2011). Those historical processes selected species with extreme dispersal traits (e.g. large body size) among the pool of species that survived in the glacial refuges. The interplay of these two processes therefore explains why fish assemblages in temperate and cold rivers of the northern hemisphere have a low functional richness, because of the low species richness, but a high functional dispersion, because morphologies promoting dispersal were the most prone to colonize new basins after glaciers retreat.

Dispersal and locomotion ability are strongly linked to body size (Griffiths 2006, Tedesco *et al*. 2012) and caudal fin size (Webb *et al*. 1984, Su *et al*. 2020, Radinger & Wolter 2014), which are strongly influential variables on PC3, PC4 and PC5 of the functional space (maximum body length and caudal peduncle throttling, see Figure S1), explaining the latitudinal gradient in functional identity observed on those PC axes. This finding parallels the patterns observed for functional dispersion, testifying that species with high dispersal abilities tend to have extreme trait values within assemblages (FDis) and in the global functional space (FIde). The functional traits of those species underline the high functional dispersal ability of fishes in

the assemblages found in previously glaciated river basins of the northern hemisphere. These results are in line with Blanchet *et al* (2010) who found that large-bodied fish species dominate in higher latitudes of the northern hemisphere. Nevertheless, this trend was not verified in the southern hemisphere, probably because of the limited surface of continental areas in the temperate and cold regions. Moreover, those southern temperate fauna are mainly composed of marine lineages that secondarily colonized freshwaters (e.g. Gobiidae and Galaxidae in southern South America, South Australia and New Zealand; McDowall 1978, Leveque *et al*. 2008). In those regions, the colonization of freshwater environments by marine fishes may be more related to physiological constraints (fish ability to deal with gradients of salinity) than on dispersal capacities. However, beyond the strong influence of latitude, the realm effect also markedly shapes the functional identity patterns (Figure 4 c-g, Figure S5), which is consistent with the result from Su *et al* (2019) that observed generally different morphological traits of fishes across realms. For instance, fish assemblages from different realms tend to have distinct traits related to feeding habits and trophic levels (See FIde PC1 on Figures 3, 4, and S1), with fish assemblages from Afrotropical and Neotropical realms having lower trophic levels than in other realms. This could be attributed to the distinct evolutionary history of fishes from different realms under distinct environmental constraints (Su *et al*. 2019). For instance, algae browsers are overrepresented in the Neotropics (mainly represented by Siluriforms from the Loricaridae family, but also by some Characiforms), explaining the strong contribution of this realm to FIde on PC1.

Our findings also suggest that variables related to climate/energy hypotheses contributed to shape the patterns of FDis and FIde. For instance, FIde on PC2 and PC3, which mostly represent the fish body elongation and shape, are most influenced by climate/energy related variables (Figure 3). The shifts of species position in the functional space along the climate/energy variables might be explained by two different mechanisms: i) divergent traits make the fishes from a given assemblage more efficient in using the highly diversified food resources present in more productive regions (e.g. with high NPP values, Wright 1983); ii) the functional traits reflect the physiological limits of the species which are constrained by climate

related variables (e.g. temperature, Hawkins *et al*. 2003).

To conclude, our study shows that the historical legacy played a significant role in the global FDis and FIde patterns of freshwater fish, opening insights into the relative role of historical and environmental determinants on the functional structure of fish assemblages. In contrast, the FRic was mainly determined by habitat size/diversity hypothesis, due to its strong dependency to species richness. More generally, our results strengthen that the independence of facets of functional diversity from the species richness makes them essential biodiversity variables (*sensu* Pereira *et al*. 2013) to understand the structure of communities and their changes through global changes. Global hotspots of species richness have been widely used to determine priority areas for conservation, while other facets of biological diversity are still under considered (but see Su et al 2021). Our results provide evidence of the distinct processes shaping taxonomic and functional facets of biodiversity and thus advocate for considering functional diversity measures in global freshwater conservation priorities.

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**Data availability statement**

Fish occurrence data can be retrieved freely from https://doi. org/10.6084/m9.figshare.c.3739145, and metadata are available in Tedesco et al. (2017). Fish functional trait data can be retrieved freely from at https://doi.org/10.6084/m9.figshare.13383170., and metadata are available in Su et al. (2021). Environmental variable data sources were given in Table S1.

**References**

Barnagaud, J. Y., Kissling, W. D., Tsirogiannis, C., Fisikopoulos, V., Villeger, S., Sekercioglu, C. H., & Svenning, J. C. (2017). Biogeographical, environmental and anthropogenic determinants of global patterns in bird taxonomic and trait turnover. *Global Ecology and Biogeography, 26*, 1190-1200.

Bellwood, D., Goatley, C., Brandl, S., & Bellwood, O. (2014). Fifty million years of herbivory on coral reefs: fossils, fish and functional innovations. *Proceedings of the Royal Society B: Biological Sciences, 281*, 20133046.

Bennie, J. J., Duffy, J. P., Inger, R., & Gaston, K. J. (2014). Biogeography of time partitioning in mammals. *Proceedings of the National Academy of Sciences, 111*, 13727-13732.

Bivand, R., & Piras, G. (2015). *Comparing implementations of estimation methods for spatial econometrics*.

Blanchet, S., Grenouillet, G., Beauchard, O., Tedesco, P. A., Leprieur, F., Dürr, H. H., . . . Brosse, S. (2010). Non-native species disrupt the worldwide patterns of freshwater fish body size: implications for Bergmann's rule. *Ecology Letters, 13*, 421-431.

Brosse, S., Beauchard, O., Blanchet, S., Dürr, H. H., Grenouillet, G., Hugueny, B., . . . Villéger, S. (2013). Fish-SPRICH: a database of freshwater fish species richness throughout the World. *Hydrobiologia, 700*, 343-349.

Bruggeman, J., Heringa, J., & Brandt, B. W. (2009). PhyloPars: estimation of missing parameter values using phylogeny. *Nucleic acids research, 37*, W179-W184.

Chang, J., Rabosky, D. L., Smith, S. A., & Alfaro, M. E. (2019). An R package and online resource for macroevolutionary studies using the ray-finned fish tree of life. *Methods in Ecology and Evolution, 10*, 1118-1124.

Collen, B., Whitton, F., Dyer, E. E., Baillie, J. E., Cumberlidge, N., Darwall, W. R., . . . Böhm, M. (2014). Global patterns of freshwater species diversity, threat and endemism. *Global Ecology and Biogeography, 23*, 40-51.

Dias, M. S., Oberdorff, T., Hugueny, B., Leprieur, F., Jézéquel, C., Cornu, J. F., . . . Tedesco, P. A. (2014). Global imprint of historical connectivity on freshwater fish biodiversity. *Ecology Letters, 17*, 1130-1140.

Elith, J., Leathwick, J. R., & Hastie, T. (2008). A working guide to boosted regression trees. *Journal of Animal Ecology, 77*, 802-813.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of statistics*, 1189-1232.

Froese, R., & Pauly, D. (2018). Fishbase. Retrieved 21 February 2018

Gaston, K. J. (2000). Global patterns in biodiversity. *Nature, 405*, 220-227.

Griffiths, D. (2006). Pattern and process in the ecological biogeography of European freshwater fish. *Journal of Animal Ecology, 75*, 734-751.

Guégan, J.-F., Lek, S., & Oberdorff, T. (1998). Energy availability and habitat heterogeneity predict global riverine fish diversity. *Nature, 391*, 382-384.

Hawkins, B. A., Field, R., Cornell, H. V., Currie, D. J., Guégan, J.-F., Kaufman, D. M., . . . O'Brien, E. M. (2003). Energy, water, and broad-scale geographic patterns of species richness. *Ecology, 84*, 3105-3117.

Jansson, R. (2003). Global patterns in endemism explained by past climatic change.

*Proceedings of the Royal Society of London. Series B: Biological Sciences, 270*, 583-590.

Kuczynski, L., Côte, J., Toussaint, A., Brosse, S., Buisson, L., & Grenouillet, G. (2018). Spatial mismatch in morphological, ecological and phylogenetic diversity, in historical and contemporary European freshwater fish faunas. *Ecography, 41*, 1665-1674.

Laliberté, E., & Legendre, P. (2010). A distance-based framework for measuring functional diversity from multiple traits. *Ecology, 91*, 299-305.

Leprieur, F., Tedesco, P. A., Hugueny, B., Beauchard, O., Dürr, H. H., Brosse, S., & Oberdorff, T. (2011). Partitioning global patterns of freshwater fish beta diversity reveals contrasting signatures of past climate changes. *Ecology Letters, 14*, 325-334.

Lévêque, C., Oberdorff, T., Paugy, D., Stiassny, M., & Tedesco, P. (2008). Global diversity of fish (Pisces) in freshwater. *Hydrobiologia, 595*, 545-567.

McDowall, R. M. (1978). Generalized tracks and dispersal in biogeography. *Systematic zoology, 27*, 88-104.

Mouchet, M. A., Villéger, S., Mason, N. W., & Mouillot, D. (2010). Functional diversity measures: an overview of their redundancy and their ability to discriminate community assembly rules. *Functional Ecology, 24*, 867-876.

Mouillot, D., Graham, N. A., Villéger, S., Mason, N. W., & Bellwood, D. R. (2013). A functional approach reveals community responses to disturbances. *Trends in Ecology & Evolution, 28*, 167-177.

Mouillot, D., Villéger, S., Scherer-Lorenzen, M., & Mason, N. W. (2011). Functional structure of biological communities predicts ecosystem multifunctionality. *PloS one, 6*, e17476.

Oberdorff, T., Hugueny, B., & Guégan, J. F. (1997). Is there an influence of historical events on contemporary fish species richness in rivers? Comparisons between Western Europe and North America. *Journal of Biogeography, 24*, 461-467.

Oberdorff, T., Tedesco, P. A., Hugueny, B., Leprieur, F., Beauchard, O., Brosse, S., & Dürr, H. H. (2011). Global and regional patterns in riverine fish species richness: a review. *International Journal of Ecology, 2011*, 967631.

Parravicini, V., Kulbicki, M., Bellwood, D., Friedlander, A., Arias-Gonzalez, J., Chabanet, P., . . . D'Agata, S. (2013). Global patterns and predictors of tropical reef fish species richness. *Ecography, 36*, 1254-1262.

Penone, C., Davidson, A. D., Shoemaker, K. T., Di Marco, M., Rondinini, C., Brooks, T. M., . . . Costa, G. C. (2014). Imputation of missing data in life-history trait datasets: which approach performs the best? *Methods in Ecology and Evolution, 5*, 961-970.

Pereira, H. M., Ferrier, S., Walters, M., Geller, G. N., Jongman, R., Scholes, R. J., . . . Cardoso, A. (2013). Essential biodiversity variables. *Science, 339*, 277-278.

Rabosky, D. L., Chang, J., Title, P. O., Cowman, P. F., Sallan, L., Friedman, M., . . . Coll, M. (2018). An inverse latitudinal gradient in speciation rate for marine fishes. *Nature, 559*, 392.

Radinger, J., & Wolter, C. (2014). Patterns and predictors of fish dispersal in rivers. *Fish and fisheries, 15*, 456-473.

Rangel, T. F., Diniz-Filho, J. A. F., & Bini, L. M. (2010). SAM: a comprehensive application for spatial analysis in macroecology. *Ecography, 33*, 46-50.

Reyjol, Y., Hugueny, B., Pont, D., Bianco, P. G., Beier, U., Caiola, N., . . . Ferreira, T. (2007). Patterns in species richness and endemism of European freshwater fish. *Global Ecology and Biogeography, 16*, 65-75.

Stewart, P. S., Voskamp, A., Biber, M. F., Hof, C., Willis, S. G., & Tobias, J. A. (2020). Global impacts of climate change on avian functional diversity. *bioRxiv*.

Su, G., Logez, M., Xu, J., Tao, S., Villéger, S., & Brosse, S. (2021). Human impacts on global freshwater fish biodiversity. *Science, 371*, 835-838.

Su, G., Villéger, S., & Brosse, S. (2019). Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread. *Global Ecology and Biogeography, 28*, 211-221.

Su, G., Villéger, S., & Brosse, S. (2020). Morphological sorting of introduced freshwater fish species within and between donor realms. *Global Ecology and Biogeography, 29*, 803-813.

Team, R. C. (2019). R: A language and environment for statistical computing v. 3.5. 1. Vienna, Austria: Foundation for Statistical Computing. In.

Tedesco, P. A., Beauchard, O., Bigorne, R., Blanchet, S., Buisson, L., Conti, L., . . . Hugueny, B. (2017). A global database on freshwater fish species occurrence in drainage basins. *Scientific Data, 4*, 170141.

Tedesco, P. A., Leprieur, F., Hugueny, B., Brosse, S., Dürr, H. H., Beauchard, O., . . . Oberdorff, T. (2012). Patterns and processes of global riverine fish endemism. *Global Ecology and Biogeography, 21*, 977-987.

Toussaint, A., Charpin, N., Beauchard, O., Grenouillet, G., Oberdorff, T., Tedesco, P. A., . . . Villéger, S. (2018). Non-native species led to marked shifts in functional diversity of the world freshwater fish faunas. *Ecology Letters, 21*, 1649-1659.

Toussaint, A., Charpin, N., Brosse, S., & Villéger, S. (2016). Global functional diversity of freshwater fish is concentrated in the Neotropics while functional vulnerability is widespread. *Scientific reports, 6*, 22125.

Villéger, S., Brosse, S., Mouchet, M., Mouillot, D., & Vanni, M. J. (2017). Functional ecology of fish: current approaches and future challenges. *Aquatic Sciences, 79*, 783-801.

Villéger, S., Mason, N. W. H., & Mouillot, D. (2008). New multidimensional functional diversity indices for a multifaceted framework in functional ecology. *Ecology, 89*, 2290-2301.

Villéger, S., Miranda, J. R., Hernandez, D. F., & Mouillot, D. (2010). Contrasting changes in taxonomic vs. functional diversity of tropical fish communities after habitat degradation. *Ecological Applications, 20*, 1512-1522.

Webb, P. (1984). Body form, locomotion and foraging in aquatic vertebrates. *American Zoologist, 24*, 107-120.

Wright, D. H. (1983). Species-energy theory: an extension of species-area theory. *Oikos*, 496-506.

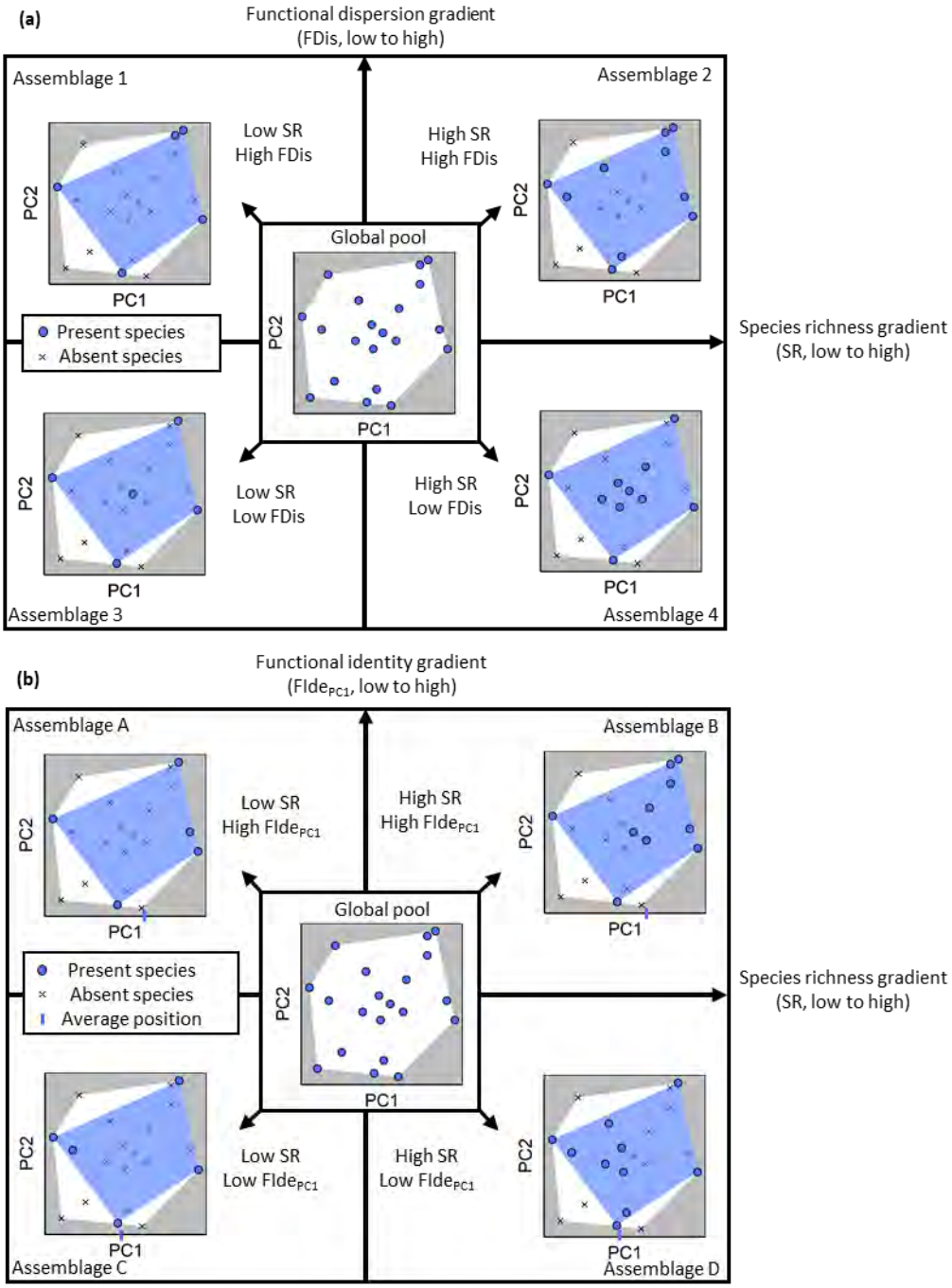



**Figure 1** Contrasted situations of functional dispersion and identity under the same functional richness and two gradients of species richness. Species are represented as blue dots, the functional space is here represented in two dimensions (PC1, PC2). The global species pool is indicated in the center of each plot, the white area represents the global functional diversity. Eight theoretical species assemblages (e.g. fish fauna from eight river basins) illustrate four contrasted situations of functional dispersion (a) and four contrasted situations of functional identity (b). They are characterized by different species richness (low on the left side, high on

the right side) but identical functional richness (blue polygons). The figure illustrates that two species assemblages with same FRic and/or species number can have contrasted FDis or FIde values. Functional dispersion (FDis) is measured as the average distance of species from the centroid of the species present in an assemblage. Functional identity (FIde) is measured as the average of species position along each functional axis (here PC1).

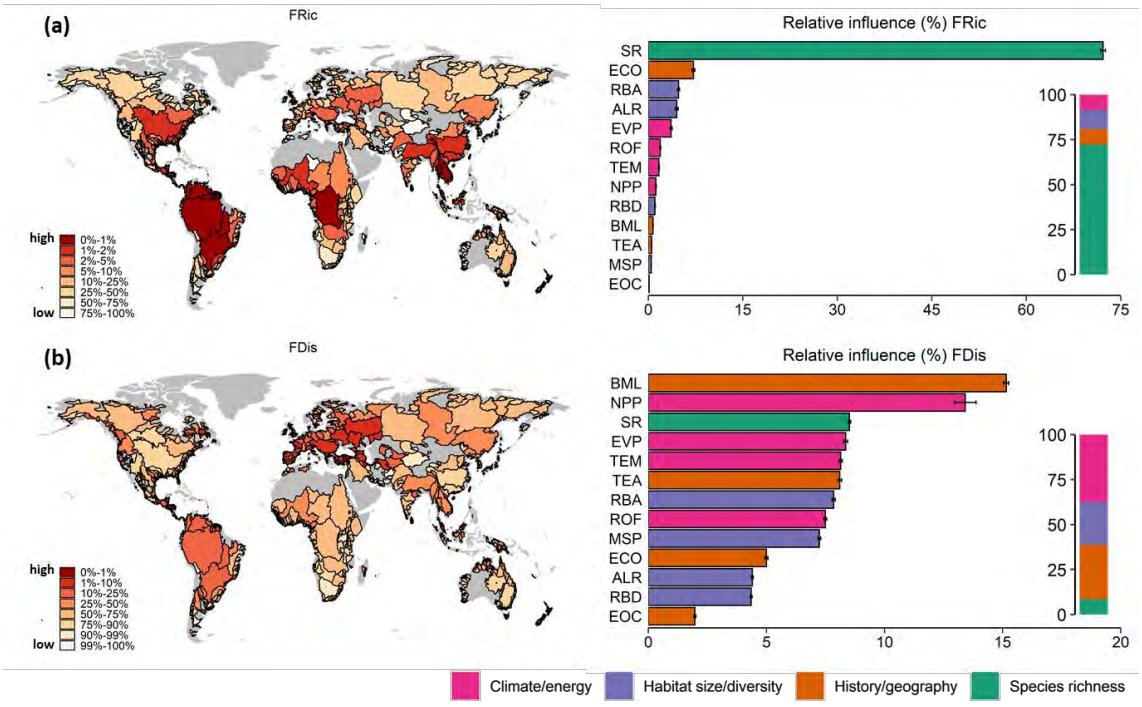



**Figure 2** Global distribution of fish functional richness and functional dispersion (left panels, gradient color shows the percentage of basins with the highest values to the lowest values) and the relative influence of the 13 predictor variables (right panels). (a) Functional richness; (b) Functional dispersion. Colors on the right panels indicate the influence of variables accounting for climate/energy, habitat size/diversity, history/geography, and species richness, on each variable (horizontal bar). The vertical bars on the right summarize the overall influence of the climate/energy, habitat size/diversity, history/geography, and species richness on FRic and FDis . TEM: temperature, EVP: evapotranspiration, ROF: runoff, NPP: net primary productivity; RBA: river basin area, ALR: altitude range, MSP: mean slope, RBD: river basin diversity; TEA: temperature anomaly, BML: basin median latitude, ECO: ecoregion, EOC: endorheic; SR: species richness.

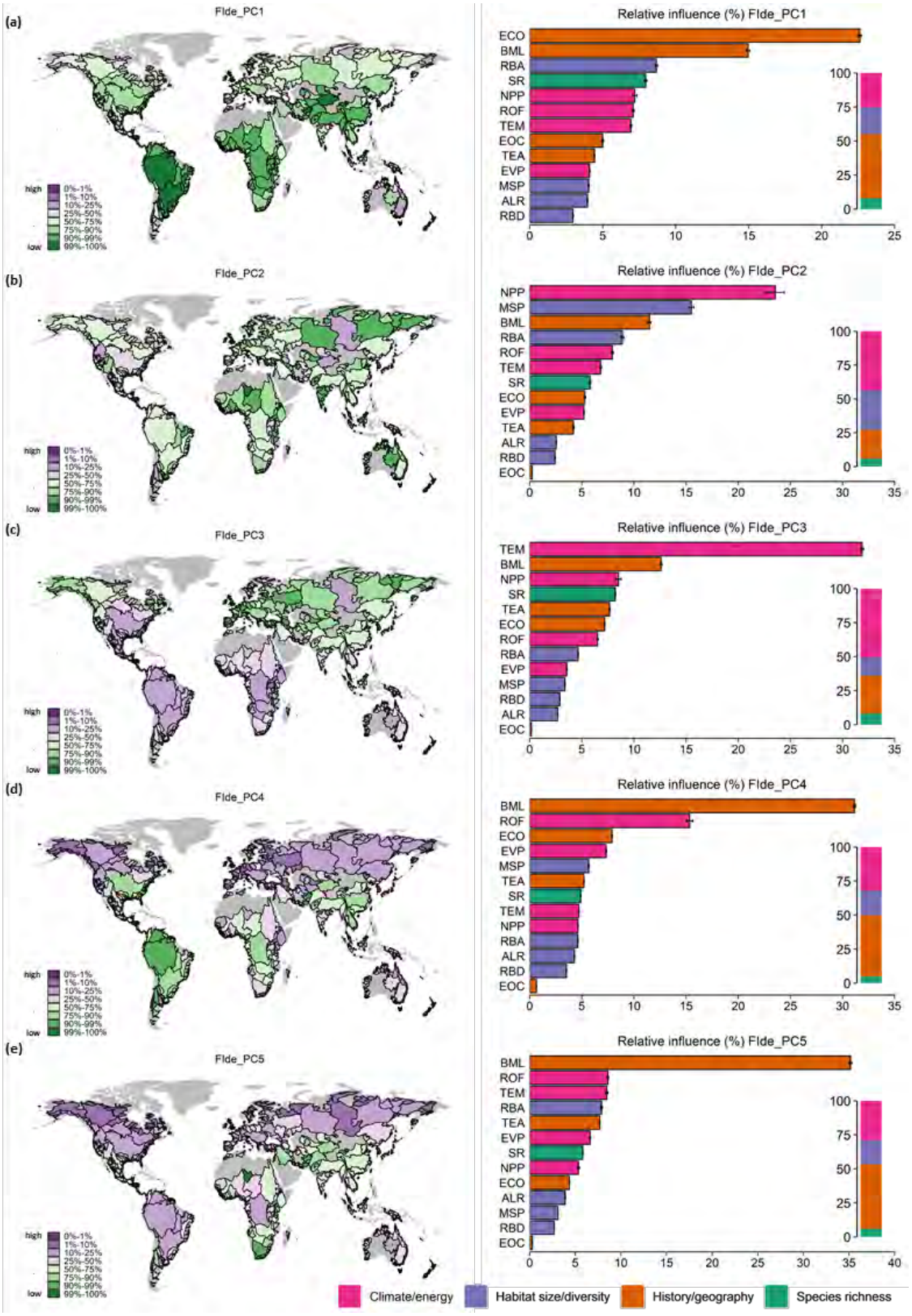



**Figure 3** Global distribution of fish functional identity on PC1 to PC5 (left panels, gradient colors from purple to green show the percentage of basins with the highest to the lowest FIde values) and the relative influence of the 13 predictor variables (right panels). (a-e) Functional

identity on PC1 to PC5 axes. Colors on the right panels indicate the influence of variables accounting for climate/energy, habitat size/diversity, history/geography, and species richness, on each Fide dimension (horizontal bars). The vertical bars on the right summarize the overall influence of the climate/energy, habitat size/diversity, history/geography, and species richness on FIde dimensions. TEM: temperature, EVP: evapotranspiration, ROF: runoff, NPP: net primary productivity; RBA: river basin area, ALR: altitude range, MSP: mean slope, RBD: river basin diversity; TEA: temperature anomaly, BML: basin median latitude, ECO: ecoregion, EOC: endorheic; SR: species richness.

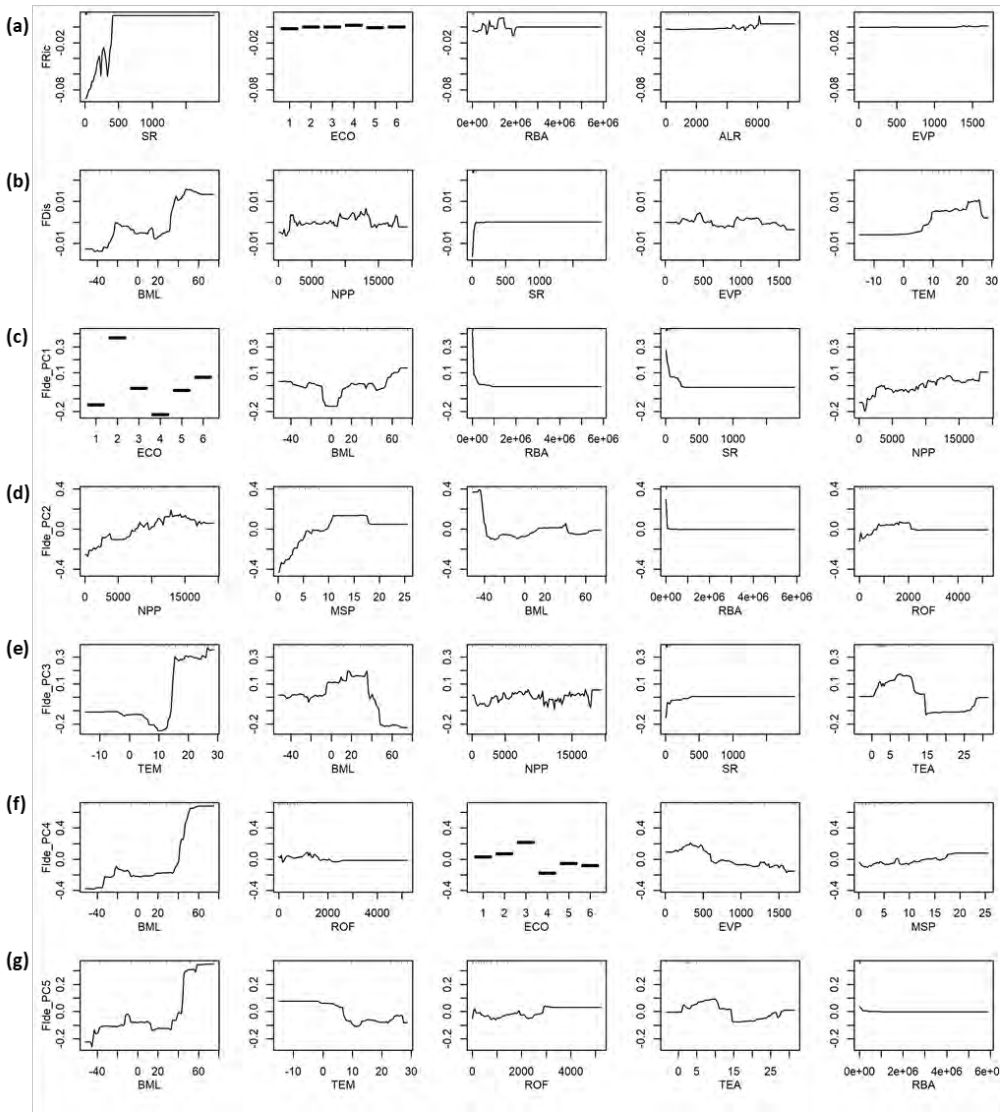



**Figure 4** Results of boosted regression trees showing the partial dependency between each functional diversity facet and the corresponding five most influential predictor variables (see Figure S5 for the remaining variables). Each row shows one functional diversity index: (a) functional richness, (b) functional dispersion, (c-g) functional identity on PC1 to PC5. SR: species richness, ECO: ecoregion, RBA: river basin area, ALR: altitude range, EVP: evapotranspiration, BML: basin median latitude, NPP: net primary productivity, TEM: temperature, TEA: temperature anomaly, ROF: runoff, MSP: mean slope. Numbers on the x axis in ECO panels correspond to the biogeographical realms (1: Afrotropical, 2: Australian, 3: Nearctic, 4: Neotropical, 5: Oriental, 6: Palearctic).

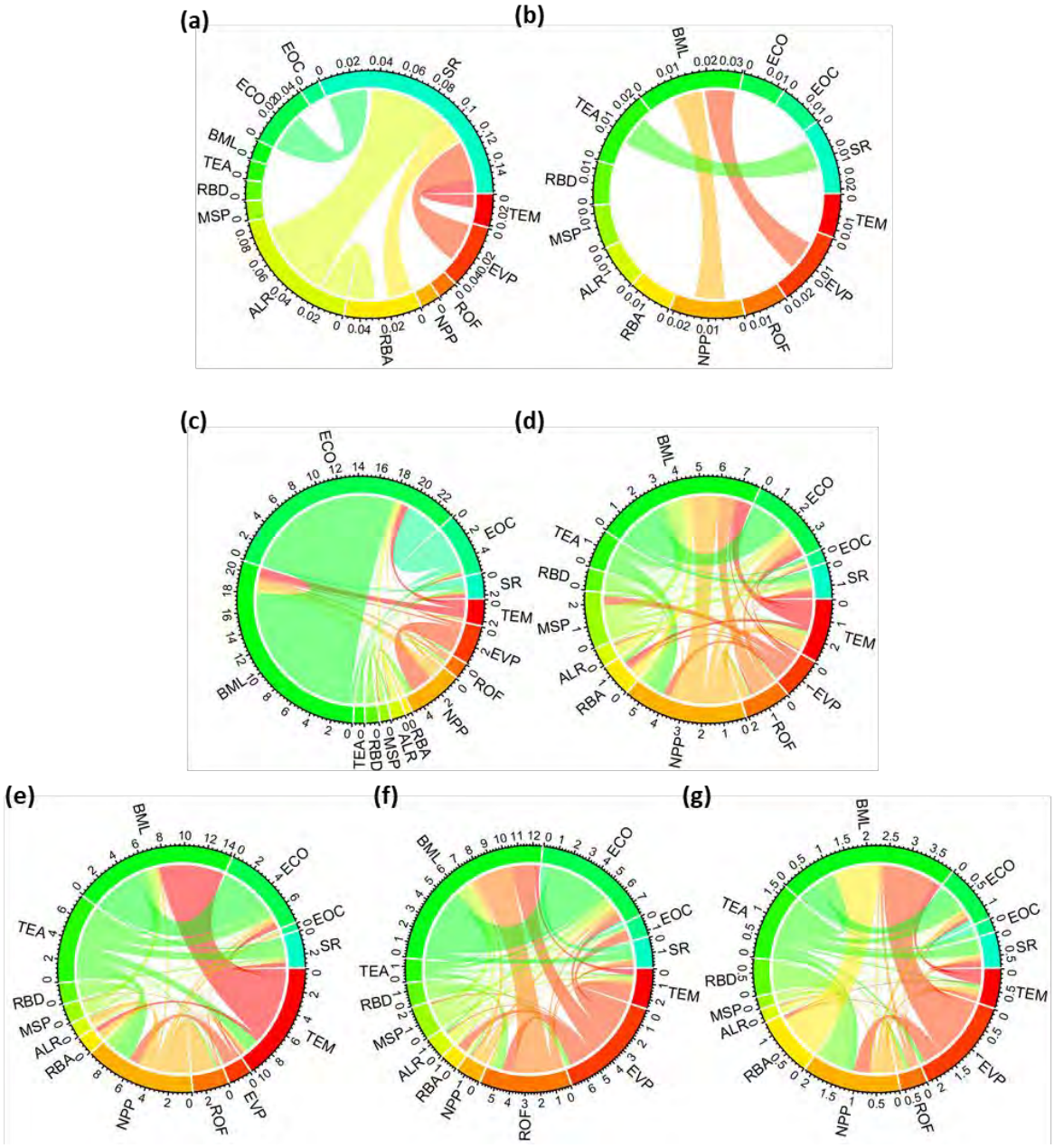



**Figure 5** Interactions between the 13 predictors for each BRT model. (A) functional richness; (B) functional dispersion; (C) to (G) functional identity on PC1 to PC5. TEM: temperature, EVP: evapotranspiration, ROF: runoff, NPP: net primary productivity; RBA: river basin area, ALR: altitude range, MSP: mean slope, RBD: river basin diversity; TEA: temperature anomaly, BML: basin median latitude, ECO: ecoregion, EOC: endorheic; SR: species richness.

**Supporting information**

# Global patterns and determinants of freshwater fish functional diversity

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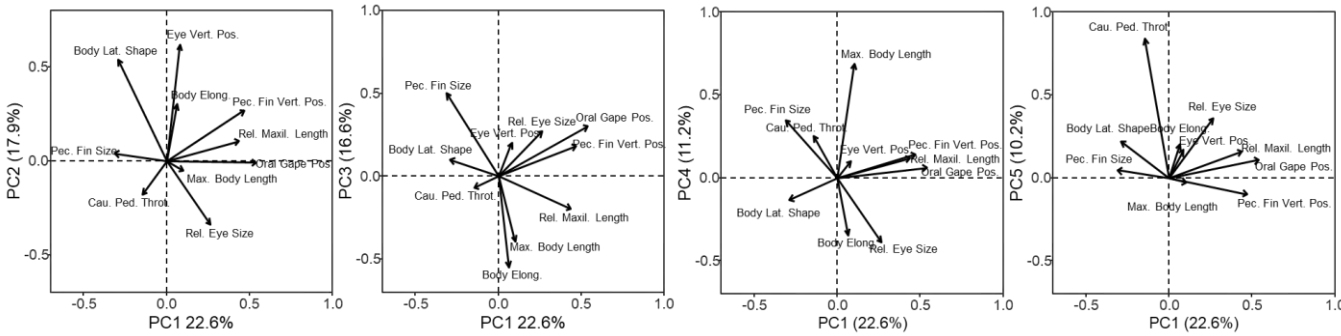
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**Figure S1.** Results of the Principal Components Analysis on morphological traits. (a) Eigenvalues and percentages of variance explained by each axis. (b) Percentage of contribution of each morphological trait to each axis.




a)

| | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|
| Eigenvalues | 2.26 | 1.79 | 1.66 | 1.12 | 1.02 |
| Proportion of Variance | 22.55 | 17.89 | 16.55 | 11.18 | 10.17 |
| Cumulative Proportion | 22.55 | 40.44 | 57.00 | 68.18 | 78.35 |

b)

| | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|
| Relative Eye size | 7.1 | 11.4 | 7.4 | 14.9 | 13.0 |
| Oral gape position | **29.1** | <1 | 9.1 | <1 | 1.2 |
| Relative maxillary length | **19.1** | 1.1 | 3.8 | 1.4 | 2.6 |
| Eye vertical position | <1 | **38.4** | 4.1 | 1.0 | 2.5 |
| Body elongation | <1 | 9.2 | **30.2** | 12.1 | 4.2 |
| Body lateral shape | 8.5 | **29.0** | 1.0 | 1.7 | 4.8 |
| Pectoral fin vertical position | **22.1** | 7.2 | 3.4 | 2.1 | <1 |
| Pectoral fin size | 9.8 | <1 | **25.1** | 12.0 | <1 |
| Caudal peduncle throttling | 2.2 | 3.2 | <1 | 6.5 | **70.4** |
| Maximum body length | 1.0 | <1 | **15.4** | **47.7** | <1 |

The contribution of each morphological trait to the first 5 axes is expressed as percentages and values higher than 15% are in bold font.

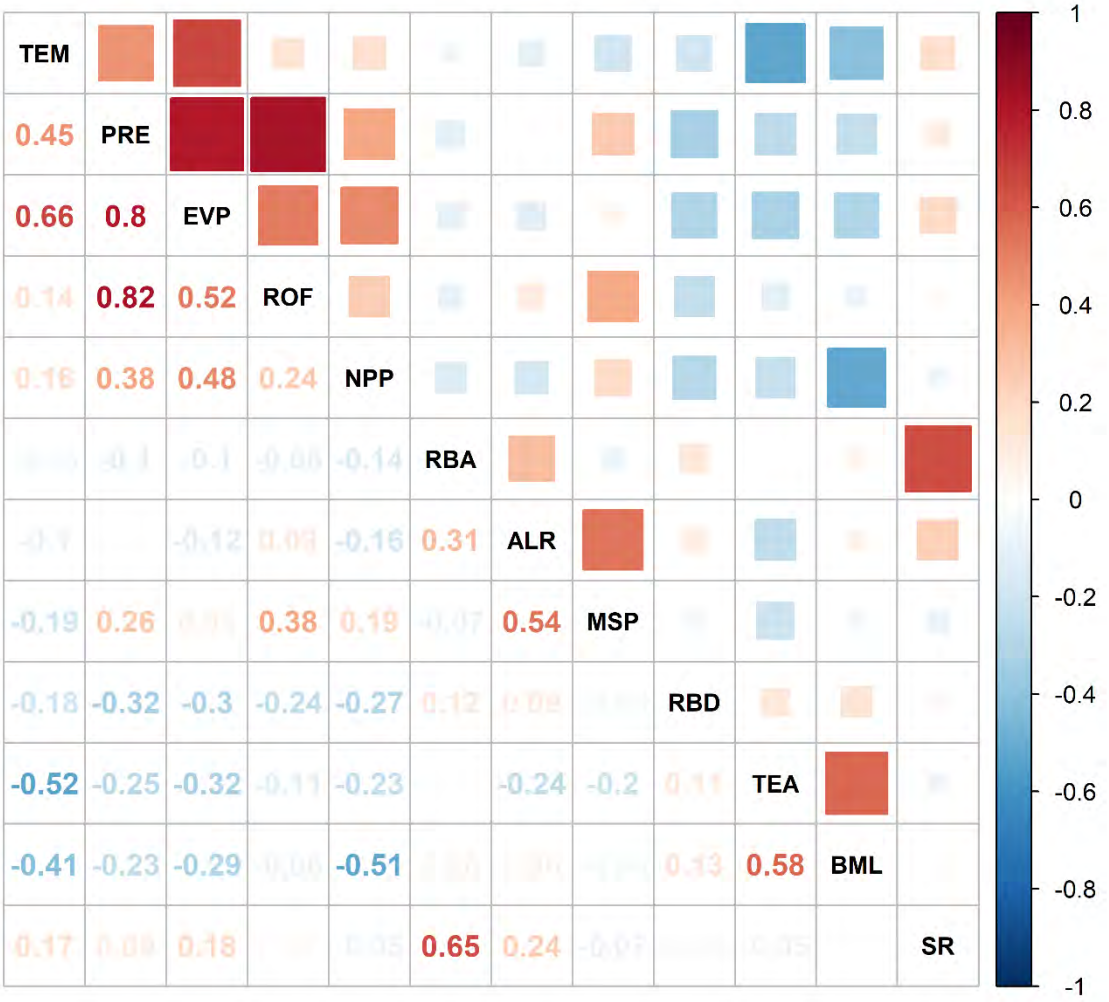



**Figure S2** Pearson correlation between the 12 continuous predictor variables. TEM: temperature, PRE: precipitation, EVP: evapotranspiration, ROF: runoff, NPP: net primary productivity; RBA: river basin area, ALR: altitude range, MSP: mean slope, RBD: river basin diversity; TEA: temperature anomaly during the quaternary period, BML: basin median latitude; SR: species richness.




**Figure S3** Pearson correlation between the 3 facets of functional diversity. FRic: functional richness, FDis: functional dispersion, FIde: functional identity.

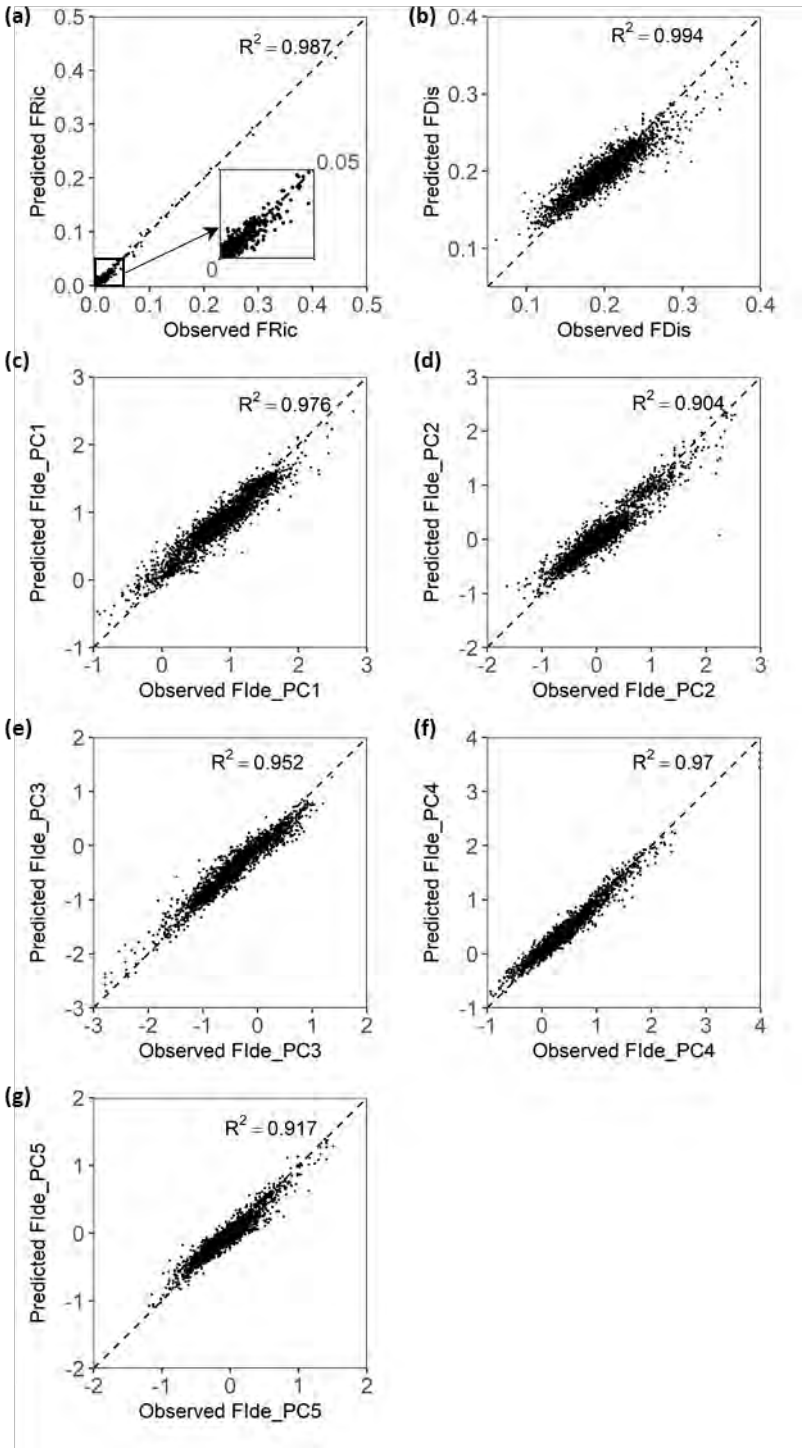
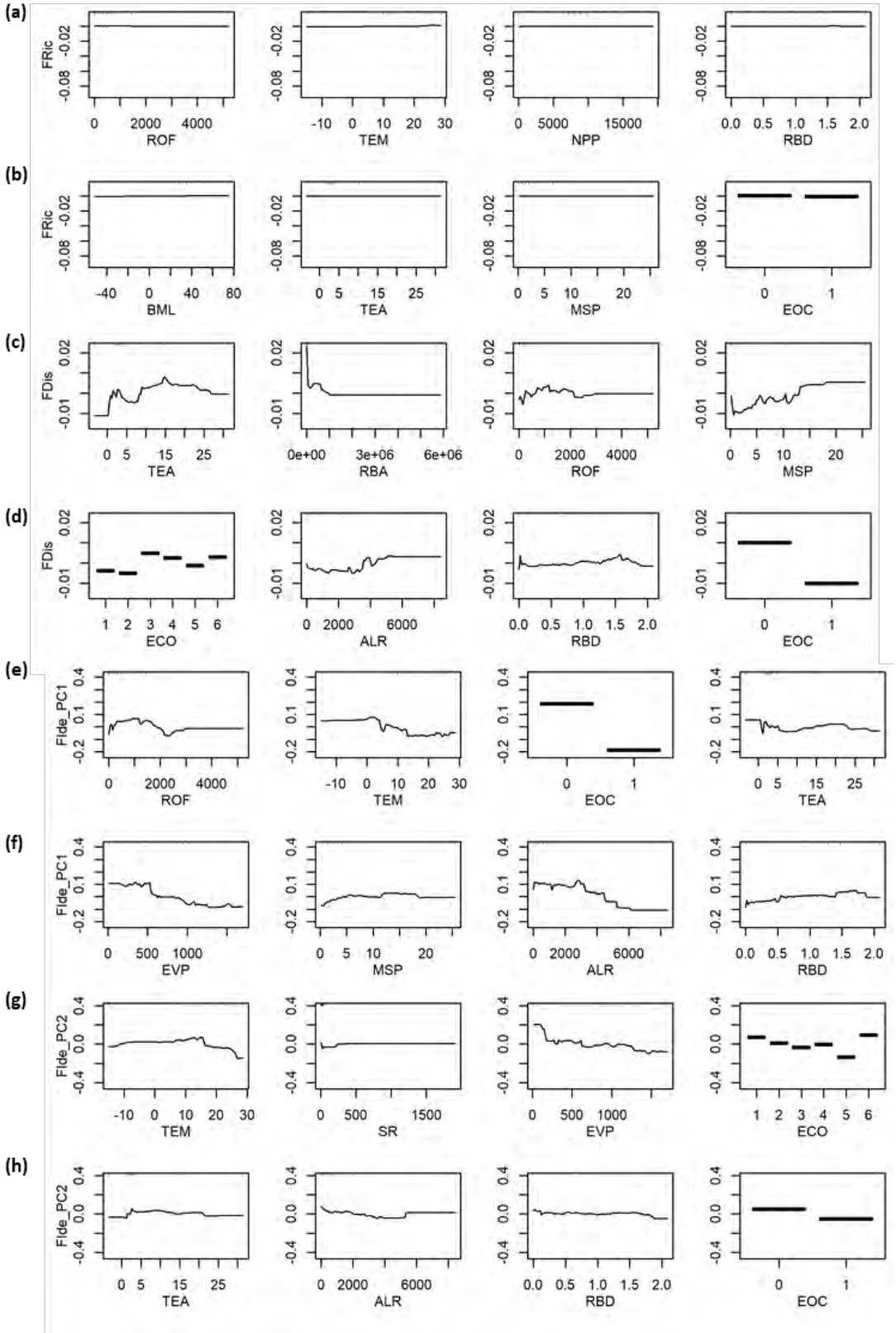



**Figure S4** Relationship between observed values and predicted values by the BRT models for each functional diversity facet in 2453 river basins. (a) functional richness (FRic), (b) functional dispersion (FDis), (c-g) functional identity (Fide) on PC1 to PC5.


(a)
FRic
ROF
TEM
NPP
RBD
(b)
FRic
BML
TEA
MSP
EOC
(c)
FDis
TEA
RBA
ROF
MSP
(d)
FDis
ECO
ALR
RBD
EOC
(e)
Fide_PC1
ROF
TEM
EOC
TEA
(f)
Fide_PC1
EVP
MSP
ALR
RBD
(g)
Fide_PC2
TEM
SR
EVP
ECO
(h)
Fide_PC2
TEA
ALR
RBD
EOC

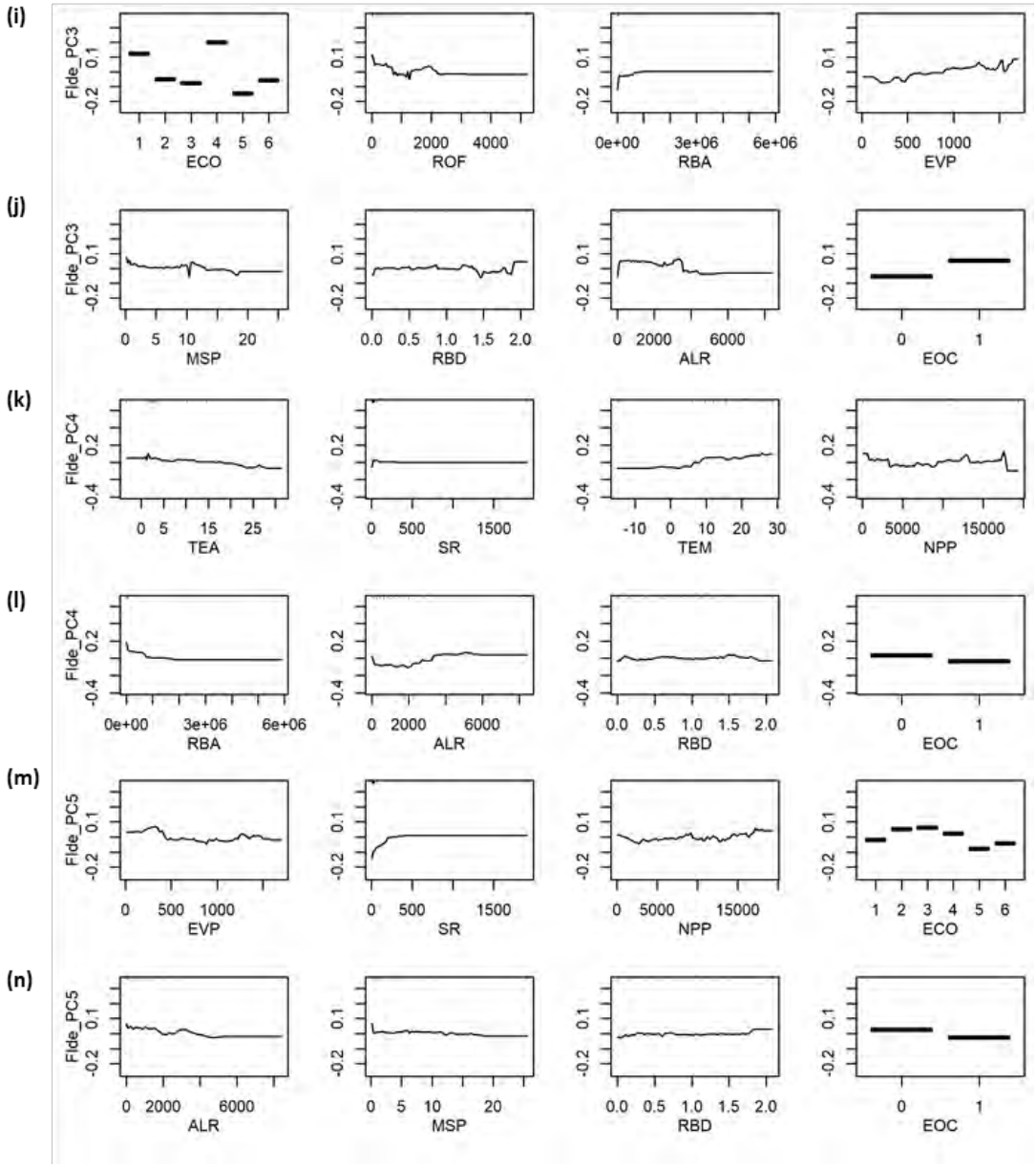



**Figure S5.** Results of boosted regression trees showing the partial dependency between each functional diversity facet and the corresponding remaining eight predictor variables. Rows (a-b) for functional richness, rows (c-d) for functional dispersion, rows (e-n) for functional identity on PC1 to PC5. TEM: temperature, EVP: evapotranspiration, ROF: runoff, NPP: net primary productivity; RBA: river basin area, ALR: altitude range, MSP: mean slope, RBD: river basin diversity; TEA: temperature anomaly, BML: basin median latitude, ECO: ecoregion, EOC: endorheic; SR: species richness. Numbers on the x axis in Ecoregion panels correspond to the biogeographical realms (1: Afrotropical, 2: Australian, 3: Nearctic, 4: Neotropical, 5: Oriental, 6: Palearctic). 0 or 1 on the x axis in Endorheic panels represents

whether it's endorheic (0: exorheic, 1: endorheic).

**Table S1.** Predictor variables considered in the analysis, their abbreviation and reference sources.

| Variables | Abbreviations | Sources |
|---|---|---|
| Mean annual temperature (°C) | TEM | https://www.worldclim.org/data/worldclim21.html |
| Mean annual precipitation (mm/year) | PRE | https://www.worldclim.org/data/worldclim21.html |
| Annual actual evapotranspiration (mm/year) | EVP | https://www.worldclim.org/data/worldclim21.html |
| Mean annual surface runoff (mm/year) | ROF | Leprieur *et al*. 2011 |
| Mean annual net primary productivity (gC/m$^2$/yr) | NPP | http://files.ntsg.umt.edu/data/NTSG_Products/MOD17/GeoTIFF/MOD17A3/GeoTIFF_30arcsec/ |
| River basin area (km$^2$) | RBA | Tedesco *et al*. 2017 |
| Altitudinal range (m) | ALR | https://www.worldclim.org/data/worldclim21.html |
| Mean slope (°) | MSP | https://www.earthenv.org/topography |
| River basin diversity (Shannon diversity index) | RBD | Diaz *et al*. 2014 |
| Temperature anomaly (difference between current and Last Glacial Maximum mean annual temperature (°C) | TEA | WorldClim; CCSM and MIROC3.2 MEDRES https://www.worldclim.org/data/worldclim21.html https://pmip.lsce.ipsl.fr/ |
| Basin median latitude (°) | BML | This study |
| Ecoregion | ECO | Brosse *et al*. 2013 |
| Endorheic | EOC | Leprieur *et al*. 2011 |
| Species richness | SR | This study |

**Table S2.** Results of the procedure used for model calibration using predictor variables. bf: bag fraction; tc: tree complexity; lr: learning rate; CV-D2: cross validated proportion of the deviance explained of 10-folds CV; nt: optimal number of trees estimated of 10-folds CV.

| FRic | | | | |
|---|---|---|---|---|
| bf | tc | lr | CV-D2 | nt |
| 0.8 | 3 | 0.01 | 0.8022 | 6150 |
| 0.8 | 3 | 0.005 | 0.8006 | 10000 |
| 0.7 | 3 | 0.01 | 0.7971 | 8300 |
| 0.8 | 5 | 0.01 | 0.795 | 6100 |
| 0.8 | 5 | 0.005 | 0.7927 | 9650 |
| 0.8 | 7 | 0.01 | 0.7859 | 6050 |
| 0.7 | 3 | 0.005 | 0.7854 | 10000 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.7 | 1 | 0.01 | 0.5443 | 650 |
| 0.5 | 1 | 0.005 | 0.5439 | 2050 |
| 0.6 | 1 | 0.01 | 0.5413 | 650 |
| 0.5 | 1 | 0.001 | 0.5412 | 5100 |
| 0.7 | 1 | 0.005 | 0.5411 | 1600 |
| 0.7 | 1 | 0.001 | 0.5408 | 5350 |
| 0.6 | 1 | 0.005 | 0.5398 | 1550 |
| 0.6 | 1 | 0.001 | 0.5387 | 5150 |
| 0.7 | 1 | 0.0005 | 0.5387 | 8900 |
| 0.5 | 1 | 0.0005 | 0.5371 | 7750 |
| 0.6 | 1 | 0.0005 | 0.5343 | 7500 |

| FDis | | | | |
|---|---|---|---|---|
| bf | tc | lr | CV-D2 | nt |
| 0.6 | 13 | 0.01 | 0.4686 | 1700 |
| 0.6 | 13 | 0.005 | 0.4662 | 3650 |
| 0.5 | 13 | 0.005 | 0.465 | 2950 |
| 0.7 | 13 | 0.005 | 0.4643 | 3500 |
| 0.7 | 13 | 0.01 | 0.4633 | 2200 |
| 0.8 | 13 | 0.01 | 0.4633 | 2200 |
| 0.5 | 13 | 0.01 | 0.4629 | 1850 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.7 | 1 | 0.005 | 0.3051 | 7650 |
| 0.8 | 1 | 0.01 | 0.3009 | 4550 |
| 0.8 | 1 | 0.005 | 0.2968 | 5750 |
| 0.5 | 1 | 0.001 | 0.2805 | 10000 |
| 0.6 | 1 | 0.001 | 0.2801 | 10000 |
| 0.7 | 1 | 0.001 | 0.2795 | 10000 |
| 0.8 | 1 | 0.001 | 0.2787 | 10000 |
| 0.5 | 1 | 0.0005 | 0.2472 | 10000 |
| 0.6 | 1 | 0.0005 | 0.2457 | 10000 |
| 0.7 | 1 | 0.0005 | 0.244 | 10000 |
| 0.8 | 1 | 0.0005 | 0.2424 | 10000 |

FIde_PC1

FIde_PC2

| bf | tc | lr | CV-D2 | nt |
|---|---|---|---|---|
| 0.7 | 13 | 0.005 | 0.6793 | 3550 |
| 0.7 | 13 | 0.01 | 0.6783 | 2250 |
| 0.8 | 13 | 0.005 | 0.6781 | 3450 |
| 0.6 | 13 | 0.01 | 0.6769 | 1950 |
| 0.6 | 13 | 0.005 | 0.6763 | 3400 |
| 0.5 | 13 | 0.01 | 0.6758 | 1750 |
| 0.7 | 11 | 0.01 | 0.6757 | 2500 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.8 | 1 | 0.01 | 0.4985 | 4750 |
| 0.7 | 1 | 0.005 | 0.4977 | 7100 |
| 0.8 | 1 | 0.005 | 0.4952 | 6750 |
| 0.5 | 1 | 0.001 | 0.4664 | 10000 |
| 0.6 | 1 | 0.001 | 0.4637 | 10000 |
| 0.7 | 1 | 0.001 | 0.4614 | 10000 |
| 0.8 | 1 | 0.001 | 0.4592 | 10000 |
| 0.5 | 1 | 0.0005 | 0.4155 | 10000 |
| 0.6 | 1 | 0.0005 | 0.4131 | 10000 |
| 0.7 | 1 | 0.0005 | 0.4109 | 10000 |
| 0.8 | 1 | 0.0005 | 0.4087 | 10000 |

| bf | tc | lr | CV-D2 | nt |
|---|---|---|---|---|
| 0.7 | 13 | 0.01 | 0.6637 | 1850 |
| 0.8 | 13 | 0.005 | 0.6637 | 3400 |
| 0.8 | 13 | 0.01 | 0.6636 | 1700 |
| 0.7 | 13 | 0.005 | 0.6636 | 3750 |
| 0.6 | 13 | 0.005 | 0.6635 | 3550 |
| 0.6 | 13 | 0.01 | 0.6634 | 1700 |
| 0.7 | 11 | 0.005 | 0.6626 | 3850 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.8 | 1 | 0.01 | 0.4687 | 7400 |
| 0.7 | 1 | 0.005 | 0.4681 | 9000 |
| 0.8 | 1 | 0.005 | 0.4597 | 8950 |
| 0.5 | 1 | 0.001 | 0.4374 | 10000 |
| 0.6 | 1 | 0.001 | 0.4337 | 10000 |
| 0.7 | 1 | 0.001 | 0.4305 | 10000 |
| 0.8 | 1 | 0.001 | 0.4276 | 10000 |
| 0.5 | 1 | 0.0005 | 0.392 | 10000 |
| 0.6 | 1 | 0.0005 | 0.389 | 10000 |
| 0.7 | 1 | 0.0005 | 0.3861 | 10000 |
| 0.8 | 1 | 0.0005 | 0.3835 | 10000 |

FIde_PC3

| bf | tc | lr | CV-D2 | nt |
|---|---|---|---|---|
| 0.8 | 13 | 0.01 | 0.6653 | 2300 |
| 0.8 | 11 | 0.01 | 0.664 | 2550 |
| 0.8 | 13 | 0.005 | 0.664 | 3750 |

FIde_PC4

| bf | tc | lr | CV-D2 | nt |
|---|---|---|---|---|
| 0.5 | 13 | 0.01 | 0.6857 | 3600 |
| 0.8 | 13 | 0.01 | 0.6857 | 2800 |
| 0.6 | 11 | 0.01 | 0.6845 | 3050 |

| | | | | |
|---|---|---|---|---|
| 0.7 | 13 | 0.005 | 0.6632 | 3950 |
| 0.8 | 9 | 0.005 | 0.6629 | 5250 |
| 0.7 | 11 | 0.005 | 0.6627 | 4550 |
| 0.6 | 13 | 0.01 | 0.6625 | 2400 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.8 | 1 | 0.01 | 0.4988 | 7400 |
| 0.7 | 1 | 0.005 | 0.4969 | 8950 |
| 0.8 | 1 | 0.005 | 0.4889 | 7500 |
| 0.5 | 1 | 0.001 | 0.468 | 10000 |
| 0.6 | 1 | 0.001 | 0.4667 | 10000 |
| 0.7 | 1 | 0.001 | 0.4655 | 10000 |
| 0.8 | 1 | 0.001 | 0.4642 | 10000 |
| 0.5 | 1 | 0.0005 | 0.4465 | 10000 |
| 0.6 | 1 | 0.0005 | 0.4451 | 10000 |
| 0.7 | 1 | 0.0005 | 0.444 | 10000 |
| 0.8 | 1 | 0.0005 | 0.4426 | 10000 |

FIde_PC5

| bf | tc | lr | CV-D2 | nt |
|---|---|---|---|---|
| 0.8 | 13 | 0.01 | 0.642 | 2200 |
| 0.7 | 13 | 0.005 | 0.6414 | 3900 |
| 0.8 | 13 | 0.005 | 0.6402 | 3850 |
| 0.7 | 13 | 0.01 | 0.6401 | 2050 |
| 0.6 | 13 | 0.005 | 0.6396 | 3800 |
| 0.8 | 11 | 0.01 | 0.6394 | 2750 |
| 0.6 | 13 | 0.01 | 0.6385 | 2150 |
| 0.8 | 13 | 0.005 | 0.6844 | 4250 |
| 0.6 | 13 | 0.01 | 0.6843 | 2700 |
| 0.5 | 13 | 0.005 | 0.6841 | 4600 |
| 0.5 | 11 | 0.01 | 0.684 | 3900 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.7 | 1 | 0.005 | 0.5439 | 9600 |
| 0.6 | 1 | 0.005 | 0.5434 | 8850 |
| 0.8 | 1 | 0.005 | 0.5434 | 10000 |
| 0.5 | 1 | 0.001 | 0.4848 | 10000 |
| 0.6 | 1 | 0.001 | 0.4828 | 10000 |
| 0.7 | 1 | 0.001 | 0.4809 | 10000 |
| 0.8 | 1 | 0.001 | 0.479 | 10000 |
| 0.5 | 1 | 0.0005 | 0.4401 | 10000 |
| 0.6 | 1 | 0.0005 | 0.4389 | 10000 |
| 0.7 | 1 | 0.0005 | 0.4379 | 10000 |
| 0.8 | 1 | 0.0005 | 0.437 | 10000 |

| | | | | |
|---|---|---|---|---|
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |
| 0.8 | 1 | 0.005 | 0.4837 | 10000 |
| 0.5 | 1 | 0.005 | 0.4833 | 9550 |
| 0.7 | 1 | 0.005 | 0.4815 | 8950 |
| 0.5 | 1 | 0.001 | 0.4329 | 10000 |
| 0.6 | 1 | 0.001 | 0.4317 | 10000 |
| 0.7 | 1 | 0.001 | 0.4303 | 10000 |
| 0.8 | 1 | 0.001 | 0.4284 | 10000 |
| 0.5 | 1 | 0.0005 | 0.4017 | 10000 |
| 0.6 | 1 | 0.0005 | 0.4008 | 10000 |
| 0.7 | 1 | 0.0005 | 0.3996 | 10000 |
| 0.8 | 1 | 0.0005 | 0.3984 | 10000 |

**Table S3**. Results of spatially explicit simultaneous autoregressive (SAR) error models for functional diversity in the global freshwater fish fauna (2 453 river basins), including coefficient estimates and the associated significant levels for each variable. The significance is given in parentheses: *** p<0.001; ** p<0.01; * p<0.05; ns p>0.05). The last two rows show the R-squared values and Moran's I statistic of the SAR model for each functional index. TEM: temperature, EVP: evapotranspiration, ROF: runoff, NPP: net primary productivity; RBA: river basin area, ALR: altitude range, MSP: mean slope, RBD: river basin diversity; TEA: temperature anomaly, BML: basin median latitude, ECO: ecoregion, EOC: endorheic; SR: species richness.

| | Coefficient estimate (significance) | | | | | | |
|---|---|---|---|---|---|---|---|
| Variables | FIde_PC1 | FIde_PC2 | FIde_PC3 | FIde_PC4 | FIde_PC5 | FRic | FDis |
| TEM | -0.047 (*) | -0.049 (*) | 0.157 (***) | -0.034 (ns) | -0.122 (***) | 0 (ns) | 0.008 (***) |
| EVP | -0.047 (*) | -0.078 (***) | 0.061 (**) | -0.108 (***) | 0.014 (ns) | -0.001 (ns) | -0.002 (ns) |
| ROF | 0.009 (ns) | 0.034 (*) | -0.061 (***) | 0.013 (ns) | 0.003 (ns) | 0 (ns) | 0.001 (ns) |
| NPP | 0.055 (***) | 0.157 (***) | -0.035 (*) | 0.001 (ns) | -0.028 (*) | 0.001 (ns) | 0.008 (***) |
| RBA | -0.004 (ns) | 0.006 (ns) | 0.018 (*) | -0.041 (***) | -0.013 (*) | -0.002 (***) | -0.002 (*) |
| ALR | -0.074 (***) | -0.021 (ns) | -0.011 (ns) | -0.016 (ns) | -0.016 (ns) | 0.001 (ns) | 0.001 (ns) |
| MSP | 0.044 (***) | 0.109 (***) | 0.025 (ns) | 0.005 (ns) | -0.024 (**) | -0.001 (*) | 0.003 (**) |
| RBD | 0.031 (***) | 0.002 (ns) | -0.019 (ns) | 0.026 (**) | -0.006 (ns) | 0 (ns) | 0.003 (***) |
| TEA | -0.03 (ns) | -0.04 (*) | -0.049 (**) | -0.058 (**) | 0.035 (**) | 0 (ns) | 0.006 (***) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| BML | -0.044 (*) | 0.028 (ns) | -0.085 (***) | 0.235 (***) | 0.058 (***) | 0 (ns) | 0.015 (***) |
| ECO | 0.188 (***) | 0.074 (***) | -0.138 (***) | 0.053 (**) | -0.042 (***) | 0 (ns) | 0 (ns) |
| EOC | -0.102 (***) | -0.029 (**) | 0.023 (*) | -0.072 (***) | -0.03 (***) | 0 (ns) | -0.006 (***) |
| SR | -0.01 (ns) | 0.001 (ns) | -0.018 (ns) | 0.032 (***) | 0.031 (***) | 0.016 (***) | 0.003 (***) |
| R-squared | 0.62 | 0.55 | 0.57 | 0.57 | 0.52 | 0.82 | 0.37 |
| Moran's I statistic | -0.055 (ns) | -0.073 (ns) | -0.043 (ns) | -0.048 (ns) | -0.07 (ns) | -0.017 (ns) | -0.025 (ns) |

# Chapter V

## Morphological sorting of introduced freshwater fish species within and between donor realms

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# Morphological sorting of introduced freshwater fish species within and between donor realms

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## Abstract

**Aim:** To determine which morphological characteristics make a fish species a good candidate for introduction and establishment, we tested whether (a) introduced species differ in morphology from non-introduced species (species only existing in native areas and not introduced to new areas) in each donor assemblage (biogeographic realm fauna); (b) within the introduced species, the morphology of established species (self-sustaining introduced species) differs from that of the non-established species; (c) within the established species, those exported out of their native realm have more extreme morphological traits than those translocated within their native realm.

**Major taxa studied:** Freshwater fish.

**Location:** Global.

**Time period:** 1960s–2010s.

**Methods:** We used a global database of freshwater fishes from the six realms. Ten morphological traits were measured on 9,150 species. Principal component analysis was conducted to combine the 10 traits into a multidimensional morphospace. We used permutational multivariate analysis of variance (PERMANOVA) and permutational analysis for the multivariate homogeneity of dispersions (PERMDISP2) to compare the distribution of species groups in the morphospace and Kolmogorov–Smirnov tests to compare their distributions on principal component (PC) axes.

**Results:** The morphology of introduced species differed from that of non-introduced species in all the six biogeographic realms. Among introduced species, established species had more extreme morphological traits than non-established species in most realms. Among established species, exported species had more extreme morphological traits than translocated species.

**Main conclusions:** Morphological differences between introduced and non-introduced species was driven by an anthropogenic trait selection for fisheries and angling, leading to the preference for the introduction of predators with large and laterally compressed bodies. Established species represent a small subset of introduced species morphologies, with established species having more extreme morphological traits than their non-established counterparts. Established fish have morphologies more adapted to lentic waters, and this trend was more marked for species exported to other realms than for species translocated within their native realm.

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KEYWORDS
biogeographic realms, establishment, exported species, invasion steps, morphological traits, morphospace, rivers, translocated species

## 1 | INTRODUCTION

Species invasion is one of the main drivers of the human-induced biodiversity crisis (Clavero & García-Berthou, 2005; Leprieur, Beauchard, Blanchet, Oberdorff, & Brosse, 2008; Vitousek, D'antonio, Loope, Rejmanek, & Westbrooks, 1997). Biological invasions have altered patterns of biodiversity from local to continental scales and resulted in both ecological and economic impacts on entire ecosystems (Simberloff et al., 2013; Vitule, Freire, & Simberloff, 2009). Those detrimental impacts are at least in part due to the ecological differences between native species and introduced species that have become established in the recipient ecosystems. Introduced species establishing in new environments are most often larger than the species from the recipient faunas (Blackburn, Cassey, & Lockwood, 2009; Blanchet et al., 2010; Roy, Jablonski, & Valentine, 2002) and perform functions distinct from those existing in recipient faunas (Marchetti, Moyle, & Levine, 2004; Toussaint et al., 2018). For instance, Olden, Poff, and Bestgen (2006) showed that introduced fishes that established in the Colorado River (USA) occupy vacant niches and have different life-history strategies compared with those of the native species. Similarly, Blackburn et al. (2009) demonstrated that the establishment success of exotic birds increases with a particular combination of ecological traits.

Since human population and economic exchanges are identified as the key drivers of species introduction and establishment in new environments (Leprieur et al., 2008; Pyšek et al., 2010), global human population growth and accelerated economic development are predicted to increase the number of candidate species for introduction as well as their establishment success (Hulme, 2009; Seebens et al., 2017, 2018). This is particularly true in freshwater ecosystems, which are among the most highly invaded ones (Moyle & Marchetti, 2006), especially because several hundreds of fish species have been transported by humans outside their native range for the three last centuries (Leprieur et al., 2008; Toussaint et al., 2018).

Fish morphology has been suggested to be related to species establishment success in both marine and freshwater ecosystems (Azzurro et al., 2014; Blanchet et al., 2010; Ribeiro, Elvira, Collares-Pereira, & Moyle, 2008). Up to now, it has been reported that established freshwater fish species (i.e. introduced species that successfully established in the new habitat) morphologically and functionally differ from the recipient fauna (e.g. Blanchet et al., 2010; Marchetti et al., 2004; Olden et al., 2006; Toussaint et al., 2018). Established species tend to be larger, with deeper bodies, and to have more piscivorous or detritivorous diets (Cucherousset, Blanchet, & Olden, 2012; Toussaint et al., 2018). However, functional diversity of fishes markedly varies between the six biogeographic realms, namely the Afrotropical, Australian, Neotropical, Nearctic, Oriental and Palaearctic realms (Su, Villéger, & Brosse, 2019; Toussaint, Charpin, Brosse, & Villéger, 2016). Therefore, to work towards a better understanding of the causes and consequences of the invasion process in the world's rivers, it is important to disentangle the relative importance of the natural functional differences between donor zones, the human preference for some functional characteristics and the capacity of each introduced species to establish (or not) in a new environment.

Tackling this challenge requires sequential consideration of the introduction and establishment steps of the species invasion process, as proposed by Blackburn et al. (2011) and Lockwood, Hoopes, and Marchetti (2013), within each realm. We therefore distinguished between non-introduced species, which only exist in native areas and have never been introduced to new areas, and introduced species, which have at least once been introduced into areas (here river basins) where they did not historically occur (Figure 1a). We here tested the hypothesis that within each realm, humans select particular morphologies for the species to be introduced (Figure 1b). Once introduced, a species can establish or not in its new environment (Figure 1a). The success of establishment is largely influenced by the match between physical (climate, habitat) and biological (competitors, predators and prey) environment and the capacities of the introduced species (Lockwood et al., 2013). A morphological difference between established and non-established species (Figure 1c) would therefore indicate trait-based filtering affecting the establishment success of introduced species (Moyle & Marchetti, 2006). Moreover, translocated fishes that were introduced to nearby areas from their native range (i.e. within same realm) are more prone to adapt to local abiotic conditions and biotic interactions than species exported to other realms, potentially facing markedly different conditions in those distant areas (Ribeiro et al., 2008). Thus, if physical and biological characteristics of the recipient environment differ between realms, we predict that the functional characteristics of the exported species will differ from those of the translocated species, exported species being therefore more morphologically extreme than translocated species (Figure 1d). We applied this framework to the world's freshwater fish faunas, considering more than 14,000 species from the 6 biogeographic realms.

## 2 | MATERIALS AND METHODS

Fish occurrence was compiled from Tedesco et al. (2017), which gives the occurrence of 14,953 species (more than 90% of the freshwater fish species) in 3,119 drainage basins, covering more than 80%

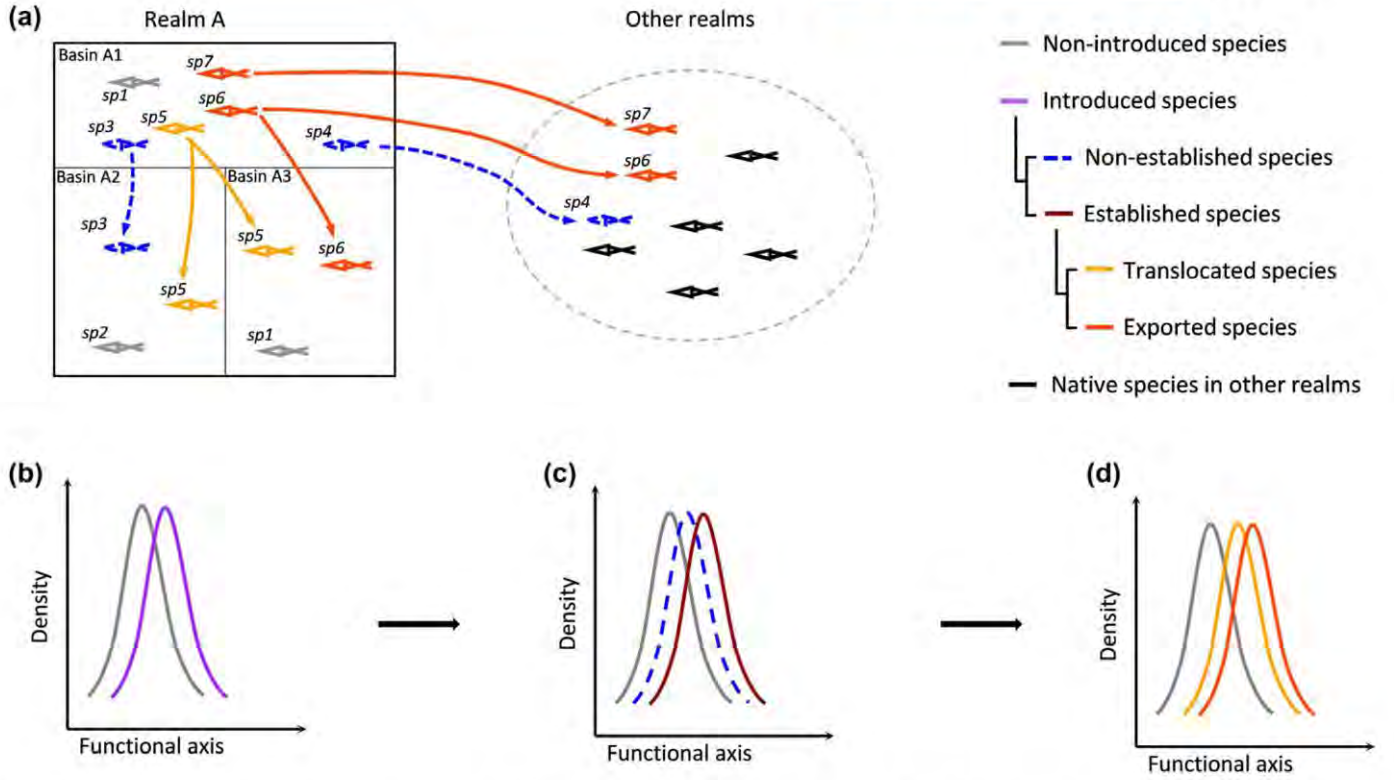



**FIGURE 1** Assessing the difference in morphological trait values between groups of freshwater fish species. (a) Conceptual representation of non-introduced, non-established, established, translocated and exported species based on their distribution in a realm (realm A) and in the world (other realms). A non-introduced species has never been introduced in a river basin where it did not historically occur. An introduced species has been introduced to river basins where it did not historically occur. An introduced species becomes established if it creates a self-sustaining population where it has been introduced. Within the established species, a translocated species has established within its native realm in one or several river basins where it did not historically occur and an exported species has been introduced and established at least in one river basin from a biogeographic realm where the species did not historically occur. Three tests of difference in the distribution of trait values were successively conducted between: (b) the non-introduced and introduced species; (c) the non-introduced, non-established and established species; (d) the non-introduced, translocated and exported species

of the Earth's surface. Each occurrence is paired with a status, either native or non-native established if the species was not historically present in the drainage river basin. As more than 98% of the species historically belong to a single realm (Tedesco et al., 2017), we used these data to determine for each realm the list of native species that have been introduced and then established within their native realm (translocated species) or in at least another realm (exported species).

Distinguishing between introduced and non-introduced species cannot be done easily because there is no global database or realm scale record of introduction attempts (only successful introductions, that is, established species, are recorded). Nevertheless, introductions of vertebrates are mainly via intentional pathways to achieve various human purposes, including farming, for ornaments, and gaming (Saul et al., 2017). This also holds for fishes and many studies have reported that freshwater fish introductions are driven by human interest (e.g. aquaculture, pest control, game fishing). Most freshwater fish species introductions are therefore intentional or at least human assisted (e.g. Gozlan, Britton, Cowx, & Copp, 2010; Leprieur et al., 2008; Marchetti et al., 2004; Padilla & Williams, 2004). We thus collected the list of species of interest to humans from FishBase (Froese & Pauly, 2018) and Blanchet et al. (2010), and these species were considered as those having a high chance of having been introduced in at least one river outside their native range (hereafter called introduced species). Within those introduced species, we then distinguished between non-established and established species using non-native occurrences from Tedesco et al. (2017). Since established species are considered as self-sustaining populations, populations of non-native species artificially maintained by constant release [e.g. rainbow trout (*Oncorhynchus mykiss*) in many European river basins, Stanković, Crivelli, & Snoj, 2015] were not considered as established (Tedesco et al., 2017).

Among the 14,953 species present in the occurrence database, 9,150 species were morphologically described using pictures and drawings from textbooks and scientific websites (Su et al., 2019). More precisely, morphology was assessed using 10 traits describing the size and shape of body parts involved in food acquisition and locomotion (Toussaint et al., 2016; Villéger, Brosse, Mouchet, Mouillot, & Vanni, 2017). The 10 traits were selected to be complementary, and they were indeed not markedly correlated to each other (Spearman test, rho < .45 for all the 45 trait comparisons).

More precisely, the fish size was described using the maximum body length (Max. Body Length) taken from FishBase (Froese & Pauly, 2018). These maximum body lengths were carefully reviewed, and irrelevant measures were corrected according to the appropriate literature. In addition to size, 11 morphological measures were assessed on side view pictures (Supporting Information Figure S1a) collected during an extensive literature review including our own field data and scientific literature sources made up of peer-reviewed articles, books, and scientific websites. We collected at least one picture (photograph or scientific drawing) per species. Only good quality pictures and scientific side view drawings of entire adult animals with confirmed species identification were kept. For species with marked sexual dimorphism, we considered male morphology, as female pictures are scarce for most species (especially for Perciformes and Cyprinodontiformes). Intraspecific morphological trait variability was not considered in this study as it hardly affects functional diversity at the large spatial resolution considered (Toussaint et al., 2018). The nine unitless traits describing the morphology of the fish head (including mouth and eye), body, pectoral and caudal fins (Supporting Information Figure S1b) were computed as ratios between 11 morphological measures using ImageJ software (http://rsb.info.nih.gov/ij/index.html). The 10 morphological traits (nine unitless ratios and body size) selected are commonly used in assessment of fish functional diversity (e.g. Bellwood, Goatley, Brandl, & Bellwood, 2014; Su et al., 2019; Toussaint et al., 2016; Villéger, Miranda, Hernandez, & Mouillot, 2010) and are linked to the feeding and locomotion functions of fish that determine their contribution to key ecosystem processes, such as controlling food webs and nutrient cycles (Villéger et al., 2017) (Supporting Information Figure S1b). Functional traits not measurable on side pictures, such as gut length, oral gape area and shape, were not included because they are currently only available for a few species in public databases. Some species have unusual morphologies (species without tails, flatfishes) that prevent the measurement of some morphological traits. We thus applied conventions as mentioned in Su et al. (2019); Toussaint et al. (2016); Villéger et al. (2010) for these few exceptions.

## 2.1 | Statistical analyses

Due to the lesion of body parts or the quality of fish pictures, some traits were not measured for some species. Overall, 17.3% of the values were missing in the raw morphological traits dataset (from 6.8% for maximum body length to 24.5% for relative maxillary length). These missing values (NA) were replaced by estimates provided by a random forest algorithm called 'missForest' (Penone et al., 2014; Stekhoven & Bühlmann, 2012). This method uses a random forest trained on the observed values of a data matrix to predict the missing values and automatically calibrates the filling values by a set of iterations (Stekhoven & Bühlmann, 2012). It can be used to impute continuous and/or categorical data and is not biased by complex interactions or nonlinear relationships (Stekhoven & Bühlmann, 2012). The efficiency of the random forest model used to fill missing values was tested on a random set of 1,000 species with complete values for all 10 traits. We randomly set 20% of the values for the 1,000 species as NA, and then used the 'missForest' to fill the NA with simulated values. We then compared the simulated values to the actual values. This procedure was repeated 999 times. Spearman's rho between actual and simulated data was used to measure the efficiency of the procedure. Spearman's rho varied from .89 to .96, testifying for the efficiency of the method. As a comparison, we also tried to fill the NA by the average value of the trait for closely related species (genera). In this case, the Spearman's rho varies from .83 to .91, which means that the 'missForest' procedure outperforms the classical imputation method of filling the gaps using the average trait value of closely related species.

We then computed a principal component analysis (PCA) using values of the 10 morphological traits for all the species (Maire, Grenouillet, Brosse, & Villéger, 2015). We used a sensitivity procedure to assess the robustness of our selected morphological traits. We tested the effect of trait identity on the distance between species (i.e. Euclidean distance in the 5D space) by rerunning the PCA based on all combinations of nine morphological traits out of ten. The distance between species was hardly affected by morphological traits accounted for (Mantel tests, $r > .900$, $p < .001$).

PERMANOVA and PERMDISP2 tests were conducted to compare the average and variance distributions of non-introduced, introduced, non-established, established, translocated and exported species in the morphological space in each realm and in the world. We then illustrated the distributions of these groups on the first five PC axes using transformed raincloud plots (Allen, Poggiali, Whitaker, Marshall, & Kievit, 2018), and compared distributions between groups using Kolmogorov–Smirnov tests (hereafter K-S tests).

All statistical analyses were performed with the R software environment version 3.3 (R Core Team, 2018). We used 'missForest' from the 'missForest' R package (Stekhoven & Bühlmann, 2012) to fill the missing trait values, the 'adonis' and 'betadisper' functions from the 'vegan' R package (Oksanen et al., 2010) to conduct the PERMANOVA and PERMDISP2 tests (Euclidean distance, permutations = 9,999), and the 'ks.test' function from the 'stats' R package (R Core Team, 2018) to conduct the K-S tests.

# 3 | RESULTS

Among the 9,150 studied species, 2,690 species have been introduced while only 418 species (including 311 translocated species and 107 exported species) successfully established in the recipient assemblages. The remaining 2,272 introduced species failed to establish. Most of the non-established species originate from the Afrotropical (877), Neotropical (651) and Oriental (475) realms. In contrast, most of the established species (translocated and exported species) originate from the Nearctic and Palaearctic realms (129 and 105 translocated species in Nearctic and Palaearctic realms,

respectively, 34 and 39 exported species in Nearctic and Palaearctic realms, respectively). The Australian realm hosts the least translocated species (18) and exported species (1) (Supporting Information Table S1). Exported species from the Australian realm were therefore not considered in the K-S test.

The first five PC axes explained 78.6% of the total variance among the world's fish morphologies (Supporting Information Figure S2). No axis was driven by a single trait and all traits contributed to the position of species in the functional space (Supporting Information Figure S2). The PC1 axis shows a strong contribution of fish mouth size and position and therefore codes for trophic level, with higher trophic levels (predators) having positive values on this axis. The PC2 axis shows a strong contribution of fish eye vertical position and body lateral shape and therefore codes for fish position in the water column, with benthic species having positive values on this axis. The PC3 axis shows a strong contribution of body elongation and fin size. It therefore codes for habitat type and locomotion, with higher dispersal ability in slow-flowing habitats having positive values on this axis. PC4 is also strongly influenced by body elongation but also by fish body size, with large and laterally flattened fish represented by positive values. The PC5 axis shows a strong contribution of caudal peduncle throttling and therefore codes for locomotion, fish with higher dispersal ability having positive values on this axis (Supporting Information Figure S2). Overall, PC1, PC2 and PC4 accounted for both nutrition and locomotion, whereas PC3 and PC5 were mainly influenced by traits related to locomotion.

Introduced species occupied 58.22% of the global species pool in the 5D morphospace, which is higher than the space occupied by non-introduced species (45.02%), although the number of introduced species is less than half that of the non-introduced species. Intriguingly, within the introduced species, the morphospace occupied by non-established species (56.58%) was almost the same as the space occupied by the introduced species, but the established species occupied only 7.15% of the global morphospace, which accounted for 1/8 of the morphospace occupied by non-established species (Figure 2). A similar pattern was observed in all the six realms, in which the space occupied by introduced species was higher than the space occupied by the non-introduced species. This pattern was more pronounced in the two arctic realms. For instance, in the Nearctic realm, introduced species represent 40% of the species but more than 65% of the morphological space of the realm. The space occupied by non-established species varied from 38.56 to 44.18%, whereas the space occupied by established species varied from 0.22 to 5.08% (Figure 3). As subgroups of established species, translocated and exported species also occupied small portions of the morphospace filled by all the species from the globe (Figure 2), and from their respective realms (Figure 3).

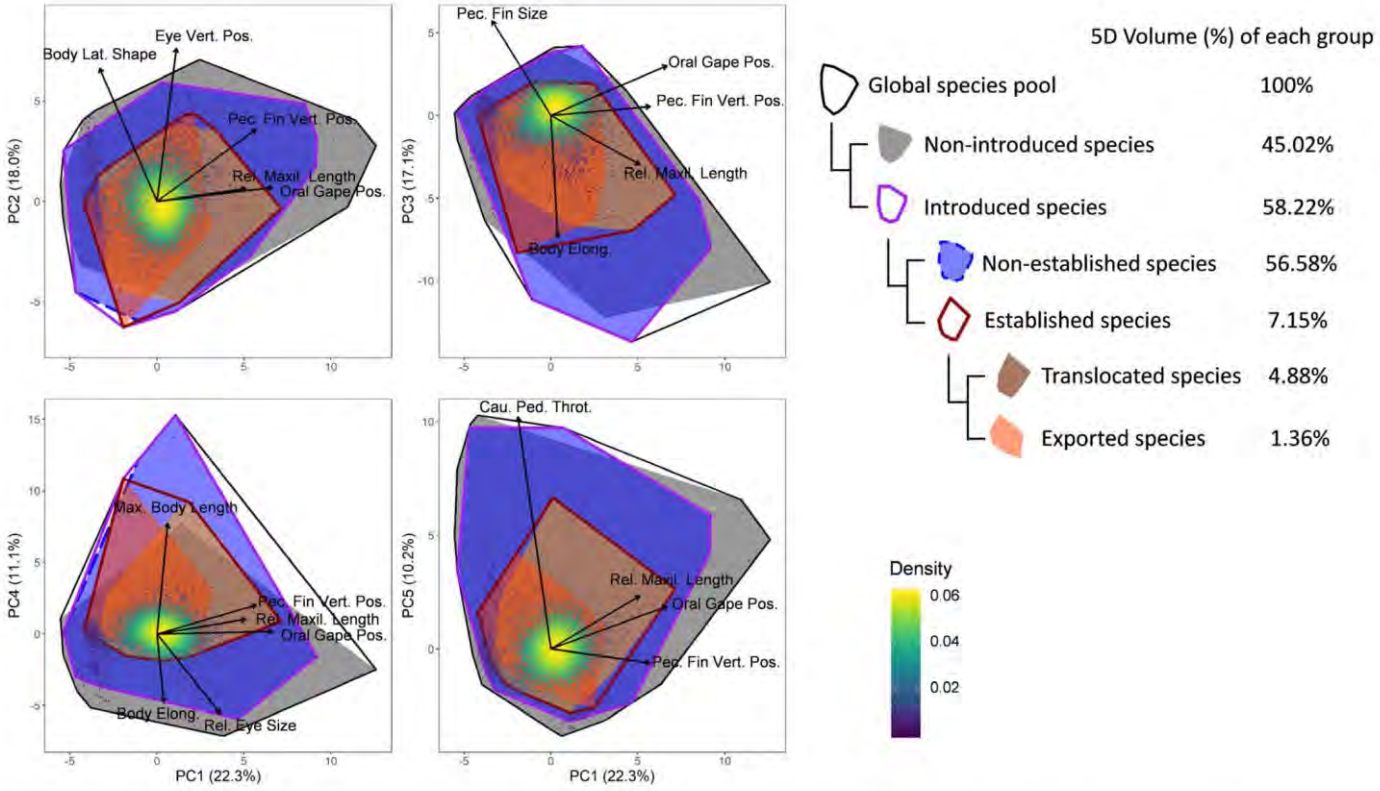
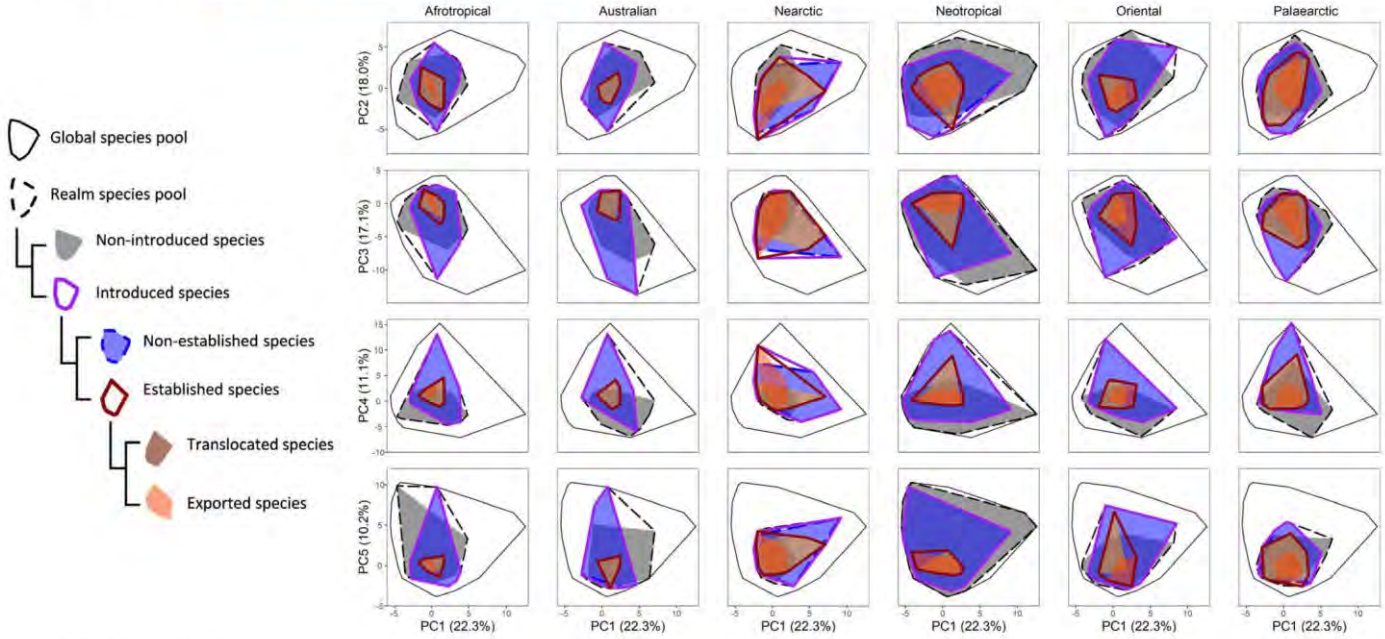



**FIGURE 2** Distribution of the different groups of species in the five dimensional morphological space of the world fish fauna. Morphological traits with high relative contribution to a principal compenent (PC) axis (> 15%) are shown (See supplementary Figure S1 for a complete view of the contribution of the 10 morphological traits to the 5 PC axes). Purple to yellow shade indicates increasing species density. Values after the legends show the percentage of 5D morphospace volume occupied by each group in the global species pool




| 5D Volume (%) of each group | Afrotropical | Australian | Nearctic | Neotropical | Oriental | Palaearctic |
|---|---|---|---|---|---|---|
| Realm species pool | 100.00% | 100.00% | 100.00% | 100.00% | 100.00% | 100.00% |
| Non-introduced species | 29.54% | 26.30% | 14.48% | 38.83% | 29.60% | 31.73% |
| Introduced species | 39.60% | 40.50% | 66.58% | 41.23% | 46.34% | 48.73% |
| Non-established species | 38.56% | 39.05% | 40.45% | 39.67% | 44.18% | 32.24% |
| Established species | 0.25% | 0.22% | 15.03% | 0.47% | 2.04% | 14.33% |
| Translocated species | 0.14% | 0.20% | 8.09% | 0.24% | 0.52% | 8.85% |
| Exported species | <0.01% | 0.00% | 1.03% | 0.08% | 0.24% | 1.56% |

**FIGURE 3** Distributions of the different groups of species in the five dimensional morphological space for each realm. Values in the table show the percentage of 5D morphospace volume occupied by each group in the realm species pool

**TABLE 1** Differences in trait distributions (average and variance) between fish groups tested using permutational multivariate analysis of variance and permutational analysis for the multivariate homogeneity of dispersions (*p*-values on left and right sides of each column, respectively; *p*-values lower than .05 are in bold)

| | Afrotropical | Australian | Nearctic | Neotropical | Oriental | Palaearctic | World |
|---|---|---|---|---|---|---|---|
| Non-introduced versus introduced | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** |
| Non-introduced versus non-established | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** |
| Non-introduced versus established | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** |
| Non-established versus established | **.022**; .241 | .724; .860 | .915; .145 | **.038**; .176 | **.014**; .052 | .189; .109 | **< .001; < .001** |
| Non-introduced versus translocated | **< .001; < .001** | **.004; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** | **< .001; < .001** |
| Non-introduced versus exported | **< .001; .014** | **.048**; NA | **< .001; < .001** | **< .001; < .001** | **< .001; .002** | **< .001; < .001** | **< .001; < .001** |
| Translocated versus exported | .651; .735 | .108; NA | **< .001; < .001** | .778; .403 | .292; .179 | .208;.884 | **.031**; .334 |

Results are given for freshwater fish species from each realm and for the world fish fauna. See text for explanations about fish groups.

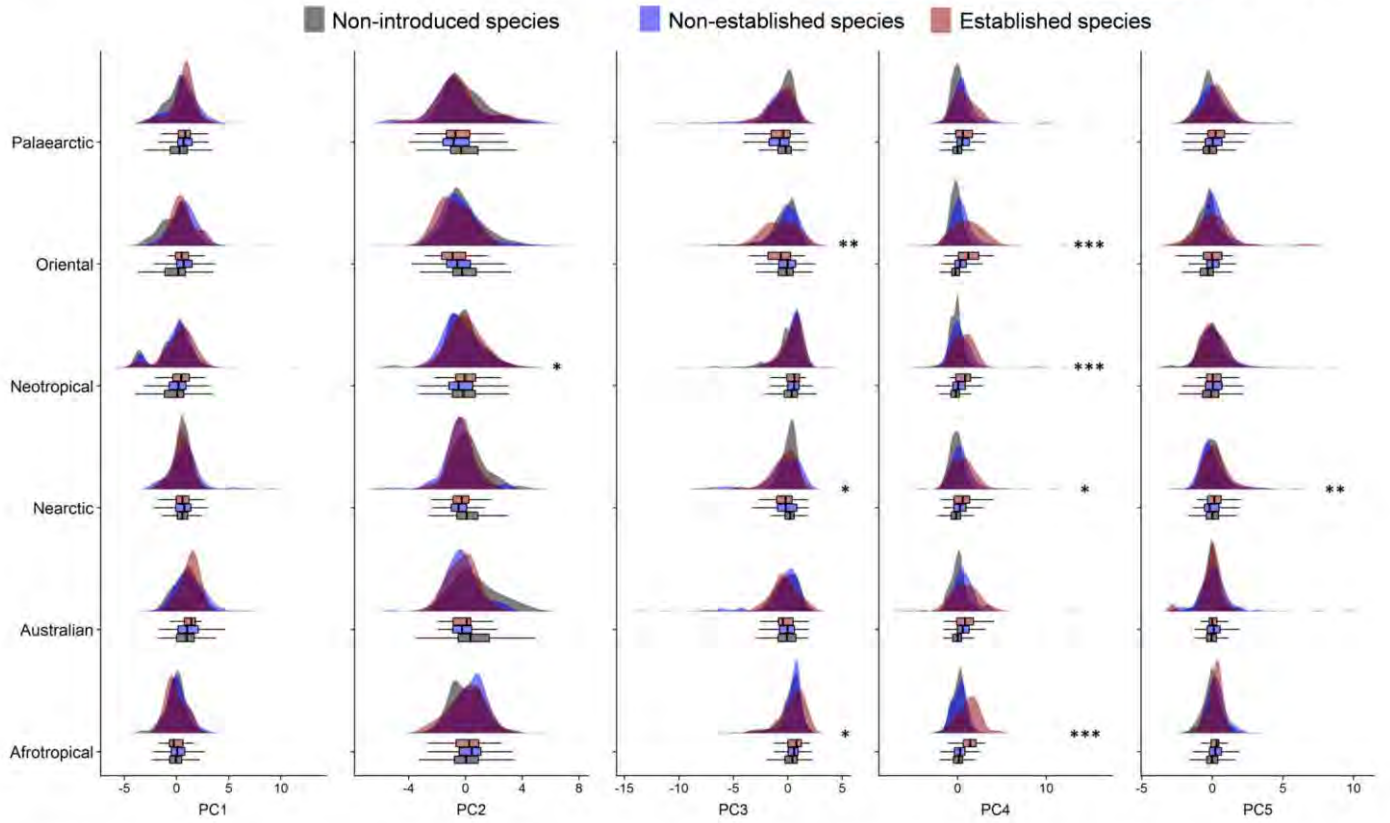



**FIGURE 4** Density distribution of species on the 5 principal compenent (PC) axes for non-introduced, non-established and established species in each realm. Boxplots and results of K-S tests between the non-established and established species are shown beside each plot item (*** $p < .001$, ** $p < .01$, *$p < .05$). (See the supplementary Table S2 for the complete K-S tests between the three groups)

At the world scale, groups of species had distinct distributions in the 5D morphospace (Figure 2), as testified by significant differences (PERMANOVA and PERMDISP2 tests) between all pairs of species groups, except for the translocated and exported species in the PERMDISP2 test (Table 1). At the realm scale, the non-introduced species were still significantly different from all the other groups belonging to the introduced species in all the six realms (Table 1). Examining those patterns for each axis confirmed this trend, with the exception of the Australian realm on PC1 and PC3, and of the Nearctic realm on PC5 (Supporting Information Figure S3).

Comparing the non-established and established species showed significant differences in the Afrotropical ($p$ = .022), Neotropical ($p$ = .038) and Oriental ($p$ = .014) realms (PERMANOVA test, Table 1). Examining PC axes one-by-one showed that established species significantly differed from the non-established species on PC3 and PC4 in both the Afrotropical and Oriental realms. These two groups of species differed significantly on PC2 and PC4 in the Neotropical realm (Figure 4). Moreover, although PERMANOVA and PERMDISP2 tests did not detect significant differences in variance of position in the morphospace between established and non-established species from the Nearctic realm (Figure 4), the K-S tests did detect significant differences in the distribution of values along PC3, PC4, and PC5 axes. In all the realms where the established species differed significantly from the non-established ones, the established species had higher values on the fourth dimension of the morphological space (PC4) than the non-established species, indicating larger and less elongated bodies (Figure 4). The established species showed the largest differences in values from the non-introduced species in all the significant comparisons, with the exception of Neotropical species on PC2 (Figure 4).

Although an overall examination of translocated versus exported species revealed a significant difference only in the Nearctic realm (PERMANOVA and PERMDISP2 tests, Table 1), an axis-by-axis examination reveals significant morphological differences on at least one out of the five PC axes in the five considered realms (the Australian realm was not considered as it has a single exported species). Generally, 9 out of the 25 comparisons between exported and translocated species in the 5 realms differed significantly. In Afrotropical and Nearctic realms, exported species were significantly different from the translocated species on PC3, PC4 and PC5. They also differed on PC3 in the Neotropical realm, on PC1 in the Oriental realm and on PC4 in the Palaearctic realm (Figure 5). In all the realms where the exported species differed significantly from the translocated ones, the exported species had larger and less elongated bodies (PC4) and higher caudal peduncle throttling values (PC5) than the translocated species (Figure 5). The exported species showed the largest difference in values from the non-introduced species in all the significant comparisons (Figure 5).

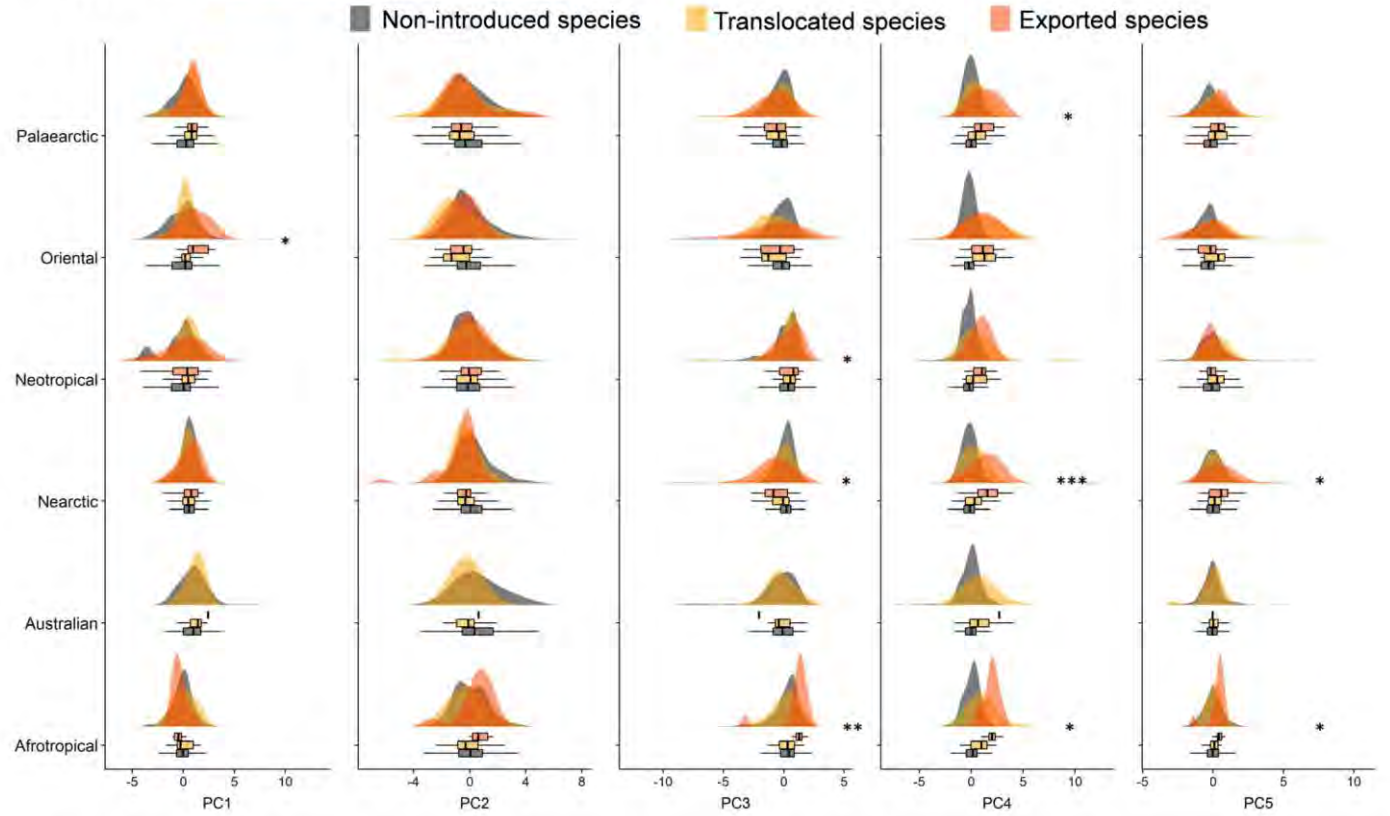



**FIGURE 5** Density distribution of species on the 5 principal compenent (PC) axes for non-introduced, translocated and exported species in each realm. Boxplots and results of K-S tests between the translocated and exported species are shown beside each plot item (*** $p < .001$, ** $p < .01$, * $p < .05$). (See the supplementary Table S2 for the complete K-S tests between the three groups)

## 4 | DISCUSSION

Across the globe, less than 20% of the known freshwater fish fauna has been used for various human purposes including fisheries, aquaculture, angling and as ornaments (Froese & Pauly, 2018). Assuming that these species of interest for humans have at least once been introduced in a river basin where they do not naturally occur appears realistic for two main reasons. First, 71.5% of the established species listed by Tedesco et al. (2017) belong to this category; and second, current records of introduction attempts, although regional and/or incomplete, report that introduced species are those having an interest for humans (Lintermans, 2004; Welcomme, 1988). The species of interest for humans here considered as introduced species have particular morphological traits distinguishing them from the non-introduced species. Moreover, the morphological space occupied by introduced species is greater than the space occupied by non-introduced species, although they account for less than a third of the number of the non-introduced species (2,272 vs. 6,460 species). Humans have therefore selected fish species with various morphological characteristics, despite a high functional redundancy of native species (Su et al., 2019). Such extended morphological selection of introduced species is consistent in all the six realms, and reflects distinct introduction purposes. The selection of large-bodied predators is associated with angling purposes, whereas laterally flattened species account for species inhabiting slow-flowing habitats that are often tolerant toward water quality and oxygen content and therefore easy to rear in aquaculture ponds (Blake, 1983; Haas, Blum, & Heins, 2010). In contrast, the selection of extreme morphologies is common in the ornamental trade (Howeth et al., 2016; Moyle & Marchetti, 2006). Compared to introduced species, established species occupy a lower volume of the world's morphospace (less than 8%), indicating that the species that have established at least in one river basin where they do not historically occur have homogeneous and similar morphologies. Established species consistently have larger and more laterally compressed bodies than the non-established species in the six realms. This pattern reflects a marked environmental filtering effect at the establishment stage, which reinforces the introduction trend toward species with larger body sizes and more lateral compressed bodies. Fishes with laterally compressed bodies have high manoeuvrability and perform sustained swimming more efficiently in slow-flowing habitats (Blake, 1983; Haas et al., 2010). Thus, the global rise of river fragmentation and dam construction, which are known to facilitate the establishment of introduced species (Havel, Lee, & Vander Zanden, 2005; Johnson, Olden, & Vander Zanden, 2008) throughout the world (Anderson et al., 2018; Winemiller et al., 2016; Zarfl, Lumsdon, Berlekamp, Tydecks, & Tockner, 2015), can partly explain the pattern of morphological trait distributions of the established species. In addition, it should be noted that established species from industrialized realms (Palaearctic and Nearctic) have more diverse morphologies than those

from the other realms (14 to 16% of the realm morphological space for Palaearctic and Nearctic realms versus less than 3% for the four other realms). This can be ascribed to a higher number of introduction events per species (i.e. propagule pressure), which is known to notably increase establishment probability (Lockwood, Cassey, & Blackburn, 2005). Among established species, the morphological differences between the translocated and the exported species were due to both differences in the species selected by humans to be introduced, and to different strengths of environmental filters, favouring the establishment of the most extreme morphologies in exported species. Indeed, most of the exported freshwater fish species result from the establishment of species intentionally introduced in one or several realms where the species do not naturally occur, whereas translocated species result from the establishment of species introduced intentionally or not. Hence, within a realm, short distance transport to nearby basins is easy and largely used by anglers to illegally spread some species of interest, but also to spread other accompanying species, such as baits. For instance, the introduction and spread of wels catfish (*Silurus glanis*) from eastern to western European rivers have largely been assisted by anglers (Cucherousset et al., 2018). Such short distance transport has also been recorded in the Neotropics for a series of harvested species including siluriforms (*Pseudoplatystoma* sp.), characiforms (*Curimata* sp.) and osteoglossiforms (*Arapaima gigas*) (Bezerra et al., 2019; Vitule et al., 2019). In contrast, exportations are more difficult, as fish must be kept alive across long distances, and therefore there is careful selection of the species to be introduced, often for commercial reasons. This discrepancy between translocation and exportation processes explains the morphological difference between the translocated and exported species, the latter being the largest and the most laterally flattened species, traits often preferred for aquaculture species. Interestingly, this trend was true for Palaearctic, Nearctic and Afrotropical species, but not for Neotropical fishes, probably because Neotropical fish account for most of the ornamental fish species, some establishing as exported species. It is, for instance, the case for the guppy, *Poecilia reticulata*, which originates from the Neotropical realm, but established in all realms following its introduction as an ornamental fish (Deacon, Ramnarine, & Magurran, 2011). Those ornamental fish have various morphologies, but are often small sized, counterbalancing therefore the trend toward the large body sized exported fishes found in the other realms. The distinct pattern between exported and translocated species observed across realms testifies for distinct human selection modes of exported and translocated species, but can also reflect distinct establishment processes between the realms, paralleling the results of Blanchet et al. (2010) demonstrating discrepancies in establishment determinants between realms. Indeed, within a realm, a translocated species will encounter similar environments to its native range and will share its new habitat with species shaped by the same regional evolutionary constraints (Colautti, Eckert, & Barrett, 2010). In contrast, an exported species has to face a fauna that evolved under a distinct regional history (Nekola & White, 1999).

To conclude, fish morphological traits are closely related to the introduction and establishment stages of the invasive process. Human selection affects the introduction stage, and selects a subset of a realm's fish morphology, with a preference towards large pelagic predators with high dispersal ability. However, environmental filtering profoundly reshapes the morphology of established species. The *c.* 400 introduced species that established (out of *c.* 2,700 introduced species) are the most laterally flattened, and inhabit slow-flowing habitats (e.g. lakes, reservoirs), therefore reinforcing that morphological selection achieved by humans at the introduction stage. Nevertheless, the recent development of the ornamental fish trade associated with the global rise of leisure activities might lead to a considerable change in human interest for particular morphological traits. Those traits will be markedly different from the traits of interest for aquaculture or angling species. That will probably lead, as initiated in the Neotropics, to the introduction of an extended range of morphologies, and will increase the pool of introduced species in the future. In the same way, the economic development of tropical countries (e.g. China in the Oriental realm and Brazil in the Neotropical realm) makes them providers of new non-native species through local aquaculture, aquarium and angling development (Vitule et al., 2009, 2019; Vitule, Skóra, & Abilhoa, 2012). This trend might thus increase in the near future the diversity of established species in the tropical realms. Among those new pools of introduced species, some might establish in the recipient environments and cause profound functional changes to the recipient communities (Capps & Flecker, 2013). Given that once established, eradicating an introduced species is often impossible, we call for particular caution against non-native species introductions (Vitule et al., 2009).

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## DATA AVAILABILITY STATEMENT

Fish occurrence data can be freely retrieved at https://doi.org/10.6084/m9.figshare.c.3739145 and metadata are available in Tedesco et al. (2017). Fish maximum body lengths were downloaded from FishBase (http://www.fishbase.org). Information about the 9,150 species with combined trait values (PC1–PC5) used in this study are available in Su et al. (2019).

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## REFERENCES

Allen, M., Poggiali, D., Whitaker, K., Marshall, T. R., & Kievit, R. (2018). Raincloud plots: A multi-platform tool for robust data visualization. *PeerJ Preprints*, *6*, e27137v1.

Anderson, E. P., Jenkins, C. N., Heilpern, S., Maldonado-Ocampo, J. A., Carvajal-Vallejos, F. M., Encalada, A. C., … Tedesco, P. A. (2018). Fragmentation of Andes-to-Amazon connectivity by hydropower dams. *Science Advances*, *4*, eaao1642. https://doi.org/10.1126/sciadv.aao1642

Azzurro, E., Tuset, V. M., Lombarte, A., Maynou, F., Simberloff, D., Rodríguez-Pérez, A., & Solé, R. V. (2014). External morphology

explains the success of biological invasions. *Ecology Letters*, *17*, 1455–1463. https://doi.org/10.1111/ele.12351

Bellwood, D., Goatley, C., Brandl, S., & Bellwood, O. (2014). Fifty million years of herbivory on coral reefs: Fossils, fish and functional innovations. *Proceedings of the Royal Society B: Biological Sciences*, *281*, https://doi.org/10.1098/rspb.2013.3046.

Bezerra, L. A. V., Freitas, M. O., Daga, V. S., Occhi, T. V. T., Faria, L., Costa, A. P. L., … Vitule, J. R. S. (2019). A network meta-analysis of threats to South American fish biodiversity. *Fish and Fisheries*, *20*, 620–639. https://doi.org/10.1111/faf.12365

Blackburn, T. M., Cassey, P., & Lockwood, J. L. (2009). The role of species traits in the establishment success of exotic birds. *Global Change Biology*, *15*, 2852–2860. https://doi.org/10.1111/j.1365-2486.2008.01841.x

Blackburn, T. M., Pyšek, P., Bacher, S., Carlton, J. T., Duncan, R. P., Jarošík, V., … Richardson, D. M. (2011). A proposed unified framework for biological invasions. *Trends in Ecology and Evolution*, *26*, 333–339. https://doi.org/10.1016/j.tree.2011.03.023

Blake, R. W. (1983). *Fish locomotion*. UK: Cambridge University Press.

Blanchet, S., Grenouillet, G., Beauchard, O., Tedesco, P. A., Leprieur, F., Dürr, H. H., … Brosse, S. (2010). Non-native species disrupt the worldwide patterns of freshwater fish body size: Implications for Bergmann's rule. *Ecology Letters*, *13*, 421–431. https://doi.org/10.1111/j.1461-0248.2009.01432.x

Capps, K. A., & Flecker, A. S. (2013). Invasive aquarium fish transform ecosystem nutrient dynamics. *Proceedings of the Royal Society B: Biological Sciences*, *280*, 20131520. https://doi.org/10.1098/rspb.2013.1520

Clavero, M., & García-Berthou, E. (2005). Invasive species are a leading cause of animal extinctions. *Trends in Ecology and Evolution*, *20*, 110.https://doi.org/10.1016/j.tree.2005.01.003

Colautti, R. I., Eckert, C. G., & Barrett, S. C. (2010). Evolutionary constraints on adaptive evolution during range expansion in an invasive plant. *Proceedings of the Royal Society B: Biological Sciences*, *277*, 1799–1806. https://doi.org/10.1098/rspb.2009.2231

Cucherousset, J., Blanchet, S., & Olden, J. D. (2012). Non-native species promote trophic dispersion of food webs. *Frontiers in Ecology and the Environment*, *10*, 406–408. https://doi.org/10.1890/12.WB.018

Cucherousset, J., Horky, P., Slavík, O., Ovidio, M., Arlinghaus, R., Boulêtreau, S., … Santoul, F. (2018). Ecology, behaviour and management of the European catfish. *Reviews in Fish Biology and Fisheries*, *28*, 177–190. https://doi.org/10.1007/s11160-017-9507-9

Deacon, A. E., Ramnarine, I. W., & Magurran, A. E. (2011). How reproductive ecology contributes to the spread of a globally invasive fish. *PLoS ONE*, *6*, e24416. https://doi.org/10.1371/journal.pone.0024416

Froese, R., & Pauly, D. (2018) Fishbase. Retrieved from www.fishbase.org

Gozlan, R. E., Britton, J., Cowx, I., & Copp, G. (2010). Current knowledge on non-native freshwater fish introductions. *Journal of Fish Biology*, *76*, 751–786. https://doi.org/10.1111/j.1095-8649.2010.02566.x

Haas, T. C., Blum, M. J., & Heins, D. C. (2010). Morphological responses of a stream fish to water impoundment. *Biology Letters*, *6*, 803–806. https://doi.org/10.1098/rsbl.2010.0401

Havel, J. E., Lee, C. E., & Vander Zanden, J. M. (2005). Do reservoirs facilitate invasions into landscapes? *BioScience*, *55*, 518–525. https://doi.org/10.1641/0006-3568(2005)055[0518:DRFIIL]2.0.CO;2

Howeth, J. G., Gantz, C. A., Angermeier, P. L., Frimpong, E. A., Hoff, M. H., Keller, R. P., … Lodge, D. M. (2016). Predicting invasiveness of species in trade: Climate match, trophic guild and fecundity influence establishment and impact of non-native freshwater fishes. *Diversity and Distributions*, *22*, 148–160. https://doi.org/10.1111/ddi.12391

Hulme, P. E. (2009). Trade, transport and trouble: Managing invasive species pathways in an era of globalization. *Journal of Applied Ecology*, *46*, 10–18. https://doi.org/10.1111/j.1365-2664.2008.01600.x

Johnson, P. T., Olden, J. D., & Vander Zanden, M. J. (2008). Dam invaders: Impoundments facilitate biological invasions into freshwaters. *Frontiers in Ecology and the Environment*, *6*, 357–363. https://doi.org/10.1890/070156

Leprieur, F., Beauchard, O., Blanchet, S., Oberdorff, T., & Brosse, S. (2008). Fish invasions in the world's river systems: When natural processes are blurred by human activities. *PLoS Biology*, *6*, e28. https://doi.org/10.1371/journal.pbio.0060322

Lintermans, M. (2004). Human-assisted dispersal of alien freshwater fish in Australia. *New Zealand Journal of Marine and Freshwater Research*, *38*, 481–501. https://doi.org/10.1080/00288330.2004.9517255

Lockwood, J. L., Cassey, P., & Blackburn, T. (2005). The role of propagule pressure in explaining species invasions. *Trends in Ecology and Evolution*, *20*, 223–228. https://doi.org/10.1016/j.tree.2005.02.004

Lockwood, J. L., Hoopes, M. F., & Marchetti, M. P. (2013). *Invasion ecology*. New York, NY: John Wiley & Sons.

Maire, E., Grenouillet, G., Brosse, S., & Villéger, S. (2015). How many dimensions are needed to accurately assess functional diversity? A pragmatic approach for assessing the quality of functional spaces. *Global Ecology and Biogeography*, *24*, 728–740. https://doi.org/10.1111/geb.12299

Marchetti, M. P., Moyle, P. B., & Levine, R. (2004). Invasive species profiling? Exploring the characteristics of non-native fishes across invasion stages in California. *Freshwater Biology*, *49*, 646–661. https://doi.org/10.1111/j.1365-2427.2004.01202.x

Moyle, P. B., & Marchetti, M. P. (2006). Predicting invasion success: Freshwater fishes in California as a model. *BioScience*, *56*, 515–524. https://doi.org/10.1641/0006-3568(2006)56[515:PISFFI]2.0.CO;2

Nekola, J. C., & White, P. S. (1999). The distance decay of similarity in biogeography and ecology. *Journal of Biogeography*, *26*, 867–878. https://doi.org/10.1046/j.1365-2699.1999.00305.x

Oksanen, J., Blanchet, F. G., Kindt, R., Legendre, P., O'hara, R., Simpson, G. L., … Wagner, H. (2010). *Vegan: Community ecology package*. R package version 2.5-4. Retrieved 10 August 2019 from http://cran.r-project.org/

Olden, J. D., Poff, N. L., & Bestgen, K. R. (2006). Life-history strategies predict fish invasions and extirpations in the Colorado River Basin. *Ecological Monographs*, *76*, 25–40. https://doi.org/10.1890/05-0330

Padilla, D. K., & Williams, S. L. (2004). Beyond ballast water: Aquarium and ornamental trades as sources of invasive species in aquatic ecosystems. *Frontiers in Ecology and the Environment*, *2*, 131–138. https://doi.org/10.1890/1540-9295(2004)002[0131:BBWAAO]2.0.CO;2

Penone, C., Davidson, A. D., Shoemaker, K. T., Di Marco, M., Rondinini, C., Brooks, T. M., … Costa, G. C. (2014). Imputation of missing data in life-history trait datasets: Which approach performs the best? *Methods in Ecology and Evolution*, *5*, 961–970. https://doi.org/10.1111/2041-210X.12232

Pysek, P., Jarosik, V., Hulme, P. E., Kuhn, I., Wild, J., Arianoutsou, M., … Winter, M. (2010). Disentangling the role of environmental and human pressures on biological invasions across Europe. *Proceedings of the National Academy of Sciences USA*, *107*, 12157–12162. https://doi.org/10.1073/pnas.1002314107

R Core Team. (2018). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. Retrieved from https://www.r-project.org/

Ribeiro, F., Elvira, B., Collares-Pereira, M. J., & Moyle, P. B. (2008). Life-history traits of non-native fishes in Iberian watersheds across several invasion stages: A first approach. *Biological Invasions*, *10*, 89–102. https://doi.org/10.1007/s10530-007-9112-2

Roy, K., Jablonski, D., & Valentine, J. W. (2002). Body size and invasion success in marine bivalves. *Ecology Letters*, *5*, 163–167. https://doi.org/10.1046/j.1461-0248.2002.00316.x

Saul, W.-C., Roy, H. E., Booy, O., Carnevali, L., Chen, H.-J., Genovesi, P., … Jeschke, J. M. (2017). Assessing patterns in introduction pathways of alien species by linking major invasion data bases. *Journal of Applied Ecology*, *54*, 657–669. https://doi.org/10.1111/1365-2664.12819

Seebens, H., Blackburn, T. M., Dyer, E. E., Genovesi, P., Hulme, P. E., Jeschke, J. M., … Essl, F. (2017). No saturation in the accumulation of alien species worldwide. *Nature Communications*, *8*, 14435. https://doi.org/10.1038/ncomms14435

Seebens, H., Blackburn, T. M., Dyer, E. E., Genovesi, P., Hulme, P. E., Jeschke, J. M., … Essl, F. (2018). Global rise in emerging alien species results from increased accessibility of new source pools. *Proceedings of the National Academy of Sciences USA*, *115*, E2264–E2273. https://doi.org/10.1073/pnas.1719429115

Simberloff, D., Martin, J.-L., Genovesi, P., Maris, V., Wardle, D. A., Aronson, J., … Vilà, M. (2013). Impacts of biological invasions: What's what and the way forward. *Trends in Ecology and Evolution*, *28*, 58–66. https://doi.org/10.1016/j.tree.2012.07.013

Stanković, D., Crivelli, A. J., & Snoj, A. (2015). Rainbow trout in Europe: Introduction, naturalization, and impacts. *Reviews in Fisheries Science and Aquaculture*, *23*, 39–71. https://doi.org/10.1080/23308249.2015.1024825

Stekhoven, D. J., & Bühlmann, P. (2012). MissForest—Non-parametric missing value imputation for mixed-type data. *Bioinformatics*, *28*, 112–118. https://doi.org/10.1093/bioinformatics/btr597

Su, G., Villéger, S., & Brosse, S. (2019). Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread. *Global Ecology and Biogeography*, *28*, 211–221. https://doi.org/10.1111/geb.12843

Tedesco, P. A., Beauchard, O., Bigorne, R., Blanchet, S., Buisson, L., Conti, L., … Oberdorff, T. (2017). A global database on freshwater fish species occurrence in drainage basins. *Scientific Data*, *4*, 170141. https://doi.org/10.1038/sdata.2017.141

Toussaint, A., Charpin, N., Beauchard, O., Grenouillet, G., Oberdorff, T., Tedesco, P. A., … Villéger, S. (2018). Non-native species led to marked shifts in functional diversity of the world freshwater fish faunas. *Ecology Letters*, *21*, 1649–1659. https://doi.org/10.1111/ele.13141

Toussaint, A., Charpin, N., Brosse, S., & Villéger, S. (2016). Global functional diversity of freshwater fish is concentrated in the Neotropics while functional vulnerability is widespread. *Scientific Reports*, *6*, 22125. https://doi.org/10.1038/srep22125

Villéger, S., Brosse, S., Mouchet, M., Mouillot, D., & Vanni, M. J. (2017). Functional ecology of fish: Current approaches and future challenges. *Aquatic Sciences*, *79*, 783–801. https://doi.org/10.1007/s00027-017-0546-z

Villéger, S., Miranda, J. R., Hernandez, D. F., & Mouillot, D. (2010). Contrasting changes in taxonomic vs. functional diversity of tropical fish communities after habitat degradation. *Ecological Applications*, *20*, 1512–1522. https://doi.org/10.1890/09-1310.1

Vitousek, P. M., D'antonio, C. M., Loope, L. L., Rejmanek, M., & Westbrooks, R. (1997). Introduced species: A significant component of human-caused global change. *New Zealand Journal of Ecology*, *21*, 1–16.

Vitule, J. R. S., Freire, C. A., & Simberloff, D. (2009). Introduction of non-native freshwater fish can certainly be bad. *Fish and Fisheries*, *10*, 98–108. https://doi.org/10.1111/j.1467-2979.2008.00312.x

Vitule, J. R. S., Occhi, T. V. T., Kang, B., Matsuzaki, S.-I., Bezerra, L. A., Daga, V. S., … Padial, A. A. (2019). Intra-country introductions unraveling global hotspots of alien fish species. *Biodiversity and Conservation*, *28*, 3037–3043. https://doi.org/10.1007/s10531-019-01815-7

Vitule, J. R. S., Skóra, F., & Abilhoa, V. (2012). Homogenization of freshwater fish faunas after the elimination of a natural barrier by a dam in Neotropics. *Diversity and Distributions*, *18*, 111–120. https://doi.org/10.1111/j.1472-4642.2011.00821.x

Welcomme, R. L. (1988). *International introductions of inland aquatic species*. FAO Fisheries Technical 294. Rome: Food and Agriculture Organization of the United Nations.

Winemiller, K. O., McIntyre, P. B., Castello, L., Fluet-Chouinard, E., Giarrizzo, T., Nam, S., … Saenz, L. (2016). Balancing hydropower and biodiversity in the Amazon, Congo, and Mekong. *Science*, *351*, 128–129. https://doi.org/10.1126/science.aac7082

Zarfl, C., Lumsdon, A. E., Berlekamp, J., Tydecks, L., & Tockner, K. (2015). A global boom in hydropower dam construction. *Aquatic Sciences*, *77*, 161–170. https://doi.org/10.1007/s00027-014-0377-0

## BIOSKETCHES

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**Sébastien Villéger** is a researcher at the Centre national de la recherche scientifique in Montpellier. He is interested in functional ecology of fishes and is particularly studying the effects of global change on the functional diversity of fish assemblages.

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## SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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Supporting information

# Morphological sorting of introduced freshwater fish species within and between donor realms

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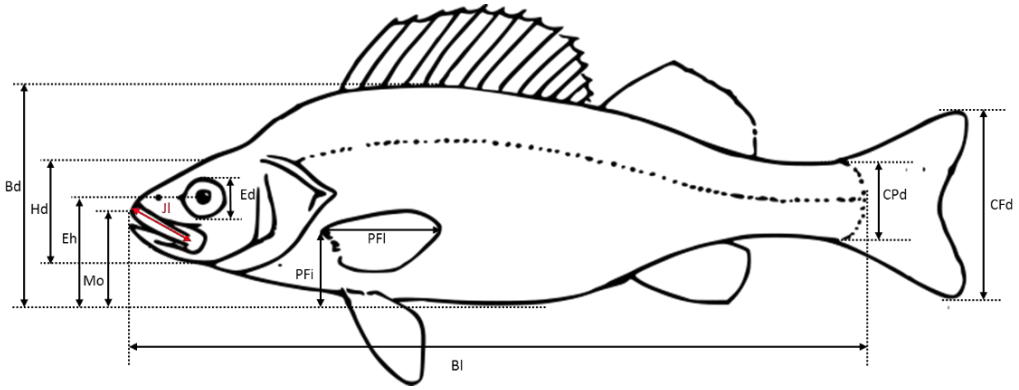
**Table S1** Number of species in each group for each realm and in the world freshwater fish fauna. The number of exported species that are also translocated species is in parentheses.

| | Non-introduced species | Introduced species | | |
|---|---|---|---|---|
| | | Non-established species | Established species | |
| | | | Translocated species | Exported species |
| Afrotropical | 1419 | 877 | 20 | 11 (6) |
| Australian | 232 | 125 | 18 | 1 (1) |
| Nearctic | 400 | 112 | 129 | 34 (29) |
| Neotropical | 2956 | 651 | 30 | 29 (12) |
| Oriental | 994 | 475 | 21 | 15 (7) |
| Palearctic | 590 | 145 | 105 | 39 (27) |
| World | 6460 | 2272 | 311 | 107 (69) |

**Table S2** Results of K-S test between the groups of species on the five PC axis for each realm. (*** $P < 0.001$, ** $P < 0.01$, *$P < 0.05$).

| | Afrotropical | Australian | Nearctic | Neotropical | Oriental | Palearctic |
|---|---|---|---|---|---|---|
| | | | Non-introduced vs Introduced | | | |
| PC1 (22.3%) | *** | 0.37 | * | *** | *** | *** |
| PC2 (18.0%) | *** | *** | *** | *** | *** | *** |
| PC3 (17.1%) | *** | 0.58 | *** | *** | *** | *** |
| PC4 (11.1%) | *** | *** | *** | *** | *** | *** |
| PC5 (10.2%) | *** | * | 0.11 | *** | *** | *** |
| | | | Non-introduced vs Non-established | | | |
| PC1 (22.3%) | *** | 0.34 | * | *** | *** | *** |
| PC2 (18.0%) | *** | *** | *** | *** | *** | *** |
| PC3 (17.1%) | *** | 0.50 | *** | *** | *** | *** |
| PC4 (11.1%) | *** | *** | *** | *** | *** | *** |
| PC5 (10.2%) | *** | * | 0.21 | *** | *** | ** |
| | | | Non-introduced vs Established | | | |
| PC1 (22.3%) | 0.17 | 0.25 | 0.42 | * | ** | *** |
| PC2 (18.0%) | 0.92 | * | *** | 0.88 | ** | ** |
| PC3 (17.1%) | * | 0.78 | *** | 0.08 | ** | * |
| PC4 (11.1%) | *** | ** | *** | *** | *** | *** |
| PC5 (10.2%) | 0.07 | 0.74 | * | 0.17 | * | *** |
| | | | Non-established vs Established | | | |
| PC1 (22.3%) | 0.06 | 0.36 | 0.33 | 0.27 | 0.53 | 0.29 |
| PC2 (18.0%) | 0.42 | 0.70 | 0.51 | * | 0.18 | 0.82 |
| PC3 (17.1%) | * | 0.38 | * | 0.87 | ** | 0.24 |
| PC4 (11.1%) | *** | 0.37 | * | *** | *** | 0.58 |
| PC5 (10.2%) | 0.42 | 0.82 | ** | 0.90 | 0.44 | 0.06 |
| | | | Non-introduced vs Translocated | | | |
| PC1 (22.3%) | 0.55 | 0.40 | 0.38 | * | 0.10 | *** |
| PC2 (18.0%) | 0.80 | * | ** | 0.82 | ** | * |
| PC3 (17.1%) | 0.95 | 0.91 | *** | 0.57 | ** | 0.13 |
| PC4 (11.1%) | *** | * | *** | ** | *** | *** |
| PC5 (10.2%) | 0.61 | 0.87 | 0.38 | 0.08 | ** | *** |
| | | | Non-introduced & Exported | | | |
| PC1 (22.3%) | 0.10 | 0.39 | * | 0.18 | * | *** |
| PC2 (18.0%) | 0.08 | 0.93 | ** | 0.48 | 0.50 | 0.28 |
| PC3 (17.1%) | *** | 0.35 | *** | * | 0.13 | 0.06 |
| PC4 (11.1%) | *** | 0.27 | *** | *** | *** | *** |
| PC5 (10.2%) | ** | 0.95 | ** | 0.62 | 0.81 | *** |
| | | | Translocated vs Exported | | | |
| PC1 (22.3%) | 0.41 | 0.11 | 0.09 | 0.66 | * | 0.92 |

| | | | | | | |
|---|---|---|---|---|---|---|
| PC2 (18.0%) | 0.07 | 0.42 | 0.44 | 0.72 | 0.23 | 0.85 |
| PC3 (17.1%) | ** | 0.21 | * | * | 0.62 | 0.89 |
| PC4 (11.1%) | * | 0.21 | *** | 0.11 | 0.98 | * |
| PC5 (10.2%) | * | 0.95 | * | 0.06 | 0.23 | 0.33 |




**a. Morphological measurements**

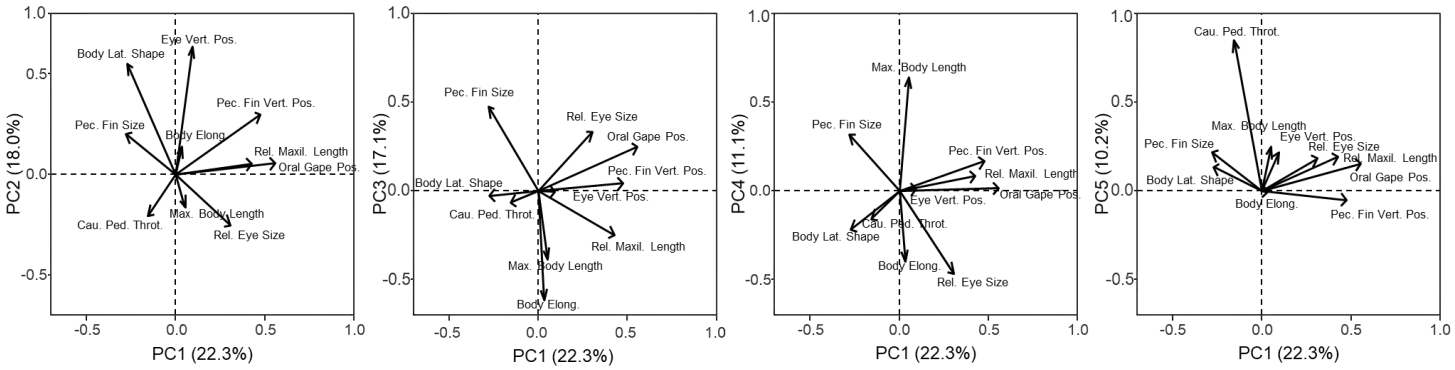
| Code | Name | Protocol for measurement |
|---|---|---|
| Blmax | Maximum Body length | Maximum adult length (obtained from FishBase (Froese & Pauly, 2018)) |
| Bl | Body length | Standard length (snout to caudal fin basis) |
| Bd | Body depth | Maximum body depth |
| Hd | Head depth | Head depth at the vertical of eye |
| CPd | Caudal peduncle depth | Minimum depth of the caudal peduncle |
| CFd | Caudal fin depth | Maximum depth of the caudal fin |
| Ed | Eye diameter | Vertical diameter of the eye |
| Eh | Eye position | Vertical distance between the centre of the eye to the bottom of the body |
| Mo | Oral gape position | Vertical distance from the top of the mouth to the bottom of the body |
| Jl | Maxillary jaw length | Length from snout to the corner of the mouth |
| PFl | Pectoral fin length | Length of the longest ray of the pectoral fin |
| PFi | Pectoral fin position | Vertical distance between the upper insertion of the pectoral fin to the bottom of the body |

All measurements were made on pictures except Blmax values, which were downloaded from Fishbase.org.

**b. Morphological traits**

| Morphological traits | Formula | Potential link with fish functions | References |
|---|---|---|---|
| Maximum body length | $BLmax$ | Size is linked to metabolism, trophic impacts, locomotion ability, nutrient cycling | Toussaint *et al.* 2016 |
| Body elongation | $\frac{Bl}{Bd}$ | Hydrodynamism | Reecht *et al.* 2013 |
| Eye vertical position | $\frac{Eh}{Bd}$ | Position of fish and/or of its prey in the water column | Winemiller 1991 |
| Relative eye size | $\frac{Ed}{Hd}$ | Visual acuity | Boyle & Horn, 2006 |
| Oral gape position | $\frac{Mo}{Bd}$ | Feeding position in the water column | Dumay *et al.* 2004, Lefcheck *et al.* 2014 |
| Relative maxillary length | $\frac{Jl}{Hd}$ | Size of mouth and strength of jaw | Toussaint *et al.* 2016 |
| Body lateral shape | $\frac{Hd}{Bd}$ | Hydrodynamism and head size | Toussaint *et al.* 2016 |
| Pectoral fin vertical position | $\frac{PFi}{Bd}$ | Pectoral fin use for swimming | Dumay *et al.* 2004 |
| Pectoral fin size | $\frac{PFl}{Bl}$ | Pectoral fin use for swimming | Fulton *et al.* 2001 |
| Caudal peduncle throttling | $\frac{CFd}{CPd}$ | Caudal propulsion efficiency through reduction of drag | Webb, 1984 |

**Fig. S1.** Morphological measurements (a) and morphological traits (b) measured on each fish species.




a)

| | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|
| Eigenvalues | 2.23 | 1.80 | 1.71 | 1.11 | 1.02 |
| Proportion of Variance | 22.3 | 17.9 | 17.1 | 11.1 | 10.2 |
| Cumulative Proportion | 22.3 | 40.3 | 57.4 | 68.5 | 78.6 |

b)

| | PC1 | PC2 | PC3 | PC4 | PC5 |
|---|---|---|---|---|---|
| Relative Eye size | 9.3 | 6.4 | 11.2 | **21.7** | 3.2 |
| Oral gape position | **30.9** | <1 | 6.3 | <1 | 2.4 |
| Relative maxillary length | **18.1** | <1 | 6.1 | <1 | 3.8 |
| Eye vertical position | <1 | **40.6** | <1 | <1 | 4.9 |
| Body elongation | <1 | 2.0 | **37.9** | **15.9** | <1 |
| Body lateral shape | 7.5 | **30.5** | <1 | 4.8 | 1.9 |
| Pectoral fin vertical position | **22.6** | 9.0 | <1 | 2.8 | <1 |
| Pectoral fin size | 7.9 | 4.1 | **22.7** | 10.3 | 4.8 |
| Caudal peduncle throttling | 2.5 | 4.3 | <1 | 2.5 | **72.4** |
| Maximum body length | <1 | 2.7 | 15.0 | **41.3** | 6.3 |

The contribution of each morphological trait to the first 5 axes is expressed as percentages and values higher than 15% are in bold font.

**Fig. S2.** Results of the Principal Components Analysis on morphological traits. (a) Eigenvalues and percentages of variance explained by each axis. (b) Percentage of contribution of each morphological trait to each axis.

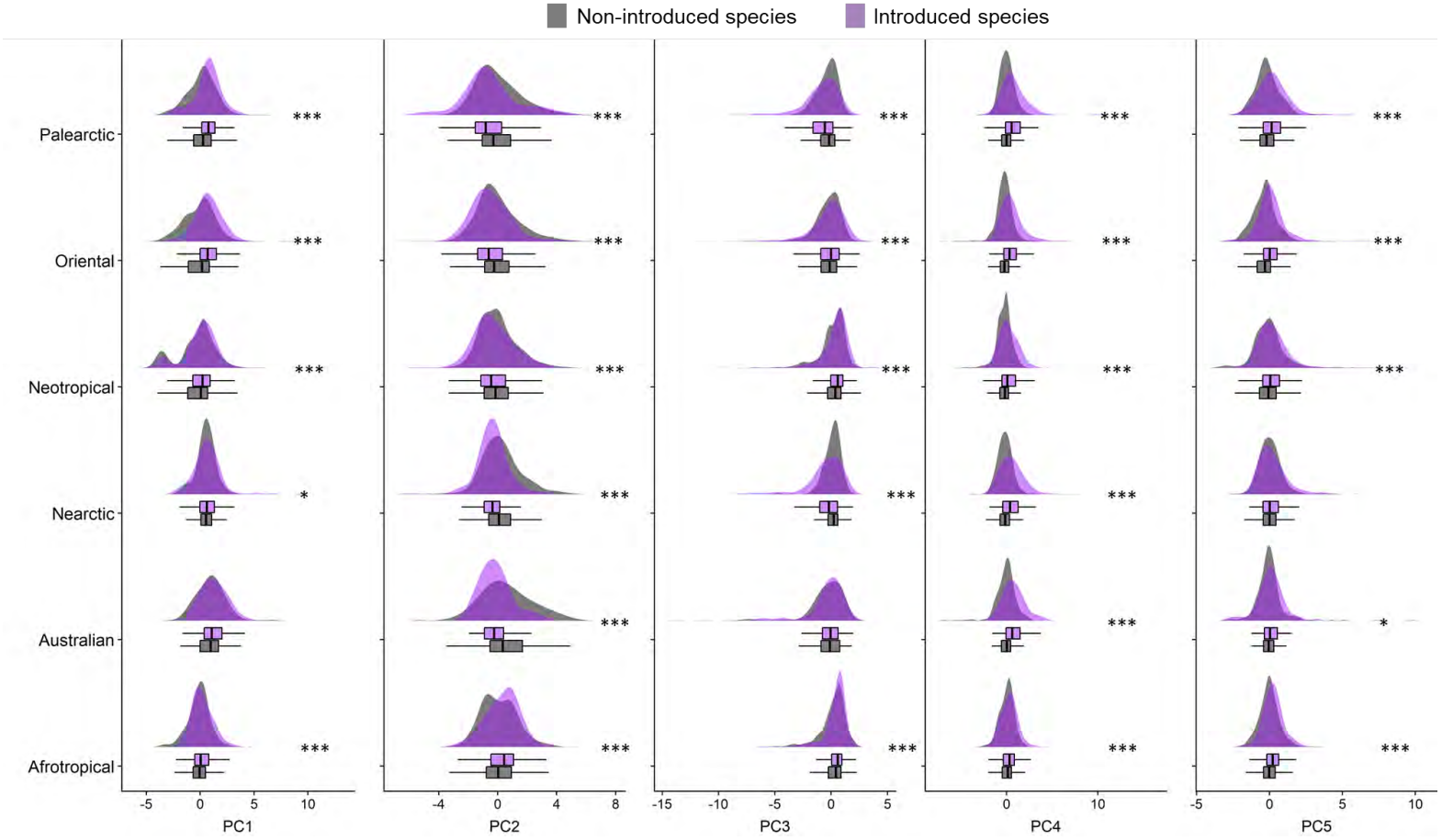



**Fig. S3.** Density distribution of species on the 5 PC axes for non-introduced, non-established and established species in each realm. Boxplots and results of K-S test between the non-introduced and introduced species are shown beside each plot item (*** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$).

# Chapter VI

## Human activities have disrupted the multiple facets of freshwater fish biodiversity

*Science, **371**, 835-838*

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CONSERVATION

# Human impacts on global freshwater fish biodiversity

Guohuan Su[1]*, Maxime Logez[2,3], Jun Xu[4,5], Shengli Tao[1], Sébastien Villéger[6]†, Sébastien Brosse[1]†

Freshwater fish represent one-fourth of the world's vertebrates and provide irreplaceable goods and services but are increasingly affected by human activities. A new index, Cumulative Change in Biodiversity Facets, revealed marked changes in biodiversity in >50% of the world's rivers covering >40% of the world's continental surface and >37% of the world's river length, whereas <14% of the world's surface and river length remain least impacted. Present-day rivers are more similar to each other and have more fish species with more diverse morphologies and longer evolutionary legacies. In temperate rivers, where the impact has been greatest, biodiversity changes were primarily due to river fragmentation and introduction of non-native species.

Rivers and lakes cover less than 1% of Earth's surface but represent substantial biodiversity, including nearly 18,000 fish species that constitute one-fourth of global vertebrates (*1*–*3*). These freshwater fishes support the functioning and stability of ecosystems through their contribution to biomass production and regulation of trophic networks and nutrient cycles (*4*). Freshwater fishes also contribute to human welfare as key food resources (*5*) and for recreative and cultural activities (*2*, *6*).

For centuries, human populations have directly affected fish biodiversity (*7*) through extraction and introduction of non-native species (*8*, *9*). Human activities have also modified the natural environment by changing land uses, altering flow regimes, fragmenting rivers by dams, polluting soil and waters, and altering climate; such actions indirectly favor extinction of native species and/or establishment of non-native species (*10*–*13*). Consequently, these direct and indirect anthropogenic impacts have led to modification of local species compositions (*8*, *9*). However, biodiversity is not restricted to purely taxonomic components but also includes functional and phylogenetic diversities. These two latter facets determine how organisms affect ecosystem functioning and stability (*14*–*18*) and are thus essential for conservation.

Here, we assess the extent to which six key facets of freshwater fish biodiversity—taxonomic, functional, and phylogenetic richness, plus the dissimilarity among river basins in each of these categories—have changed over the past two

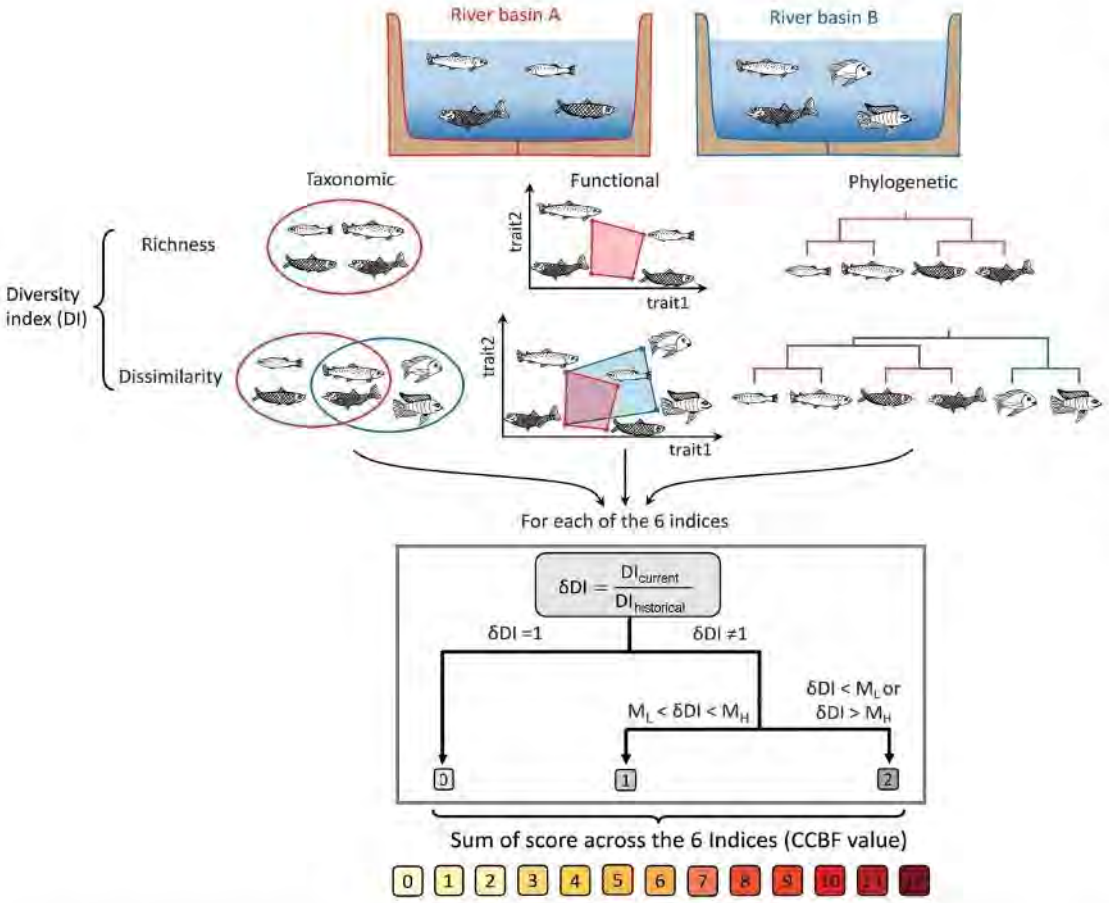



**Fig. 1. Framework to measure the cumulative change in biodiversity facets (CCBF).** δDI represents the change of a single diversity index among the six considered (taxonomic richness, functional richness, and phylogenetic richness within each river basin; taxonomic dissimilarity, functional dissimilarity, and phylogenetic dissimilarity between pairs of basins). $M_L$ is the median of all the values lower than 1; $M_H$ is the median of all the values higher than 1. The score is used to compute the CCBF index.

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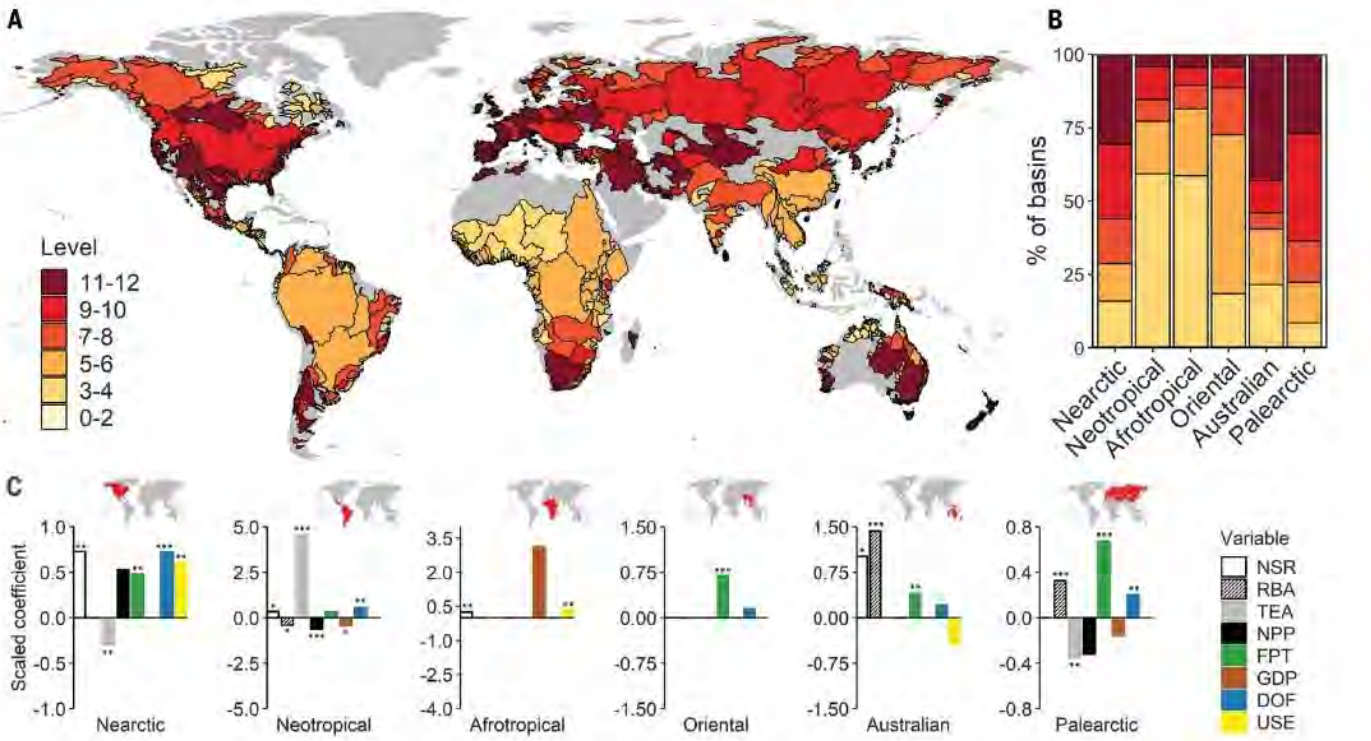



**Fig. 2. Cumulative change in biodiversity of freshwater fish faunas.** A cumulative index accounts for the sum of the changes in the six diversity indices: three facets (taxonomic, functional, and phylogenetic) measured at two scales (local and regional). (**A**) Map of the changes in 2456 river basins. (**B**) Percentage of river basins for six intensities of change in each biogeographic realm. (**C**) Scaled coefficient of the eight drivers of biodiversity change in an autoregressive error model in each realm (NSR, native species richness; RBA, river basin area; TEA, temperature anomaly since the last glacial maximum; NPP, net primary productivity; FPT, human footprint; GDP, gross domestic product; DOF, degree of fragmentation; USE, consumptive water use). Number of river basins used in the models: Afrotropical, $n$ = 198; Australian, $n$ = 525; Nearctic, $n$ = 241; Neotropical, $n$ = 350; Oriental, $n$ = 292; Palearctic, $n$ = 729. ***$P$ < 0.001, **$P$ < 0.01, *$P$ < 0.05.

centuries in 2456 river basins, covering almost the entire continental surface of Earth (excluding deserts and poles) and hosting >14,000 species (>80% of the global freshwater fish pool) (*19*). We computed an index of Cumulative Change in Biodiversity Facets (CCBF), which ranges from 0 to 12, with higher scores depicting stronger changes across more facets. A score higher than 6 indicates either changes in all six biodiversity facets or changes higher than the median in more than three facets (Fig. 1) (*20*).

More than half of the river basins (52.8%; 1297 basins representing 40.2% of the world continental surface and 37.3% of the world river length) had CCBF scores higher than 6 (Fig. 2), revealing deep and spatially distributed anthropogenic impacts on fish biodiversity. In contrast, one-third of the river basins (35.7%; 878 rivers) did not experience changes in local richness but only changes in dissimilarity relative to assemblages from the same realm (fig. S1). Those least-impacted river basins were mostly small-sized, occupying only 13.4% of the world river basin surface, and supported 3876 species, or only 21.7% of the world fish fauna. Moreover, the least-impacted rivers were overrepresented in Afrotropical and Australian regions, whereas the Neotropics, although being the richest in species, functional, and phylogenetic diversity (*21*, *22*), accounted for less than 6% of the "least impacted" category (fig. S1).

Fish assemblages from the temperate regions of Nearctic, Palearctic, and Australian realms experienced the largest biodiversity changes, with more than 60% of the rivers reaching a CCBF score higher than 6 (Fig. 2, A and B). Overall biodiversity changes in temperate regions (CCBF = 8.6 ± 0.1, mean ± SE) were higher than in tropical rivers (CCBF = 5.1 ± 0.1). For instance, large temperate rivers such as the Mississippi, Danube, or Murray-Darling had CCBF scores higher than 8, whereas large tropical rivers, such as the Amazon, Congo, or Mekong, were less impacted (CCBF = 6, fig. S2). Such a spatial pattern is consistent with previous studies on changes in taxonomic richness and dissimilarity of freshwater fishes (*8*, *9*) and with historical reports on anthropogenic degradation of ecosystems (*23*) but is not consistent with changes observed in other taxa, for which changes in biodiversity were the highest in tropical regions [e.g., for marine biome (*24*) and forest (*25*)].

Mapping the patterns of changes across the six diversity components revealed discrepancies between richness facets (Fig. 3). Except for a few rivers in the northern part of the Palearctic and Nearctic realms, fish biodiversity did not decline in most of the rivers (Fig. 3, A to C). This markedly differs from recent results documenting the decline in freshwater living resources at the local scale (i.e., >1 to 10 km of river stretch) within some of these river basins (*12*, *26*). Interestingly, we report an inverse trend in freshwater fish for local taxonomic, functional, and phylogenetic richness in more than half of the world's rivers (Fig. 3, A to C, and Fig. 4). This increase in local diversity is primarily explained by anthropogenic species introductions that compensate for or even exceed extinctions in most rivers (*27*). Among the 10,682 fish species considered, 170 fish species went extinct in a river basin, but this number might be underestimated because of the time lag between effective extinction and published extinction reports (*28*). In addition, 23% of freshwater fish species are currently considered to be threatened (*29*), and some of these might become extinct in the near future (*26*).

In addition to the overall increase in richness of fish assemblages in river basins, biotic homogenization—a general declining trend

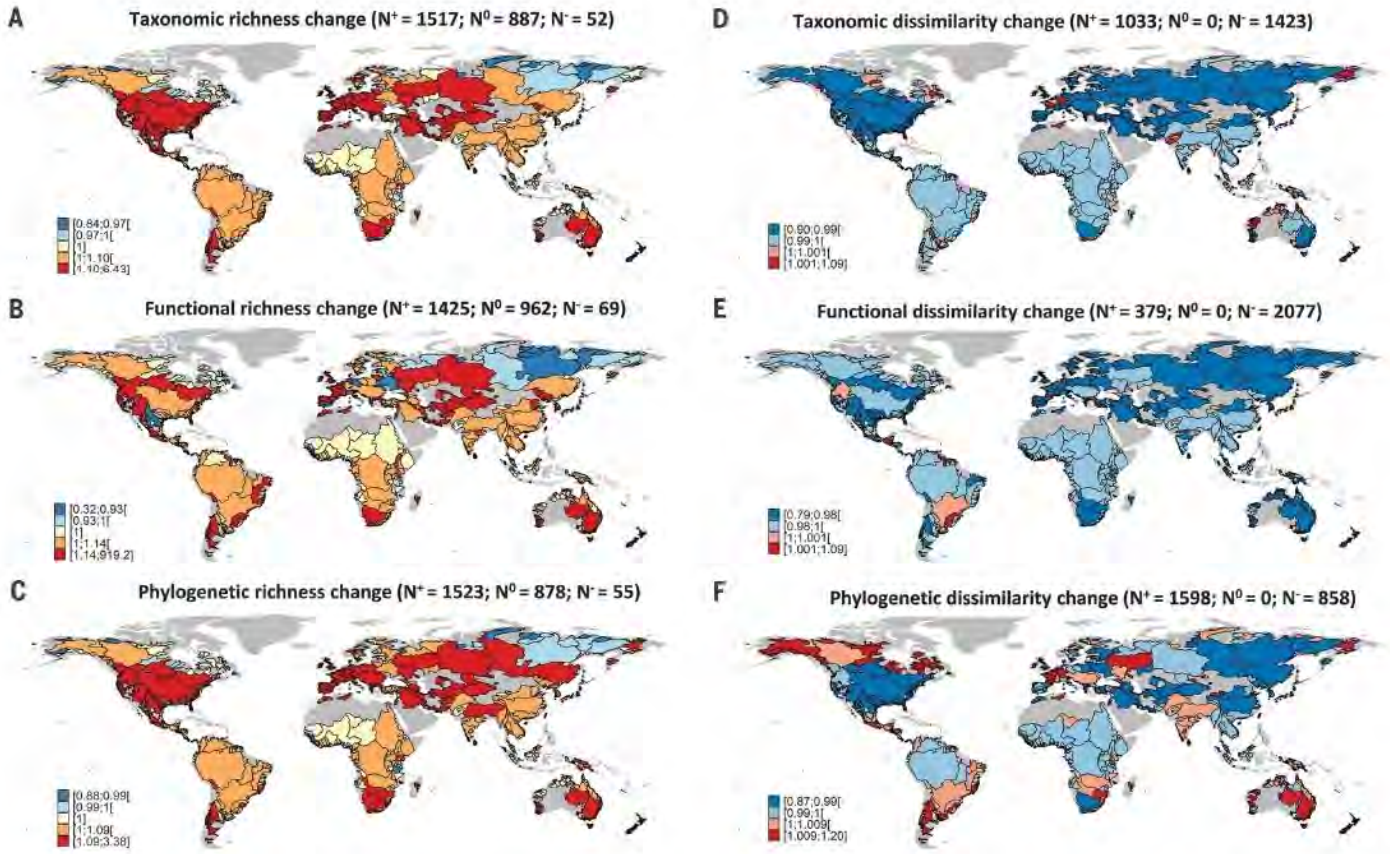



**Fig. 3. Changes in each of the six biodiversity indices for the world freshwater fish assemblages (2456 river basins).** (**A**) Taxonomic richness change; (**B**) functional richness change; (**C**) phylogenetic richness change; (**D**) taxonomic dissimilarity change; (**E**) functional dissimilarity change; (**F**) phylogenetic dissimilarity change. Legend values are the original ratio $DI_{current}/DI_{historical}$. Numbers of basins where fish diversity increased ($N^+$), remained unchanged ($N^0$), or decreased ($N^-$) are shown at the top of each panel.

in biological dissimilarity between river basins—appears pervasive throughout the world's rivers (Fig. 3, D to F). Functional dissimilarity was the most impacted facet, with a decrease in 84.6% of the rivers, whereas taxonomic dissimilarity and phylogenic dissimilarity decreased in only 58% and 35% of the rivers, respectively (Fig. 3, D to F). The discrepancy between change in functional diversity and changes in taxonomic and phylogenetic diversity (Fig. 4) primarily stems from the origin of non-native species introduced into rivers. Species translocated from a river to nearby basins promote losses of dissimilarity because they often already occur as natives in many rivers of the realm and are often functionally and phylogenetically close to other native species (*9*, *30*). In contrast, exotic species (i.e., originating from other realms) are less frequently introduced, and their divergent evolutionary history with native species led to increased phylogenetic dissimilarity of their recipient rivers (*30*). For instance, the exotic species introduced in only a few rivers of Europe (e.g., the mosquitofish, *Gambusia affinis*, established in southwestern Europe, or the brook trout, *Salvelinus fontinalis*, established in cold-water ecosystems) (*30*), markedly enhanced the phylogenetic dissimilarity between those rivers. However, exotic species even from distinct evolutionary lineages could share functional traits with some native species, thereby leading to increased phylogenetic dissimilarity but decreased functional dissimilarity (Fig. 3). For instance, European trout (*Salmo trutta*) and Pacific salmon (*Oncorhynchus mykiss*) belong to an order (Salmoniformes) absent from the Australian realm, but those exotic salmonids are functionally similar to some native Australian fishes such as the spotted mountain trout (*Galaxias truttaceus*; Osmeriformes) (*31*).

The CCBF score was positively linked to human activities related to industrialization and economic development, such as human footprint [FPT (*23*)], with an increase of biodiversity changes with increased FPT in all the industrialized and populated realms. River fragmentation by dams, represented by the degree of fragmentation [DOF (*32*)] index, was also a widespread disturbance in the Nearctic and Palearctic realms (Fig. 2C) that experienced intensive damming for more than a century (*33*). Fragmentation by dams was also a significant driver of biodiversity change in the Neotropics, probably due to the rise of hydropower dam construction in this realm (*34*). Higher DOF values were reached in small- or medium-sized rivers, whereas the largest and most diverse rivers such as the Amazon, Orinoco, or Congo remain mostly free-flowing (*32*), although the current rise of dam construction on those rivers (*35*, *36*) will constitute a major threat to their biodiversity. Apart from river fragmentation, consumptive water use for agriculture and industry [USE (*32*)] was a significant driver of CCBF increase in the Nearctic realm, where water withdrawal for agriculture is intense (*32*, *37*) and acts in synergy with increasing DOF. In the Afrotropics, USE was the only significant human driver of CCBF, due to marked consumptive water use in regions with marked seasonal aridity (*32*, *37*). In addition, the CCBF score was positively correlated to richness in native species in most of the realms, indicating that the most speciose rivers are also the most impacted by biodiversity changes. Moreover, no negative associations between species richness and the CCBF were

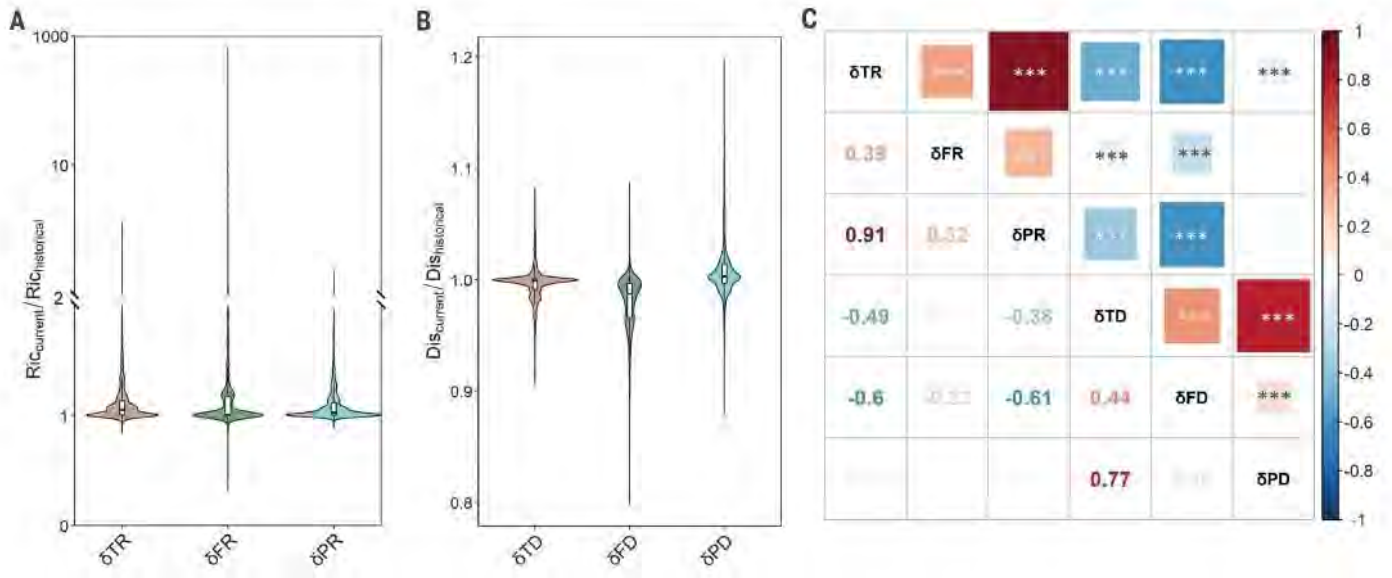



**Fig. 4. Changes in biodiversity from historical period to current period.** (**A** and **B**) Violin plots show the distribution of values of the three richness change indices (A) and the three dissimilarity change indices (B). (**C**) Pearson correlation between the changes in diversity indices (river basin number = 2456, ***$P < 0.001$).

observed, providing little support for the hypothesis of biotic resistance that assumes a higher resistance of species-rich assemblages against disturbances (*38*–*40*).

Conserving freshwater fish diversity in the least-impacted rivers (accounting for 13.4% of the world basin surface) will remain below the target to protect at least 30% of Earth's surface by 2030, as proposed by a broad coalition of environmental organizations (*41*, *42*). This result suggests that reaching the freshwater fish target must involve consideration of not only the least impacted rivers but also areas where biodiversity has already been eroded by human activity. Moreover, conservation has moved toward systematically identifying regions in need of protection (*43*). The discrepancy in biodiversity erosion we report between freshwater ecosystems and marine and terrestrial ecosystems (*24*, *25*) demonstrates that current measures of biodiversity erosion, derived from marine and terrestrial organisms, do not apply to freshwaters, and thus underscores the need to develop freshwater-focused conservation priorities. In addition, the mismatches between changes in taxonomic, functional, and phylogenetic dissimilarities among the world freshwater fish fauna highlight the risk of evaluation based on change in a single facet as a surrogate of the changes in other facets. More important, our results highlight the need to consider the cumulative and synergistic effects of multiple human activities on the complementary facets of biodiversity. The CCBF index we propose presents a holistic measure of multiple measures of biodiversity change and offers potential for prioritizing and informing adaptive management and global conservation targets. Future studies and planning need to expand the focus from simple loss of species to integrated changes in facets of biodiversity resulting from interactions between synergetic human activities.

## REFERENCES AND NOTES


1. G. H. Allen, T. M. Pavelsky, *Science* **361**, 585–588 (2018).
2. D. Dudgeon *et al.*, *Biol. Rev. Camb. Philos. Soc.* **81**, 163–182 (2006).
3. R. van der Laan, *Freshwater Fish List* (Almere, Netherlands, ed. 30, 2020).
4. S. Villéger, S. Brosse, M. Mouchet, D. Mouillot, M. J. Vanni, *Aquat. Sci.* **79**, 783–801 (2017).
5. R. Hassan, R. Scholes, N. Ash, *Ecosystems and Human Well-Being: Current State and Trends* (Island, 2005).
6. R. Arlinghaus *et al.*, *Proc. Natl. Acad. Sci. U.S.A.* **116**, 5209–5213 (2019).
7. R. C. Hoffmann, *Am. Hist. Rev.* **101**, 631–669 (1996).
8. F. Leprieur, O. Beauchard, S. Blanchet, T. Oberdorff, S. Brosse, *PLOS Biol.* **6**, e28 (2008).
9. S. Villéger, S. Blanchet, O. Beauchard, T. Oberdorff, S. Brosse, *Proc. Natl. Acad. Sci. U.S.A.* **108**, 18003–18008 (2011).
10. H. M. Pereira *et al.*, *Science* **330**, 1496–1501 (2010).
11. C. J. Vörösmarty *et al.*, *Nature* **467**, 555–561 (2010).
12. A. J. Reid *et al.*, *Biol. Rev. Camb. Philos. Soc.* **94**, 849–873 (2019).
13. D. Tickner *et al.*, *Bioscience* **70**, 330–342 (2020).
14. D. Mouillot, N. A. Graham, S. Villéger, N. W. Mason, D. R. Bellwood, *Trends Ecol. Evol.* **28**, 167–177 (2013).
15. S. Naeem, J. E. Duffy, E. Zavaleta, *Science* **336**, 1401–1406 (2012).
16. D. Craven *et al.*, *Nat. Ecol. Evol.* **2**, 1579–1587 (2018).
17. P. Brun *et al.*, *Nat. Commun.* **10**, 5691 (2019).
18. C. Pimiento *et al.*, *Sci. Adv.* **6**, eaay7650 (2020).
19. P. A. Tedesco *et al.*, *Sci. Data* **4**, 170141 (2017).
20. See supplementary materials.
21. A. Toussaint, N. Charpin, S. Brosse, S. Villéger, *Sci. Rep.* **6**, 22125 (2016).
22. D. L. Rabosky, *J. Biogeogr.* **47**, 1207–1217 (2020).
23. O. Venter *et al.*, *Sci. Data* **3**, 160067 (2016).
24. S. A. Blowes *et al.*, *Science* **366**, 339–345 (2019).
25. M. C. Hansen *et al.*, *Science* **342**, 850–853 (2013).
26. J. S. Albert *et al.*, *Ambio* **50**, 85–94 (2021).
27. A. Toussaint *et al.*, *Ecol. Lett.* **21**, 1649–1659 (2018).
28. S. T. Jackson, D. F. Sax, *Trends Ecol. Evol.* **25**, 153–160 (2010).
29. IUCN, *The IUCN Red List of Threatened Species* Version 2017-3 (2018); www.iucnredlist.org.
30. S. Villéger, G. Grenouillet, S. Brosse, *Glob. Ecol. Biogeogr.* **23**, 1450–1460 (2014).
31. S. Burgin, *Sustainability* **9**, 280 (2017).
32. G. Grill *et al.*, *Nature* **569**, 215–221 (2019).
33. B. Lehner *et al.*, *Front. Ecol. Environ.* **9**, 494–502 (2011).
34. C. Zarfl, A. E. Lumsdon, J. Berlekamp, L. Tydecks, K. Tockner, *Aquat. Sci.* **77**, 161–170 (2015).
35. E. P. Anderson *et al.*, *Sci. Adv.* **4**, eaao1642 (2018).
36. K. O. Winemiller *et al.*, *Science* **351**, 128–129 (2016).
37. P. Döll, K. Fiedler, J. Zhang, *Hydrol. Earth Syst. Sci.* **13**, 2413–2432 (2009).
38. J. E. Duffy, J. S. Lefcheck, R. D. Stuart-Smith, S. A. Navarrete, G. J. Edgar, *Proc. Natl. Acad. Sci. U.S.A.* **113**, 6230–6235 (2016).
39. M. Loreau *et al.*, *Science* **294**, 804–808 (2001).
40. J. M. Levine, *Science* **288**, 852–854 (2000).
41. E. Dinerstein *et al.*, *Sci. Adv.* **5**, eaaw2869 (2019).
42. S. L. Pimm, C. N. Jenkins, B. V. Li, *Sci. Adv.* **4**, eaat2616 (2018).
43. C. R. Margules, R. L. Pressey, *Nature* **405**, 243–253 (2000).


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## SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S4
Tables S1 to S3
References (*44*–*81*)

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Science

# Human impacts on global freshwater fish biodiversity

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**No waters left untouched**

We are increasingly aware of human impacts on biodiversity across our planet, especially in terrestrial and marine systems. We know less about fresh waters, including large rivers. Su *et al.* looked across such systems globally, focusing on several key measures of fish biodiversity. They found that half of all river systems have been heavily affected by human activities, with only very large tropical river basins receiving the lowest levels of change. Fragmentation and non-native species have also led to the homogenization of rivers, with many now containing similar species and fewer specialized lineages.

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# Supplementary Materials for

## Human impacts on global freshwater fish biodiversity

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**This PDF file includes:**

Materials and Methods
Figs. S1 to S4
Table S1
Captions for tables S2 and S3
References

**Other supplementary material for this manuscript includes:**

Tables S2 and S3 (Excel format)

## Materials and methods

**Occurrence data:** We used the most comprehensive database of freshwater fish species distributions across the world [(*19*) available at http://data.freshwaterbiodiversity.eu]. The fish occurrence database gathers the occurrence of 14,953 species (more than 83% of the freshwater fish species) in 3,119 drainage basins, covering more than 80% of the Earth's surface (*19*). Fish occurrence in each river basin accounts for all the freshwater fish species inhabiting the entire river network of each basin, from 1$^{st}$ order streams to the sea. Each occurrence is paired with a status, either native or non-native established if the species was not historically present in the river basin. Each river basin was assigned to one of the six terrestrial biogeographic realms [i.e., Afrotropical, Australian (including Oceania), Nearctic, Neotropical, Oriental and Palearctic] according to Lévêque et al. (*44*) and Brosse et al. (*45*). Historical fish assemblages composition in the river basins refers to only native species, and thus roughly corresponds to the preindustrial period (i.e., before the 18$^{th}$ century), from when industrialization began and fish introductions for aquaculture, fishing, and ornamental purposes sharply increased (*8, 46, 47*). Similarly, the current sixth mass extinction rises from the beginning of the industrial period (*48*). Therefore, despite a few human mediated species introductions and extinctions occurred before the 18$^{th}$ century (e.g., common carp, *Cyprinus carpio*, introduction in Western Europe), most are more recent. Current fish assemblages composition refers to the non-native species and excludes the local extirpated or extinct native ones. Extirpations refer to the extinction of a fish species within a river basin and data were extracted from the literature reviews of Brosse et al. (*45*) and Diaz et al. (*49*). We then updated these data using IUCN Red lists (*29*). Species with "extinct" or

"extinct in the wild" status in the IUCN Red list were thus considered as extinct in their native river basins (Table S2). Although species extirpations/extinctions are probably underestimated for a number of reasons such as the time lag between local report of species extinction and validation of extinction over an entire river basin, we here used the most comprehensive and updated information on fish extinction at the river basin scale.

**Functional traits:** Among the 14,953 species present in the occurrence database, 10,705 species were morphologically described using pictures and drawings from textbooks and scientific websites. Morphology was assessed using ten traits describing the size and shape of body parts involved in food acquisition and locomotion (*19, 21*). The fish size was described using the maximum body length (Max. Body Length) taken from (*50*). Those maximum body lengths were carefully reviewed, and irrelevant measures have been corrected according to appropriate literature. In addition to size, 11 morphological measures were assessed on side view pictures (Fig. S4a) collected during an extensive literature review including our field data and scientific literature sources made of peer-reviewed articles, books, and scientific websites. We collected at least one picture (photograph or scientific drawing) per species. Only good quality pictures and scientific side view drawings of entire adult animals, with confirmed species identification, were kept. For species with marked sexual dimorphism, we considered male morphology, as female pictures are scarce for most species (especially for Perciformes and Cyprinodontiformes). Intraspecific morphological traits variability was not considered in this study as it hardly affects functional diversity at the large spatial resolution considered (*27*). The nine unitless traits describing the morphology of the fish head (including mouth and eye), body, pectoral and caudal fins (Fig. S4b) were computed as ratios between 11 morphological

measures done using ImageJ software (http://rsb.info.nih.gov/ij/index.html). The 10 morphological traits (9 unitless ratios and body size) selected are commonly used in assessment of fish functional diversity [e.g., (*21, 51-53*)] and are linked to the feeding and locomotion functions of fish that themselves determine their contribution to key ecosystem processes such as controlling food webs and nutrient cycles (*4*) (Fig. S4b). The 10 traits were not markedly correlated to each other (Spearman's correlation coefficient, |rho|<0.45 for all the 45 pairwise comparisons). Functional traits not measurable on side pictures, such as gut length, oral gape area and shape, were not included because they are currently only available for a few species in public databases.

Some species have unusual morphologies (species without tails, flatfishes) that prevent from measuring some morphological traits. We thus applied conventions as mentioned in Su et al. (*51*), Toussaint et al. (*21*) and Villéger et al. (*53*) for these few exceptions. Due to the lesion of body parts or the quality of fish pictures, some traits have not been measured for some species. Overall, 24.1% of the values were missing in the raw morphological traits dataset (from 6.9% for maximum body length to 31.4% for relative maxillary length). Those missing values were filled using a phylogenetic generalized linear model (*54, 55*). We then computed a principal component analysis (PCA) using values of the 10 morphological traits for all the species. We selected the first four PCA axes, which explained 68.2% of the total variance among the world's fish functions, to compute the functional diversity indices.

**Phylogenetic diversity:** Phylogenetic distances between all species were computed on the tree from Rabosky et al. (*56*), including 31,526 marine and freshwater ray-finned fishes. This dataset is based on 11,638 species whose position was estimated from genetic data; the remaining

19,888 species were placed in the tree using stochastic polytomy resolution (*56*).

**Environmental and Anthropogenic variables:** We selected four environmental and four anthropogenic variables as proxies of the main processes responsible from native biodiversity and impacts of human activities on freshwater ecosystems: (i) NSR: native species richness. (ii) RBA: river basin area; (iii) NPP: net primary productivity; (iv) TEA: temperature anomaly from the Last Glacial Maximum to the present; (v) DOF: degree of fragmentation; (vi) FPT: human footprint; (vii) GDP: gross domestic product; and (viii) USE: consumptive water use. These 8 metrics were overall independent of each other (|Pearson's r| < 0.5), to the exception of NSR and RBA (Fig. S3).

NSR accounts for the biotic resistance hypothesis, which assumes a higher resistance of species rich assemblages against disturbances (*40*). RBA, NPP and TEA were from Tedesco et al. (*19*), and account for the three main hypotheses explaining biodiversity, namely the species-area hypothesis that predicts a positive relationship between river basin area and biodiversity; the species-energy hypothesis that predicts higher biodiversity in energy rich areas, and the historical contingency, which has largely been influenced by the last glacial events in freshwater fish (*57-60*).

FPT is a comprehensive representation of anthropogenic threats to biodiversity, which cumulatively accounts for eight human pressures—built environments, croplands, pasture lands, human population density, night lights, railways, major roadways and navigable waterways (*23*). FPT dataset (resolution: 1 $km^2$) was taken from Venter et al. (*23*). GDP measures the size of the economy and is defined as the market value of all final goods and services produced within a region in a given period (*61, 62*). GDP dataset (1 square degree resolution) was taken from

Nordhaus & Chen (*61*).

DOF accounts for the degree to which river networks are fragmented longitudinally by infrastructure, such as hydropower and irrigation dams (*32*). DOF dataset (resolution: 500 $m^2$) was taken from Grill et al. (*32*).

USE accounts for water consumption for irrigation, industry, municipal uses and water transfer to other river systems. USE (resolution: 1 $km^2$) for each river basin was calculated by using $100*(d_{nat}-d_{ant})/d_{nat}$, where $d_{nat}$ represents the total amount of long-term discharge without human influences in each river basin and $d_{ant}$ represents the total amount of average long-term discharge after human extractions and use in each river basin. $d_{ant}$ and $d_{nat}$ were both taken from the WaterGAP model (*32, 63*).

We mapped FTP, GDP, DOF and USE by their relative resolution grid data over the basin-scale map and then calculated the mean value of all the cells covered by each basin. Here we considered the 2,335 river basins (out of the 3,119) with available values for both CCBF (see below) and the eight environmental and anthropogenic variables.

**Measuring temporal changes in biodiversity of freshwater fishes**: Among the 3,119 river basins with fish occurrence data, diversity indices were measured for all basins with more than 5 fish species to meet the requirements of functional diversity calculation, leading to consider a total of 2,456 river basins. 10,682 species were obtained after matching the occurrence, functional and phylogenetic databases. We assessed the 6 facets of biodiversity (Fig. 1) for fish assemblage inhabiting each of the 2,456 river basins for the current and historical period: taxonomic richness (TR) measured as the number of species, functional richness (FR) measured as the volume of the functional space occupied by an assemblage [i.e., the volume of the

minimum convex hull in the functional space which includes all the species in the assemblage, (*64*)], and phylogenetic richness (PR) as the total length of branches linking all species from the assemblage on the phylogenetic tree (*65*). In addition to these indices describing diversity within each assemblage (i.e., alpha-diversity), we also accounted for the dissimilarity among assemblages (i.e., beta-diversity). More specifically, we quantified taxonomic, functional and phylogenetic dissimilarity between each pair of fish assemblages from the same realm as the proportion of total richness in the pair that is not shared by the assemblages, [i.e., Jaccard-index for taxonomic dissimilarity (*66*), beta-FRic for functional dissimilarity (*64*) and UniFrac for phylogenetic dissimilarity (*67*)]. Then the average value of dissimilarity between a fish assemblage and all the other assemblages from the same realm was computed to get for each river basin values of taxonomic dissimilarity (TD), functional dissimilarity (FD), and phylogenetic dissimilarity (PD).

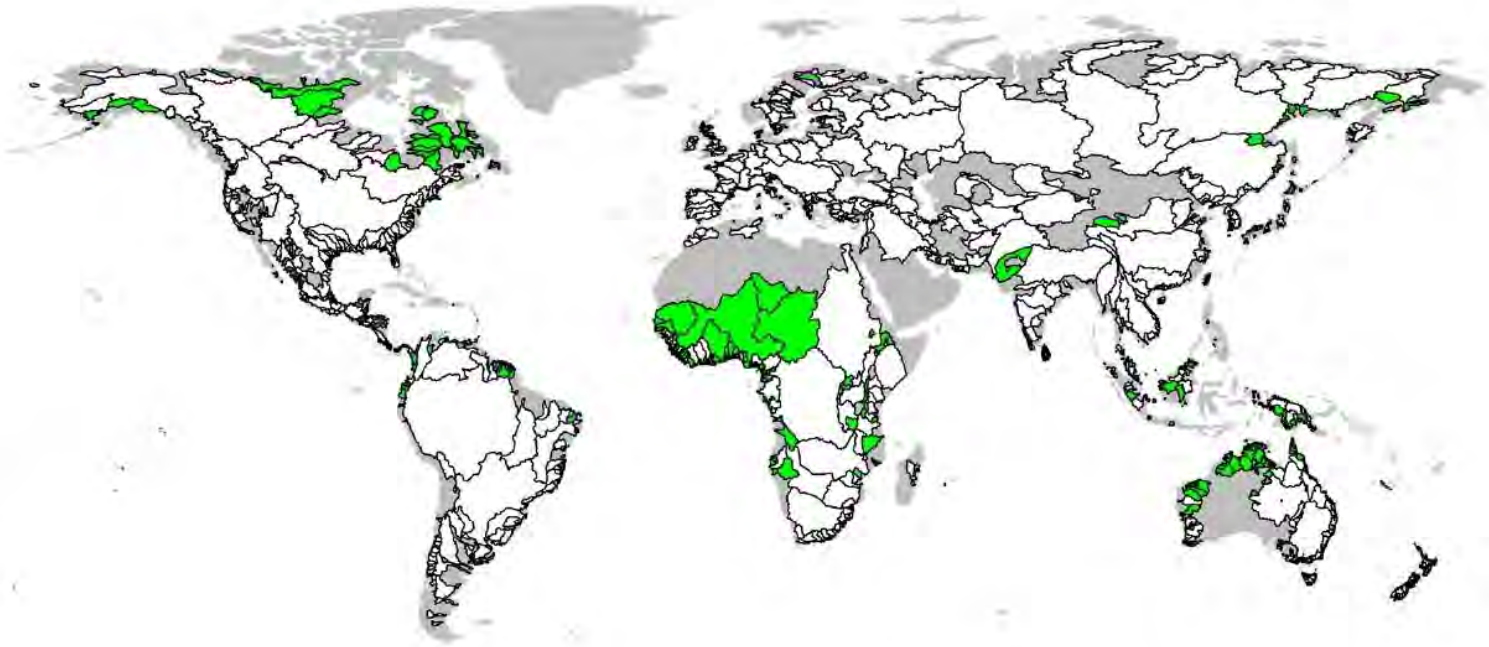
We then calculated for each of these six diversity indices the temporal change (δDI) as the ratio: $DI_{current} / DI_{historical}$. We then computed score for each δDI according to its values: If δDI = 1, it scores 0; if δDI is higher than the median of all the values lower than 1 and lower than the median of all the values higher than 1, it scores 1; and if δDI is lower than the median of all the values lower than 1 or higher than the median of all the values higher than 1, it scores 2. Then we sum up the scores of the six δDI for each basin to get the index of cumulative changes in biodiversity facets which ranges from 0 to 12 (CCBF, Fig. 1).

Thus, our cumulative index accounts for all marked changes in biodiversity facets, not only species loss. Null values are possible only if taxonomic, functional and phylogenetic composition of all assemblages from a realm remained unchanged because otherwise all

dissimilarity indices are changed. CCBF scores from 0 to 6 account for moderate changes in biodiversity for all the six facets (all the 6 facets scoring 0 or 1) or strong changes for no more than half of the facets (no more than 3 facets scoring 2). Such CCBF values are considered as moderate changes in biodiversity. CCBF scores from 7 to 12 account for strong changes in biodiversity with all 6 facets changes or more than half of the facets scoring 2. See table S3 for the six diversity indices and CCBF scores for the 2,456 basins.

**Statistical analyses:** To assess how environmental processes and human activities contributed to the observed change in biodiversity in each realm, we quantified the relative contribution of NSR, RBA, TEA, NPP, FPT, GDP, DOF and USE to the CCBF values of the 2,335 river basins for which all variables were available, using spatial simultaneous autoregressive error models ($SAR_{error}$). These eight variables were previously scaled to a zero mean and unit variance to ensure equal weighting in the models. We first ran the null model (intercept-only) with none of the variables as a reference. Then we used stepwise regression to select the best models by AIC (Akaike's Information Criterion). We eventually selected the model with the lowest AIC (*68*) (Table S2). We used Nagelkerke's R2 (*69*) as the pseudo R-squared to qualify the final models' performance. After model fitting, we checked for broad spatial autocorrelation in model residuals by computing the Moran's *I* statistic (*70*).

All statistical analyses were performed with the R software environment version 3.3 (*R Core Team*), including the library 'RPhylopars' (*55*) for filling the missing values in the trait database, 'betapart' for computing dissimilarity indices (*71, 72*), 'spatialreg' and 'spdep' for developing $SAR_{error}$ models (*73*) and performing Moran's *I* tests (*74*).



| | Nearctic | Neotropical | Afrotropical | Oriental | Australian | Palearctic | World |
|---|---|---|---|---|---|---|---|
| Number of basin without changes in richness | 76 | 225 | 116 | 97 | 198 | 166 | 878 |
| Percentage of basin number (%) | 31.54 | 60 | 57.43 | 28.87 | 37.08 | 21.61 | 35.75 |
| Percentage of basin area (%) | 11.45 | 5.6 | 34.9 | 10.06 | 26.65 | 1.94 | 13.4 |
| Percentage of river length (%) | 12.77 | 6.02 | 34.7 | 11.15 | 27.21 | 2.05 | 13.42 |

**Fig. S1.**

River basins (green color) where the three richness diversity facets remained unchanged from historical to current period.




| Basin names | δTR | δFR | δPR | δTD | δFD | δPD | CCBF |
|---|---|---|---|---|---|---|---|
| Amazon | 1.0026 | 1.0003 | 1.0036 | 0.9997 | 0.9980 | 0.9991 | 6 |
| Congo | 1.0073 | 1.0061 | 1.0122 | 0.9991 | 0.9952 | 0.9989 | 6 |
| Mekong | 1.0346 | 1.0061 | 1.0394 | 0.9977 | 0.9958 | 0.9973 | 6 |
| Danube | 1.2247 | 1.1238 | 1.1865 | 0.9797 | 0.9937 | 1.0051 | 9 |
| Mississippi | 1.1523 | 1.0727 | 1.1863 | 0.9802 | 0.9955 | 0.9885 | 10 |
| Murray-Darling | 1.2642 | 1.1458 | 1.2800 | 0.9906 | 0.9714 | 1.0189 | 11 |

**Fig. S2.**

Changes in freshwater fish biodiversity for 6 representative river basins over the world. δTR: taxonomic richness change; δFR: functional richness

change; δPR: phylogenetic richness change; δTD: taxonomic dissimilarity change; δFD: functional dissimilarity change; δPD: phylogenetic dissimilarity change. CCBF is the index of cumulative change in biodiversity facets.

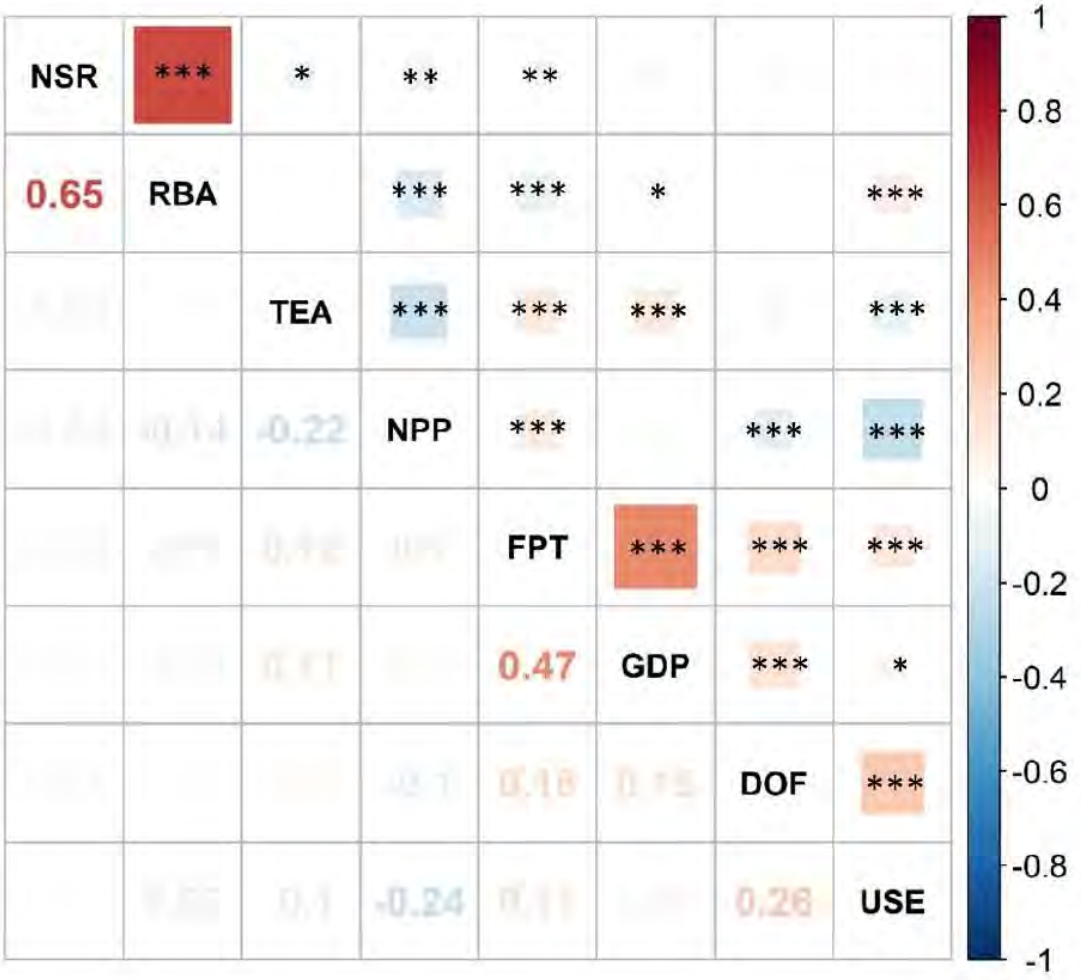
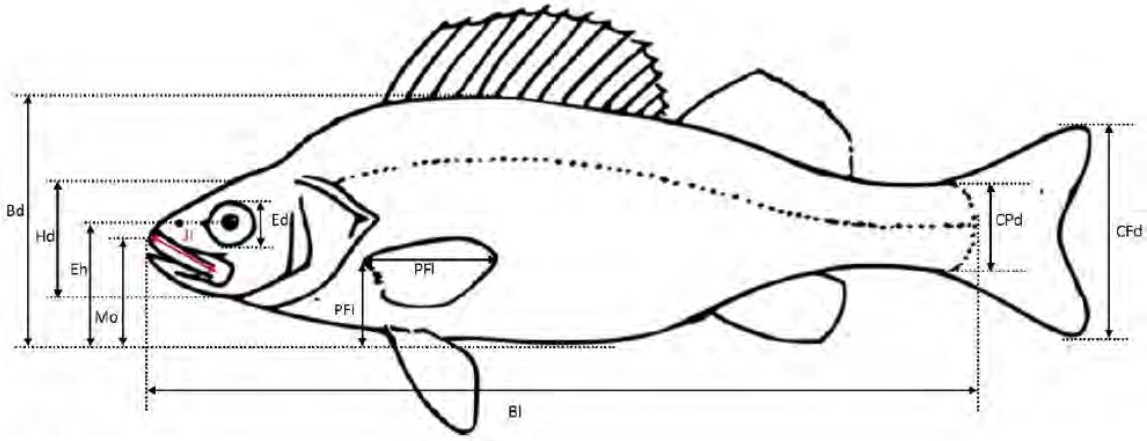



**Fig. S3.**

Pearson correlation between the eight environmental and human activity variables of the world river basins (n = 2,335). NSR: native species richness; RBA: river basin area; TEA: temperature anomaly since the last glacial maximum; NPP: net primary productivity; FPT: human footprint; GDP: gross domestic product; DOF: degree of fragmentation; USE: consumptive water use. (*** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$)




a. Morphological measurements

| Code | Name | Protocol for measurement |
|---|---|---|
| Blmax | Maximum Body length | Maximum adult length |
| Bl | Body length | Standard length (snout to caudal fin basis) |
| Bd | Body depth | Maximum body depth |
| Hd | Head depth | Head depth at the vertical of eye |
| CPd | Caudal peduncle depth | Minimum depth of the caudal peduncle |
| CFd | Caudal fin depth | Maximum depth of the caudal fin |
| Ed | Eye diameter | Vertical diameter of the eye |
| Eh | Eye position | Vertical distance between the centre of the eye to the bottom of the body |
| Mo | Oral gape position | Vertical distance from the top of the mouth to the bottom of the body |
| Jl | Maxillary jaw length | Length from snout to the corner of the mouth |
| PFl | Pectoral fin length | Length of the longest ray of the pectoral fin |
| PFi | Pectoral fin position | Vertical distance between the upper insertion of the pectoral fin to the bottom of the body |

All measurements were made on pictures except Blmax values, which were downloaded from Fishbase.org

b. Morphological traits

| Morphological traits | Formula | Potential link with fish functions | References |
|---|---|---|---|
| Maximum body length | BLmax | Size is linked to metabolism, trophic impacts, locomotion ability, nutrient cycling | (*21*) |
| Body elongation | $\frac{Bl}{Bd}$ | Hydrodynamism | (*75*) |
| Eye vertical position | $\frac{Eh}{Bd}$ | Position of fish and/or of its prey in the water column | (*76*) |
| Relative eye size | $\frac{Ed}{Hd}$ | Visual acuity | (*77*) |
| Oral gape position | $\frac{Mo}{Bd}$ | Feeding position in the water column | (*78, 79*) |
| Relative maxillary length | $\frac{Jl}{Hd}$ | Size of mouth and strength of jaw | (*21*) |
| Body lateral shape | $\frac{Hd}{Bd}$ | Hydrodynamism and head size | (*21*) |
| Pectoral fin vertical position | $\frac{PFi}{Bd}$ | Pectoral fin use for swimming | (*78*) |
| Pectoral fin size | $\frac{PFl}{Bl}$ | Pectoral fin use for swimming | (*80*) |
| Caudal peduncle throttling | $\frac{CFd}{CPd}$ | Caudal propulsion efficiency through reduction of drag | (*81*) |

**Fig. S4.**

Morphological measurements (a) and morphological traits (b) measured on each fish species. For each morphological trait, the potential link with food acquisition and locomotion and associated references are provided.

**Table S1.**

Results of the spatial simultaneous autoregressive error models ($SAR_{error}$) showing the coefficients of the selected optimal model in each realm**.** Model with the lowest AIC was selected for each realm. (NSR: native species richness; RBA: river basin area; TEA: temperature anomaly since the last glacial maximum; NPP: net primary productivity; FPT: human footprint; GDP: gross domestic product; DOF: degree of fragmentation; USE: consumptive water use; AIC: Akaike's Information Criterion). The Moran's *I* value represents the remaining autocorrelation on the residuals of the model for the first distance class (i.e., neighbor drainages) in each realm.

**Nearctic (n=241)**

| | | | |
|---|---|---|---|
| NULL Model | | | AIC = 1416.66 |
| Initial Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF+USE | | | AIC = 1116.66 |
| **Optimal Model:NSR+TEA+NPP+FPT+DOF+USE** | | | **AIC = 1113.68** |
| Statistics in Optimal Model | coefficient (SE) | z-value | *P* |
| NSR | 0.73(0.3) | 2.433 | **0.015** |
| TEA | -0.297(0.139) | -2.139 | **0.0325** |
| NPP | 0.526(0.297) | 1.769 | 0.0768 |
| FPT | 0.482(0.187) | 2.581 | **0.0099** |
| DOF | 0.722(0.197) | 3.667 | **0.0002** |
| USE | 0.604(0.196) | 3.077 | **0.0021** |
| **Nagelkerke's R2** | 0.353 | | |
| **Moran's I** | -0.006 | | n.s. |

| **Neotropical (n=350)** | | | |
|---|---|---|---|
| NULL Model | | | AIC = 1578.34 |
| Initial Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF+USE | | | AIC = 1546.92 |
| **Optimal Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF** | | | **AIC = 1544.92** |
| Statistics in Optimal Model | coefficient (SE) | z-value | *P* |
| NSR | 0.344(0.142) | 2.421 | **0.0155** |
| RBA | -0.412(0.168) | -2.45 | **0.0143** |
| TEA | 4.54(1.166) | 3.893 | **0.0001** |
| NPP | -0.63(0.189) | -3.331 | **0.0009** |
| FPT | 0.32(0.189) | 1.692 | 0.0907 |
| GDP | -0.434(0.192) | -2.265 | **0.0235** |
| DOF | 0.535(0.203) | 2.632 | **0.0085** |
| **Nagelkerke's R2** | 0.336 | | |
| **Moran's I** | -0.03 | | n.s. |

| **Afrotropical (n=198)** | | | |
|---|---|---|---|
| NULL Model | | | AIC = 831.50 |
| Initial Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF+USE | | | AIC = 827.53 |
| **Optimal Model: NSR+GDP+USE** | | | **AIC = 821.11** |
| Statistics in Optimal Model | coefficient (SE) | z-value | *P* |
| NSR | 0.247(0.09) | 2.749 | **0.006** |
| GDP | 3.108(1.598) | 1.945 | 0.0518 |
| USE | 0.355(0.127) | 2.799 | **0.0051** |
| **Nagelkerke's R2** | 0.444 | | |
| **Moran's I** | -0.044 | | n.s. |

| **Oriental (n=292)** | | | |
|---|---|---|---|
| NULL Model | | | AIC = 1154.37 |
| Initial Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF+USE | | | AIC = 1146.19 |
| **Optimal Model: FPT+DOF** | | | **AIC = 1137.89** |
| Statistics in Optimal Model | coefficient (SE) | z-value | *P* |
| FPT | 0.695(0.149) | 4.675 | **<0.0001** |
| DOF | 0.144(0.083) | 1.739 | 0.082 |
| **Nagelkerke's R2** | 0.246 | | |
| **Moran's I** | -0.003 | | n.s. |

| **Australian (n=525)** | | | |
|---|---|---|---|
| NULL Model | | | AIC = 2382.61 |
| Initial Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF+USE | | | AIC = 2361.76 |
| **Optimal Model: NSR+RBA+FPT+DOF+USE** | | | **AIC = 2357.49** |
| Statistics in Optimal Model | coefficient (SE) | z-value | *P* |
| NSR | 1.015(0.474) | 2.143 | **0.0321** |
| RBA | 1.43(0.328) | 4.365 | **<0.0001** |
| FPT | 0.4(0.182) | 2.194 | **0.0282** |
| DOF | 0.208(0.122) | 1.701 | 0.0889 |
| USE | -0.43(0.289) | -1.489 | 0.1366 |
| **Nagelkerke's R2** | 0.529 | | |
| **Moran's I** | -0.12 | | n.s. |

**Palearctic (n=729)**

| | | | |
|---|---|---|---|
| NULL Model | | | AIC = 3161.83 |
| Initial Model: NSR+RBA+TEA+NPP+FPT+GDP+DOF+USE | | | AIC = 3101.15 |
| **Optimal Model: RBA+TEA+NPP+FPT+GDP+DOF** | | | **AIC = 3097.47** |
| Statistics in Optimal Model | coefficient (SE) | z-value | *P* |
| RBA | 0.324(0.069) | 4.673 | **<0.0001** |
| TEA | -0.356(0.126) | -2.835 | **0.0046** |
| NPP | -0.319(0.181) | -1.761 | 0.0783 |
| FPT | 0.671(0.112) | 5.999 | **<0.0001** |
| GDP | -0.155(0.086) | -1.808 | 0.0706 |
| DOF | 0.206(0.063) | 3.28 | **0.001** |
| **Nagelkerke's R2** | 0.38 | | |
| **Moran's I** | -0.047 | | n.s. |

**Table S2. (separate file)**

Species extirpations/extinctions and introductions in each of the 2456 river basins.

**Table S3. (separate file)**

The changes in the six diversity indices and the index of cumulative change in biodiversity facets for the fish faunas in 2,456 basins. δTR: change in taxonomic richness; δFR: change in functional richness; δPR: change in phylogenetic richness; δTD: change in taxonomic dissimilarity; δFD: change in functional dissimilarity; δPD: change in phylogenetic dissimilarity; CCBF: index of cumulative change in biodiversity facets.

Tables S2 and S3 are also provided in a public online repository (figshare.com). Here is the private link: https://figshare.com/s/5fadc2c14cbb1f39c25c.

**References**


1. G. H. Allen, T. M. Pavelsky, Global extent of rivers and streams. *Science* **361**, 585–588 (2018). doi:10.1126/science.aat0636 Medline

2. D. Dudgeon, A. H. Arthington, M. O. Gessner, Z. Kawabata, D. J. Knowler, C. Lévêque, R. J. Naiman, A.-H. Prieur-Richard, D. Soto, M. L. J. Stiassny, C. A. Sullivan, Freshwater biodiversity: Importance, threats, status and conservation challenges. *Biol. Rev. Camb. Philos. Soc.* **81**, 163–182 (2006). doi:10.1017/S1464793105006950 Medline

3. R. van der Laan, *Freshwater Fish List* (Almere, Netherlands, ed. 30, 2020).

4. S. Villéger, S. Brosse, M. Mouchet, D. Mouillot, M. J. Vanni, Functional ecology of fish: Current approaches and future challenges. *Aquat. Sci.* **79**, 783–801 (2017). doi:10.1007/s00027-017-0546-z

5. R. Hassan, R. Scholes, N. Ash, *Ecosystems and Human Well-Being: Current State and Trends* (Island, 2005).

6. R. Arlinghaus, J. K. Abbott, E. P. Fenichel, S. R. Carpenter, L. M. Hunt, J. Alós, T. Klefoth, S. J. Cooke, R. Hilborn, O. P. Jensen, M. J. Wilberg, J. R. Post, M. J. Manfredo, Opinion: Governing the recreational dimension of global fisheries. *Proc. Natl. Acad. Sci. U.S.A.* **116**, 5209–5213 (2019). doi:10.1073/pnas.1902796116 Medline

7. R. C. Hoffmann, Economic development and aquatic ecosystems in medieval Europe. *Am. Hist. Rev.* **101**, 631–669 (1996). doi:10.2307/2169418

8. F. Leprieur, O. Beauchard, S. Blanchet, T. Oberdorff, S. Brosse, Fish invasions in the world's river systems: When natural processes are blurred by human activities. *PLOS Biol.* **6**, e28 (2008). doi:10.1371/journal.pbio.0060028 Medline

9. S. Villéger, S. Blanchet, O. Beauchard, T. Oberdorff, S. Brosse, Homogenization patterns of the world's freshwater fish faunas. *Proc. Natl. Acad. Sci. U.S.A.* **108**, 18003–18008 (2011). doi:10.1073/pnas.1107614108 Medline

10. H. M. Pereira, P. W. Leadley, V. Proença, R. Alkemade, J. P. W. Scharlemann, J. F. Fernandez-Manjarrés, M. B. Araújo, P. Balvanera, R. Biggs, W. W. L. Cheung, L. Chini, H. D. Cooper, E. L. Gilman, S. Guénette, G. C. Hurtt, H. P. Huntington, G. M. Mace, T. Oberdorff, C. Revenga, P. Rodrigues, R. J. Scholes, U. R. Sumaila, M. Walpole, Scenarios for global biodiversity in the 21st century. *Science* **330**, 1496–1501 (2010). doi:10.1126/science.1196624 Medline

11. C. J. Vörösmarty, P. B. McIntyre, M. O. Gessner, D. Dudgeon, A. Prusevich, P. Green, S. Glidden, S. E. Bunn, C. A. Sullivan, C. R. Liermann, P. M. Davies, Global threats to human water security and river biodiversity. *Nature* **467**, 555–561 (2010). doi:10.1038/nature09440 Medline

12. A. J. Reid, A. K. Carlson, I. F. Creed, E. J. Eliason, P. A. Gell, P. T. J. Johnson, K. A. Kidd, T. J. MacCormack, J. D. Olden, S. J. Ormerod, J. P. Smol, W. W. Taylor, K. Tockner, J. C. Vermaire, D. Dudgeon, S. J. Cooke, Emerging threats and persistent conservation challenges for freshwater biodiversity. *Biol. Rev. Camb. Philos. Soc.* **94**, 849–873 (2019). doi:10.1111/brv.12480 Medline

13. D. Tickner, J. J. Opperman, R. Abell, M. Acreman, A. H. Arthington, S. E. Bunn, S. J. Cooke, J. Dalton, W. Darwall, G. Edwards, I. Harrison, K. Hughes, T. Jones, D. Leclère, A. J. Lynch, P. Leonard, M. E. McClain, D. Muruven, J. D. Olden, S. J. Ormerod, J. Robinson, R. E. Tharme, M. Thieme, K. Tockner, M. Wright, L. Young, Bending the curve of global freshwater biodiversity loss: An emergency recovery plan. *Bioscience* **70**, 330–342 (2020). doi:10.1093/biosci/biaa002 Medline

14. D. Mouillot, N. A. Graham, S. Villéger, N. W. Mason, D. R. Bellwood, A functional approach reveals community responses to disturbances. *Trends Ecol. Evol.* **28**, 167–177 (2013). doi:10.1016/j.tree.2012.10.004 Medline

15. S. Naeem, J. E. Duffy, E. Zavaleta, The functions of biological diversity in an age of extinction. *Science* **336**, 1401–1406 (2012). doi:10.1126/science.1215855 Medline

16. D. Craven, N. Eisenhauer, W. D. Pearse, Y. Hautier, F. Isbell, C. Roscher, M. Bahn, C. Beierkuhnlein, G. Bönisch, N. Buchmann, C. Byun, J. A. Catford, B. E. L. Cerabolini, J. H. C. Cornelissen, J. M. Craine, E. De Luca, A. Ebeling, J. N. Griffin, A. Hector, J. Hines, A. Jentsch, J. Kattge, J. Kreyling, V. Lanta, N. Lemoine, S. T. Meyer, V. Minden, V. Onipchenko, H. W. Polley, P. B. Reich, J. van Ruijven, B. Schamp, M. D. Smith, N. A. Soudzilovskaia, D. Tilman, A. Weigelt, B. Wilsey, P. Manning, Multiple facets of biodiversity drive the diversity-stability relationship. *Nat. Ecol. Evol.* **2**, 1579–1587 (2018). doi:10.1038/s41559-018-0647-7 Medline

17. P. Brun, N. E. Zimmermann, C. H. Graham, S. Lavergne, L. Pellissier, T. Münkemüller, W. Thuiller, The productivity-biodiversity relationship varies across diversity dimensions. *Nat. Commun.* **10**, 5691 (2019). doi:10.1038/s41467-019-13678-1 Medline

18. C. Pimiento, F. Leprieur, D. Silvestro, J. S. Lefcheck, C. Albouy, D. B. Rasher, M. Davis, J.-C. Svenning, J. N. Griffin, Functional diversity of marine megafauna in the Anthropocene. *Sci. Adv.* **6**, eaay7650 (2020). doi:10.1126/sciadv.aay7650 Medline

19. P. A. Tedesco, O. Beauchard, R. Bigorne, S. Blanchet, L. Buisson, L. Conti, J.-F. Cornu, M. S. Dias, G. Grenouillet, B. Hugueny, C. Jézéquel, F. Leprieur, S. Brosse, T. Oberdorff, A global database on freshwater fish species occurrence in drainage basins. *Sci. Data* **4**, 170141 (2017). doi:10.1038/sdata.2017.141 Medline

20. See supplementary materials.

21. A. Toussaint, N. Charpin, S. Brosse, S. Villéger, Global functional diversity of freshwater fish is concentrated in the Neotropics while functional vulnerability is widespread. *Sci. Rep.* **6**, 22125 (2016). doi:10.1038/srep22125 Medline

22. D. L. Rabosky, Speciation rate and the diversity of fishes in freshwaters and the oceans. *J. Biogeogr.* **47**, 1207–1217 (2020). doi:10.1111/jbi.13839

23. O. Venter, E. W. Sanderson, A. Magrach, J. R. Allan, J. Beher, K. R. Jones, H. P. Possingham, W. F. Laurance, P. Wood, B. M. Fekete, M. A. Levy, J. E. M. Watson, Global terrestrial Human Footprint maps for 1993 and 2009. *Sci. Data* **3**, 160067 (2016). doi:10.1038/sdata.2016.67 Medline

24. S. A. Blowes, S. R. Supp, L. H. Antão, A. Bates, H. Bruelheide, J. M. Chase, F. Moyes, A. Magurran, B. McGill, I. H. Myers-Smith, M. Winter, A. D. Bjorkman, D. E. Bowler, J. E. K. Byrnes, A. Gonzalez, J. Hines, F. Isbell, H. P. Jones, L. M. Navarro, P. L. Thompson, M. Vellend, C. Waldock, M. Dornelas, The geography of biodiversity change in marine and terrestrial assemblages. *Science* **366**, 339–345 (2019). doi:10.1126/science.aaw1620 Medline

25. M. C. Hansen, P. V. Potapov, R. Moore, M. Hancher, S. A. Turubanova, A. Tyukavina, D. Thau, S. V. Stehman, S. J. Goetz, T. R. Loveland, A. Kommareddy, A. Egorov, L. Chini, C. O. Justice, J. R. G. Townshend, High-resolution global maps of 21st-century forest cover change. *Science* **342**, 850–853 (2013). doi:10.1126/science.1244693 Medline

26. J. S. Albert, G. Destouni, S. M. Duke-Sylvester, A. E. Magurran, T. Oberdorff, R. E. Reis, K. O. Winemiller, W. J. Ripple, Scientists' warning to humanity on the freshwater biodiversity crisis. *Ambio* **50**, 85–94 (2021). doi:10.1007/s13280-020-01318-8 Medline

27. A. Toussaint, N. Charpin, O. Beauchard, G. Grenouillet, T. Oberdorff, P. A. Tedesco, S. Brosse, S. Villéger, Non-native species led to marked shifts in functional diversity of the world freshwater fish faunas. *Ecol. Lett.* **21**, 1649–1659 (2018). doi:10.1111/ele.13141 Medline

28. S. T. Jackson, D. F. Sax, Balancing biodiversity in a changing environment: Extinction debt, immigration credit and species turnover. *Trends Ecol. Evol.* **25**, 153–160 (2010). doi:10.1016/j.tree.2009.10.001 Medline

29. IUCN, *The IUCN Red List of Threatened Species* Version 2017-3 (2018); www.iucnredlist.org.

30. S. Villéger, G. Grenouillet, S. Brosse, Functional homogenization exceeds taxonomic homogenization among European fish assemblages. *Glob. Ecol. Biogeogr.* **23**, 1450–1460 (2014). doi:10.1111/geb.12226

31. S. Burgin, Indirect consequences of recreational fishing in freshwater ecosystems: An exploration from an Australian perspective. *Sustainability* **9**, 280 (2017). doi:10.3390/su9020280

32. G. Grill, B. Lehner, M. Thieme, B. Geenen, D. Tickner, F. Antonelli, S. Babu, P. Borrelli, L. Cheng, H. Crochetiere, H. Ehalt Macedo, R. Filgueiras, M. Goichot, J. Higgins, Z. Hogan, B. Lip, M. E. McClain, J. Meng, M. Mulligan, C. Nilsson, J. D. Olden, J. J. Opperman, P. Petry, C. Reidy Liermann, L. Sáenz, S. Salinas-Rodríguez, P. Schelle, R. J. P. Schmitt, J. Snider, F. Tan, K. Tockner, P. H. Valdujo, A. van Soesbergen, C. Zarfl, Mapping the world's free-flowing rivers. *Nature* **569**, 215–221 (2019). doi:10.1038/s41586-019-1111-9 Medline

33. B. Lehner, C. R. Liermann, C. Revenga, C. Vörösmarty, B. Fekete, P. Crouzet, P. Döll, M. Endejan, K. Frenken, J. Magome, C. Nilsson, J. C. Robertson, R. Rödel, N. Sindorf, D. Wisser, High-resolution mapping of the world's reservoirs and dams for sustainable river-flow management. *Front. Ecol. Environ.* **9**, 494–502 (2011). doi:10.1890/100125

34. C. Zarfl, A. E. Lumsdon, J. Berlekamp, L. Tydecks, K. Tockner, A global boom in hydropower dam construction. *Aquat. Sci.* **77**, 161–170 (2015). doi:10.1007/s00027-014-0377-0

35. E. P. Anderson, C. N. Jenkins, S. Heilpern, J. A. Maldonado-Ocampo, F. M. Carvajal-Vallejos, A. C. Encalada, J. F. Rivadeneira, M. Hidalgo, C. M. Cañas, H. Ortega, N. Salcedo, M. Maldonado, P. A. Tedesco, Fragmentation of Andes-to-Amazon connectivity by hydropower dams. *Sci. Adv.* **4**, eaao1642 (2018). doi:10.1126/sciadv.aao1642

36. K. O. Winemiller, P. B. McIntyre, L. Castello, E. Fluet-Chouinard, T. Giarrizzo, S. Nam, I. G. Baird, W. Darwall, N. K. Lujan, I. Harrison, M. L. J. Stiassny, R. A. M. Silvano, D. B. Fitzgerald, F. M. Pelicice, A. A. Agostinho, L. C. Gomes, J. S. Albert, E. Baran, M. Petrere Jr., C. Zarfl, M. Mulligan, J. P. Sullivan, C. C. Arantes, L. M. Sousa, A. A.

Koning, D. J. Hoeinghaus, M. Sabaj, J. G. Lundberg, J. Armbruster, M. L. Thieme, P. Petry, J. Zuanon, G. Torrente Vilara, J. Snoeks, C. Ou, W. Rainboth, C. S. Pavanelli, A. Akama, A. van Soesbergen, L. Sáenz, Balancing hydropower and biodiversity in the Amazon, Congo, and Mekong. *Science* **351**, 128–129 (2016). doi:10.1126/science.aac7082 Medline

37. P. Döll, K. Fiedler, J. Zhang, Global-scale analysis of river flow alterations due to water withdrawals and reservoirs. *Hydrol. Earth Syst. Sci.* **13**, 2413–2432 (2009). doi:10.5194/hess-13-2413-2009

38. J. E. Duffy, J. S. Lefcheck, R. D. Stuart-Smith, S. A. Navarrete, G. J. Edgar, Biodiversity enhances reef fish biomass and resistance to climate change. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 6230–6235 (2016). doi:10.1073/pnas.1524465113 Medline

39. M. Loreau, S. Naeem, P. Inchausti, J. Bengtsson, J. P. Grime, A. Hector, D. U. Hooper, M. A. Huston, D. Raffaelli, B. Schmid, D. Tilman, D. A. Wardle, Biodiversity and ecosystem functioning: Current knowledge and future challenges. *Science* **294**, 804–808 (2001). doi:10.1126/science.1064088 Medline

40. J. M. Levine, Species diversity and biological invasions: Relating local process to community pattern. *Science* **288**, 852–854 (2000). doi:10.1126/science.288.5467.852 Medline

41. E. Dinerstein, C. Vynne, E. Sala, A. R. Joshi, S. Fernando, T. E. Lovejoy, J. Mayorga, D. Olson, G. P. Asner, J. E. M. Baillie, N. D. Burgess, K. Burkart, R. F. Noss, Y. P. Zhang, A. Baccini, T. Birch, N. Hahn, L. N. Joppa, E. Wikramanayake, A global deal for nature: Guiding principles, milestones, and targets. *Sci. Adv.* **5**, eaaw2869 (2019). doi:10.1126/sciadv.aaw2869 Medline

42. S. L. Pimm, C. N. Jenkins, B. V. Li, How to protect half of Earth to ensure it protects sufficient biodiversity. *Sci. Adv.* **4**, eaat2616 (2018). doi:10.1126/sciadv.aat2616 Medline

43. C. R. Margules, R. L. Pressey, Systematic conservation planning. *Nature* **405**, 243–253 (2000). doi:10.1038/35012251 Medline

44. C. Lévêque, T. Oberdorff, D. Paugy, M. Stiassny, P. Tedesco, Global diversity of fish (Pisces) in freshwater. *Hydrobiologia* **595**, 545–567 (2008). doi:10.1007/s10750-007-9034-0

45. S. Brosse, O. Beauchard, S. Blanchet, H. H. Dürr, G. Grenouillet, B. Hugueny, C. Lauzeral, F. Leprieur, P. A. Tedesco, S. Villéger, T. Oberdorff, Fish-SPRICH: A

database of freshwater fish species richness throughout the World. *Hydrobiologia* **700**, 343–349 (2013). doi:10.1007/s10750-012-1242-6

46. H. Seebens, T. M. Blackburn, E. E. Dyer, P. Genovesi, P. E. Hulme, J. M. Jeschke, S. Pagad, P. Pyšek, M. Winter, M. Arianoutsou, S. Bacher, B. Blasius, G. Brundu, C. Capinha, L. Celesti-Grapow, W. Dawson, S. Dullinger, N. Fuentes, H. Jäger, J. Kartesz, M. Kenis, H. Kreft, I. Kühn, B. Lenzner, A. Liebhold, A. Mosena, D. Moser, M. Nishino, D. Pearman, J. Pergl, W. Rabitsch, J. Rojas-Sandoval, A. Roques, S. Rorke, S. Rossinelli, H. E. Roy, R. Scalera, S. Schindler, K. Štajerová, B. Tokarska-Guzik, M. van Kleunen, K. Walker, P. Weigelt, T. Yamanaka, F. Essl, No saturation in the accumulation of alien species worldwide. *Nat. Commun.* **8**, 14435 (2017). doi:10.1038/ncomms14435 Medline

47. S. Blanchet, G. Grenouillet, O. Beauchard, P. A. Tedesco, F. Leprieur, H. H. Dürr, F. Busson, T. Oberdorff, S. Brosse, Non-native species disrupt the worldwide patterns of freshwater fish body size: Implications for Bergmann's rule. *Ecol. Lett.* **13**, 421–431 (2010). doi:10.1111/j.1461-0248.2009.01432.x Medline

48. G. Ceballos, P. R. Ehrlich, R. Dirzo, Biological annihilation via the ongoing sixth mass extinction signaled by vertebrate population losses and declines. *Proc. Natl. Acad. Sci. U.S.A.* **114**, E6089–E6096 (2017). doi:10.1073/pnas.1704949114 Medline

49. M. S. Dias, P. A. Tedesco, B. Hugueny, C. Jézéquel, O. Beauchard, S. Brosse, T. Oberdorff, Anthropogenic stressors and riverine fish extinctions. *Ecol. Indic.* **79**, 37–46 (2017). doi:10.1016/j.ecolind.2017.03.053

50. R. Froese, D. Pauly, FishBase (2018); www.fishbase.org.

51. G. Su, S. Villéger, S. Brosse, Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread. *Glob. Ecol. Biogeogr.* **28**, 211–221 (2019). doi:10.1111/geb.12843

52. D. R. Bellwood, C. H. Goatley, S. J. Brandl, O. Bellwood, Fifty million years of herbivory on coral reefs: Fossils, fish and functional innovations. *Proc. R. Soc. B* **281**, 20133046 (2014). doi:10.1098/rspb.2013.3046 Medline

53. S. Villéger, J. Ramos Miranda, D. Flores Hernández, D. Mouillot, Contrasting changes in taxonomic vs. functional diversity of tropical fish communities after habitat degradation. *Ecol. Appl.* **20**, 1512–1522 (2010). doi:10.1890/09-1310.1 Medline

54. C. Penone, A. D. Davidson, K. T. Shoemaker, M. Di Marco, C. Rondinini, T. M. Brooks, B. E. Young, C. H. Graham, G. C. Costa, Imputation of missing data in life-history

trait datasets: Which approach performs the best? *Methods Ecol. Evol.* **5**, 961–970 (2014). doi:10.1111/2041-210X.12232

55. J. Bruggeman, J. Heringa, B. W. Brandt, PhyloPars: Estimation of missing parameter values using phylogeny. *Nucleic Acids Res.* **37** (suppl. 2), W179–W184 (2009). doi:10.1093/nar/gkp370 Medline

56. D. L. Rabosky, J. Chang, P. O. Title, P. F. Cowman, L. Sallan, M. Friedman, K. Kaschner, C. Garilao, T. J. Near, M. Coll, M. E. Alfaro, An inverse latitudinal gradient in speciation rate for marine fishes. *Nature* **559**, 392–395 (2018). doi:10.1038/s41586-018-0273-1 Medline

57. F. Leprieur, P. A. Tedesco, B. Hugueny, O. Beauchard, H. H. Dürr, S. Brosse, T. Oberdorff, Partitioning global patterns of freshwater fish beta diversity reveals contrasting signatures of past climate changes. *Ecol. Lett.* **14**, 325–334 (2011). doi:10.1111/j.1461-0248.2011.01589.x Medline

58. D. J. Currie, G. G. Mittelbach, H. V. Cornell, R. Field, J.-F. Guegan, B. A. Hawkins, D. M. Kaufman, J. T. Kerr, T. Oberdorff, E. O'Brien, J. R. G. Turner, Predictions and tests of climate-based hypotheses of broad-scale variation in taxonomic richness. *Ecol. Lett.* **7**, 1121–1134 (2004). doi:10.1111/j.1461-0248.2004.00671.x

59. M. Dynesius, R. Jansson, Evolutionary consequences of changes in species' geographical distributions driven by Milankovitch climate oscillations. *Proc. Natl. Acad. Sci. U.S.A.* **97**, 9115–9120 (2000). doi:10.1073/pnas.97.16.9115 Medline

60. J.-F. Guégan, S. Lek, T. Oberdorff, Energy availability and habitat heterogeneity predict global riverine fish diversity. *Nature* **391**, 382–384 (1998). doi:10.1038/34899

61. W. Nordhaus, X. Chen, Global gridded geographically based economic data (G-econ), version 4 (NASA Socioeconomic Data and Applications Center, 2016).

62. W. D. Nordhaus, Geography and macroeconomics: New data and new findings. *Proc. Natl. Acad. Sci. U.S.A.* **103**, 3510–3517 (2006). doi:10.1073/pnas.0509842103 Medline

63. P. Döll, S. Siebert, Global modeling of irrigation water requirements. *Water Resour. Res.* **38**, 8-1 (2002). doi:10.1029/2001WR000355

64. S. Villéger, N. W. H. Mason, D. Mouillot, New multidimensional functional diversity indices for a multifaceted framework in functional ecology. *Ecology* **89**, 2290–2301 (2008). doi:10.1890/07-1206.1 Medline

65. D. P. Faith, Conservation evaluation and phylogenetic diversity. *Biol. Conserv.* **61**, 1–10 (1992). doi:10.1016/0006-3207(92)91201-3

66. A. Baselga, Partitioning the turnover and nestedness components of beta diversity. *Glob. Ecol. Biogeogr.* **19**, 134–143 (2010). doi:10.1111/j.1466-8238.2009.00490.x

67. C. Lozupone, R. Knight, UniFrac: A new phylogenetic method for comparing microbial communities. *Appl. Environ. Microbiol.* **71**, 8228–8235 (2005). doi:10.1128/AEM.71.12.8228-8235.2005 Medline

68. K. P. Burnham, D. R. Anderson, *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach* (Springer, ed. 2, 2002).

69. N. J. Nagelkerke, A note on a general definition of the coefficient of determination. *Biometrika* **78**, 691–692 (1991). doi:10.1093/biomet/78.3.691

70. A. D. Cliff, J. K. Ord, *Spatial Processes: Models and Applications* (Taylor & Francis, 1981).

71. A. Baselga, D. Orme, S. Villeger, J. De Bortoli, F. Leprieur, Partitioning beta diversity into turnover and nestedness components. Package betapart, Version 1.4-1 (2017).

72. A. Baselga, C. D. L. Orme, betapart: An R package for the study of beta diversity. *Methods Ecol. Evol.* **3**, 808–812 (2012). doi:10.1111/j.2041-210X.2012.00224.x

73. R. Bivand, G. Piras, Comparing implementations of estimation methods for spatial econometrics. *J. Stat. Softw.* **63**, 1–36 (2015).

74. R. S. Bivand, D. W. Wong, Comparing implementations of global and local indicators of spatial association. *Test* **27**, 716–748 (2018). doi:10.1007/s11749-018-0599-x

75. Y. Reecht, M. J. Rochet, V. M. Trenkel, S. Jennings, J. K. Pinnegar, Use of morphological characteristics to define functional groups of predatory fishes in the Celtic Sea. *J. Fish Biol.* **83**, 355–377 (2013). doi:10.1111/jfb.12177 Medline

76. K. O. Winemiller, Ecomorphological diversification in lowland freshwater fish assemblages from five biotic regions. *Ecol. Monogr.* **61**, 343–365 (1991). doi:10.2307/2937046

77. K. S. Boyle, M. H. Horn, Comparison of feeding guild structure and ecomorphology of intertidal fish assemblages from central California and central Chile. *Mar. Ecol. Prog. Ser.* **319**, 65–84 (2006). doi:10.3354/meps319065

78. O. Dumay, P. Tari, J. Tomasini, D. Mouillot, Functional groups of lagoon fish species in Languedoc Roussillon, southern France. *J. Fish Biol.* **64**, 970–983 (2004). doi:10.1111/j.1095-8649.2004.00365.x

79. J. S. Lefcheck, A. Buchheister, K. M. Laumann, M. A. Stratton, K. L. Sobocinski, S. T. C. Chak, T. R. Clardy, P. L. Reynolds, R. J. Latour, J. E. Duffy, Dimensions of

biodiversity in Chesapeake Bay demersal fishes: Patterns and drivers through space and time. *Ecosphere* **5**, 14 (2014). doi:10.1890/ES13-00284.1

80. C. Fulton, D. Bellwood, P. Wainwright, The relationship between swimming ability and habitat use in wrasses (Labridae). *Mar. Biol.* **139**, 25–33 (2001). doi:10.1007/s002270100565

81. P. Webb, Body form, locomotion and foraging in aquatic vertebrates. *Am. Zool.* **24**, 107–120 (1984). doi:10.1093/icb/24.1.107

# Chapter VII: Conclusions and Perspectives

## VII.1 Conclusions

Freshwater ecosystems have been altered by human activities for decades (Tickner *et al.* 2020). To assess the impact of human activities on the global freshwater fish biodiversity, especially on its functional facet, in my thesis, I have measured the spatial patterns of morphology-based functional traits, and unravel their drivers. I have also assessed how functional traits were sorted by humans across the invasion process (i.e. introduction, and establishment stages).

***1) High variation of global freshwater fish morphology***

The first important result of my thesis is that the high species diversity of fish supports a high degree of variation in their morphological characteristics. This result was verified in all the ten measured morphological traits, which provides a high probability for the species to act distinct roles in the ecosystems. Moreover, we also testified that the distributions of morphological traits are significantly different between the six biogeographic realms, and those differences are attributed to the evolutionary divergence between realm fish faunas and distinct environmental constraints between the six realms. Besides this, we also found that a small part of morphologically extreme species supported the variety of morphological space and widely distributed across the six realms. To conclude, these results will constitute a benchmark for future studies on fish morphological diversity across the world and will help in setting conservation strategies. (Chapter III).

***2) Different hypotheses explained the patterns of global fish functional diversity***

We then mapped the spatial patterns of three complementary facets of functional diversity across the world river basins. We found values of the three facets of functional diversity highly varied among river basins between and within the six realms. We then disentangled the contributions of the three main ecological hypotheses explaining biological diversity (habitat

size/diversity; climate/energy; history/geography) to the functional diversity patterns. We found that functional richness was mainly driven by the species richness, thus largely explained by the habitat size/diversity hypothesis. In contrast, functional dispersion and functional identity were mainly explained by the history/geography and climate/energy hypotheses. Our results highlight the influence of historical processes on shaping the patterns of functional diversity, and thus provide new insights into the role of historical and environmental determinants on the functional structure of fish assemblages, revealing the relationship between the functions of species and ecosystem processes (Chapter IV).

***3) Distinctiveness of morphologies between introduced species and non-introduced species***

Introduction of non-native species is a crucial component of the current biodiversity crisis, especially for freshwater fish species. To evaluate the morphological difference between the non-introduced species and introduced species at different invasive status, we identified 2 690 introduced species from the 9 150 morphologically measured species pool, among which 418 species were successfully established in the recipient assemblages while the remaining 2 272 species failed to establish. We found that introduced species are morphologically different from the non-introduced species, implying that people preferred to introduce large pelagic predators with high dispersal ability to satisfy their interests. Moreover, the particular morphology of established species, which are characterized by a laterally flattened body, suggest their successful establishment in slow flowing habitats. These findings highlight the morphological differences of introduced fish species in different invasive stages, and thus provide new insights into the species introduction policy, since our results might be used to profile future introductions and help to better develop strategies to prevent future invasions (Chapter V).

***4) River fragmentation and regulation drive change in global freshwater fish biodiversity***

We designed a new cumulative index to combine the changes in multi-faceted biodiversity to assess the impact of various human activities on freshwater fish biodiversity. This index has the main interest to combine taxonomic, functional and phylogenetic information considered

both at a locality scale (here river basin, alpha-diversity), and between localities (between river basins, beta-diversity). Using this index, we demonstrated that more than half of the world rivers experienced marked and multifaceted changes in their fish biodiversity, with the most pronounced changes occurring in temperate rivers with a high level of river fragmentation by dams and of water use, testifying for deep and spatially extended impacts of water regulation on fish biodiversity. However, fish faunas from a few rivers remain weakly affected and could thus represent biodiversity refuges and key opportunities for biodiversity conservation. Our results hence highlight the need to consider the cumulative and synergistic effects of multiple human activities on the changes in the complementary facets of biodiversity. Finally, the index here applied to the world freshwater fish fauna, can be used on any taxa or ecosystems, it therefore represents a cutting-edge contribution to our current understanding of the impacts of humans on global biodiversity (Chapter VI).

## VII.2 Perspectives

### *1) Predictions of future freshwater fish invasions*

The introduction and establishment of alien species are among the greatest threats to biodiversity and ecosystem services (Blackburn, Bellard, & Ricciardi, 2019) which profoundly affect the native communities, ecosystem functioning, human health and economy (Seebens *et al.* 2017). However, the number of established non-native species is far from saturated, and more non-native species are expected to establish in the future (Seebens *et al.* 2017). Therefore, a major task for ecologists is to predict future biological invasions including distributions, identities, numbers, and the economic loss they cause (Diagne *et al.* 2020, Howeth *et al.* 2015, Seebens *et al.* 2020). Most previous studies attempted to predict the future distributions of existing alien species or the identities of future alien species based on the characteristics of existing alien species (Azzurro *et al.* 2014, Fournier *et al.* 2019, Moyle & Marchetti 2006, Olden *et al.* 2006). Indeed, including our results in chapter IV, numerous studies are verifying that introduced or alien species from most taxa are significantly different from their counterparts in donor or recipient places (e.g. plants, Drenovsky *et al.* 2012; fish, Su *et al.* 2020; birds, Shirley & Kark 2009; mammals, Capellini *et al.* 2015). This provides fundamental theoretical support for predicting forthcoming alien species and subsequent invasions, as recently shown by Seebens *et al.* (2020), who developed models to predict the dynamics of alien species numbers for many taxonomic groups. This work provides a quantitative baseline to assess future biological invasions. Moreover, Diagne *et al.* (2020) built up a database of the economic costs of biological invasions for diverse taxonomic groups (InvaCost). Combining our results on non-native species profiling, together with Seebens *et al.* (2020) predictions and Diagne *et al.* (2020) measures of the invasion costs would make it possible to predict the future economic cost of biological invasions. Besides the fish information included in the Seebens *et al.* (2020) and Diagne *et al.* (2020) databases, we could apply the models they used to finer grained studies (e.g. basin or subbasin), and thus develop insightful studies on local species invasions and associated economic costs. Together with the predictions of distribution and identities, these three dimensions of future freshwater fish invasions will provide us a holistic

view of future fish invasions, as well as a novel useful information for the policymakers and managers.

***2) Enriching the current functional trait database with more species and traits***

Among the 18 000 described freshwater fish species (Fricke 2020, Van Der Laan 2020), 14 953 benefits from information on their spatial distribution (Tedesco *et al.* 2017). However, due to the insufficient information for a large part of these species, at the beginning of my Ph.D. project, 9 150 species matching the occurrence dataset have been morphologically measured, leaving near 5 000 fishes unmeasured. This dataset allows me to analyze the morphological variation between fishes from the six biogeographic realms as well as the morphological variation between different types of introduced species and non-introduced species. Then, we enriched the morphological dataset to 10 682 species, which supported my two other studies, i.e. the pattern and determinants of fish functional diversity and the multifaceted change in global fish biodiversity. Although, the dataset covers more than half of the freshwater fish species and is the most comprehensive fish morphological dataset so far, morphological measurement on the remaining and newly discovered fish species should be permanently updated to complete the current dataset. Besides, although morphological traits are easy to obtain and measure, some traits do not represent a clear function, thus the relationship between them should be further tested.

Moreover, benefit from the statistical and computational advances, machine learning approaches have opened new ways of data collection and integration rather than a time-consuming and laborious manual collection, which makes it possible and efficient for the collection of other functional traits of fishes, such as life-history, physiological, phenological and behavioral traits. We expect a new and more complete database with more fish species and functional traits will be established in the near future.

3) ***Towards a refined fish occurrence database***

The fish occurrence data we accessed from Tedesco *et al.* (2013) is currently the most

comprehensive spatial distribution database of the global freshwater fishes. This database compiles the occurrence status of 14 953 fish species in 3119 river basins covering more than 80% of the Earth surface. This database at the basin scale helped us to explore and testify lots of questions and hypotheses in the macroecological and biogeographic frameworks, such as how the historical and environmental variables shape the fish biodiversity patterns or how human activities change the fish biodiversity across different continents. However, occurrence data at this scale is not appropriate to handle species interactions and community assembly rules. This data thus does not permit us to test for niche vs neutral theories or to investigate the relationships between biodiversity and ecosystem stability. Moreover, river networks are very complex systems with a total of 35.9 million km of rivers globally (Linke *et al.* 2019) and some key questions remain to be solved under this context, such as the relationship between freshwater fish distribution and river network geometry and hydrology (Muneepeerakul *et al.* 2018). We, therefore, appeal for a more local grained fish occurrence database for the global fishes. Fortunately, owing to several long-term research projects on freshwater fishes, such local scale data sets have been developed for some large river basins throughout the world, such as the Amazon River basin (Oberdorff *et al.* 2019, Jézéquel *et al.* 2020), Yangtze River basin (Kang *et al.* 2018), and Mississippi River basin (Muneepeerakul *et al.* 2008). Furthermore, Carvajal-Quintero *et al.* (2019) reported a global occurrence database of 9 075 freshwater fish species at the subbasin scale, which will also accelerate the establishment of the refined occurrence database of global freshwater fishes. Although most large-scale studies have not considered the species abundance or biomass (but see Stuart-Smith *et al.* 2013), ecological effects of species are generally proportional to their abundance or biomass (Grime 1998), making the abundance-weighted functional diversity reflecting community functional structure more accurately than diversity metrics based on only the occurrence database (Stuart-Smith *et al.* 2013). Thus, we also expect species abundance or biomass information will be collected and incorporated into the refined fish occurrence database for future functional diversity studies.

***4) Functional diversity: shed new light on the diversity-stability relationship.***

The relationship between biodiversity and ecosystem stability has been a major subject of ecological research for decades (McNaughton 1978, Ives & Carpenter 2007). However, until now there is no uniform conclusion on the diversity-stability debate (McCann 2000, Pennekamp *et al.* 2018) and their relationships can be negative, neutral or positive in different situations (Pennekamp *et al.* 2018). Ecosystem stability consists of several properties, including temporal variability, resistance to environmental change and rate of recovery after perturbation (resilience) (Donohue *et al.* 2016). One major reason is that most previous empirical or field studies focused on the number of species as the diversity metric to assess the stability–diversity relationship (but see Craven *et al.* 2018), which might be insufficient to reveal the underlying mechanisms. Functional diversity measures the diversity in the traits of species that influence their performance in the ecosystems and thus impact the ecosystem functioning (Diaz & Cabido, 2001). Therefore, functional diversity reflects the position of species in the functional space and the role they play in the ecosystems, which could also reflect the species interactions (e.g. prey-predator) in the communities and their response to the perturbations. Moreover, functional diversity contains many complementary components, which reflect the extent and structure of the functional space. To assess the relationships between the ecosystem stability and the complementary facets of functional diversity would shed new light on the diversity-stability debate. For instance, one could expect that between two communities with the same number of species, the stability will be lower for the community that has larger functional space but most of the species gathering in a small part of functional space than for the community that has smaller functional space but all the species evenly distributing in the functional space. Indeed, Mouillot *et al.* (2013b) revealed that in the highly diverse ecosystems (e.g. coral reef fishes), vulnerable functions, which are overwhelmingly supported by rare species, make communities and hence ecosystem processes more unstable in the face of environmental changes, implying the so-called ecosystem insurance (Yachi & Loreau 1999) is only true for a few redundant functions. Su *et al.* (2019) showed a similar phenomenon in freshwater fishes, in which a few species represent most of the functional space. Moreover, the ongoing high rate of species invasions and extinctions among global freshwater fish faunas provide a good opportunity to

test the diversity-stability relationship by comparing the functional diversity (e.g. functional richness, functional evenness, and functional density, Villeger *et al.* 2008, Cormona *et al.* 2016) of fish communities with different extent of invasions and extinctions. However, this analysis also needs a smaller scale fish occurrence database rather than the current basin scale database. It would allow us to consider species assemblages that co-occur in a location at a given time. Therefore, developing smaller scale occurrence database of the global freshwater fish fauna (or at least fish faunas over some river basins of the world) is a prerequisite for developing such diversity/stability approaches.

***5) Toward a global fish functional database considering marine and freshwater faunas altogether***

There are currently more than 35 600 valid fish species recorded (Fricke 2020), almost half of them inhabit freshwater ecosystems, and the other half inhabit marine ecosystems. These numerous fishes are remarkably diverse in morphology, physiology, habit, diet and life-history strategy (Helfman 2009, Arthington *et al.* 2016). Furthermore, the distinctiveness of evolution history between the freshwater and marine fishes, as well as the different environmental conditions between freshwater and marine ecosystems make the patterns of fish diversity (e.g. phylogenetic and genetic diversity) from the two ecosystems quite different (Thomaz *et al.* 2016, Rabosky *et al.* 2020, Manel *et al.* 2020). Recent studies also investigated the differences in speciation rates between freshwater fish and marine fish (Miller 2020, Rabosky *et al.* 2020). Similar to the status of freshwater fishes in inland water ecosystems (illustrated in the thesis introduction), marine fishes, especially the coral reef fishes inhabit coastal areas, also play key roles in the marine ecosystems due to their high biodiversity (Leprieur *et al.* 2016a, 2016b, Pellissier *et al.* 2014) and face severe human threats (Mora *et al.* 2011b). However, until now no studies focused on the comparisons between their morphological traits, let alone the difference of functional diversity between the fishes in the two ecosystems. A key reason is that although there are plenty of independent studies related to the functional diversity of either global freshwater (Toussaint *et al.* 2016b, 2018; Su *et al.* 2019, 2020) or marine fishes (Claverie

& Wainwright 2014, Mouillot *et al.* 2014, Borstein *et al.* 2019), there is no uniform morphological/functional trait database for both the freshwater and marine fishes so far. In addition, even if the same functions are investigated, it is still out of reach to combine data sets from different studies due to different measurement standards, (e.g. morphological traits used in both Toussaint *et al.* 2016b and Borstein *et al.* 2019). Another reason is that most of the studies did not provide the raw functional trait data in their published papers, which makes the goal more difficult to realize by combining different studies. Nevertheless, in the era of big data, collaboration is becoming a consensus in the research of macroecology and biogeography (Huang *et al.* 2020). For instance, the establishment of a database of global plant traits (i.e. TRY, Kattge *et al.* 2011) is the result of a long-term collaboration between numerous research groups throughout the world. Furthermore, with the increasing number of studies on the trait-based analysis of freshwater and marine fishes, I expect a unique functional trait database, including at least morphological traits (but maybe more), for global freshwater and marine fishes will be available soon. This achievement will help us to better understand the evolutionary divergence between marine and freshwater faunas. It would also permit us to investigate how morphological traits were shaped to adapt to distinct environments and how such distinct functional properties are eroded by global changes.

# References

Abell, R., Thieme, M. L., Revenga, C., Bryer, M., Kottelat, M., Bogutskaya, N., . . . Bussing, W. (2008). Freshwater ecoregions of the world: a new map of biogeographic units for freshwater biodiversity conservation. *Bioscience, 58*, 403-414.

Agresti, A. (2003). *Categorical data analysis* (Vol. 482): John Wiley & Sons.

Akaike, H. (1973). *Information theory as an extension of the maximum likelihood principle–In: Second International Symposium on Information Theory (Eds) BN Petrov, F*. Budapest: Academiai Kiado.

Albouy, C., Guilhaumon, F., Villéger, S., Mouchet, M., Mercier, L., Culioli, J., . . . Mouillot, D. (2011). Predicting trophic guild and diet overlap from functional traits: statistics, opportunities and limitations for marine ecology. *Marine Ecology Progress Series, 436*, 17-28.

Allen, G. H., & Pavelsky, T. M. (2018). Global extent of rivers and streams. *Science, 361*, 585-588.

Allen, M., Poggiali, D., Whitaker, K., Marshall, T. R., & Kievit, R. (2018). Raincloud plots: a multi-platform tool for robust data visualization. *PeerJ Preprints, 6*, e27137v27131.

Anderson, E. P., Jenkins, C. N., Heilpern, S., Maldonado-Ocampo, J. A., Carvajal-Vallejos, F. M., Encalada, A. C., . . . Ortega, H. (2018). Fragmentation of Andes-to-Amazon connectivity by hydropower dams. *Science advances, 4*, eaao1642.

Arlinghaus, R., Abbott, J. K., Fenichel, E. P., Carpenter, S. R., Hunt, L. M., Alós, J., . . . Jensen, O. P. (2019). Opinion: Governing the recreational dimension of global fisheries. *Proceedings of the National Academy of Sciences, 116*, 5209-5213.

Arthington, A. H., Dulvy, N. K., Gladstone, W., & Winfield, I. J. (2016). Fish conservation in freshwater and marine realms: status, threats and management. *Aquatic Conservation: Marine and Freshwater Ecosystems, 26*, 838-857.

Azzurro, E., Tuset, V. M., Lombarte, A., Maynou, F., Simberloff, D., Rodríguez-Pérez, A., & Solé, R. V. (2014). External morphology explains the success of biological invasions. *Ecology Letters, 17*, 1455-1463.

Barnagaud, J. Y., Kissling, W. D., Tsirogiannis, C., Fisikopoulos, V., Villeger, S., Sekercioglu, C. H., & Svenning, J. C. (2017). Biogeographical, environmental and anthropogenic determinants of global patterns in bird taxonomic and trait turnover. *Global Ecology and Biogeography, 26*, 1190-1200.

Barnosky, A. D., Matzke, N., Tomiya, S., Wogan, G. O., Swartz, B., Quental, T. B., . . . Maguire, K. C. (2011). Has the Earth/'s sixth mass extinction already arrived? *Nature, 471*, 51-

57.

Baselga, A. (2010). Partitioning the turnover and nestedness components of beta diversity. *Global Ecology and Biogeography, 19*, 134-143.

Baselga, A., & Orme, C. D. L. (2012). betapart: an R package for the study of beta diversity. *Methods in Ecology and Evolution, 3*, 808-812.

Baselga, A., Orme, D., Villeger, S., De Bortoli, J., & Leprieur, F. (2017). Partitioning beta diversity into turnover and nestedness components. *Package betapart, Version*, 1.4-1.

Bellwood, D., Goatley, C., Brandl, S., & Bellwood, O. (2014). Fifty million years of herbivory on coral reefs: fossils, fish and functional innovations. *Proceedings of the Royal Society B: Biological Sciences, 281*, 20133046.

Bennie, J. J., Duffy, J. P., Inger, R., & Gaston, K. J. (2014). Biogeography of time partitioning in mammals. *Proceedings of the National Academy of Sciences, 111*, 13727-13732.

Betancur-R, R., Wiley, E. O., Arratia, G., Acero, A., Bailly, N., Miya, M., . . . Orti, G. (2017). Phylogenetic classification of bony fishes. *BMC evolutionary biology, 17*, 162.

Bezerra, L. A. V., Freitas, M. O., Daga, V. S., Occhi, T. V. T., Faria, L., Costa, A. P. L., . . . Vitule, J. R. S. (2019). A network meta-analysis of threats to South American fish biodiversity. *Fish and fisheries, 20*, 620-639.

Bivand, R., & Piras, G. (2015). *Comparing implementations of estimation methods for spatial econometrics*.

Bivand, R. S., & Wong, D. W. (2018). Comparing implementations of global and local indicators of spatial association. *Test, 27*, 716-748.

Blackburn, T. M., Bellard, C., & Ricciardi, A. (2019). Alien versus native species as drivers of recent extinctions. *Frontiers in Ecology and the Environment, 17*, 203-207.

Blackburn, T. M., Cassey, P., & Lockwood, J. L. (2009). The role of species traits in the establishment success of exotic birds. *Global change biology, 15*, 2852-2860.

Blackburn, T. M., Pyšek, P., Bacher, S., Carlton, J. T., Duncan, R. P., Jarošík, V., . . . Richardson, D. M. (2011). A proposed unified framework for biological invasions. *Trends in Ecology & Evolution, 26*, 333-339.

Blake, R. W. (1983). *Fish locomotion*. Cambridge, UK: Cambridge University Press.

Blanchet, S., Grenouillet, G., Beauchard, O., Tedesco, P. A., Leprieur, F., Dürr, H. H., . . . Brosse, S. (2010). Non-native species disrupt the worldwide patterns of freshwater fish body size: implications for Bergmann's rule. *Ecology Letters, 13*, 421-431.

Blowes, S. A., Supp, S. R., Antão, L. H., Bates, A., Bruelheide, H., Chase, J. M., . . . Myers-Smith, I. H. (2019). The geography of biodiversity change in marine and terrestrial assemblages. *Science, 366*, 339-345.

Borowsky, R. (2018). Cavefishes. *Current Biology, 28*, R51-R65.

Borstein, S. R., Fordyce, J. A., O'Meara, B. C., Wainwright, P. C., & McGee, M. D. (2019). Reef fish functional traits evolve fastest at trophic extremes. *Nature ecology & evolution, 3*, 191-199.

Bowler, D. E., Bjorkman, A. D., Dornelas, M., Myers-Smith, I. H., Navarro, L. M., Niamir, A., . . . Vellend, M. (2020). Mapping human pressures on biodiversity across the planet uncovers anthropogenic threat complexes. *People and Nature, 2*, 380– 394.

Boyle, K. S., & Horn, M. H. (2006). Comparison of feeding guild structure and ecomorphology of intertidal fish assemblages from central California and central Chile. *Marine Ecology Progress Series, 319*, 65-84.

Brooks, T. M., Mittermeier, R. A., da Fonseca, G. A., Gerlach, J., Hoffmann, M., Lamoreux, J. F., . . . Rodrigues, A. S. (2006). Global biodiversity conservation priorities. *Science, 313*, 58-61.

Brosse, S., Beauchard, O., Blanchet, S., Dürr, H. H., Grenouillet, G., Hugueny, B., . . . Villéger, S. (2013). Fish-SPRICH: a database of freshwater fish species richness throughout the World. *Hydrobiologia, 700*, 343-349.

Brown, J., & Lomolino, M. (1998). Biogeography Sinauer. *Sunderland, MA*, 1-624.

Bruggeman, J., Heringa, J., & Brandt, B. W. (2009). PhyloPars: estimation of missing parameter values using phylogeny. *Nucleic acids research, 37*, W179-W184.

Brun, P., Zimmermann, N. E., Graham, C. H., Lavergne, S., Pellissier, L., Münkemüller, T., & Thuiller, W. (2019). The productivity-biodiversity relationship varies across diversity dimensions. *Nature communications, 10*, 5691.

Buckley, L. B., & Jetz, W. (2007). Environmental and historical constraints on global patterns of amphibian richness. *Proceedings of the Royal Society B: Biological Sciences, 274*, 1167-1173. doi:10.1098/rspb.2006.0436

Burgin, S. (2017). Indirect consequences of recreational fishing in freshwater ecosystems: an exploration from an Australian perspective. *Sustainability, 9*, 280.

Butler, E. E., Datta, A., Flores-Moreno, H., Chen, M., Wythers, K. R., Fazayeli, F., . . . Amiaud, B. (2017). Mapping local and global variability in plant trait distributions. *Proceedings of the National Academy of Sciences*, 201708984.

Cambray, J. A. (2003). Impact on indigenous species biodiversity caused by the globalisation of alien recreational freshwater fisheries. *Hydrobiologia, 500*, 217-230.

Capellini, I., Baker, J., Allen, W. L., Street, S. E., & Venditti, C. (2015). The role of life history traits in mammalian invasion success. *Ecology Letters, 18*, 1099-1107.

Capps, K. A., & Flecker, A. S. (2013). Invasive aquarium fish transform ecosystem nutrient dynamics. *Proceedings of the Royal Society B: Biological Sciences, 280*, 20131520.

Carmona, C. P., De Bello, F., Mason, N. W., & Lepš, J. (2016). Traits without borders: integrating functional diversity across scales. *Trends in Ecology & Evolution, 31*, 382-394.

Carpio, A. J., De Miguel, R. J., Oteros, J., Hillström, L., & Tortosa, F. (2019). Angling as a source of non-native freshwater fish: a European review. *Biological Invasions, 21*, 3233-3248.

Carvajal-Quintero, J., Villalobos, F., Oberdorff, T., Grenouillet, G., Brosse, S., Hugueny, B., . . . Tedesco, P. A. (2019). Drainage network position and historical connectivity explain global patterns in freshwater fishes' range size. *Proceedings of the National Academy of Sciences, 116*, 13434-13439.

Ceballos, G., Ehrlich, P. R., & Dirzo, R. (2017). Biological annihilation via the ongoing sixth mass extinction signaled by vertebrate population losses and declines. *Proceedings of the National Academy of Sciences, 114*, E6089-E6096.

Chang, J., Rabosky, D. L., Smith, S. A., & Alfaro, M. E. (2019). An R package and online resource for macroevolutionary studies using the ray-finned fish tree of life. *Methods in Ecology and Evolution, 10*, 1118-1124.

Cilleros, K., Allard, L., Grenouillet, G., & Brosse, S. (2016). Taxonomic and functional diversity patterns reveal different processes shaping European and Amazonian stream fish assemblages. *Journal of Biogeography, 43*, 1832-1843.

Claverie, T., & Wainwright, P. C. (2014). A morphospace for reef fishes: elongation is the dominant axis of body shape evolution. *PloS one, 9*, e112732.

Clavero, M., & García-Berthou, E. (2005). Invasive species are a leading cause of animal extinctions. *Trends in Ecology & Evolution, 20*, 110.

Cliff, A. D., & Ord, J. K. (1981). *Spatial processes: models & applications*: Taylor & Francis.

Colautti, R. I., Eckert, C. G., & Barrett, S. C. (2010). Evolutionary constraints on adaptive evolution during range expansion in an invasive plant. *Proceedings of the Royal Society B: Biological Sciences, 277*, 1799-1806.

Collen, B., Whitton, F., Dyer, E. E., Baillie, J. E., Cumberlidge, N., Darwall, W. R., . . . Böhm, M. (2014). Global patterns of freshwater species diversity, threat and endemism. *Global Ecology and Biogeography, 23*, 40-51.

Covain, R., & Fisch-Muller, S. (2007). The genera of the Neotropical armored catfish subfamily Loricariinae (Siluriformes: Loricariidae): a practical key and synopsis. *Zootaxa, 1462*, 1-40.

Cox, C. B., Moore, P. D., & Ladle, R. J. (2016). *Biogeography: an ecological and evolutionary approach*: John Wiley & Sons.

Craven, D., Eisenhauer, N., Pearse, W. D., Hautier, Y., Isbell, F., Roscher, C., . . . Buchmann,

N. (2018). Multiple facets of biodiversity drive the diversity–stability relationship. *Nature ecology & evolution, 2*, 1579-1587.

Cucherousset, J., Blanchet, S., & Olden, J. D. (2012). Non-native species promote trophic dispersion of food webs. *Frontiers in Ecology and the Environment, 10*, 406-408.

Cucherousset, J., Horky, P., Slavík, O., Ovidio, M., Arlinghaus, R., Boulêtreau, S., . . . Santoul, F. (2018). Ecology, behaviour and management of the European catfish. *Reviews in Fish Biology and Fisheries, 28*, 177-190.

Currie, D. J. (1991). Energy and large-scale patterns of animal-and plant-species richness. *The American Naturalist, 137*, 27-49.

Currie, D. J., Mittelbach, G. G., Cornell, H. V., Field, R., Guégan, J. F., Hawkins, B. A., . . . O'Brien, E. (2004). Predictions and tests of climate-based hypotheses of broad-scale variation in taxonomic richness. *Ecology Letters, 7*, 1121-1134.

Deacon, A. E., Ramnarine, I. W., & Magurran, A. E. (2011). How reproductive ecology contributes to the spread of a globally invasive fish. *PloS one, 6*, e24416.

Devictor, V., Mouillot, D., Meynard, C., Jiguet, F., Thuiller, W., & Mouquet, N. (2010). Spatial mismatch and congruence between taxonomic, phylogenetic and functional diversity: the need for integrative conservation strategies in a changing world. *Ecology Letters, 13*, 1030-1040.

Diagne, C., Leroy, B., Gozlan, R., Vaissière, A.-C., Assailly, C., Nuninger, L., . . . Courchamp, F. (2020). InvaCost, a public database of the economic costs of biological invasions worldwide. *Scientific Data, 7*, 227.

Dias, M. S., Oberdorff, T., Hugueny, B., Leprieur, F., Jézéquel, C., Cornu, J. F., . . . Tedesco, P. A. (2014). Global imprint of historical connectivity on freshwater fish biodiversity. *Ecology Letters, 17*, 1130-1140.

Díaz, S., & Cabido, M. (2001). Vive la différence: plant functional diversity matters to ecosystem processes. *Trends in Ecology & Evolution, 16*, 646-655.

Díaz, S., Kattge, J., Cornelissen, J. H., Wright, I. J., Lavorel, S., Dray, S., . . . Prentice, I. C. (2016). The global spectrum of plant form and function. *Nature, 529*, 167-171.

Díaz, S., Pascual, U., Stenseke, M., Martín-López, B., Watson, R. T., Molnár, Z., . . . Brauman, K. A. (2018). Assessing nature's contributions to people. *Science, 359*, 270-272.

Díaz, S., Settele, J., Brondízio, E., Ngo, H., Guèze, M., Agard, J., . . . Butchart, S. (2019). IPBES (2019): Summary for policymakers of the global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. *IPBES Secretariat: Bonn, Germany*, 56.

Dinerstein, E., Vynne, C., Sala, E., Joshi, A. R., Fernando, S., Lovejoy, T. E., . . . Baillie, J. (2019). A global deal for nature: guiding principles, milestones, and targets. *Science*

*advances, 5*, eaaw2869.

Dirzo, R., Young, H. S., Galetti, M., Ceballos, G., Isaac, N. J., & Collen, B. (2014). Defaunation in the Anthropocene. *Science, 345*, 401-406.

Donohue, I., Hillebrand, H., Montoya, J. M., Petchey, O. L., Pimm, S. L., Fowler, M. S., . . . McClean, D. (2016). Navigating the complexity of ecological stability. *Ecology Letters, 19*, 1172-1185.

Dornelas, M., Gotelli, N. J., McGill, B., Shimadzu, H., Moyes, F., Sievers, C., & Magurran, A. E. (2014). Assemblage time series reveal biodiversity change but not systematic loss. *Science, 344*, 296-299.

Douglas, M. E., & Matthews, W. J. (1992). Does morphology predict ecology? Hypothesis testing within a freshwater stream fish assemblage. *Oikos*, 213-224.

Drenovsky, R. E., Grewell, B. J., D'antonio, C. M., Funk, J. L., James, J. J., Molinari, N., . . . Richards, C. L. (2012). A functional trait perspective on plant invasion. *Annals of botany, 110*, 141-153.

Dudgeon, D. (2008). *Tropical stream ecology*: Academic Press.

Dudgeon, D. (2019). Multiple threats imperil freshwater biodiversity in the Anthropocene. *Current Biology, 29*, R960-R967.

Dudgeon, D., Arthington, A. H., Gessner, M. O., Kawabata, Z.-I., Knowler, D. J., Lévêque, C., . . . Stiassny, M. L. (2006). Freshwater biodiversity: importance, threats, status and conservation challenges. *Biological reviews, 81*, 163-182.

Duffy, J. E., Lefcheck, J. S., Stuart-Smith, R. D., Navarrete, S. A., & Edgar, G. J. (2016). Biodiversity enhances reef fish biomass and resistance to climate change. *Proceedings of the National Academy of Sciences, 113*, 6230-6235.

Dumay, O., Tari, P., Tomasini, J., & Mouillot, D. (2004). Functional groups of lagoon fish species in Languedoc Roussillon, southern France. *Journal of Fish Biology, 64*, 970-983.

Dynesius, M., & Jansson, R. (2000). Evolutionary consequences of changes in species' geographical distributions driven by Milankovitch climate oscillations. *Proceedings of the National Academy of Sciences, 97*, 9115-9120.

Elith, J., Leathwick, J. R., & Hastie, T. (2008). A working guide to boosted regression trees. *Journal of Animal Ecology, 77*, 802-813.

Faith, D. P. (2008). Phylogenetic diversity and conservation. *Conservation biology: evolution in action*, 99-115.

Felsenstein, J. (1985). Phylogenies and the comparative method. *The American Naturalist, 125*, 1-15.

Fournier, A., Penone, C., Pennino, M. G., & Courchamp, F. (2019). Predicting future invaders and future invasions. *Proceedings of the National Academy of Sciences*, 201803456.

Fricke, R. (2020). Eschmeyer's catalog of fishes: genera, species, references, online version. Updated 3 August 2020. In.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of statistics*, 1189-1232.

Froese, R., & Pauly, D. (2018). FishBase. Retrieved from www.fishbase.org.

Fulton, C., Bellwood, D., & Wainwright, P. (2001). The relationship between swimming ability and habitat use in wrasses (Labridae). *Marine Biology, 139*, 25-33.

Funk, J. L., Cleland, E. E., Suding, K. N., & Zavaleta, E. S. (2008). Restoration through reassembly: plant traits and invasion resistance. *Trends in Ecology & Evolution, 23*, 695-703.

Gallien, L., Münkemüller, T., Albert, C. H., Boulangeat, I., & Thuiller, W. (2010). Predicting potential distributions of invasive species: where to go from here? *Diversity and Distributions, 16*, 331-342.

Garnier, E., Navas, M.-L., & Grigulis, K. (2016). *Plant functional diversity: Organism traits, community structure, and ecosystem properties*: Oxford University Press.

Gaston, K. J. (2000). Global patterns in biodiversity. *Nature, 405*, 220-227.

Gaston, K. J., & Spicer, J. I. (2004). *Biodiversity: an introduction*: John Wiley & Sons.

Gatz, A. J. (1979). Community organization in fishes as indicated by morphological features. *Ecology, 60*, 711-718.

Gossner, M. M., Lewinsohn, T. M., Kahl, T., Grassein, F., Boch, S., Prati, D., . . . Wubet, T. (2016). Land-use intensification causes multitrophic homogenization of grassland communities. *Nature, 540*, 266-269.

Gozlan, R. E., Britton, J., Cowx, I., & Copp, G. (2010). Current knowledge on non-native freshwater fish introductions. *Journal of Fish Biology, 76*, 751-786.

Griffiths, D. (2006). Pattern and process in the ecological biogeography of European freshwater fish. *Journal of Animal Ecology, 75*, 734-751.

Grill, G., Lehner, B., Thieme, M., Geenen, B., Tickner, D., Antonelli, F., . . . Crochetiere, H. (2019). Mapping the world's free-flowing rivers. *Nature, 569*, 215.

Grime, J. (1998). Benefits of plant diversity to ecosystems: immediate, filter and founder effects. *Journal of ecology, 86*, 902-910.

Grooten, M., & Almond, R. (2018). Living planet report-2018: aiming higher. *Living planet report-2018: aiming higher.*

Guégan, J.-F., Lek, S., & Oberdorff, T. (1998). Energy availability and habitat heterogeneity predict global riverine fish diversity. *Nature, 391*, 382-384.

Haas, T. C., Blum, M. J., & Heins, D. C. (2010). Morphological responses of a stream fish to water impoundment. *Biology letters, 6*, 803-806.

Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., . . . Loveland, T. R. (2013). High-resolution global maps of 21st-century forest cover change. *Science, 342*, 850-853.

Hasnain, S. S., Shuter, B. J., & Minns, C. K. (2013). Phylogeny influences the relationships linking key ecological thermal metrics for North American freshwater fish species. *Canadian Journal of Fisheries and Aquatic Sciences, 70*, 964-972.

Hassan, R., Scholes, R., & Ash, N. (2005). *Ecosystems and human well-being: current state and trends*. Washington, DC: Island Press.

Havel, J. E., Lee, C. E., & Vander Zanden, J. M. (2005). Do reservoirs facilitate invasions into landscapes? *Bioscience, 55*, 518-525.

Hawkins, B. A., Field, R., Cornell, H. V., Currie, D. J., Guégan, J.-F., Kaufman, D. M., . . . O'Brien, E. M. (2003). Energy, water, and broad-scale geographic patterns of species richness. *Ecology, 84*, 3105-3117.

Heino, J., Schmera, D., & Erős, T. (2013). A macroecological perspective of trait patterns in stream communities. *Freshwater Biology, 58*, 1539-1555.

Helfman, G., Collette, B. B., Facey, D. E., & Bowen, B. W. (2009). *The diversity of fishes: biology, evolution, and ecology*: John Wiley & Sons.

Hocutt, C. H., & Wiley, E. O. (1986). *The zoogeography of North American freshwater fishes*: Wiley.

Hoffmann, R. C. (1996). Economic development and aquatic ecosystems in medieval Europe. *The American Historical Review, 101*, 631-669.

Howeth, J. G., Gantz, C. A., Angermeier, P. L., Frimpong, E. A., Hoff, M. H., Keller, R. P., . . . Romagosa, C. M. (2016). Predicting invasiveness of species in trade: climate match, trophic guild and fecundity influence establishment and impact of non-native freshwater fishes. *Diversity & Distributions, 22*, 148-160.

Huang, T.-Y., Downs, M. R., Ma, J., & Zhao, B. (2020). Collaboration across Time and Space in the LTER Network. *Bioscience, 70*, 353-364.

Hugueny, B. (1989). West African rivers as biogeographic islands: species richness of fish communities. *Oecologia, 79*, 236-243.

Hugueny, B., & Pouilly, M. (1999). Morphological correlates of diet in an assemblage of West African freshwater fishes. *Journal of Fish Biology, 54*, 1310-1325.

Hulme, P. E. (2009). Trade, transport and trouble: managing invasive species pathways in an era of globalization. *Journal of applied ecology, 46*, 10-18.

IUCN. (2018). *The IUCN red list of threatened species, Version 2017-3, http://www.iucnredlist.org*.

Ives, A. R., & Carpenter, S. R. (2007). Stability and diversity of ecosystems. *Science, 317*, 58-62.

Jackson, J. B. (2008). Ecological extinction and evolution in the brave new ocean. *Proceedings of the National Academy of Sciences, 105*, 11458-11465.

Jackson, S. T., & Sax, D. F. (2010). Balancing biodiversity in a changing environment: extinction debt, immigration credit and species turnover. *Trends in Ecology & Evolution, 25*, 153-160.

Jansson, R. (2003). Global patterns in endemism explained by past climatic change. *Proceedings of the Royal Society of London. Series B: Biological Sciences, 270*, 583-590.

Jansson, R., & Davies, T. J. (2008). Global variation in diversification rates of flowering plants: energy vs. climate change. *Ecology Letters, 11*, 173-183.

Jézéquel, C., Tedesco, P. A., Bigorne, R., Maldonado-Ocampo, J. A., Ortega, H., Hidalgo, M., . . . Acosta, A. (2020). A database of freshwater fish species of the Amazon Basin. *Scientific Data, 7*, 96.

Johnson, P. T., Olden, J. D., & Vander Zanden, M. J. (2008). Dam invaders: impoundments facilitate biological invasions into freshwaters. *Frontiers in Ecology and the Environment, 6*, 357-363.

Kang, B., Huang, X., Yan, Y., Yan, Y., & Lin, H. (2018). Dataset on the distribution location and biological traits of freshwater fishes in the Yangtze River Basin. *Data in brief, 21*, 1066-1070.

Kattge, J., Diaz, S., Lavorel, S., Prentice, I., Leadley, P., Bönisch, G., . . . Wright, I. (2011). TRY–a global database of plant traits. *Global change biology, 17*, 2905-2935.

Kent, M. (2005). Biogeography and macroecology. *Progress in Physical Geography, 29*, 256-264.

Kreft, H., & Jetz, W. (2007). Global patterns and determinants of vascular plant diversity. *Proceedings of the National Academy of Sciences, 104*, 5925-5930. doi:10.1073/pnas.0608361104

Kuczynski, L., Côte, J., Toussaint, A., Brosse, S., Buisson, L., & Grenouillet, G. (2018). Spatial mismatch in morphological, ecological and phylogenetic diversity, in historical and contemporary European freshwater fish faunas. *Ecography, 41*, 1665-1674.

Laliberté, E., & Legendre, P. (2010). A distance-based framework for measuring functional

diversity from multiple traits. *Ecology, 91*, 299-305.

Lamouroux, N., Poff, N. L., & Angermeier, P. L. (2002). Intercontinental convergence of stream fish community traits along geomorphic and hydraulic gradients. *Ecology, 83*, 1792-1807.

Lande, R. (1996). Statistics and partitioning of species diversity, and similarity among multiple communities. *Oikos*, 5-13.

Lefcheck, J. S., Buchheister, A., Laumann, K. M., Stratton, M. A., Sobocinski, K. L., Chak, S. T., . . . Duffy, J. E. (2014). Dimensions of biodiversity in Chesapeake Bay demersal fishes: patterns and drivers through space and time. *Ecosphere, 5*, 14.

Leitão, R. P., Zuanon, J., Mouillot, D., Leal, C. G., Hughes, R. M., Kaufmann, P. R., . . . de Paula, F. R. (2018). Disentangling the pathways of land use impacts on the functional structure of fish assemblages in Amazon streams. *Ecography, 41*, 219-232.

Leprieur, F., Beauchard, O., Blanchet, S., Oberdorff, T., & Brosse, S. (2008). Fish invasions in the world's river systems: when natural processes are blurred by human activities. *Plos Biology, 6*, e28.

Leprieur, F., Brosse, S., Garcia-Berthou, E., Oberdorff, T., Olden, J., & Townsend, C. (2009). Scientific uncertainty and the assessment of risks posed by non-native freshwater fishes. *Fish and fisheries, 10*, 88-97.

Leprieur, F., Tedesco, P. A., Hugueny, B., Beauchard, O., Dürr, H. H., Brosse, S., & Oberdorff, T. (2011). Partitioning global patterns of freshwater fish beta diversity reveals contrasting signatures of past climate changes. *Ecology Letters, 14*, 325-334.

Lévêque, C., Oberdorff, T., Paugy, D., Stiassny, M., & Tedesco, P. (2008). Global diversity of fish (Pisces) in freshwater. *Hydrobiologia, 595*, 545-567.

Levine, J. M. (2000). Species diversity and biological invasions: relating local process to community pattern. *Science, 288*, 852-854.

Linke, S., Lehner, B., Dallaire, C. O., Ariwi, J., Grill, G., Anand, M., . . . Moidu, H. (2019). Global hydro-environmental sub-basin and river reach characteristics at high spatial resolution. *Scientific Data, 6*, 1-15.

Lintermans, M. (2004). Human-assisted dispersal of alien freshwater fish in Australia. *New Zealand Journal of Marine and Freshwater Research, 38*, 481-501.

Lockwood, J. L., Cassey, P., & Blackburn, T. (2005). The role of propagule pressure in explaining species invasions. *Trends in Ecology & Evolution, 20*, 223-228.

Lockwood, J. L., Hoopes, M. F., & Marchetti, M. P. (2013). *Invasion ecology*: John Wiley & Sons.

Loreau, M., Naeem, S., Inchausti, P., Bengtsson, J., Grime, J., Hector, A., . . . Schmid, B. (2001). Biodiversity and ecosystem functioning: current knowledge and future challenges.

*Science, 294*, 804-808.

Lozupone, C., & Knight, R. (2005). UniFrac: a new phylogenetic method for comparing microbial communities. *Appl. Environ. Microbiol., 71*, 8228-8235.

MacArthur, R. H., & Wilson, E. O. (1963). An equilibrium theory of insular zoogeography. *Evolution, 17*, 373-387.

MacArthur, R. H., & Wilson, E. O. (2001). *The theory of island biogeography* (Vol. 1): Princeton university press.

Maire, E., Grenouillet, G., Brosse, S., & Villéger, S. (2015). How many dimensions are needed to accurately assess functional diversity? A pragmatic approach for assessing the quality of functional spaces. *Global Ecology and Biogeography, 24*, 728-740.

Manel, S., Guerin, P.-E., Mouillot, D., Blanchet, S., Velez, L., Albouy, C., & Pellissier, L. (2020). Global determinants of freshwater and marine fish genetic diversity. *Nature communications, 11*, 1-9.

Marchetti, M. P., Moyle, P. B., & Levine, R. (2004). Invasive species profiling? Exploring the characteristics of non-native fishes across invasion stages in California. *Freshwater Biology, 49*, 646-661.

Margules, C. R., & Pressey, R. L. (2000). Systematic conservation planning. *Nature, 405*, 243.

Maron, J. L., & Vilà, M. (2001). When do herbivores affect plant invasion? Evidence for the natural enemies and biotic resistance hypotheses. *Oikos, 95*, 361-373.

May, R. M. (2010). Tropical arthropod species, more or less? *Science, 329*, 41-42.

McCann, K. S. (2000). The diversity–stability debate. *Nature, 405*, 228.

McGill, B. J., Enquist, B. J., Weiher, E., & Westoby, M. (2006). Rebuilding community ecology from functional traits. *Trends in Ecology & Evolution, 21*, 178-185.

McIntyre, P. B., Liermann, C. A. R., & Revenga, C. (2016). Linking freshwater fishery management to global food security and biodiversity conservation. *Proceedings of the National Academy of Sciences, 113*, 12880-12885.

McNaughton, S. (1978). Stability and diversity of ecological communities. *Nature, 274*, 251-253.

Miller, E. C. (2020). Comparing speciation rates in lakes, rivers, and the sea. *bioRxiv*.

Mora, C., Aburto-Oropeza, O., Bocos, A. A., Ayotte, P. M., Banks, S., Bauman, A. G., . . . Brokovich, E. (2011b). Global human footprint on the linkage between biodiversity and ecosystem functioning in reef fishes. *Plos Biology, 9*.

Mora, C., Tittensor, D. P., Adl, S., Simpson, A. G., & Worm, B. (2011a). How many species are there on Earth and in the ocean? *PLoS Biol, 9*, e1001127.

Mouchet, M. A., Villéger, S., Mason, N. W., & Mouillot, D. (2010). Functional diversity measures: an overview of their redundancy and their ability to discriminate community assembly rules. *Functional Ecology, 24*, 867-876.

Mouillot, D., Bellwood, D. R., Baraloto, C., Chave, J., Galzin, R., Harmelin-Vivien, M., . . . Mouquet, N. (2013b). Rare species support vulnerable functions in high-diversity ecosystems. *Plos Biology, 11*, e1001569.

Mouillot, D., Graham, N. A., Villéger, S., Mason, N. W., & Bellwood, D. R. (2013). A functional approach reveals community responses to disturbances. *Trends in Ecology & Evolution, 28*, 167-177.

Mouillot, D., Villéger, S., Parravicini, V., Kulbicki, M., Arias-González, J. E., Bender, M., . . . Vigliola, L. (2014). Functional over-redundancy and high functional vulnerability in global fish faunas on tropical reefs. *Proceedings of the National Academy of Sciences, 111*, 13757-13762.

Moyle, P. B., & Marchetti, M. P. (2006). Predicting invasion success: freshwater fishes in California as a model. *Bioscience, 56*, 515-524.

Muneepeerakul, R., Bertuzzo, E., Lynch, H. J., Fagan, W. F., Rinaldo, A., & Rodriguez-Iturbe, I. (2008). Neutral metacommunity models predict fish diversity patterns in Mississippi–Missouri basin. *Nature, 453*, 220-222.

Myers, N., Mittermeier, R. A., Mittermeier, C. G., Da Fonseca, G. A., & Kent, J. (2000). Biodiversity hotspots for conservation priorities. *Nature, 403*, 853.

Naeem, S., Duffy, J. E., & Zavaleta, E. (2012). The functions of biological diversity in an age of extinction. *Science, 336*, 1401-1406.

Nagelkerke, N. J. (1991). A note on a general definition of the coefficient of determination. *Biometrika, 78*, 691-692.

Nekola, J. C., & White, P. S. (1999). The distance decay of similarity in biogeography and ecology. *Journal of Biogeography, 26*, 867-878.

Nelson, J. S., Grande, T. C., & Wilson, M. V. (2016). *Fishes of the World*: John Wiley & Sons.

Nordhaus, W., & Chen, X. (2016). Global gridded geographically based economic data (G-econ), version 4. *NASA Socioeconomic Data and Applications Center (SEDAC)*.

Nordhaus, W. D. (2006). Geography and macroeconomics: New data and new findings. *Proceedings of the National Academy of Sciences, 103*, 3510-3517.

Oberdorff, T., Dias, M. S., Jézéquel, C., Albert, J. S., Arantes, C. C., Bigorne, R., . . . Hidalgo, M. (2019). Unexpected fish diversity gradients in the Amazon basin. *Science advances, 5*, eaav8681.

Oberdorff, T., Hugueny, B., & Guégan, J. F. (1997). Is there an influence of historical events on contemporary fish species richness in rivers? Comparisons between Western Europe and

North America. *Journal of Biogeography, 24*, 461-467.

Oberdorff, T., Tedesco, P. A., Hugueny, B., Leprieur, F., Beauchard, O., Brosse, S., & Dürr, H. H. (2011). Global and regional patterns in riverine fish species richness: a review. *International Journal of Ecology, 2011*, 967631.

Oksanen, J., Blanchet, F. G., Kindt, R., Legendre, P., O'hara, R., Simpson, G. L., . . . Wagner, H. (2010). Vegan: community ecology package. R package version 2.5-4. *Available at: http://cran.r-project.org/ (accessed 10 August 2019).*

Olden, J. D., Hogan, Z. S., & Zanden, M. J. V. (2007). Small fish, big fish, red fish, blue fish: size-biased extinction risk of the world's freshwater and marine fishes. *Global Ecology & Biogeography, 16*, 694–701.

Olden, J. D., Poff, N. L., & Bestgen, K. R. (2006). Life-history strategies predict fish invasions and extirpations in the Colorado River Basin. *Ecological Monographs, 76*, 25-40.

Olden, J. D., Vitule, J. R. S., Cucherousset, J., & Kennard, M. J. (2020). There's more to Fish than Just Food: Exploring the Diverse Ways that Fish Contribute to Human Society. *Fisheries, 45*, 453-464. doi:10.1002/fsh.10443

Padilla, D. K., & Williams, S. L. (2004). Beyond ballast water: aquarium and ornamental trades as sources of invasive species in aquatic ecosystems. *Frontiers in Ecology and the Environment, 2*, 131-138.

Pagel, M. (1999). Inferring the historical patterns of biological evolution. *Nature, 401*, 877.

Parravicini, V., Kulbicki, M., Bellwood, D., Friedlander, A., Arias-Gonzalez, J., Chabanet, P., . . . D'Agata, S. (2013). Global patterns and predictors of tropical reef fish species richness. *Ecography, 36*, 1254-1262.

Parravicini, V., Villéger, S., McClanahan, T. R., Arias-González, J. E., Bellwood, D. R., Belmaker, J., . . . Guilhaumon, F. (2014). Global mismatch between species richness and vulnerability of reef fish assemblages. *Ecology Letters, 17*, 1101-1110.

Pellissier, L., Leprieur, F., Parravicini, V., Cowman, P. F., Kulbicki, M., Litsios, G., . . . Mouillot, D. (2014). Quaternary coral reef refugia preserved fish diversity. *Science, 344*, 1016-1019.

Pennekamp, F., Pontarp, M., Tabi, A., Altermatt, F., Alther, R., Choffat, Y., . . . Griffiths, J. I. (2018). Biodiversity increases and decreases ecosystem stability. *Nature, 563*, 109-112.

Penone, C., Davidson, A. D., Shoemaker, K. T., Di Marco, M., Rondinini, C., Brooks, T. M., . . . Costa, G. C. (2014). Imputation of missing data in life-history trait datasets: which approach performs the best? *Methods in Ecology and Evolution, 5*, 961-970.

Pereira, H. M., Ferrier, S., Walters, M., Geller, G. N., Jongman, R., Scholes, R. J., . . . Cardoso, A. (2013). Essential biodiversity variables. *Science, 339*, 277-278.

Pereira, H. M., Leadley, P. W., Proença, V., Alkemade, R., Scharlemann, J. P., Fernandez-

Manjarrés, J. F., . . . Cheung, W. W. (2010). Scenarios for global biodiversity in the 21st century. *Science, 330*, 1496-1501.

Pereira, H. M., Navarro, L. M., & Martins, I. S. (2012). Global biodiversity change: the bad, the good, and the unknown. *Annual Review of Environment and Resources, 37*, 25-50.

Petchey, O. L., & Gaston, K. J. (2006). Functional diversity: back to basics and looking forward. *Ecology Letters, 9*, 741-758.

Pimentel, D., Lach, L., Zuniga, R., & Morrison, D. (2000). Environmental and economic costs of nonindigenous species in the United States. *Bioscience, 50*, 53-65.

Pimiento, C., Leprieur, F., Silvestro, D., Lefcheck, J., Albouy, C., Rasher, D., . . . Griffin, J. (2020). Functional diversity of marine megafauna in the Anthropocene. *Science advances, 6*, eaay7650.

Pimm, S. L., Jenkins, C. N., & Li, B. V. (2018). How to protect half of Earth to ensure it protects sufficient biodiversity. *Science advances, 4*, eaat2616.

Pörtner, H. O., & Knust, R. (2007). Climate change affects marine fishes through the oxygen limitation of thermal tolerance. *Science, 315*, 95-97.

Puth, L. M., & Post, D. M. (2005). Studying invasion: have we missed the boat? *Ecology Letters, 8*, 715-721.

Pyšek, P., Jarošík, V., Hulme, P. E., Kühn, I., Wild, J., Arianoutsou, M., . . . Essl, F. (2010). Disentangling the role of environmental and human pressures on biological invasions across Europe. *Proceedings of the National Academy of Sciences, 107*, 12157-12162.

R Core Team. (2017). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. Retrieved from https://www.R-project.org. In.

R Core Team. (2018). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. Retrieved from https://www.R-project.org. In.

Rabosky, D. L. (2020). Speciation rate and the diversity of fishes in freshwaters and the oceans. *Journal of Biogeography, 47*, 1207– 1217.

Rabosky, D. L., Chang, J., Title, P. O., Cowman, P. F., Sallan, L., Friedman, M., . . . Coll, M. (2018). An inverse latitudinal gradient in speciation rate for marine fishes. *Nature, 559*, 392.

Radinger, J., & Wolter, C. (2014). Patterns and predictors of fish dispersal in rivers. *Fish and fisheries, 15*, 456-473.

Rangel, T. F., Diniz-Filho, J. A. F., & Bini, L. M. (2010). SAM: a comprehensive application for spatial analysis in macroecology. *Ecography, 33*, 46-50.

Reecht, Y., Rochet, M. J., Trenkel, V. M., Jennings, S., & Pinnegar, J. K. (2013). Use of morphological characteristics to define functional groups of predatory fishes in the Celtic Sea. *Journal of Fish Biology, 83*, 355-377.

Reyjol, Y., Hugueny, B., Pont, D., Bianco, P. G., Beier, U., Caiola, N., . . . Ferreira, T. (2007). Patterns in species richness and endemism of European freshwater fish. *Global Ecology and Biogeography, 16*, 65-75.

Ribeiro, F., Elvira, B., Collares-Pereira, M. J., & Moyle, P. B. (2008). Life-history traits of non-native fishes in Iberian watersheds across several invasion stages: a first approach. *Biological Invasions, 10*, 89-102.

Ricklefs, R. E. (1987). Community diversity: relative roles of local and regional processes. *Science, 235*, 167-171.

Ricklefs, R. E. (2012). Species richness and morphological diversity of passerine birds. *Proceedings of the National Academy of Sciences, 109*, 14482-14487.

Ripley, B., Venables, B., Bates, D. M., Hornik, K., Gebhardt, A., Firth, D., & Ripley, M. B. (2013). Package 'mass'. *Cran R, 538*.

Ripple, W. J., Wolf, C., Newsome, T. M., Hoffmann, M., Wirsing, A. J., & McCauley, D. J. (2017). Extinction risk is most acute for the world's largest and smallest vertebrates. *Proceedings of the National Academy of Sciences, 114*, 10678-10683.

Rodrigues-Filho, C. A., Leitão, R. P., Zuanon, J., Sánchez-Botero, J. I., & Baccaro, F. B. (2018). Historical stability promoted higher functional specialization and originality in Neotropical stream fish assemblages. *Journal of Biogeography*.

Romanuk, T. N., Hayward, A., & Hutchings, J. A. (2011). Trophic level scales positively with body size in fishes. *Global Ecology and Biogeography, 20*, 231-240.

Roy, K., Jablonski, D., & Valentine, J. W. (2002). Body size and invasion success in marine bivalves. *Ecology Letters, 5*, 163-167.

Safi, K., Cianciaruso, M. V., Loyola, R. D., Brito, D., Armour-Marshall, K., & Diniz-Filho, J. A. F. (2011). Understanding global patterns of mammalian functional and phylogenetic diversity. *Philosophical Transactions of the Royal Society B: Biological Sciences, 366*, 2536-2544.

Sala, O. E., Chapin, F. S., Armesto, J. J., Berlow, E., Bloomfield, J., Dirzo, R., . . . Kinzig, A. (2000). Global biodiversity scenarios for the year 2100. *Science, 287*, 1770-1774.

Sarkar, S. (2002). Defining "biodiversity"; assessing biodiversity. *The Monist, 85*, 131-155.

Saul, W. C., Roy, H. E., Booy, O., Carnevali, L., Chen, H. J., Genovesi, P., . . . Pergl, J. (2017). Assessing patterns in introduction pathways of alien species by linking major invasion data bases. *Journal of applied ecology, 54*, 657-669.

Schleuter, D., Daufresne, M., Massol, F., & Argillier, C. (2010). A user's guide to functional

diversity indices. *Ecological Monographs, 80*, 469-484.

Secretariat, C. o. B. D. (2006). *Global biodiversity outlook 2*.

Seebens, H., Bacher, S., Blackburn, T. M., Capinha, C., Dawson, W., Dullinger, S., . . . Kühn, I. (2020). Projecting the continental accumulation of alien species through to 2050. *Global change biology*.

Seebens, H., Blackburn, T. M., Dyer, E. E., Genovesi, P., Hulme, P. E., Jeschke, J. M., . . . Winter, M. (2018). Global rise in emerging alien species results from increased accessibility of new source pools. *Proceedings of the National Academy of Sciences, 115*, E2264-E2273.

Seebens, H., Blackburn, T. M., Dyer, E. E., Genovesi, P., Hulme, P. E., Jeschke, J. M., . . . Arianoutsou, M. (2017). No saturation in the accumulation of alien species worldwide. *Nature communications, 8*, 14435.

Shirley, S. M., & Kark, S. (2009). The role of species traits and taxonomic patterns in alien bird impacts. *Global Ecology and Biogeography, 18*, 450-459.

Simberloff, D., Martin, J.-L., Genovesi, P., Maris, V., Wardle, D. A., Aronson, J., . . . Pascal, M. (2013). Impacts of biological invasions: what's what and the way forward. *Trends in Ecology & Evolution, 28*, 58-66.

Soulé, M. E. (1980). *Conservation biology an evolutionary ecological perspective*. Retrieved from

Stanković, D., Crivelli, A. J., & Snoj, A. (2015). Rainbow trout in Europe: introduction, naturalization, and impacts. *Reviews in Fisheries Science & Aquaculture, 23*, 39-71.

Stekhoven, D. J., & Bühlmann, P. (2012). MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics, 28*, 112-118.

Sternberg, D., & Kennard, M. J. (2014). Phylogenetic effects on functional traits and life history strategies of Australian freshwater fish. *Ecography, 37*, 54-64.

Stone, R. (2007). The Last of the Leviathans. *Science, 316*, 1684-1688.

Stork, N. E. (2009). Chapter 21 - Biodiversity. In V. H. Resh & R. T. Cardé (Eds.), *Encyclopedia of Insects (Second Edition)* (pp. 75-80). San Diego: Academic Press.

Strecker, A. L., Olden, J. D., Whittier, J. B., & Paukert, C. P. (2011). Defining conservation priorities for freshwater fishes according to taxonomic, functional, and phylogenetic diversity. *Ecological Applications, 21*, 3002-3013.

Stuart-Smith, R. D., Bates, A. E., Lefcheck, J. S., Duffy, J. E., Baker, S. C., Thomson, R. J., . . . Airoldi, L. (2013). Integrating abundance and functional traits reveals new global hotspots of fish diversity. *Nature, 501*, 539-542.

Su, G., Logez, M., Xu, J., Tao, S., Villéger, S., & Brosse, S. (2021). Human impacts on global

freshwater fish biodiversity. *Science, 371*, 835-838.

Su, G., Villéger, S., & Brosse, S. (2019). Morphological diversity of freshwater fishes differs between realms, but morphologically extreme species are widespread. *Global Ecology and Biogeography, 28*, 211-221.

Su, G., Villéger, S., & Brosse, S. (2020). Morphological sorting of introduced freshwater fish species within and between donor realms. *Global Ecology and Biogeography, 29*, 803-813.

Taylor, B. W., & Irwin, R. E. (2004). Linking economic activities to the distribution of exotic plants. *Proceedings of the National Academy of Sciences, 101*, 17725-17730.

Team, R. C. (2019). R: A language and environment for statistical computing v. 3.5. 1. Vienna, Austria: Foundation for Statistical Computing. Retrieved from https ://www.r-project.org/.

Tedesco, P. A., Beauchard, O., Bigorne, R., Blanchet, S., Buisson, L., Conti, L., . . . Hugueny, B. (2017). A global database on freshwater fish species occurrence in drainage basins. *Scientific Data, 4*, 170141.

Tedesco, P. A., Leprieur, F., Hugueny, B., Brosse, S., Dürr, H. H., Beauchard, O., . . . Oberdorff, T. (2012). Patterns and processes of global riverine fish endemism. *Global Ecology and Biogeography, 21*, 977-987.

Thomas, C. D., Cameron, A., Green, R. E., Bakkenes, M., Beaumont, L. J., Collingham, Y. C., . . . Hannah, L. (2004). Extinction risk from climate change. *Nature, 427*, 145.

Thomaz, A. T., Christie, M. R., & Knowles, L. L. (2016). The architecture of river networks can drive the evolutionary dynamics of aquatic populations. *Evolution, 70*, 731-739.

Tickner, D., Opperman, J. J., Abell, R., Acreman, M., Arthington, A. H., Bunn, S. E., . . . Edwards, G. (2020). Bending the curve of global freshwater biodiversity loss: an emergency recovery plan. *Bioscience, 70*, 330-342.

Tidwell, J. H., & Allan, G. L. (2001). Fish as food: aquaculture's contribution. *EMBO reports, 2*, 958-963.

Tittensor, D. P., Mora, C., Jetz, W., Lotze, H. K., Ricard, D., Berghe, E. V., & Worm, B. (2010). Global patterns and predictors of marine biodiversity across taxa. *Nature, 466*, 1098-1101. doi:10.1038/nature09329

Toussaint, A., Beauchard, O., Oberdorff, T., Brosse, S., & Villéger, S. (2016a). Worldwide freshwater fish homogenization is driven by a few widespread non-native species. *Biological Invasions, 18*, 1295-1304.

Toussaint, A., Charpin, N., Beauchard, O., Grenouillet, G., Oberdorff, T., Tedesco, P. A., . . . Villéger, S. (2018). Non-native species led to marked shifts in functional diversity of the world freshwater fish faunas. *Ecology Letters, 21*, 1649-1659.

Toussaint, A., Charpin, N., Brosse, S., & Villéger, S. (2016b). Global functional diversity of freshwater fish is concentrated in the Neotropics while functional vulnerability is widespread. *Scientific reports, 6*, 22125.

Turner, J. R., Gatehouse, C. M., & Corey, C. A. (1987). Does solar energy control organic diversity? Butterflies, moths and the British climate. *Oikos*, 195-205.

van der Laan, R. (2020). *Freshwater fish list* (30th ed.). Almere, the Netherlands.

Venter, O., Sanderson, E. W., Magrach, A., Allan, J. R., Beher, J., Jones, K. R., . . . Fekete, B. M. (2016). Global terrestrial Human Footprint maps for 1993 and 2009. *Scientific Data, 3*, 160067.

Villéger, S., Blanchet, S., Beauchard, O., Oberdorff, T., & Brosse, S. (2011). Homogenization patterns of the world's freshwater fish faunas. *Proceedings of the National Academy of Sciences, 108*, 18003-18008.

Villéger, S., Blanchet, S., Beauchard, O., Oberdorff, T., & Brosse, S. (2015). From current distinctiveness to future homogenization of the world's freshwater fish faunas. *Diversity and Distributions, 21*, 223-235.

Villéger, S., Brosse, S., Mouchet, M., Mouillot, D., & Vanni, M. J. (2017). Functional ecology of fish: current approaches and future challenges. *Aquatic Sciences, 79*, 783-801.

Villéger, S., Grenouillet, G., & Brosse, S. (2013). Decomposing functional β-diversity reveals that low functional β-diversity is driven by low functional turnover in E uropean fish assemblages. *Global Ecology and Biogeography, 22*, 671-681.

Villéger, S., Grenouillet, G., & Brosse, S. (2014). Functional homogenization exceeds taxonomic homogenization among European fish assemblages. *Global Ecology and Biogeography, 23*, 1450-1460.

Villéger, S., Mason, N. W. H., & Mouillot, D. (2008). New multidimensional functional diversity indices for a multifaceted framework in functional ecology. *Ecology, 89*, 2290-2301.

Villéger, S., Miranda, J. R., Hernandez, D. F., & Mouillot, D. (2010). Contrasting changes in taxonomic vs. functional diversity of tropical fish communities after habitat degradation. *Ecological Applications, 20*, 1512-1522.

Violle, C., Navas, M. L., Vile, D., Kazakou, E., Fortunel, C., Hummel, I., & Garnier, E. (2007). Let the concept of trait be functional! *Oikos, 116*, 882-892.

Violle, C., Reich, P. B., Pacala, S. W., Enquist, B. J., & Kattge, J. (2014). The emergence and promise of functional biogeography. *Proceedings of the National Academy of Sciences, 111*, 13690-13696.

Vitousek, P. M., D'antonio, C. M., Loope, L. L., Rejmanek, M., & Westbrooks, R. (1997). Introduced species: a significant component of human-caused global change. *New*

*Zealand Journal of Ecology, 21*, 1-16.

Vitule, J. R., Agostinho, A. A., Azevedo-Santos, V. M., Daga, V. S., Darwall, W. R., Fitzgerald, D. B., . . . Magalhães, A. L. (2017). We need better understanding about functional diversity and vulnerability of tropical freshwater fishes. *Biodiversity and Conservation, 26*, 757-762.

Vitule, J. R., Occhi, T. V., Kang, B., Matsuzaki, S.-I., Bezerra, L. A., Daga, V. S., . . . Padial, A. A. (2019). Intra-country introductions unraveling global hotspots of alien fish species. *Biodiversity and Conservation, 28*, 3037-3043.

Vitule, J. R. S., Freire, C. A., & Simberloff, D. (2009). Introduction of non-native freshwater fish can certainly be bad. *Fish & Fisheries, 10*, 98-108.

Vitule, J. R. S., Skóra, F., & Abilhoa, V. (2012). Homogenization of freshwater fish faunas after the elimination of a natural barrier by a dam in Neotropics. *Diversity and Distributions, 18*, 111-120.

Vörösmarty, C. J., McIntyre, P. B., Gessner, M. O., Dudgeon, D., Prusevich, A., Green, P., . . . Liermann, C. R. (2010). Global threats to human water security and river biodiversity. *Nature, 467*, 555.

Ward, A. B., & Mehta, R. S. (2010). Axial elongation in fishes: using morphological approaches to elucidate developmental mechanisms in studying body shape. *Integrative and Comparative Biology, 50*, 1106-1119.

Watson, D., & Balon, E. (1984). Ecomorphological analysis of fish taxocenes in rainforest streams of northern Borneo. *Journal of Fish Biology, 25*, 371-384.

Webb, C. O., Ackerly, D. D., McPeek, M. A., & Donoghue, M. J. (2002). Phylogenies and community ecology. *Annual review of ecology and systematics, 33*, 475-505.

Webb, P. (1984). Body form, locomotion and foraging in aquatic vertebrates. *American Zoologist, 24*, 107-120.

Weiss, D. J., Nelson, A., Gibson, H., Temperley, W., Peedell, S., Lieber, A., . . . Fullman, N. (2018). A global map of travel time to cities to assess inequalities in accessibility in 2015. *Nature, 553*, 333.

Welcomme, R. L. (1988). *International introductions of inland aquatic species*: Food & Agriculture Org.

Whittaker, R. (1977). Evolution of species diversity in land communities. *Evolutionary Biology, 10*, pp.1-67.

Wilcove, D. S., Rothstein, D., Dubow, J., Phillips, A., & Losos, E. (1998). Quantifying threats to imperiled species in the United States. *Bioscience, 48*, 607-615.

Wilson, E. O. (1988). *Biodiversity*. Washington, DC: National Academies Press.

Winemiller, K. O. (1991). Ecomorphological diversification in lowland freshwater fish assemblages from five biotic regions. *Ecological Monographs, 61*, 343-365.

Winemiller, K. O., McIntyre, P. B., Castello, L., Fluet-Chouinard, E., Giarrizzo, T., Nam, S., . . . Harrison, I. (2016). Balancing hydropower and biodiversity in the Amazon, Congo, and Mekong. *Science, 351*, 128-129.

Winter, M., Schweiger, O., Klotz, S., Nentwig, W., Andriopoulos, P., Arianoutsou, M., . . . Hejda, M. (2009). Plant extinctions and introductions lead to phylogenetic and taxonomic homogenization of the European flora. *Proceedings of the National Academy of Sciences, 106*, 21721-21725.

Wright, D. H. (1983). Species-energy theory: an extension of species-area theory. *Oikos*, 496-506.

Yachi, S., & Loreau, M. (1999). Biodiversity and ecosystem productivity in a fluctuating environment: the insurance hypothesis. *Proceedings of the National Academy of Sciences, 96*, 1463-1468.

Zarfl, C., Lumsdon, A. E., Berlekamp, J., Tydecks, L., & Tockner, K. (2015). A global boom in hydropower dam construction. *Aquatic Sciences, 77*, 161-170.

# APPENDIX

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ORIGINAL ARTICLE

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# Cross-taxon congruence of multiple diversity facets of freshwater assemblages is determined by large-scale processes across China

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## Abstract

1. An intensively debated issue in ecology is whether variability in the patterns of diversity of different groups of organisms is congruent in space, a phenomenon referred to as cross-taxon congruence. Whereas this has been previously mainly tested in terms of taxonomic dissimilarity, the role of ecological processes in determining the congruence of multiple diversity facets (i.e. α- and β-diversity, taxonomic, and functional) remains poorly understood.
2. We used a data set of observation records for 469 macrophyte and 543 fish taxa at the catchment-scale from the existing literature and data bases to test the existence of multi-faceted congruence patterns and investigate the variables driving them across 214 catchments covering the whole Chinese mainland.
3. We found cross-taxon congruence of multiple diversity aspects between fish and macrophyte communities. The *energy* (i.e. diversity is limited by energy availability), *area/environmental heterogeneity* (i.e. diversity is higher in larger and more heterogeneous areas), and *dispersal* (i.e. diversity is driven by dispersal) hypotheses were all significantly attributed to the cross-taxon congruence, suggesting the existence of key repeated mechanisms underlying assemblage organisation.
4. Our study provides new evidence that can further our understanding of the factors and underlying processes explaining cross-taxon congruence patterns at broad spatial scales in the freshwater realm, the studies of which significantly lag those in the marine and terrestrial realms. The present findings also provide important baseline information for freshwater conservation initiatives.

**KEYWORDS**
cross-taxon congruence, dissimilarity, fishes, functional diversity, macrophytes

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## 1 | INTRODUCTION

Assessing the degree to which different groups of organisms show similar patterns of species diversity across space, i.e. cross-taxon congruence, is one of the most significant challenges facing ecologists, biogeographers, and conservation biologists (Özkan et al., 2014; Rooney & Azeria, 2015; Su, Debinski, Jakubauskas, & Kindscher, 2004; Tisseuil et al., 2013). Cross-taxon congruence is not only central to understanding the scale-dependent mechanisms driving spatial patterns of biodiversity (Wolters, Bengtsson, & Zaitsev, 2006), thus contributing to the improvement of general models for predicting assemblage patterns at multiple spatial scales (Qian & Kissling, 2010; Westgate, Barton, Lane, & Lindenmayer, 2014), but it is also of considerable conservation interest. The existence of spatial congruence in diversity patterns allows the possibility for using surrogate taxa when monitoring and managing less well-known taxonomic groups (Lewandowski, Noss, & Parsons, 2010). The existence of common patterns in the spatial arrangement of biodiversity is a recurrent feature in nature (Gaston, 2000; Tittensor et al., 2010), yet consensus regarding the generality of the underlying processes driving spatial variation in biodiversity remains elusive.

Three main mechanisms have been proposed to explain the existence of congruent diversity patterns among different groups of organisms. First, cross-taxon congruence may be generated by similar yet independent responses of taxa to environmental factors (Hawkins et al., 2003; Mittelbach et al., 2007; Willig, Kaufman, & Stevens, 2003), thus making spatial changes in assemblages of different taxa predictable from those in the environment (Gioria, Bacaro, & Feehan, 2011; Heino, 2010; Oertli, Müller, Steiner, Breitenstein, & Dorn, 2005). Second, congruence may be determined by a shared biogeographic history, whereby different organismal groups have exhibited similar evolutionary histories or dispersal routes (Gouveia, Hortal, Cassemiro, Rangel, & Diniz-Filho, 2013; Gove et al., 2013; Ricklefs & Schluter, 1993). Third, the diversity of one group of species may partly determine that of another group of species, as in the case of hosts and their parasites or prey and their specialist predators (Burkle & Alarcón, 2011; Qian & Kissling, 2010; Sandom et al., 2013).

Here, we focus on the environmental factors and dispersal history underlying cross-taxon congruence. This mechanism has gained previous support as an explanation for the diversity gradients observed in many groups of organisms at broad spatial scales (Field et al., 2009; Ricklefs, 2004; Whittaker, Willis, & Field, 2001). Specifically, the expectation of common, independent cross-taxa responses to environmental variation has been formulated via three non-exclusive hypotheses: the *energy*, *area/environmental heterogeneity*, and *dispersal* hypotheses. The energy hypothesis, a climatically based hypothesis, claims that species richness within an area is limited by energy availability (Clarke & Gaston, 2006). In general, species richness tends to be higher in areas where the energy supply is higher. In contrast, the heterogeneity hypothesis states that species richness is higher in larger and more heterogeneous areas through increasing habitat availability and diversity (Whittaker et al., 2001). Finally, the dispersal hypothesis, which includes many variants, proposes that diversity patterns are related to differential speciation or extinction rates, coupled with dispersal limitation related to historical contingency (Ricklefs, 2004, 2007). These three hypotheses have received increasing support in explaining cross-taxon congruence in multiple terrestrial and marine taxa, including mammals, birds, insects, and vascular plants (Cabra-García, Bermúdez-Rivas, Osorio, & Chacón, 2012; Miranda & Parrini, 2015; Oertli et al., 2005). Evidence on freshwater taxa is, however, comparatively much more scarce, and it is unclear to what extent the same mechanisms affect cross-taxon congruence in the freshwater realm, because knowledge from other systems is not directly transferable to freshwaters, especially at broad spatial scales (Heino, 2010, 2011).

Although patterns of taxonomic and functional diversities have been characterised for many biological groups (Gioria et al., 2011; Miranda & Parrini, 2015; Rooney & Azeria, 2015), we still lack a general understanding of how cross-taxon relationships vary among the multiple facets of species diversity. At broad spatial scales, much emphasis has been given to patterns of species richness and the ecological and evolutionary processes underlying the origin and maintenance of the associated cross-taxon congruence (Gioria et al., 2011; Hawkins et al., 2003; Qian & Kissling, 2010; Clement Tisseuil et al., 2013). However, biodiversity is a multi-faceted concept, and these facets are complementary and pertain not only to the species comprising local assemblages (α-diversity), but also to the variation in species among assemblages (β-diversity) and not only to their taxonomic but also to their functional traits and phylogenetic relatedness (Baselga, 2010; Dobrovolski, Melo, Cassemiro, & Diniz-Filho, 2012; Heino & Tolonen, 2017; Leprieur et al., 2011). Therefore, it is now widely accepted that taxonomic richness is only a partial descriptor of wholesale biodiversity, and a multi-faceted approach is needed to gain a robust understanding of the mechanisms structuring species assemblages in different ecosystems and geographic regions.

To fill these knowledge gaps, we examined the multi-faceted (i.e. taxonomic and functional α- and β-diversity) cross-taxon congruence of freshwater fishes and macrophytes across the entire catchment network in China. Given the important habitat that macrophytes provide for fish (e.g. refuge, food, breeding), we anticipated the existence of cross-taxon congruence between them (Cook, 1974; Zhang, García Molinos, Zhang, & Xu, 2018). We were particularly interested in testing for: (1) the existence of spatial correlation between cross-taxon patterns of taxonomic and functional richness (Cabra-García et al., 2012; Miranda & Parrini, 2015; Oertli et al., 2005) and of taxonomic and functional dissimilarity (Baselga, 2010; Villéger, Grenouillet, & Brosse, 2014); (2) whether existing cross-taxon congruence patterns for macrophyte and fish assemblages are driven by variations in environmental conditions through any of the described mechanisms (i.e. *energy*, *area/environmental heterogeneity*, and *dispersal* hypotheses; Field et al., 2009); and (3) whether they are consistently congruent for the different diversity facets.

To our knowledge, our study represents the first one testing cross-taxon congruence patterns of multiple facets of taxonomic and functional diversity against variations in environmental conditions at broad scales. This knowledge should contribute to a better understanding of the potential effects of global change, the identification of vulnerable ecosystem structures and functions, and to tests on the degree to which common ecological mechanisms underlie patterns of cross-taxon congruence (Petchey & Gaston, 2002; Qian & Kissling, 2010).

## 2 | MATERIAL AND METHODS

### 2.1 | Data acquisition and key definitions

We compiled location records on the native freshwater macrophyte and fish assemblages from within 214 catchments across China (National Remote Sensing Center of China, http://www.nrscc.gov.cn/nrscc/cyyfw/geosjgx2/gxfw/; Figure S1). These catchments have been delimited by the National Council of China as a part of the National Water Resources Strategic Plan, they represent subdivisions of main river basins based on different hydrologic and ecological criteria (e.g. river order, landscape, and climate), and they provide an important basis for the development, utilisation, conservation, and environmental management of hydrological resources in China.

The term *assemblage* refers here to the total number of species and their composition recorded from ponds, lakes, and rivers within a given catchment (i.e. our spatial unit). Pooling data sets into meaningful, large spatial working units such as catchments or ecoregions is a practical, compromise solution frequently used for the analysis of macroscale diversity patterns with spatially sparse data sets such as ours (Veech & Crist, 2007). The multiple factors influencing richness gradients are scale-dependent relative to both the spatial extent of the analysis and the grain size of the sample resolution (Willis & Whittaker, 2002). We focus on catchments defining subdivisions (mean area of 104 $km^2$) of main river basins across a vast territory (*c.* $3.7 \times 10^6$ $km^2$), encompassing a range of climatic, geographical and hydroecological conditions over which richness may vary, and seek to understand the forces driving macroecological diversity congruence across China. Catchments are also relatively independent entities separated by natural barriers (i.e. mountains), within which there is a high degree of connectivity between habitats and less variation in environmental parameters (Cai, Xu, Zhang, Wang, & Heino, 2019; Cai, Zhang, Xu, & Heino, 2018; Reyjol et al., 2007; Schleuter et al., 2012; Tisseuil et al., 2013). Therefore, they can be effectively used in a comparative analysis to explore the factors driving assemblage patterns amongst them.

Although there are different definitions of *macrophyte* in the literature, we followed that of Cook (1974) and considered a macrophyte as any plant visible to the naked eye "whose photosynthetically active parts are permanently or, at least, for several months each year submerged in freshwater or floating on the water surface". This definition includes all higher aquatic plants, vascular cryptograms and bryophytes, together with groups of algae that can be seen to be composed predominantly of a single species. Based on this definition, we first made a detailed literature review of aquatic plant species in China. The distribution of different aquatic plant species in China was determined from published (1960–2010) records, most of which are concerned with lakes and rivers and, occasionally, seasonal agricultural ponds. Documented sources included original research papers and monographs relevant to the distribution and ecology of aquatic plants or to their habitats and status, together with the scientific database of China plant species (http://db.kib.ac.cn/eflora/Default.aspx), the database of invasive alien species in China (http://www.chinaias.cn/wjPart/index.aspx), the Chinese species information system (http://monkey.ioz.ac.cn/bwg-cciced/english/cesis/csispage.htm), and grey research reports. This exhaustive literature review provided information on a total of 992 aquatic plant species. We then prepared a data matrix including the taxonomic information and functional traits for the species. To ensure a good quality of the data set, we used five quality-control rules. (1) We used the definition for macrophytes in Water Plants of the World (Cook, 1974) and the records in Flora of China to exclude non-macrophyte species. (2) Scientific names were standardised, and synonyms were removed according to the Chinese Virtual Herbarium (http://www.cvh.org.cn/cms/en/). (3) The varieties were treated as the same species. (4) The habitat traits of the species were corrected following the Flora of China. (5) Non-freshwater species were excluded. The application of these rules resulted in a total of 469 species being kept for analysis, including 93 submerged species, 40 floating-leaved species, 25 free-floating species and 311 emergent species.

The information on the distribution of fishes follows a similar exhaustive literature review as that of macrophytes, comprising the scientific and grey literature (1960–2010), as well as the Chinese species database and biodiversity information system. Of the total of 1,331 fish species initially documented from the literature, we retained 543 species after the revision using a similar set of quality-control rules. These included the following: (1) all records not determined to the species level or not geo-referenced to county and lower spatial levels were excluded; (2) scientific names were standardised according to the Fishbase (http://www.fishbase.org); (3) infraspecific taxa were merged to the species level; (4) multiple entries referring to the same specimen were removed; (5) records of species native to China but occurring outside their natural range were excluded from the database; and (6) catadromous and brackish water species were excluded from the database.

### 2.2 | Functional traits considered in this study

We gathered information on five categorical functional traits to calculate the functional distance between macrophyte species according to the Flora of China, including life form (i.e. submerged, floating-leaved, emergent, free-floating), life cycle (i.e. annual, perennial), morphology (i.e. turion, stem, roset, leafy), sexual propagation (monoecism, dioecy), and habitat preference (i.e. lake, agriculture, river, estuary; Zhang, García Molinos, Su, Zhang, & Xu, 2019; Zhang et al., 2018).

Fish species were functionally characterised using seven traits commonly used in studies on fish functional diversity (Matsuzaki, Sasaki, & Akasaka, 2013). Two of these traits were continuous numeric variables (maximum total body length and maturation), and the remainder (mouth position, morphology, dietary traits, diet breadth, and vertical position in the water column) were categorical traits (Table S1). All the functional traits were taken from the ichthyography (Chu & Chen, 1989, 1990) and Fishbase (Froese & Pauly, 2010).

## 2.3 | Environmental variables

We selected seven variables to quantify the environmental conditions in each catchment. These variables were used as proxies for testing the *energy*, *area/environmental heterogeneity*, and *dispersal* hypotheses (Clarke & Gaston, 2006; Currie et al., 2004; Ricklefs, 2004, 2007; Whittaker et al., 2001) in relation to the ecological processes responsible for cross-taxon congruence between macrophytes and fishes. These variables are often associated with regional- to global-scale diversity patterns (Clarke & Gaston, 2006; Currie et al., 2004; Ricklefs, 2004, 2007; Whittaker et al., 2001).

To test the energy hypothesis, we used the annual normalised difference vegetation index (NDVI), mean annual temperature (MAT), mean annual precipitation (MAP), solar radiation (SOLAR), and annual run-off within each catchment. NDVI is an index of terrestrial plant photosynthetic activity and is one of the most commonly used terrestrial vegetation indices (Crippen, 1990). Here, we used NDVI as an indicator of the processes of weathering, leaching, and input of terrestrial organic material into waterbodies, which can be a very important energy source of aquatic ecosystems (Correa & Winemiller, 2018; Griffith, Martinko, Whistler, & Price, 2002). These variables are used as surrogates for energy entering each catchment. Indeed, energy can regulate the biodiversity of producers and consumers by means of two very different processes: (1) resource availability (i.e. productive energy) and (2) the physiological limits of the species (i.e. ambient energy; Clarke & Gaston, 2006; Hawkins et al., 2003).

To test for the area/environmental heterogeneity hypothesis, we used the total surface area of each catchment (AREA), altitudinal range (ALTVAR), land cover heterogeneity (LANDVAR), and climate heterogeneity (i.e. spatial climatic ranges of MAT, MAP, and SOLAR), including MATVAR, MAPVAR, and SOLARVAR. LANDVAR is measured as the Shannon diversity index based on the proportions of land cover classes (forest, grass, farm, urban, water, and desert) within each catchment (Tisseuil et al., 2013). ALTVAR is a proxy for topographic heterogeneity, calculated as the range between the maximum and minimum altitude for each catchment (Astorga, Heino, Luoto, & Muotka, 2011). These variables are recognised as important factors shaping biodiversity through increasing habitat heterogeneity and availability, thus favouring speciation while reducing species extinction rates (Tisseuil et al., 2013; Whittaker et al., 2001).

To test for the *dispersal* hypothesis, we used spatial location, including longitude, latitude, and altitude. Biodiversity patterns are often associated with latitudinal (Hof, Brandle, & Brandl, 2008) and altitudinal (Davies et al., 2007; Rahbek, 1995) gradients. Geographical distance is also important in relation to the dispersal of species among catchments.

All data sets had a nominal horizontal resolution of 0.5 × 0.5° and were obtained from the Data Sharing Infrastructure of the Earth System Science, National Science & Technology Infrastructure Center (http://www.nstic.gov.cn/) and the Data Center of the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences. Environmental variables were calculated for each catchment as the average value of the variable for all 0.5° cells comprising the catchment (i.e. having their cell centroid within the catchment boundary).

## 2.4 | Measurement of different facets of diversity

Taxonomic and functional richness were calculated, respectively, as the number of species comprising an assemblage and the value of the convex hull volume filled by a community in multidimensional functional space defined by the species traits (Villéger, Mason, & Mouillot, 2008). Functional richness thus equals the total branch length needed to join all species in an assemblage and standardised to range between 0 (assemblages composed of one species) and 1 (Villéger et al., 2008). Functional richness aims to quantify resource use complementary, and it has been suggested that it generally performs better as a predictor of ecosystem functioning than alternative methods (Petchey & Gaston, 2002).

Taxonomic dissimilarity, defined as the proportion of species present in only one assemblage within a pair of assemblages, was calculated using the Jaccard index (Villéger et al., 2014). Similarly, functional dissimilarity was estimated as the non-overlap in functional space between assemblage pairs defined by the intersection of their convex hulls. Both taxonomic (Baselga, 2012) and functional (Villéger, Grenouillet, & Brosse, 2013) diversity were partitioned into their turnover and nestedness components to assess the relative contribution of the different processes leading to changes in assemblage composition, namely, species replacement (turnover) and changes in species numbers without replacement (nestedness). At both extremes of the gradient, a high dissimilarity between two assemblages can result from them sharing no species, despite having equal numbers (complete turnover without nestedness), or from the assemblages having very different numbers of species despite sharing them, i.e. the species-poor being a subset of, or nested in, the species-rich assemblage.

## 2.5 | Statistical analyses

Functional richness and dissimilarity were computed on the functional space made by the first three principal axes of the principal coordinates analysis conducted on the species functional distance matrix using Gower's distance (Villéger et al., 2008). The resulting three-dimensional functional space provided an accurate representation of the functional dissimilarity between species, explaining 65% and 71% of the total functional space for macrophytes and fish, respectively, thus achieving the necessary trade-off between information

quality and computation time (Maire, Grenouillet, Brosse, & Villéger, 2015; Villéger et al., 2013, 2008). Indices of taxonomic and functional richness, dissimilarity and contribution of turnover and nestedness to dissimilarity between each pair of fish and macrophyte assemblages were computed according to recently developed methodology (Baselga, 2010, 2012; Villéger et al., 2008, 2014).

We used Spearman's rank correlation to test for significant cross-taxon congruence in the form of taxonomic and functional richness correlations between fish and macrophyte assemblages across catchments. Similarly, cross-taxon correlations for taxonomic and functional total dissimilarity and its constituent nestedness and turnover components were tested using Mantel tests, a permutation technique that estimates the resemblance between two proximity matrices computed for the same objects (Mantel, 1967).

To test for relationships between changes in taxonomic and functional richness along environmental gradients (i.e. related to *energy*, *area/environmental heterogeneity*, and *dispersal* hypotheses), we first applied a principal component analysis (PCA) on the data matrix related to the suite of environmental covariates used as proxies for each of the hypotheses. This was performed to reduce the multidimensionality and to eliminate collinearity between predictors. We retained the first two PCA components as synthetic predictors (Figure S4), which accounted for at least 70% of the total variability in the original variables in all cases (Figure S4). We then used multiple linear regression with Gaussian errors, where the response variables were the taxonomic or functional richness of fish and macrophytes and the PCA components were the predictor variables. Predictor variables were logarithmic or square-root transformed as required to achieve normality, and they were standardised to a common scale between 0 and 1 before PCA.

To test how taxonomic and functional total dissimilarity and its turnover and nestedness components varied along environmental gradients, we used permuted multiple regression on distance matrices (Lichstein, 2007) with changes of environmental gradients (i.e. related to the *energy*, *area/environmental heterogeneity*, and *dispersal* hypotheses) as predictor Euclidean distance matrices. Predictor variables were also normalised and standardised before the analysis.

The *p*-values (0.05 significance level) for Mantel's permutation test and the multiple regression on distance matrices models were obtained by comparing each observed regression coefficient with a distribution of 10,000 permuted values. All statistical analyses (Table 1) and plotting were produced in R 3.1.0 (R Development Core Team, 2014) using the built-in functions and those in the packages reshape2 (Wickham, 2007), FD (Laliberté, Legendre, Shipley, & Laliberté, 2014), betapart (Baselga & Orme, 2012), lme4 (Bates, Maechler, & Bolker, 2012), ecodist (Goslee & Urban, 2007), vegan (Oksanen et al., 2014), sjPlot (Lüdecke, 2015) and ggplot2 (Wickham, 2009).

## 3 | RESULTS

Across catchments, macrophyte assemblages (200 ± 47 species, mean ± 1 standard deviation) were, on average, richer in species than fish assemblages (30 ± 51). Although the overall species pool was higher for fish (543 recorded species across China) than macrophytes (468), the representation of the regional species pool at the individual catchment level was, on average, much higher for macrophytes (46%) than for fish assemblages (10%; Figure 1a,e), reflecting a higher degree of differentiation among fish assemblages. This trend was also observed for functional richness (Figure 2a,e), with individual catchments containing, on average, 56% and 34% of the regional macrophyte and fish functional diversity, respectively. The higher representation of regional functional richness relative to taxonomic richness within individual catchments suggests some degree of functional redundancy among species in both groups.

**TABLE 1** Statistical methods used in this study

| Method | Purpose | Response variable | Predictor variable[a] |
|---|---|---|---|
| Spearman's rank correlation | Correlation between taxonomic and functional richness | Taxonomic and functional richness | Taxonomic and functional richness |
| Mantel's permutation test | Correlation between taxonomic and functional (total, turnover, nestedness) dissimilarity | Taxonomic and functional total dissimilarity and its constituent nestedness and turnover components | Taxonomic and functional total dissimilarity and its constituent nestedness and turnover components |
| Principal component analysis (PCA) | Reduction of the multidimensionality and elimination of collinearity between environmental predictors | – | – |
| Multiple linear regression with Gaussian errors | Effects of environment on richness | Taxonomic and functional richness | PCA components related to energy, heterogeneity and dispersal hypotheses |
| Multiple regression on distance matrices | Effects of environment on dissimilarity | Taxonomic and functional total dissimilarity and its constituent nestedness and turnover components | Euclidean distance matrices of PCA components related to energy, heterogeneity, and dispersal hypotheses |

[a]All predictors were logarithmic or square root transformed as required to achieve normality and standardised to a scale between 0 and 1 before PCA.

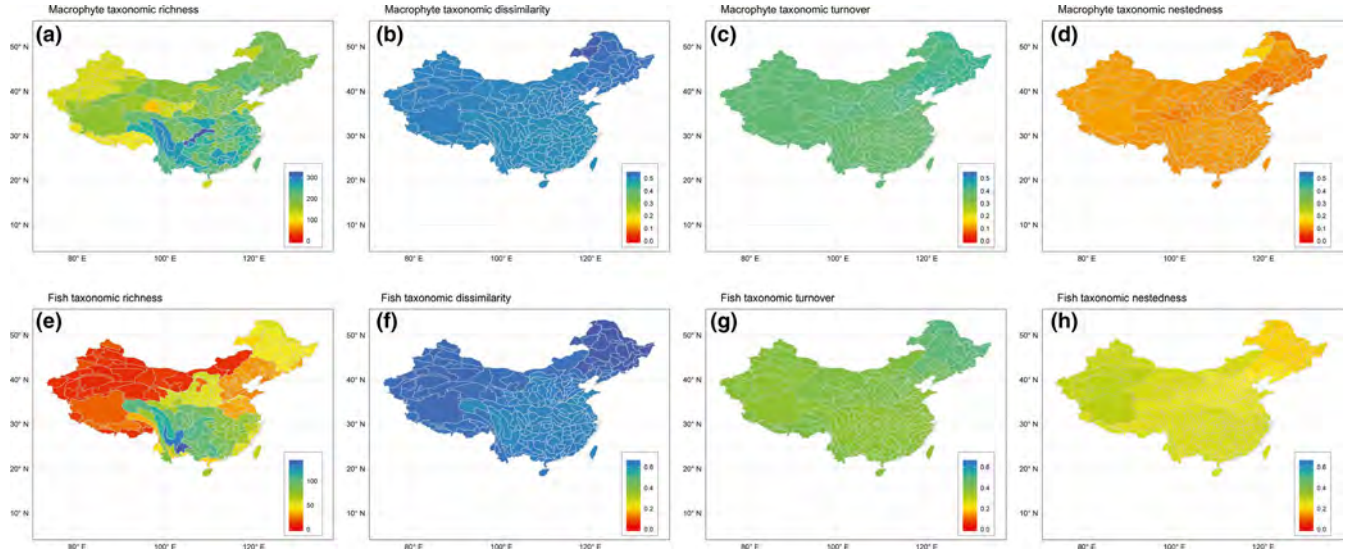



**FIGURE 1** Spatial patterns of taxonomic diversity across the 214 catchments. Shown are macrophyte (a–d) taxonomic richness, dissimilarity and its turnover and nestedness components, and fish (e–h) taxonomic richness, dissimilarity and its turnover and nestedness components. Index values for dissimilarity and its turnover and nestedness components correspond to the average of the pairwise comparisons between that assemblage and the other assemblages

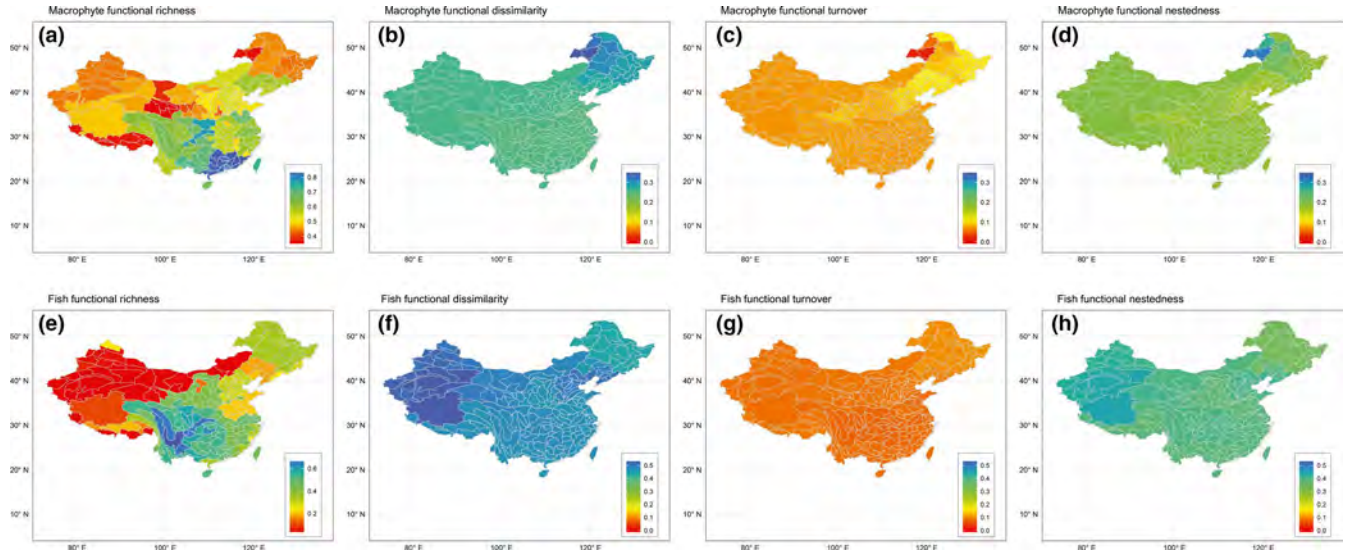



**FIGURE 2** Spatial patterns of functional diversity across the 214 catchments. Shown are macrophyte (a–d) functional richness, dissimilarity and its turnover and nestedness components, and fish (e–h) functional richness, dissimilarity and its turnover and nestedness components. Index values for dissimilarity and its turnover and nestedness components correspond to the average of the pairwise comparisons between that assemblage and the other assemblages

The total taxonomic dissimilarity among macrophyte (Figure 1b) and fish (Figure 1f) assemblages was relatively high (0.51 ± 0.01 and 0.65 ± 0.02, respectively), being mainly driven by the species turnover among assemblages, with a relative contribution towards a total dissimilarity of 0.4 ± 0.03 for both groups (Figure 1c,g), rather than by the isolated effect of species losses or gains (i.e. nestedness; 0.11 ± 0.01 and 0.25 ± 0.03; Figure 1d,h). In contrast, functional dissimilarity was lower for both groups (macrophyte: 0.26 ± 0.02, fish: 0.48 ± 0.02; Figure 2b,f) and had a much larger contribution from nestedness (0.19 ± 0.02 and 0.39 ± 0.02; Figure 2d,h) than turnover (0.09 ± 0.01 for both groups; Figure 2c,g). These common patterns between the two taxonomic groups are also reflected by the significant correlations found for all taxonomic and functional diversity facets (Figures S2 and S3). Spatially, the variability in taxonomic and functional richness was relatively high and followed the same patterns for both macrophytes and fish assemblages, with higher richness concentrated in catchments from central-southern China (Figures 1a,e and 2a,e). In contrast, taxonomic and functional

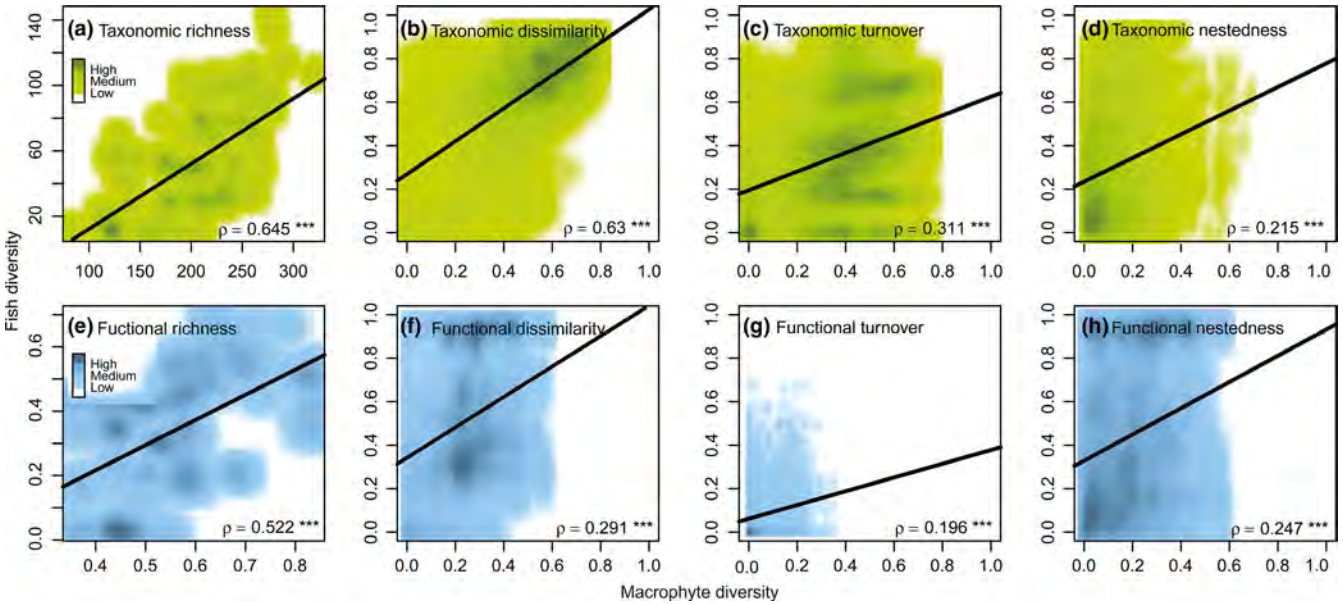



**FIGURE 3** Cross-taxon congruence tests for taxonomic and functional diversity facets between macrophyte and fish assemblages across the studied catchments. Panels show smoothed colour density representations of observed paired assemblage values for each facet: (a–d) taxonomic richness, taxonomic dissimilarity, and its turnover and nestedness components; (e–h) functional richness, functional dissimilarity and its turnover and nestedness components. Values displayed in each panel correspond to the (a, e) Spearman rank correlation coefficient and (b–d, f–h) Mantel statistic (***$p < 0.001$). Colour density is obtained through a two-dimensional kernel density estimate, and low, medium, and high density represent 25%, 50%, and 75% of maximum density. Black lines in the panels represent the linear regression trends

dissimilarity showed very little spatial variation among catchments (Figures 1b–d,f–h and 2b–d, f–h).

Cross-taxon congruence between fish and macrophyte assemblages across catchments was highly significant ($p < 0.001$) for all diversity facets, including taxonomic and functional richness (Spearman's correlation coefficient $\rho = 0.645$ and $\rho = 0.522$, respectively; Figure 3a,e); taxonomic total dissimilarity, turnover and nestedness components ($\rho = 0.63$, 0.311, and 0.215; Figure 3b–d); and functional total dissimilarity, turnover and nestedness components ($\rho = 0.291$, 0.196 and 0.247; Figure 3f–h). Multiple linear regression models based on the full set of predictors explained a significant proportion of the variability of the response variables around its mean (Figure 4). These models provided strong supporting evidence of a combined role of all three hypotheses (i.e. *energy*, *area/environmental heterogeneity*, and *dispersal*) in driving observed taxonomic and functional diversity patterns of fish and macrophyte communities. However, the mean effects of each individual predictor varied across diversity facets and taxonomic groups. Specifically, macrophyte taxonomic richness was higher in more-productive environments associated with higher precipitation and temperature regimes (first $PCA_{ENE}$ axis, Figure 4a,e and Figure S4a), although elevated run-off (second $PCA_{ENE}$ axis, Figure 4e and Figure S4a) appeared to have a significant negative effect on macrophyte functional richness. Similar positive effects of energy availability on taxonomic and functional richness were observed for fish communities, although, in contrast to macrophytes, increased annual run-off had a significantly positive effect on them. Richness patterns for both taxonomic groups also varied significantly along geographical gradients, where lower macrophyte and fish richness was found towards higher latitudes (second $PCA_{DIS}$ axis; Figure 4a,e and Figure S4c) as well as higher altitudes and lower longitudes (first $PCA_{DIS}$ axis). Interestingly, habitat heterogeneity in the form of higher altitudinal, solar and temperature variability (first $PCA_{HET}$ axis; Figure 4a,e and Figure S4b) was associated with significantly taxonomically and functionally richer macrophyte communities, an effect that was not observed for fish communities (Figure 4a,e).

As with richness, energy availability significantly promoted more taxonomically (Figure 4b,c) and functionally (Figure 4f–h) diversified fish and macrophyte communities among catchments. Increasing distance among catchments also tended to result in significantly more-dissimilar communities (i.e. higher β-diversity) for both taxonomic groups. These effects were consistent overall for the turnover and nestedness components of diversity. However, we found opposite significant effects of habitat heterogeneity on β diversity, where more-heterogeneous habitats promoted taxonomic and functional dissimilarity among catchment fish communities but homogenised macrophyte communities. The results for components showed habitat complexity significantly decreased fish and macrophyte turnover (Figure 4c,g) and increased nestedness (Figure 4d,h), although there was a negative effect on functional macrophyte nestedness.

## 4 | DISCUSSION

Our analysis adds novel insights into previous studies analysing patterns of cross-taxon congruence for three reasons. First, we considered for the first time cross-taxon congruence across six aspects of

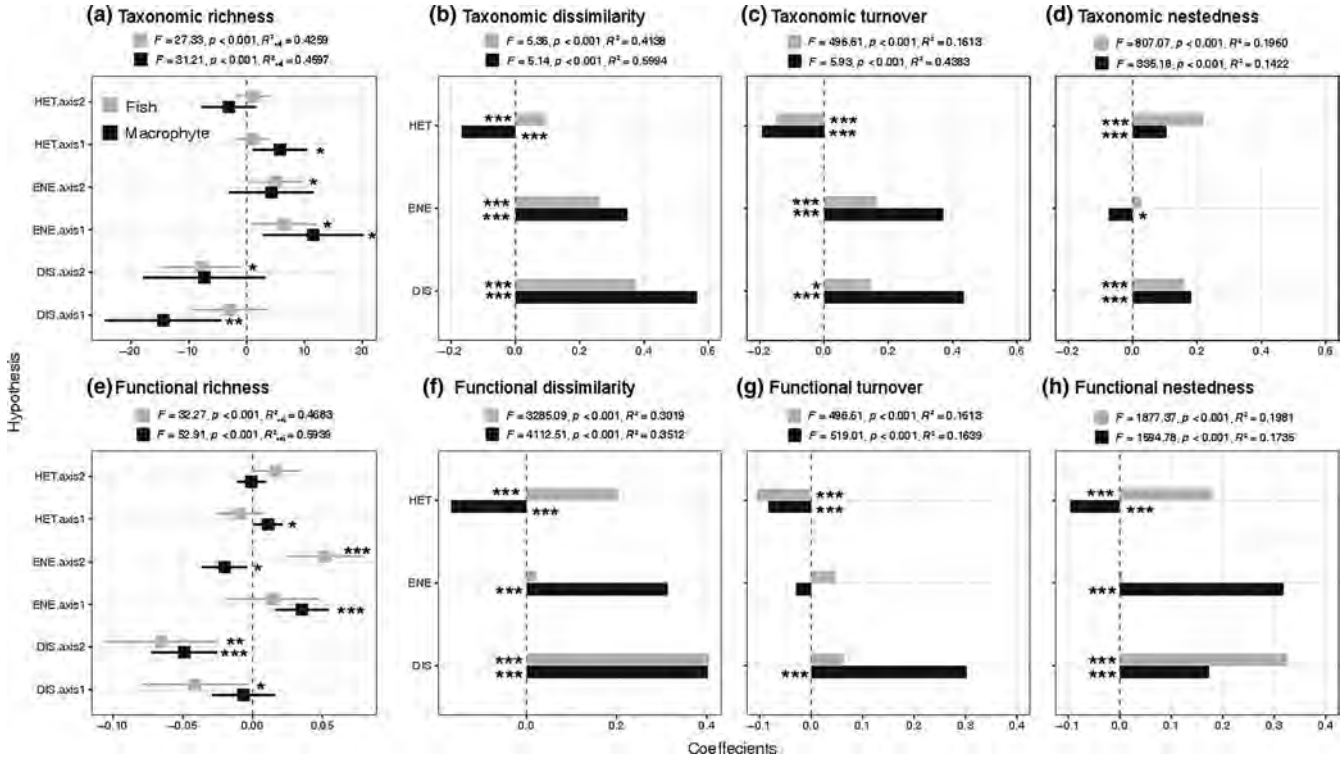



**FIGURE 4** Effects of the three alternative hypotheses on fish and macrophyte diversity patterns across catchments. The plots show the mean effects and associated confidence intervals associated with each hypothesis—*energy* (ENE), *area/environmental heterogeneity* (HET), and *dispersal* (DIS) hypotheses—resulting from multiple regression models on (a) taxonomic richness, (e) functional richness, and (b–d) taxonomic and (f–h) functional (b, f) total dissimilarity and its corresponding (c, g) turnover and (d, h) nestedness components (*$p$ < 0.05, **$p$ < 0.01 & ***$p$ < 0.001). Response variables for the richness model (a, e) correspond to the first two axes of a PCA on each group of environmental variables (see Section 2 and Figure S3). Response variables for the dissimilarity model (b–d, f–h) correspond to the average value of all pairwise comparisons between each assemblage with all its peers. See Section 2 for a detailed description of the models

diversity, including taxonomic and functional α- and β-diversity, along with their turnover and nestedness components. Second, whereas most of the literature on cross-taxon congruence focusses on local-to-regional scales (Lários et al., 2017), this is the first study examining cross-taxon congruence between fish and macrophyte communities at broad spatial scales encompassing all of China. Cross-taxon congruence can be driven by mechanisms operating at very different spatial scales such as differences in environmental conditions, biotic interactions, dispersal limitations and speciation (Wolters et al., 2006). Such scale-dependent effects have also implications for diversity-related ecosystem functionality (Pasari, Levi, Zavaleta, & Tilman, 2013). Consideration to multiple scales is therefore necessary for disentangling difference in the relative contribution of large and small-scale processes across taxa (Burrascano et al., 2018). Lastly, we used different environmental variables as proxies for testing different plausible driving mechanisms for the observed cross-taxon congruence patterns. Thus, our study helps researchers to understand the factors and underlying processes explaining cross-taxon congruence patterns at broad spatial scales in the freshwater realm, thus bridging the gap in the field with respect to marine and terrestrial studies (Collen et al., 2014; Leprieur et al., 2011; Tisseuil et al., 2013).

In general, the biogeography of the Chinese freshwater fish and macrophytes is determined to some extent by the legacy of the last glacial maximum. For example, the current distribution of the species richness of freshwater macrophytes and fishes follows a decreasing gradient from the Yunnan-Guizhou region in southwestern China, a former Pleistocene glacial refugia and centre of speciation, to more westerly regions (Li et al., 2011), a pattern identifiable from our data set (Figure 1a,e). Despite the small observed differences, the Yunnan–Guizhou Plateau is the region with the lowest contribution of taxonomic turnover to fish and macrophyte total taxonomic β-diversity (Figure 1c,g). This agrees with recent studies conducted on different animal groups stressing a tendency for macroecological β-diversity patterns to be dominated by nested species losses in areas affected by strong historical environmental disturbance (e.g. Pleistocene glaciations) and by species replacements in areas that experienced less-severe paleoclimatic changes (Baselga, Gómez-Rodríguez, & Lobo, 2012; Hortal et al., 2011; Leprieur et al., 2011). It is nonetheless interesting that the overwhelming contribution of macrophyte and fish species turnover to taxonomic dissimilarity generalised across all Chinese catchments, relative to that of isolated species extirpations or invasions (i.e. nestedness), was reversed for functional dissimilarity, for which nestedness had a much larger relative contribution (Figures 1 and 2). This suggests that the frequent species replacements occurring in both macrophyte and fish assemblages are mostly between species that are functionally

redundant, including species with common trait combinations. The generalised low level of functional turnover in both taxonomic groups and the relatively low functional dissimilarity might result from either functional convergence among these assemblages (Heino & Tolonen, 2017; Logez, Pont, & Ferreira, 2010) or allopatric speciation resulting from dispersal limitation (Baselga et al., 2012). The consistent significant cross-taxon congruence across all facets of diversity found between fishes and macrophytes (Figure 3) and their consistent overall patterns of response to variation in environmental conditions (Figure 4) strongly support the concept that under comparable environmental conditions, the structure of phylogenetically unrelated communities should vary similarly (Oertli et al., 2005; Qian & Kissling, 2010; Rooney & Azeria, 2015; Westgate et al., 2014). This hypothesis holds the deterministic view that assemblage structure can be predicted at least partly from the environment, to which organisms respond in a predictable way both in ecological and evolutionary terms. In our case, the observed significance of multifaceted cross-taxon congruence between fishes and macrophytes confirms the generality of assemblage patterns and the processes causing those patterns at broad spatial scales (Heino, 2010; Qian & Kissling, 2010; Su et al., 2004).

With some exceptions, the strong and consistent overall positive effects of energy availability and environmental heterogeneity on both fishes and macrophytes could be expected. Whereas higher energy levels should lead to higher biomass or abundance and, thereby, higher species richness (Currie, 1991; Currie et al., 2004), heterogeneous habitats and refuges should facilitate the occurrence of larger populations, higher speciation rates and lower extinction rates (Ricklefs, 2007; Ricklefs & Schluter, 1993; Tisseuil et al., 2013).

These effects were not without exceptions. For example, the fact that elevated run-off (ENE axis2) had a significant negative effect on macrophyte functional richness might be related to some types of macrophytes being absent from rivers and lakes that are prone to flooding or that are perhaps under irregular hydrological regimes (i.e. high variability in water levels; Friedman & Auble, 2000; Paillex, Doledec, Castella, & Merigoux, 2009). Similarly, highly heterogeneous landscapes tended to facilitate stronger selective extinction, the main mechanism influencing fish assemblages. This result in assemblages within catchments that is a subset of the regional pool and, hence, show increased nestedness but less turnover (Cutler, 1991; Wright & Reeves, 1992). In contrast, macrophytes showed a significant negative association of overall dissimilarity with more-heterogeneous environments across catchments. The opposite response of fishes and macrophytes to environmental heterogeneity may be explained by a more variable environment allowing more macrophyte species to persist (Figure 4a), increasing catchment species richness and causing lower compositional turnover through time. However, as much as 65% of the macrophytes comprising the regional species pool are cosmopolitan, with 25% of all species being Asian endemic species. Selective colonisation may hence play less of a role in macrophyte distributions at the national scale than it does for fish (Ricklefs, 2007; Ricklefs & Schluter, 1993; Tisseuil et al., 2013).

Dispersal limitation was the most consistent and strongest determinant of variation in taxonomic and functional total dissimilarity, including its turnover and nestedness components. Assemblages of strict freshwater species with low dispersal capacities via riverine corridors, such as fish, receive new colonists so rarely that immigration and speciation processes often occur on similar time scales and can be considered as specific to each catchment (Cai et al., 2018; Reyjol et al., 2007; Schleuter et al., 2012; Tisseuil et al., 2013). Macrophytes, despite having high dispersal capacities mediated by flows, birds, or winds, have been demonstrated to have important historical levels of taxonomic dissimilarity between assemblages at regional scales (Zhang et al., 2018). Therefore, our results, which hold for both fishes and macrophytes, despite major differences in their dispersal capacities and other biological traits, stress the importance of dispersal limitation in explaining cross-taxon congruence patterns (Baselga, 2010; Leprieur et al., 2011; Qian & Ricklefs, 2012) and reinforce recent evidence on the role of distance decay relationships in understanding diversity patterns in freshwater systems (Leprieur et al., 2011; Villéger et al., 2014). Furthermore, our findings also highlight the role of connectivity in highly spatially structured systems, such as hydrological networks, where long-distance dispersal may be easily disrupted, preventing freshwater taxa from tracking other environmental gradients (Heino et al., 2015; Tonkin et al., 2018), in contrast to many marine and terrestrial taxa (Dobrovolski et al., 2012).

Locally, aquatic plants can affect fish diversity by altering the abundance and composition of prey, by adjusting prey behaviour in refugia as a response to fish, by directly influencing predator–prey interactions (herbivory by fish and its prey), and by increasing the reproductive success of some fish species through providing attachment of eggs and protection for young fish from predation (Diehl & Kornijów, 1998; Maceina, Cichra, & Betsill, 1992). However, energy and environmentally-related factors still can play an important role in driving patterns of diversity at broad geographic scales (Hawkins et al., 2003). In this study, we found that broad-scale climatic and environmental factors, either directly or indirectly via productivity, together with dispersal-related geographic gradients were important in explaining existing macroecological congruence patterns in species richness and composition across China. A recent study on forest communities found that when compared with the effect of fine-scale determinants directly or indirectly related to forest structure (e.g. microclimatic conditions, occurrence of specific elements), the inclusion of a wider extent and, consequentially, broader ecological gradients, increased congruence for all pairs of taxa (Burrascano et al., 2018). Thus, the congruence we observed suggests that fish diversity may be predicted by aquatic plant diversity at broad scales, but the local ecological preferences of the species should also be assessed to preserve their habitats and ecological interactions between macrophytes and fish.

To conclude, our analysis strongly suggests that the processes associated with the *energy*, *area/environmental heterogeneity*, and *dispersal* hypotheses have all played important roles in determining the diversity patterns of fish and macrophyte assemblages across

the catchments examined in this study. Nonetheless, our results also provide strong evidence of the existence of consistent cross-taxon congruence patterns in multiple facets of diversity between fishes and macrophytes, driven by underlying variation in environmental drivers. Cross-taxon congruence studies can provide valuable information for predicting assemblage patterns across multiple taxonomic groups at broad spatial scales. Such studies should be of interest to ecologists, conservation biologists and environmental managers aiming to understand anthropogenic effects on species diversity and to prioritise large-scale conservation efforts.

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## REFERENCES

Astorga, A., Heino, J., Luoto, M., & Muotka, T. (2011). Freshwater biodiversity at regional extent: Determinants of macroinvertebrate taxonomic richness in headwater streams. *Ecography*, *34*, 705–713. https://doi.org/10.1111/j.1600-0587.2010.06427.x

Baselga, A. (2010). Partitioning the turnover and nestedness components of beta diversity. *Global Ecology and Biogeography*, *19*, 134–143. https://doi.org/10.1111/j.1466-8238.2009.00490.x

Baselga, A. (2012). The relationship between species replacement, dissimilarity derived from nestedness, and nestedness. *Global Ecology and Biogeography*, *21*, 1223–1232. https://doi.org/10.1111/j.1466-8238.2011.00756.x

Baselga, A., Gómez-Rodríguez, C., & Lobo, J. M. (2012). Historical legacies in world amphibian diversity revealed by the turnover and nestedness components of beta diversity. *PLoS ONE*, *7*, e32341. https://doi.org/10.1371/journal.pone.0032341

Baselga, A., & Orme, C. D. L. (2012). betapart: An R package for the study of beta diversity. *Methods in Ecology and Evolution*, *3*, 808–812. https://doi.org/10.1111/j.2041-210x.2012.00224.x

Bates, D., Maechler, M., & Bolker, B. (2012). lme4: Linear mixed-effects models using S4 classes.

Burkle, L. A., & Alarcón, R. (2011). The future of plant–pollinator diversity: Understanding interaction networks across time, space, and global change. *American Journal of Botany*, *98*, 528–538. https://doi.org/10.3732/ajb.1000391

Burrascano, S., de Andrade, R. B., Paillet, Y., Oder, P., Antonini, G., Bouget, C., … Blasi, C. (2018). Congruence across taxa and spatial scales: Are we asking too much of species data? *Global Ecology and Biogeography*, *27*, 980–990. https://doi.org/10.1111/geb.12766

Cabra-García, J., Bermúdez-Rivas, C., Osorio, A. M., & Chacón, P. (2012). Cross-taxon congruence of α and β diversity among five leaf litter arthropod groups in Colombia. *Biodiversity and Conservation*, *21*, 1493–1508. https://doi.org/10.1007/s10531-012-0259-5

Cai, Y., Xu, J., Zhang, M., Wang, J., & Heino, J. (2019). Different roles for geography, energy and environment in determining three facets of freshwater molluscan beta diversity across China. *Science of the Total Environment*, *659*, 451–462. https://doi.org/10.1016/j.scitotenv.2018.1012.1373

Cai, Y., Zhang, M., Xu, J., & Heino, J. (2018). Geographical gradients in the biodiversity of Chinese freshwater molluscs: Implications for conservation. *Diversity and Distributions*, *24*, 485–496. https://doi.org/10.1111/ddi.12695

Chu, X., & Chen, Y. (1989). *The fishes of Yunnan, China, Part I*. Beijing, China: Science Press.

Chu, X., & Chen, Y. (1990). *The fishes of Yunnan, China, Part II*. Beijing, China: Science Press.

Clarke, A., & Gaston, K. J. (2006). Climate, energy and diversity. *Proceedings of the Royal Society of London B: Biological Sciences*, *273*, 2257–2266. https://doi.org/10.1098/rspb.2006.3545

Collen, B., Whitton, F., Dyer, E. E., Baillie, J. E., Cumberlidge, N., Darwall, W. R., … Böhm, M. (2014). Global patterns of freshwater species diversity, threat and endemism. *Global Ecology and Biogeography*, *23*, 40–51. https://doi.org/10.1111/geb.12096

Cook, C. D. K. (1974). *Water plants of the world: A manual for the identification of the genera of freshwater macrophytes*. The Hague, the Netherlands: Dr. W. Junk Publishers.

Correa, S. B., & Winemiller, K. (2018). Terrestrial–aquatic trophic linkages support fish production in a tropical oligotrophic river. *Oecologia*, *186*, 1069–1078. https://doi.org/10.1007/s00442-018-4093-7

Crippen, R. E. (1990). Calculating the vegetation index faster. *Remote Sensing of Environment.*, *34*, 71–73. https://doi.org/10.1016/0034-4257(90)90085-z

Currie, D. J. (1991). Energy and large-scale patterns of animal-and plant-species richness. *The American Naturalist*, *137*, 27–49. https://doi.org/10.1086/285144

Currie, D. J., Mittelbach, G. G., Cornell, H. V., Field, R., Guegan, J.-F., Hawkins, B. A., … Turner, J. R. G. (2004). Predictions and tests of climate-based hypotheses of broad-scale variation in taxonomic richness. *Ecology Letters*, *7*, 1121–1134. https://doi.org/10.1086/285144

Cutler, A. (1991). Nested faunas and extinction in fragmented habitats. *Conservation Biology*, *5*, 496–504. https://doi.org/10.1111/j.1523-1739.1991.tb00357.x

Davies, R. G., Orme, C. D. L., Storch, D., Olson, V. A., Thomas, G. H., Ross, S. G., … Owens, I. P. (2007). Topography, energy and the global distribution of bird species richness. *Proceedings of the Royal Society B: Biological Sciences*, *274*, 1189–1197. https://doi.org/10.1098/rspb.2006.0061

Diehl, S., & Kornijów, R. (1998). *Influence of submerged macrophytes on trophic interactions among fish and macroinvertebrates*. New York, NY: Springer. https://doi.org/10.1007/978-1-4612-0695-8

Dobrovolski, R., Melo, A. S., Cassemiro, F. A., & Diniz-Filho, J. A. F. (2012). Climatic history and dispersal ability explain the relative importance of turnover and nestedness components of beta

diversity. *Global Ecology and Biogeography*, *21*, 191–197. https://doi.org/10.1111/j.1466-8238.2011.00671.x

Field, R., Hawkins, B. A., Cornell, H. V., Currie, D. J., Diniz-Filho, J. A. F., Guégan, J.-F., & Turner, J. R. G. (2009). Spatial species-richness gradients across scales: A meta-analysis. *Journal of Biogeography*, *36*, 132–147. https://doi.org/10.1111/j.1365-2699.2008.01963.x

Friedman, J. M., & Auble, G. T. (2000). Floods, flood control, and bottomland vegetation. In E. E. Wohl (Ed.), *Inland flood hazards: Human, riparian and aquatic communities* (pp. 219–237). Cambridge, UK: Cambridge University Press. https://doi.org/10.1017/cbo9780511529412

Froese, R., & Pauly, D. (2010). FishBase: International Center for Living Aquatic Resources Management.

Gaston, K. J. (2000). Global patterns in biodiversity. *Nature*, *405*, 220–227. https://doi.org/10.1038/35012228

Gioria, M., Bacaro, G., & Feehan, J. (2011). Evaluating and interpreting cross-taxon congruence: Potential pitfalls and solutions. *Acta Oecologica*, *37*, 187–194. https://doi.org/10.1016/j.actao.2011.02.001

Goslee, S. C., & Urban, D. L. (2007). The ecodist package for dissimilarity-based analysis of ecological data. *Journal of Statistical Software*, *22*, 1–19. https://doi.org/10.18637/jss.v022.i07

Gouveia, S. F., Hortal, J., Cassemiro, F. A., Rangel, T. F., & Diniz-Filho, J. A. F. (2013). Nonstationary effects of productivity, seasonality, and historical climate changes on global amphibian diversity. *Ecography*, *36*, 104–113. https://doi.org/10.1111/j.1600-0587.2012.07553.x

Gove, A. D., Dunn, R. R., Fitzpatrick, M. C., Sanders, N. J., Normand, S., Svenning, J.-C., & Ferrier, S. (2013). Environmental and historical imprints on beta diversity. *Proceedings of the Royal Society B: Biological Sciences*, *280*, 1–8. https://doi.org/10.1098/rspb.2013.1201

Griffith, J. A., Martinko, E. A., Whistler, J. L., & Price, K. P. (2002). Interrelationships among landscapes, NDVI, and stream water quality in the U.S. central plains. *Ecological Applications*., *12*, 1702–1718. https://doi.org/10.1890/1051-0761(2002) 012[1702:IALNAS]2.0.CO;2

Hawkins, B. A., Field, R., Cornell, H. V., Currie, D. J., Guégan, J.-F., Kaufman, D. M., … O'Brien, E. M. (2003). Energy, water, and broad-scale geographic patterns of species richness. *Ecology*, *84*, 3105–3117. https://doi.org/10.1098/rspb.2013.1201

Heino, J. (2010). Are indicator groups and cross-taxon congruence useful for predicting biodiversity in aquatic ecosystems? *Ecological Indicators*, *10*, 112–117. https://doi.org/10.1016/j.ecolind.2009.04.013

Heino, J. (2011). A macroecological perspective of diversity patterns in the freshwater realm. *Freshwater Biology*, *56*, 1703–1722. https://doi.org/10.1111/j.1365-2427.2011.02610.x

Heino, J., Melo, A. S., Siqueira, T., Soininen, J., Valanko, S., & Bini, L. M. (2015). Metacommunity organisation, spatial extent and dispersal in aquatic systems: Patterns, processes and prospects. *Freshwater Biology*, *60*, 845–869. https://doi.org/10.1111/fwb.12533

Heino, J., & Tolonen, K. T. (2017). Ecological drivers of multiple facets of beta diversity in a lentic macroinvertebrate metacommunity. *Limnology and Oceanography*, *62*, 2431–2444. https://doi.org/10.1002/lno.10577

Hof, C., Brandle, M., & Brandl, R. (2008). Latitudinal variation of diversity in European freshwater animals is not concordant across habitat types. *Global Ecology and Biogeography*, *17*, 539–546. https://doi.org/10.1111/j.1466-8238.2008.00394.x

Hortal, J., Dinizfilho, J. A. F., Bini, L. M., Rodriguez, M. A., Baselga, A., Noguesbravo, D., … Lobo, J. M. (2011). Ice age climate, evolutionary constraints and diversity patterns of European dung beetles. *Ecology Letters*, *14*, 741–748. https://doi.org/10.1111/j.1461-0248.2011.01634.x

Laliberté, E., Legendre, P., Shipley, B., & Laliberté, M. E. (2014). *Package 'FD'. Measuring functional diversity from multiple traits, and other tools for functional ecology*. https://cran.r-project.org/web/packages/FD/

Lários, M. C., da Cunha, C. N., Penha, J., Landeiro, V. L., Pinho, J. B., Aragona, M., … Chupel, T. F. (2017). Evidence of cross-taxon congruence in Neotropical wetlands: Importance of environmental and spatial factors. *Global Ecology and Conservation*, *12*, 108–118. https://doi.org/10.1016/j.gecco.2017.09.003

Leprieur, F., Tedesco, P. A., Hugueny, B., Beauchard, O., Dürr, H. H., Brosse, S., & Oberdorff, T. (2011). Partitioning global patterns of freshwater fish beta diversity reveals contrasting signatures of past climate changes. *Ecology Letters*, *14*, 325–334. https://doi.org/10.1111/j.1461-0248.2011.01589.x

Lewandowski, A. S., Noss, R. F., & Parsons, D. R. (2010). The effectiveness of surrogate taxa for the representation of biodiversity. *Conservation Biology*, *24*, 1367–1377. https://doi.org/10.1111/j.1523-1739.2010.01513.x

Li, Y., Zhai, S.-N., Qiu, Y.-X., Guo, Y.-P., Ge, X.-J., & Comes, H. P. (2011). Glacial survival east and west of the 'Mekong–Salween Divide' in the Himalaya-Hengduan Mountains region as revealed by AFLPs and cpDNA sequence variation in *Sinopodophyllum hexandrum* (Berberidaceae). *Molecular Phylogenetics and Evolution*, *59*, 412–424. https://doi.org/10.1016/j.ympev.2011.01.009

Lichstein, J. W. (2007). Multiple regression on distance matrices: A multivariate spatial analysis tool. *Plant Ecology*, *188*, 117–131. https://doi.org/10.1007/s11258-006-9126-3

Logez, M., Pont, D., & Ferreira, M. T. (2010). Do Iberian and European fish faunas exhibit convergent functional structure along environmental gradients? *Journal of the North American Benthological Society*, *29*, 1310–1323. https://doi.org/10.1899/09-125.1

Lüdecke, D. (2015). sjPlot: Data Visualization for Statistics in Social Science. Retrieved from http://CRAN.R-project.org/package=sjPlot

Maceina, M. J., Cichra, M. F., & Betsill, R. K. (1992). Limnological changes in a large reservoir following vegetation removal by grass carp. *Journal of Freshwater Ecology*, *7*(1), 81–95. https://doi.org/10.1080/02705060.1992.9664673

Maire, E., Grenouillet, G., Brosse, S., & Villéger, S. (2015). How many dimensions are needed to accurately assess functional diversity? A pragmatic approach for assessing the quality of functional spaces. *Global Ecology and Biogeography*, *24*(6), 728–740. https://doi.org/10.1111/geb.12299

Mantel, N. (1967). The detection of disease clustering and a generalized regression approach. *Cancer Research*, *27*, 209–220.

Matsuzaki, S. i. S., Sasaki, T., & Akasaka, M. (2013). Consequences of the introduction of exotic and translocated species and future extirpations on the functional diversity of freshwater fish assemblages. *Global Ecology and Biogeography*, *22*, 1071–1082. https://doi.org/10.1111/geb.12067

Miranda, M., & Parrini, F. (2015). Congruence between species phylogenetic and trophic distinctiveness. *Biodiversity and Conservation*, *24*, 355–369. https://doi.org/10.1007/s10531-014-0813-4

Mittelbach, G. G., Schemske, D. W., Cornell, H. V., Allen, A. P., Brown, J. M., Bush, M. B., … Lessios, H. A. (2007). Evolution and the latitudinal diversity gradient: Speciation, extinction and biogeography. *Ecology Letters*, *10*, 315–331. https://doi.org/10.1111/j.1461-0248.2007.01020.x

Oertli, S., Müller, A., Steiner, D., Breitenstein, A., & Dorn, S. (2005). Cross-taxon congruence of species diversity and community similarity among three insect taxa in a mosaic landscape. *Biological Conservation*, *126*, 195–205. https://doi.org/10.1016/j.biocon.2005.05.014

Oksanen, J., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'hara, R., … Wagner, H. (2014). vegan: Community Ecology Package. R package version 2.4-1. Retrieved from https://cran.r-project.org/web/packages/vegan/index.html

Özkan, K., Jeppesen, E., Davidson, T. A., Søndergaard, M., Lauridsen, T. L., Bjerring, R., & Svenning, J.-C. (2014). Cross-taxon congruence in lake plankton largely independent of environmental gradients. *Ecology*, *95*, 2778–2788. https://doi.org/10.1890/13-2141.1

Paillex, A., Doledec, S., Castella, E., & Merigoux, S. (2009). Large river floodplain restoration: Predicting species richness and trait responses to the restoration of hydrological connectivity. *Journal of Applied Ecology*, *46*, 250–258. https://doi.org/10.1111/j.1365-2664.2008.01593.x

Pasari, J. R., Levi, T., Zavaleta, E. S., & Tilman, D. (2013). Several scales of biodiversity affect ecosystem multifunctionality. *Proceedings of the National Academy of Sciences of the United States of America*, *110*, 10219–10222. https://doi.org/10.1073/pnas.1220333110

Petchey, O. L., & Gaston, K. J. (2002). Functional diversity (FD), species richness and community composition. *Ecology Letters*, *5*, 402–411. https://doi.org/10.1046/j.1461-0248.2002.00339.x

Qian, H., & Kissling, W. D. (2010). Spatial scale and cross-taxon congruence of terrestrial vertebrate and vascular plant species richness in China. *Ecology*, *91*, 1172–1183. https://doi.org/10.1890/09-0620.1

Qian, H., & Ricklefs, R. E. (2012). Disentangling the effects of geographic distance and environmental dissimilarity on global patterns of species turnover. *Global Ecology and Biogeography*, *21*, 341–351. https://doi.org/10.1111/j.1466-8238.2011.00672.x

R Development Core Team. (2014). R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing. Retrieved from http://www.R-project.org/

Rahbek, C. (1995). The elevational gradient of species richness: A uniform pattern? *Ecography*, *18*, 200–205. https://doi.org/10.1111/j.1600-0587.1995.tb00341.x

Reyjol, Y., Hugueny, B., Pont, D., Bianco, P. G., Beier, U., Caiola, N., … Ferreira, T. (2007). Patterns in species richness and endemism of European freshwater fish. *Global Ecology and Biogeography*, *16*, 65–75. https://doi.org/10.1111/j.1466-8238.2006.00264.x

Ricklefs, R. E. (2004). A comprehensive framework for global patterns in biodiversity. *Ecology Letters*, *7*, 1–15. https://doi.org/10.1046/j.1461-0248.2003.00554.x

Ricklefs, R. E. (2007). History and diversity: Explorations at the intersection of ecology and evolution. *The American Naturalist*, *170*, S56–S70. https://doi.org/10.1086/519402

Ricklefs, R. E., & Schluter, D. (1993). *Species diversity in ecological communities: Historical and geographical perspectives*. Chicago, IL: University of Chicago Press.

Rooney, R. C., & Azeria, E. T. (2015). The strength of cross-taxon congruence in species composition varies with the size of regional species pools and the intensity of human disturbance. *Journal of Biogeography*, *42*, 439–451. https://doi.org/10.1111/jbi.12400

Sandom, C., Dalby, L., Fløjgaard, C., Kissling, W. D., Lenoir, J., Sandel, B., … Svenning, J.-C. (2013). Mammal predator and prey species richness are strongly linked at macroscales. *Ecology*, *94*, 1112–1122. https://doi.org/10.1890/12-1342.1

Schleuter, D., Daufresne, M., Veslot, J., Mason, N. W., Lanoiselée, C., Brosse, S., … Argillier, C. (2012). Geographic isolation and climate Govern the functional diversity of native fish communities in European drainage basins. *Global Ecology and Biogeography*, *21*, 1083–1095. https://doi.org/10.1111/j.1466-8238.2012.00763.x

Su, J. C., Debinski, D. M., Jakubauskas, M. E., & Kindscher, K. (2004). Beyond species richness: Community similarity as a measure of cross-taxon congruence for coarse-filter conservation. *Conservation Biology*, *18*, 167–173. https://doi.org/10.1111/j.1523-1739.2004.00337.x

Tisseuil, C., Cornu, J. F., Beauchard, O., Brosse, S., Darwall, W., Holland, R., … Oberdorff, T. (2013). Global diversity patterns and cross-taxa convergence in freshwater systems. *Journal of Animal Ecology*, *82*, 365–376. https://doi.org/10.1111/1365-2656.12018

Tittensor, D. P., Mora, C., Jetz, W., Lotze, H. K., Ricard, D., Berghe, E. V., & Worm, B. (2010). Global patterns and predictors of marine biodiversity across taxa. *Nature*, *466*, 1098–1101. https://doi.org/10.1038/nature09329

Tonkin, J. D., Altermatt, F., Finn, D., Heino, J., Olden, J. D., Pauls, S. U., & Lytle, D. A. (2018). The role of dispersal in river network metacommunities: Patterns, processes, and pathways. *Freshwater Biology*, *63*, 141–163. https://doi.org/10.1111/fwb.13037

Veech, J. A., & Crist, T. O. (2007). Habitat and climate heterogeneity maintain beta-diversity of birds among landscapes within ecoregions. *Global Ecology and Biogeography*, *16*, 650–656. https://doi.org/10.1111/j.1466-8238.2007.00315.x

Villéger, S., Grenouillet, G., & Brosse, S. (2013). Decomposing functional β-diversity reveals that low functional β-diversity is driven by low functional turnover in European fish assemblages. *Global Ecology and Biogeography*, *22*, 671–681. https://doi.org/10.1111/geb.12021

Villéger, S., Grenouillet, G., & Brosse, S. (2014). Functional homogenization exceeds taxonomic homogenization among European fish assemblages. *Global Ecology and Biogeography*, *23*, 1450–1460. https://doi.org/10.1111/geb.12226

Villéger, S., Mason, N. W., & Mouillot, D. (2008). New multidimensional functional diversity indices for a multifaceted framework in functional ecology. *Ecology*, *89*, 2290–2301. https://doi.org/10.1890/07-1206.1

Westgate, M. J., Barton, P. S., Lane, P. W., & Lindenmayer, D. B. (2014). Global meta-analysis reveals low consistency of biodiversity congruence relationships. *Nature Communications*, *5*, 3899. https://doi.org/10.1038/ncomms4899

Whittaker, R. J., Willis, K. J., & Field, R. (2001). Scale and species richness: Towards a general, hierarchical theory of species diversity. *Journal of Biogeography*, *28*, 453–470. https://doi.org/10.1046/j.1365-2699.2001.00563.x

Wickham, H. (2007). Reshaping data with the reshape package. *Journal of Statistical Software*, *21*, 1–20. https://doi.org/10.18637/jss.v021.i12

Wickham, H. (2009). *ggplot2: Elegant graphics for data analysis*. New York, NY: Springer. https://doi.org/10.1007/978-0-387-98141-3

Willig, M. R., Kaufman, D. M., & Stevens, R. D. (2003). Latitudinal gradients of biodiversity: Pattern, process, scale, and synthesis. *Annual Review of Ecology, Evolution, and Systematics*, *34*, 273–309. https://doi.org/10.1146/annurev.ecolsys.34.012103.144032

Willis, K. J., & Whittaker, R. J. (2002). Species diversity—Scale matters. *Science*, *295*, 1245–1247. https://doi.org/10.1126/science.1067335

Wolters, V., Bengtsson, J., & Zaitsev, A. S. (2006). Relationship among the species richness of different taxa. *Ecology*, *87*, 1886–1895. https://doi.org/10.1890/0012-9658(2006) 87[1886:RATSRO]2.0.CO;2

Wright, D. H., & Reeves, J. H. (1992). On the meaning and measurement of nestedness of species assemblages. *Oecologia*, *92*, 416–428. https://doi.org/10.1007/bf00317469

Zhang, M., García Molinos, J., Su, G., Zhang, H., & Xu, J. (2019). Spatially structured environmental variation plays a prominent role on the biodiversity of freshwater macrophytes across China. *Frontiers in Plant Science*, *10*, 161. https://doi.org/10.3389/fpls.2019.00161

Zhang, M., García Molinos, J., Zhang, X., & Xu, J. (2018). Functional and taxonomic differentiation of macrophyte assemblages across the Yangtze River floodplain under human impacts. *Frontier in Plant Science*, *9*, 387. https://doi.org/10.3389/fpls.2018.00387

## SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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# Spatially Structured Environmental Variation Plays a Prominent Role on the Biodiversity of Freshwater Macrophytes Across China

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Different non-mutually exclusive mechanisms interactively shape large-scale diversity patterns. However, our understanding of multi-faceted diversity and their determinants in aquatic ecosystems is far from complete compared to terrestrial ones. Here, we use variation partitioning based on redundancy analysis to analyze the relative contribution of environmental and spatial variables to the patterns of phylogenetic, taxonomic, and functional diversity in macrophyte assemblages across 214 Chinese watersheds. We found extremely high spatial congruence among most aspects of biodiversity, with some important exceptions. We then used variation partitioning to estimate the proportions of variation in macrophyte biodiversity explained by environmental and spatial variables. All diversity facets were optimally explained by spatially structured environmental variables, not the pure environment effect, implying that macrophyte are taxonomically, phylogenetically, and functionally clustered in space, which might be the result of the interaction of environmental and/or evolutionary drives. We demonstrate that macrophytes might face extensive dispersal limitations across watersheds such as topography and habitat fragmentation and availability.

**Keywords: freshwater macrophyte, functional diversity, spatial congruence, species richness, taxonomic distinctness, phylogenetic diversity**

## INTRODUCTION

Macrophytes support many structural and functional aspects of freshwater ecosystems and their ecological properties have been extensively studied over (Westlake, 1975; Jeppesen et al., 1998; Wiegleb et al., 2015). Highly productive, macrophyte communities have a key role in carbon and nutrient fluxes, serving as sinks for organic material and sources of nutrients to the water (Jeppesen et al., 1998; Chambers et al., 2008; Hansson et al., 2010). They provide structurally complex habitat for a large diversity of organisms such as macroinvertebrate, fish and birds (Westlake, 1975; Jeppesen et al., 1998; Hansson et al., 2010). Consequently, a significant change in the structure

and composition of macrophyte communities can have important knock on effects on freshwater ecosystems with important management conservation implications (Hellsten and Riihimäki, 1996; Hansson et al., 2010; Wiegleb et al., 2016; Zhang et al., 2019). The development of local macrophyte assemblages strongly depends on a variety of abiotic and biotic factors, e.g., nutrient concentrations, flow velocity, light condition, and trophic interaction (Westlake, 1975; Dodkins et al., 2005; Zhang et al., 2019). However, consensus regarding the generality of large-scale processes driving spatial variation in biodiversity of macrophytes remains elusive (Chappuis et al., 2014; Wiegleb et al., 2015).

Most macrophyte species are regarded as cosmopolitan, their broad distribution traditionally explained by common life history traits such as long-distance hydrochory, anemochory, and zoochory seed dispersal (Santamaria, 2002; Chambers et al., 2008). Such strong dispersal capacity has facilitated the evolution of their plastic response and ecological tolerances to local environmental change. For example, many macrophytes are very resilient because of their fast asexual reproduction abilities such as clonal growth and abundant propagules (Chambers et al., 2008). Consequently, under natural conditions, regional-scale taxonomic richness is generally high and relatively stable across space (Alahuhta et al., 2018). However, high environmental heterogeneity may promote species selection (environmental filtering) and persistence (niche diversification), increasing the spatial turnover of species at landscape scales (Xu et al., 2015; Alahuhta et al., 2018). Therefore, regional spatial diversity patterns in aquatic taxa may be governed by the interplay between their dispersal capacity and the spatiotemporal heterogeneity (Shmida and Wilson, 1985; Shurin, 2007; Gianuca et al., 2017; Cai et al., in press).

Three aspects of species diversity patterns (i.e., taxonomic, phylogenetic, and functional diversity) are the main focus of macroecological studies. Taxonomic diversity, the most straightforward and commonly used measurement of biodiversity in broad-scale studies (Tisseuil et al., 2013; Heino et al., 2015; Cai et al., 2018), treats all species as functionally equivalent and phylogenetically independent. However, phylogenetic diversity and functional trait variation can exert a much stronger control on biodiversity effects on ecosystem functions, such as production or nutrient cycling, than taxonomic diversity (Cadotte et al., 2011; Flynn et al., 2011). Phylogenetic diversity, incorporating evolutionary relationships between species, provides also promising way of interpreting the role of biogeographic history in community ecology (Webb et al., 2002). Therefore, studies considering all these complementary facets can provide a more complete understanding of the mechanistic links between ecological processes and evolutionary history in shaping biodiversity patterns and the provision of ecosystem services (Devictor et al., 2010; Tucker and Cadotte, 2013; Pardo et al., 2017). Here, we assess the relative contribution of spatial and environmental variables to multifaceted biodiversity patterns in macrophyte assemblages across 214 tributary drainage basins (hereafter watersheds) covering the whole China. While the existence of spatial congruence among patterns in diversity facets is an open debate, our previous work in the Yangtze River has suggested high spatial congruence of macrophyte taxonomic and functional diversity at the catchment level (Zhang et al., 2018). Whereas these patterns hold at larger, regional scales across catchments remains to be tested. On the other hand, environmental variables are themselves spatially structured (Legendre and Legendre, 2012), making it possible for spatially structured environmental variation to drive the observed variability in spatial patterns within and among diversity facets.

# MATERIALS AND METHODS

## Data Acquisition and Key Definitions

Data on freshwater macrophyte species compiled from published sources (see below) was grouped at the watershed-scale (214 watersheds) across China (National Remote Sensing Center of China, as delimited by the National Council of China under the National Water Resources Strategic Plan[1]. These watersheds represent subdivisions of main river basins based on different ecohydrological criteria (e.g., river order, landscape, climate), and provide a spatial basis for the development, utilization, conservation, and management of hydrological resources in China (Cai et al., 2018). Pooling data sets into meaningful, large spatial working units such as watersheds (Pool et al., 2014) or ecoregions (Veech and Crist, 2007) is a practical, compromise solution frequently used for the analysis of macroscale diversity patterns with spatially sparse data sets (Cai et al., in press).

Although different definitions of "macrophyte" are available in the literature, we follow Cook (1974) by considering any aquatic plant that is visible to the naked eye including all higher aquatic plants, vascular cryptogams, bryophytes, and groups of algae that can be seen to be dominated by a single species. Based on this definition, we made a detailed literature review of macrophyte species in China from published (1960–2010) records related to lakes, rivers, and seasonal agricultural ponds. Documented sources included research articles and monographs together with the Scientific Database of China Plant Species[2], the Database of Invasive Alien species in China[3], Chinese Species Information System[4], and gray research reports. This exhaustive literature review provided information for a total of 992 aquatic plant species. We then prepared a data matrix covering taxonomic information and functional traits of the species. To guarantee consistency across the data set we used five quality-control rules: (1) non-macrophyte species were filtered according to the Cook's definition for macrophytes (Cook, 1974) and the records in Flora of China, (2) scientific names were standardized and synonyms were removed on the basis of the Chinese Virtual Herbarium[4], (3) varieties were treated as a single species, (4) the distribution traits of the species were corrected according to the Flora of China, and (5) non-freshwater species were excluded. The

[1]http://www.nrscc.gov.cn/nrscc/

[2]http://db.kib.ac.cn/eflora/Default.aspx

[3]http://www.chinaias.cn/wjPart/index.aspx

[4]http://db.kib.ac.cn/

application of these rules resulted in a total of 469 species from 214 watersheds retained for analysis, including 93 submerged species, 40 floating-leaved species, 25 free-floating species, and 311 emergent species.

## Measurements of Multiple Facets of Biodiversity

We defined taxonomic richness (TRic) as the number of species recorded in each watershed over the study period. Given the lack of sufficient and consistent phylogeny information about all included macrophytes in our data set, phylogeny diversity was assessed using taxonomic hierarchies as a proxy for phylogenetic relationships (Schweiger et al., 2008; Heino et al., 2015). Taxonomic distinctness (TDis) measures the mean taxonomic (i.e., phylogenetic) distances between species (Clarke and Warwick, 1998), which was calculated by giving equal branch lengths and six supra-species taxonomic levels (i.e., genus, family, order, subclass, class, and phylum). Hence, watershed with low TDis value indicates low phylogenetic diversity, and vice versa. TDis was calculated using the functions *"taxondive"* and *"taxa2dist"* in the R package *"vegan"* (Oksanen et al., 2016).

To calculate functional distance between macrophyte species, we gathered information on four categorical functional traits from the Flora of China, namely life form (i.e., free-floating, emergent, floating-leaved, submerged), life cycle (i.e., annual, perennial), morphology (i.e., turion, stem, rosette, leafy), sexual propagation (monoecism, dioecy), and species' mean adult weight (Zhang et al., 2018). Morphology was defined qualitatively based on the plant description. Leafy plants typically have more lamina, often the parts concentrating the majority of photosynthesis; stem plants are those with stem and easy to propagate due to broken branches; rosette plants have a shortened stem axis and relatively large projection area that facilitates light competition; turion plants produce winter/overwintering buds as dormant storage organs in response to unfavorable ecological conditions (Cook, 1974; Adamec, 2018). Although we acknowledge that quantitative morphological traits such as shoot height, stem diameter, specific leaf area, or leaf dry mass content, available from some local studies (Fu et al., 2014, 2018), would provide a more precise assessment, these were not available given the nature of our data set and the large scale of our study area.

Four multidimensional functional diversity indices (i.e., functional richness, functional evenness, functional divergence, and functional dispersion) were computed for each assemblage. Functional richness (FRic) describes the convex hull volume filled by a community in the multidimensional functional trait space and used as a measure of the functional richness. Functional evenness (FEve) describes the evenness of the distribution of species in a community over the functional trait space by using the minimum spanning tree. Feve quantifies the regularity with which the functional space is filled by species. Functional divergence (FDiv) describes how species distribute within the volume of functional trait space. For presence/absence data, FDiv is the highest when all the species are on the convex hull and at equal distance to its center of gravity (i.e., if the center of gravity of the convex hull is also a center of symmetry of the functional space). Finally, functional dispersion (FDis) describes the mean distance of each individual species to the centroid of all other species in the assemblage (Anderson et al., 2006), which was calculated using the function *"dbFD"* in the R package *"FD"* (Laliberté et al., 2014).

## Environmental Factors

We include several major environmental factors (**Supplementary Table S1**) used in previous similar macroecological studies (Whittaker et al., 2001; Tisseuil et al., 2013; Cai et al., 2018), namely mean annual precipitation (MAP), mean annual temperature (MAT), solar radiation (SOLAR), and total annual runoff (RUN), as surrogates for the energy input in each watershed, together with total surface area of each watershed (AREA), and spatial variation of altitude, MAT, MAP, SOLAR, and the Shannon diversity index based on the proportions of land cover classes (forest, grass, farm, urban, water, and desert) (i.e., ALTVAR, MATVAR, MAPVAR, SOLARVAR, and LANDVAR) within each watershed representing important factors shaping biodiversity through increasing habitat diversity and availability (Whittaker et al., 2001; Tisseuil et al., 2013). ALTVAR is used as a proxy for topographic heterogeneity, calculated as the range between the maximum and minimum altitude for each watershed. The selected factors should cover the environmental drivers of macrogeographic distributions of species at the watershed scale.

All environmental data were extracted from open-access databases such as Data Sharing Infrastructure of Earth System Science, National Science and Technology Infrastructure Center[5] and the Data Center of Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences. All parameters were calculated as the average of all cell values with centroids falling within each watershed at a spatial resolution of $0.5^\circ \times 0.5^\circ$ over the study period (1960–2010). Prior to the analyses, variables were logarithm or square root-transformed to improve normalization when necessary and standardized to have a mean of 0 and a variance of 1. We also computed the values of variance inflation factor (VIF) for each predictor variable before analyses to assess collinearity (Cai et al., 2018).

## Spatial Structure

Spatial structure in natural communities can be simultaneously generated by spatial autocorrelation in species assemblages and forcing (explanatory) variables, such as environmental and biotic controls or life history events (Borcard et al., 2004). In order to account for the contribution of the spatial structure of watershed configuration to observed variability in diversity facets, we used multiscale principal coordinates of neighbor matrices (PCNM; Borcard and Legendre, 2002) applied to a geographic distance

[5]http://zgswz.lifescience.com.cn/

matrix computed from the watershed centroid coordinates. In essence, distance between watersheds was first represented as a Euclidean distance matrix calculated from the watershed centroid coordinates. Principal coordinates analysis (PCoA) was then conducted on this matrix after truncation using a threshold distance equivalent to the minimum spanning tree of the distance matrix, defining what is considered to be "large" distances.

The resulting eigenvectors with positive eigenvalues (distance-based Moran's Eigenvector Maps; dbMEMs), provide a spectral decomposition of any possible spatial relationships between the watersheds (Borcard and Legendre, 2002). That is, each eigenvector captures the dominant spatial structures, i.e., low-order eigenvectors (dbMEM1-3 in **Supplementary Figure S1**), associated with large eigenvalues, represent country-scale groupings whereas high-order eigenvectors (dbMEM69-71 in **Supplementary Figure S1**), associated with small eigenvalues, represent more regional-scale groupings. Following a constrained analysis, those positive eigenvectors found to significantly explain variation in macrophyte diversity were selected as independent explanatory variables for further analysis. dbMEMs were calculated using the function *"eigenmap"* with default values in the package *"codep"* in R (Dray et al., 2012).

## Data Analysis

Spatial congruence of watersheds with high diversity was assessed as the amount of concordance between the top 10% of watersheds (i.e., highest diversity values) for all six diversity indices (Pool et al., 2014). That is, the 21 watersheds with the highest FRic values (i.e., the top 10%) were compared to the 21 watersheds with the highest FEve values to calculate how many watersheds were shared within both groups. Subsequently, spatial congruence of pairwise diversity was assessed at the successive 10% intervals. A randomization procedure was performed (999 iterations) for all six diversity indices to determine whether watershed congruence was greater than the random expectations.

We used variation partitioning based on redundancy analysis (RDA) models, with significance assessed using 999 Monte Carlo permutations, to reveal the partial effects of environmental variables and spatial structure on each diversity index (Peres-Neto et al., 2006). This procedure decomposes the total variation in the response dataset into a pure spatial component (S|E), a pure environmental component (E|S), a component of the spatial structured environmental variation (E∩S) and the unexplained variation. Only significant predictor variables were used for variation partitioning as identified from multiple linear regression models by using forward selection procedure and two stopping rules: either exceeding the critical $p$-value ($P = 0.05$) or the adjusted $R^2$ value of the reduced model against the global model based on 999 random permutations (Blanchet et al., 2008). We ran the variation partitioning analyses by using function *"varpart"* in the R package *"vegan"* (Oksanen et al., 2016). We then computed the Moran's $I$ correlograms to evaluate the degree of spatial autocorrelation of the diversity indices and the residuals from the linear models (Diniz-Filho and Bini, 2005) by using function *"correlog"* in the R package *"pgirmess"* (Giraudoux, 2016).

Given the nature of our data set, sampling bias might be expected among the 214 studied watersheds. For example, watersheds that are easy to access or largely populated may be expected to be more intensively sampled. However, Pearson correlation analysis between the number of literature sources (NOL, ranging from 4 to 23; **Supplementary Figures S2a,b**) compiled for each watershed, used as proxy for the total sampling effort, and species richness showed no significant correlation (**Supplementary Figure S2c**). Furthermore, results from a sensitivity analysis conducted by repeating the analysis only on watersheds with NOL >8 (25th quantile; $n = 160$) remain largely invariant to those obtained from the complete data set (**Supplementary Figure S2d**). Therefore, we report results for all watersheds.

# RESULTS

Higher values of taxonomic richness and most functional indices concentrated in watersheds from central-southern China (**Figures 1A–F**). In contrast, higher values of macrophyte taxonomic distinctness and functional evenness appeared in watersheds from western China (**Figures 1B–D**). Among all six diversity indices, taxonomic, and functional richness presented the highest range of variation across watersheds (**Figures 1A–C**), with mean values representing, respectively, 62 and 70% of the total pool of 469 documented species and functional richness across China. At the other extreme, taxonomic distinctness of individual watersheds was very homogeneous across China (**Figure 1B**), but accounted on average for 96% of the total distinctness in the study region.

Spatial congruence for the top 20% of each type of diversity indices presented two distinct groups with high (>40%) and low (<20%) pairwise congruence (**Figure 2**). The highest and lowest congruence occurred, respectively, for FDiv and FDis (58.6%) and FRic and FEve (0%). At this level, all high-congruence pairs were significantly more congruent than random expectations (999 iterations; $p < 0.001$), and showed a significant strong positive correlation (**Figure 3**). On the contrary, incongruent pairs showed mainly weak or strong negative (e.g., FEve – TRic and FEve – FRic) correlations (**Figure 3**).

Based on the selected linear regression model interpreting the variation of the diversity indices (**Table 1**), the percentage of variation explained varied from 40% for FEve to 81% for FDis and FDiv (**Figure 4**). Within each diversity index, the proportion of variation explained by shared fractions (spatially structured environmental gradients) was significantly higher compared to the proportion by unique fractions (pure effects), although the pure spatial component was responsible for relatively high amounts of variation in TRic, FRic, FEve, and FDiv (**Figure 4**). Geographic variations of the six diversity indices were strongly spatially autocorrelated along with a steady decreasing of Moran's I coefficient across distances (**Supplementary Figure S3**). Most of the residuals of the models showed weak spatial patterns, with the exception of the significantly positive autocorrelation at short distances, indicating that our models captured well the

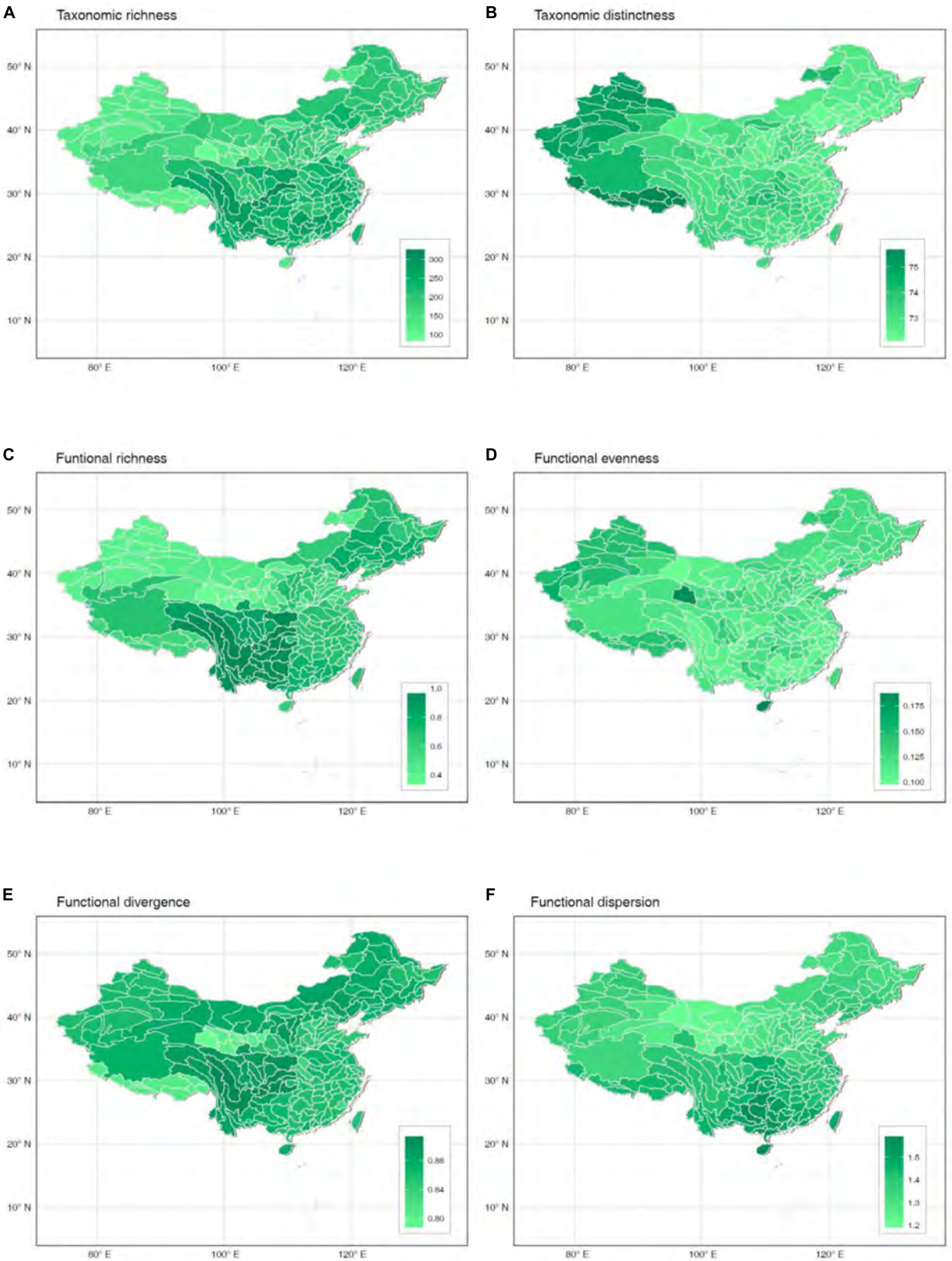



**FIGURE 1 |** Spatial patterns of taxonomic richness **(A)**, taxonomic distinctness **(B)**, functional richness **(C)**, functional evenness **(D)**, functional divergence **(E)**, and functional dispersion **(F)** of freshwater macrophyte assemblages across Chinese watersheds.

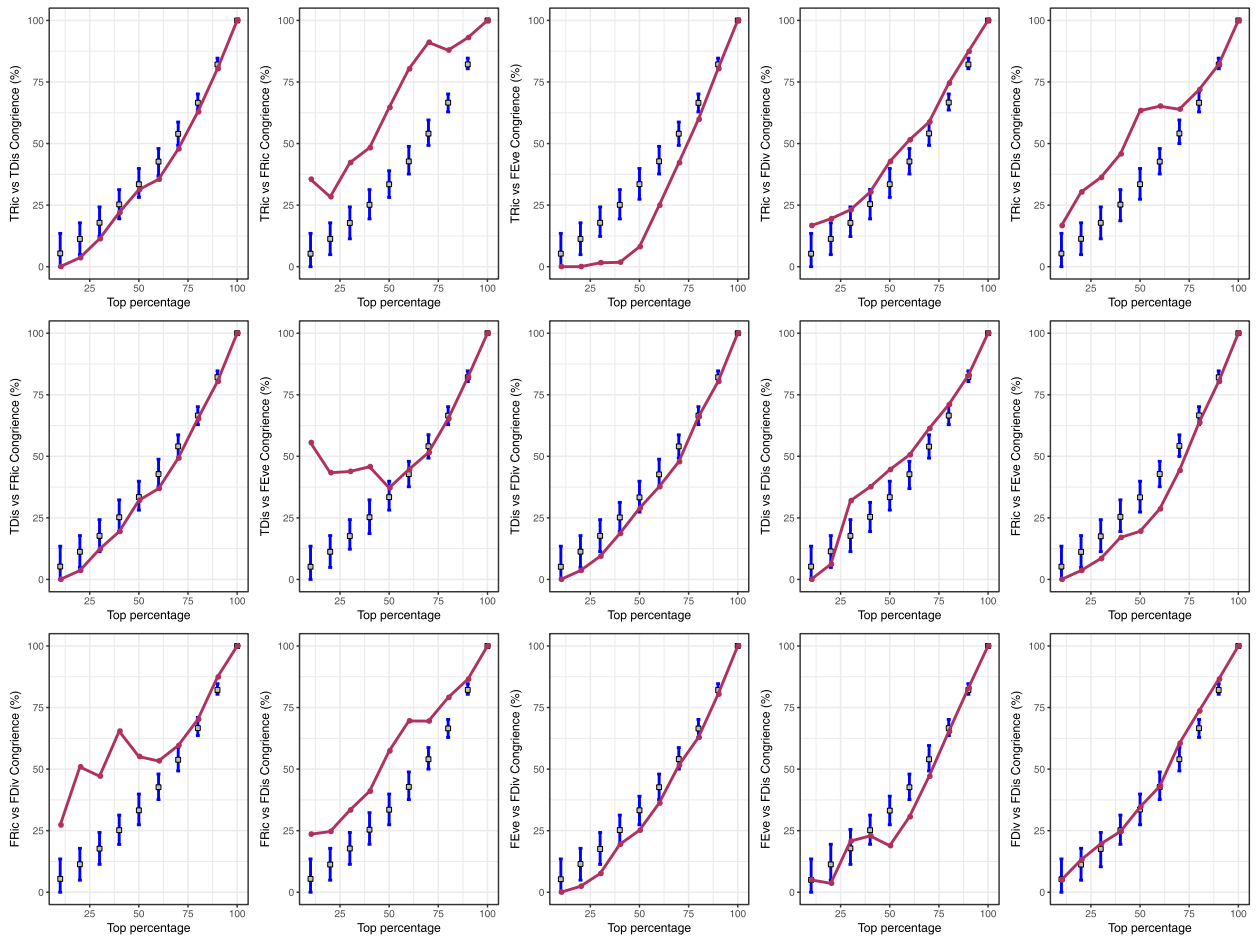



**FIGURE 2 |** Pairwise congruence between taxonomic richness (TRic), taxonomic distinctness (TDis), functional richness (FRic), functional evenness (FEve), functional divergence (FDiv), and functional dispersion (FDis) of freshwater macrophyte assemblages across Chinese watersheds. Congruence was assessed by comparing the spatial concordance for each pair of biodiversity facets across watersheds grouped by percentiles (10% intervals).

major ecological factors underlying geographic gradients of the diversity facets.

## DISCUSSION

We found a relatively high spatial congruence between many, but not all, biodiversity facets (i.e., TRic and FRic, FRic and FDis, TDis and FEve, FDiv and FDis, and FRic and FDiv). Given the strong effects of spatially structured environmental factors on shaping the biodiversity patterns (**Figure 1**), this high spatial congruence can presumably be explained by topography-related dispersal limitation affecting specific functional groups and, consequently, species. Our results corroborate previous evidence on strong correlations between different components of diversity. For instance, Heino et al. (2008) found that species richness was highly correlated with functional richness in stream macroinvertebrate assemblages. Likewise, Strecker et al. (2011) and Pool et al. (2014) reported high spatial congruence among three aspects of species diversity patterns of freshwater fish assemblages. Meynard et al. (2011) found evidence that hypotheses generated for local and regional taxonomic diversity were equally applicable to both phylogenetic diversity and functional diversity. These findings suggest the possibility of using a single diversity measurement as a surrogate for other facets to optimize conservation planning. Given resources are often limited, pinpointing conservation priorities and simultaneously protecting multiple diversity facets is highly desirable (Devictor et al., 2010).

On the other hand, the extremely low spatial congruence found between FEve and both TRic and FRic (**Figure 2**), and the low spatial congruence between TDis and both TRic and FRic may be related to the definition of the measurements, whereby increases in taxonomic diversity and functional diversity can only cause small changes in phylogenetic diversity (Schweiger et al., 2008; Pool et al., 2014; Cai et al., 2018). The scatter of the negative correlation indicates that taxonomic and functional richness increase with decreasing phylogenetic and functional diversity (**Figure 3**). This suggests a stronger effect of species identity relative to that of taxonomic

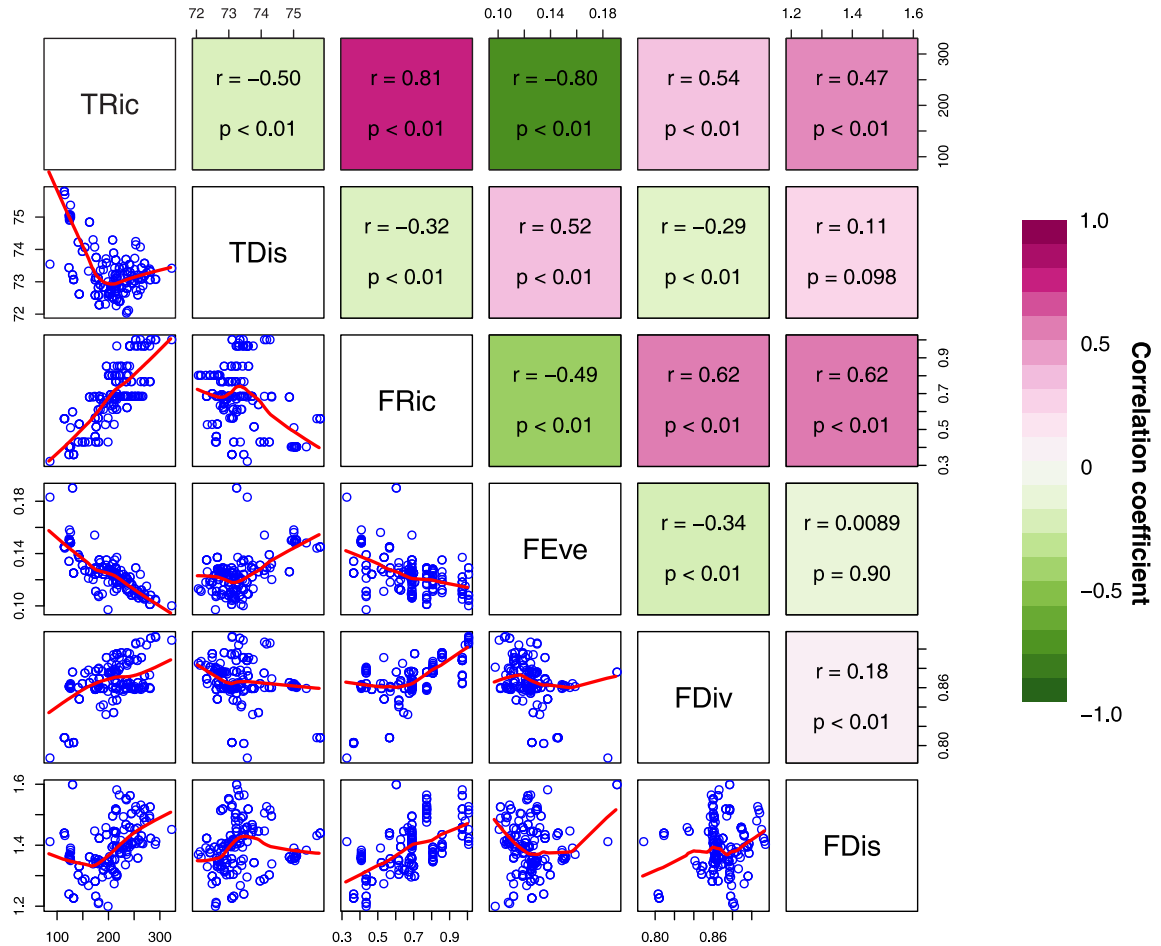



**FIGURE 3 |** Paired scatterplots and Pearson correlations between diversity indices of freshwater macrophyte assemblages across Chinese watersheds (Abbreviations as in **Figure 2**). Red and blue represents positive and negative correlations, respectively.

and functional richness. For instance, watersheds from central-southern China show highest TRic and FRic but relatively low FEve and TDis, whereas watersheds from western China show relatively high FEve and TDis but low TRic and FRic. Macrophytes from watersheds in central-southern China were phylogenetically and biologically more closely related, whereas those in western China were more distantly related to each other. An increasing number of studies reveal spatial mismatches among the three aspects of species diversity patterns (Heino et al., 2005; de Carvalho and Tejerina-Garro, 2015), suggesting that species occurring locally may originate from regional species pools with distinct biogeographic and evolutionary processes, respectively (Losos, 2008; Devictor et al., 2010). In other cases, species in the same region might respond differently to environmental variables affecting spatial TDis patterns and result in mismatches among different facets of diversity (Tucker and Cadotte, 2013). Thus, it is not surprising that great longitudinal gradients in China from west to east are associated with distinct environmental filtering conditions and dispersal limitations affecting functional traits and phylogenetic diversity of macrophytes.

Our results showed that both environmental and spatial factors influence the different facets of macrophyte biodiversity. In particular, spatially structured environmental gradients, rather than pure environmental effects, shaped the different facets of macrophyte biodiversity in watersheds across China (**Figure 4**). Pure spatial factors had a significant role in shaping several facets of the macrophyte biodiversity patterns suggesting that dispersal limitations exert a strong effect on macrophyte assemblage structure across the different diversity facets.

We found significant spatial autocorrelations among all six diversity metrics, whereas the selected spatially structured environmental variables optimally explained the spatial structure of all diversity facets. This results highlights the role of spatially structured environmental gradients, over and above the effect of environmental factors *per se*, as a major driver of biodiversity, which feeds into the debate about the effects of environmental heterogeneity and dispersal limitations on species distributions (Field et al., 2009). Our results also highlight the dominant role of climatic gradients in driving spatially structured patterns of all facets of diversity across watersheds. Climatic gradients at large spatial scales can influence biodiversity patterns through

**TABLE 1 |** Variables retained, adjusted $R^2$, and significance values from the forward-selected multiple regression models examining the effect of environmental and spatial factors on six aspects of macrophyte biodiversity.

| Functional diversity index | Factors | Selected variables | $R^2_{adj}$ | *F* | *P* |
|---|---|---|---|---|---|
| Taxonomic richness | Environment | MAP + MAPVAR − ALTVAR + SOLARVAR + AREA | 0.494 | 42.66 | <0.001 |
| | Space | dbMEM4 − dbMEM1 + dbMEM2 − dbMEM34 + dbMEM6 − dbMEM5 − dbMEM15 + dbMEM23 + dbMEM49 − dbMEM3 − dbMEM46 + dbMEM10 + dbMEM38 | 0.618 | 27.56 | <0.001 |
| Taxonomic distinctness | Environment | ALTVAR + MATVAR − MAPVAR | 0.321 | 34.62 | <0.001 |
| | Space | −dbMEM4 + dbMEM2 + dbMEM1 + dbMEM5 + dbMEM6 − dbMEM3 + dbMEM7 + dbMEM15 + dbMEM21 + dbMEM9 | 0.661 | 42.56 | <0.001 |
| Functional richness | Environment | MAP + MAPVAR + SOLAR | 0.612 | 110.4 | <0.001 |
| | Space | dbMEM2 + dbMEM4 − dbMEM1 − dbMEM3 + dbMEM6 + dbMEM15 | 0.695 | 81.88 | <0.001 |
| Functional evenness | Environment | MAP + ALTVAR + SOLARVAR + MAPVAR | 0.331 | 34.62 | <0.001 |
| | Space | −dbMEM4 − dbMEM3 + dbMEM1 + dbMEM34 − dbMEM38 + dbMEM56 − dbMEM44 − dbMEM49 + dbMEM5 + dbMEM59 − dbMEM69 | 0.381 | 12.94 | <0.001 |
| Functional divergence | Environment | MAP + SOLARVAR + MAT | 0.567 | 94.14 | <0.001 |
| | Space | dbMEM2 − dbMEM1 + dbMEM6 − dbMEM3 + dbMEM9 − dbMEM13 | 0.794 | 137.4 | <0.001 |
| Functional dispersion | Environment | MAP + MAT + SOLARVAR + SOLAR | 0.901 | 150.3 | <0.001 |
| | Space | −dbMEM1 + dbMEM2 + dbMEM6 − dbMEM3 − dbMEM5 + dbMEM4 + dbMEM17 − dbMEM21 − dbMEM13 − dbMEM40 + dbMEM39 + dbMEM9 | 0.614 | 85.9 | <0.001 |

*For each factor, forward selection was applied to identify which variables best described variation in functional diversity index using an inclusion threshold of alpha = 0.05.*

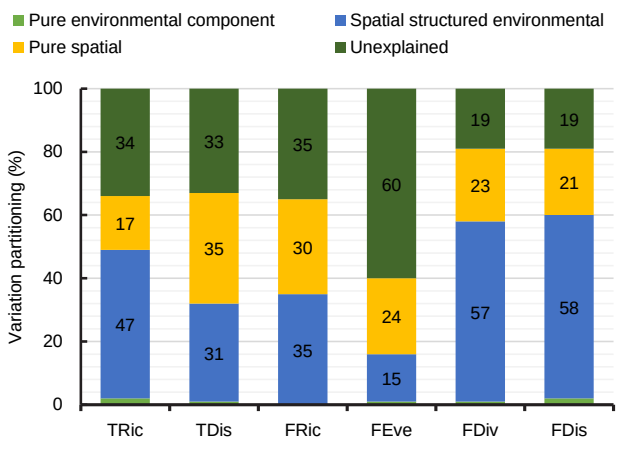



**FIGURE 4 |** Variation partitioning for diversity indices of freshwater macrophyte assemblages across Chinese watersheds (Abbreviations as in **Figure 2**).

multiple mechanisms related to the physiology, energetic demand and dispersal limitations of species (Fu et al., 2018; Li et al., 2018; Zhang et al., 2019). Spatial structure in the diversity facets of TRic, FRic, FEve, and FDiv was significantly explained by broad-scale dbMEMs (**Supplementary Figure S1**). Macrophytes seem to generally confront dispersal limitations although they are often recognized as good dispersers (Santamaria, 2002; Chambers et al., 2008). Our findings suggest that, given the presence of mountain ranges, habitat variability, and other obstacles across the studied watersheds, macrophytes assemblages across China for most diversity aspects are strongly structured by dispersal limitation. Such a pattern of species distributions is consistent with a number of previous studies for other aquatic plant assemblages at regional scales (Capers et al., 2010; Mikulyuk et al., 2011; Alahuhta and Heino, 2013; Viana et al., 2014).

## CONCLUSION

Our study on variation-partitioning analysis demonstrates that macrophyte diversity patterns in watersheds across China are not always congruent and mainly driven by spatially structured environmental determinism. This finding implies that macrophyte are taxonomically, phylogenetically, and functionally clustered in space, which might be the result of environmental and/or evolutionary forces.

## AUTHOR CONTRIBUTIONS

JX and MZ conceived the ideas and compiled the data. MZ, GS, and JX analyzed the data. MZ, JGM, GS, HZ, and JX wrote the manuscript. All authors contributed to the final manuscript.

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## SUPPLEMENTARY MATERIAL

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fpls.2019.00161/full#supplementary-material

## REFERENCES

Adamec, L. (2018). Ecophysiological characteristics of turions of aquatic plants: a review. *Aquat. Bot.* 148, 64–77. doi: 10.1016/j.aquabot.2018.04.011

Alahuhta, J. and Heino, J. (2013). Spatial extent, regional specificity and metacommunity structuring in lake macrophytes. *J. Biogeogr.* 40, 1572–1582. doi: 10.1111/jbi.12089

Alahuhta, J., Lindholm, M., Bove, C. P., Chappuis, E., Clayton, J., De Winton, M., et al. (2018). Global patterns in the metacommunity structuring of lake macrophytes: regional variations and driving factors. *Oecologia* 188, 1167–1182. doi: 10.1007/s00442-018-4294-0

Anderson, M. J., Ellingsen, K. E., and McArdle, B. H. (2006). Multivariate dispersion as a measure of beta diversity. *Ecol. Lett.* 9, 683–693. doi: 10.1111/j.1461-0248.2006.00926.x

Blanchet, F. G., Legendre, P., and Borcard, D. (2008). Forward selection of explanatory variables. *Ecology* 89, 2623–2632. doi: 10.1890/07-0986.1

Borcard, D., and Legendre, P. (2002). All-scale spatial analysis of ecological data by means of principal coordinates of neighbour matrices. *Ecol. Model.* 153, 51–68. doi: 10.1016/S0304-3800(01)00501-4

Borcard, D., Legendre, P., Avoisjacquet, C., and Tuomisto, H. (2004). Dissecting the spatial structures of ecologial data at multiple scales. *Ecology* 85, 1826–1832. doi: 10.1890/03-3111

Cadotte, M. W., Carscadden, K. A., and Mirotchnick, N. (2011). Beyond species: functional diversity and the maintenance of ecological processes and services. *J. Appl. Ecol.* 48, 1079–1087. doi: 10.1111/j.1365-2664.2011.02048.x

Cai, Y., Xu, J., Zhang, M., Wang, J., and Heino, J. (in press). Different roles for geography, energy and environment in determining three facets of freshwater molluscan beta diversity across China. *Sci. Total Environ.* doi: 10.1016/j.scitotenv.2018.1012.1373

Cai, Y., Zhang, M., Xu, J., and Heino, J. (2018). Geographical gradients in the biodiversity of Chinese freshwater molluscs: implications for conservation. *Divers. Distribut.* 24, 485–496. doi: 10.1111/ddi.12695

Capers, R. S., Selsky, R., and Bugbee, G. J. (2010). The relative importance of local conditions and regional processes in structuring aquatic plant communities. *Freshw. Biol.* 55, 952–966. doi: 10.1111/j.1365-2427.2009.02328.x

Chambers, P. A., Lacoul, P., Murphy, K. J., and Thomaz, S. M. (2008). Global diversity of aquatic macrophytes in freshwater. *Hydrobiologia* 595, 9–26. doi: 10.1007/s10750-007-9154-6

Chappuis, E., Gacia, E., and Ballesteros, E. (2014). Environmental factors explaining the distribution and diversity of vascular aquatic macrophytes in a highly heterogeneous Mediterranean region. *Aquat. Bot.* 113, 72–82. doi: 10.1016/j.aquabot.2013.11.007

Clarke, K., and Warwick, R. (1998). A taxonomic distinctness index and its statistical properties. *J. Appl. Ecol.* 35, 523–531. doi: 10.1046/j.1365-2664.1998.3540523.x

Cook, C. D. K. (1974). *Water Plants of the World: A Manual for the Identification of the Genera of Freshwater Macrophytes*. Berlin: Springer.

de Carvalho, R. A., and Tejerina-Garro, F. L. (2015). Relationships between taxonomic and functional components of diversity: implications for conservation of tropical freshwater fishes. *Freshw. Biol.* 60, 1854–1862. doi: 10.1111/fwb.12616

Devictor, V., Mouillot, D., Meynard, C., Jiguet, F., Thuiller, W., and Mouquet, N. (2010). Spatial mismatch and congruence between taxonomic, phylogenetic and functional diversity: the need for integrative conservation strategies in a changing world. *Ecol. Lett.* 13, 1030–1040. doi: 10.1111/j.1461-0248.2010.01493.x

Diniz-Filho, J. A. F., and Bini, L. M. (2005). Modelling geographical patterns in species richness using eigenvector-based spatial filters. *Glob. Ecol. Biogeogr.* 14, 177–185. doi: 10.1111/j.1466-822X.2005.00147.x

Dodkins, I., Rippey, B., and Hale, P. (2005). An application of canonical correspondence analysis for developing ecological quality assessment metrics for river macrophytes. *Freshw. Biol.* 50, 891–904. doi: 10.1111/j.1365-2427.2005.01360.x

Dray, S., Pélissier, R., Couteron, P., Fortin, M. J., Legendre, P., Peres-Neto, P. R., et al. (2012). Community ecology in the age of multivariate multiscale spatial analysis. *Ecol. Monogr.* 82, 257–275. doi: 10.1890/11-1183.1

Field, R., Hawkins, B. A., Cornell, H. V., Currie, D. J., Diniz-Filho, J. A. F., Guégan, J.-F., et al. (2009). Spatial species-richness gradients across scales: a meta-analysis. *J. Biogeogr.* 36, 132–147. doi: 10.1111/ele.12277

Flynn, D. F. B., Mirotchnick, N., Jain, M., Palmer, M. I., and Naeem, S. (2011). Functional and phylogenetic diversity as predictors of biodiversity–ecosystem-function relationships. *Ecology* 92, 1573–1581. doi: 10.1890/10-1245.1

Fu, H., Yuan, G., Luo, Q., Dai, T., Xu, J., Cao, T., et al. (2018). Functional traits mediated cascading effects of water depth and light availability on temporal stability of a macrophyte species. *Ecol. Indicat.* 89, 168–174. doi: 10.1016/j.ecolind.2018.02.010

Fu, H., Zhong, J., Yuan, G., Ni, L., Xie, P., and Cao, T. (2014). Functional traits composition predict macrophytes community productivity along a water depth gradient in a freshwater lake. *Ecol. Evol.* 4, 1516–1523. doi: 10.1002/ece3.1022

Gianuca, A. T., Declerck, S. A. J., Lemmens, P., and De Meester, L. (2017). Effects of dispersal and environmental heterogeneity on the replacement and nestedness components of β-diversity. *Ecology* 98, 525–533. doi: 10.1002/ecy.1666

Giraudoux, P. (2016). *Package 'Pgirmess': Data Analysis in Ecology. R package Version 1.6.5*. Available at: https://cran.r-project.org/web/packages/pgirmess/index.html

Hansson, L.-A., Nicolle, A., Bronmark, C., Hargeby, A., Lindstrom, A., and Andersson, G. (2010). Waterfowl, macrophytes, and the clear water state of shallow lakes. *Hydrobiologia* 646, 101–109. doi: 10.1007/s004420050121

Heino, J., Alahuhta, J., and Fattorini, S. (2015). Phylogenetic diversity of regional beetle faunas at high latitudes: patterns, drivers and chance along ecological gradients. *Biodivers. Conserv.* 24, 2751–2767. doi: 10.1007/s10531-015-0963-z

Heino, J., Mykrä, H., and Kotanen, J. (2008). Weak relationships between landscape characteristics and multiple facets of stream macroinvertebrate biodiversity in a boreal drainage basin. *Landscape Ecol.* 23, 417–426. doi: 10.1007/s10980-008-9199-6

Heino, J., Soininen, J., Lappalainen, J., and Virtanen, R. (2005). The relationship between species richness and taxonomic distinctness in freshwater organisms. *Limnol. Oceanogr.* 30, 978–986. doi: 10.4319/lo.2005.50.3.0978

Hellsten, S., and Riihimäki, J. (1996). Effects of lake water level regulation on the dynamics of aquatic macrophytes in northern Finland. *Hydrobiologia* 340, 85–92. doi: 10.1007/BF00012738

Jeppesen, E., Søndergaard, M., Søndergaard, M., and Christoffersen, K. (1998). *The Structuring role of Submerged Macrophytes in Lakes*. New York, NY: Springer-Verlag. doi: 10.1007/978-1-4612-0695-8

Laliberté, E., Legendre, P., and Shipley, B. (2014). *FD: Measuring Functional Diversity From Multiple Traits, and Other Tools for Functional ecology*. Available at: http://www.elaliberte.info/publications

Legendre, P., and Legendre, L. (2012). *Numerical Ecology*, 3rd Edn. Amsterdam: Elsevier.

Li, C., Wang, T., Zhang, M., and Xu, J. (2018). Maternal Environment Effect of Warming and Eutrophication on the Emergence of Curled Pondweed, *Potamogeton crispus* L. *Water* 10:1285. doi: 10.3390/w10091285

Losos, J. B. (2008). Phylogenetic niche conservatism, phylogenetic signal and the relationship between phylogenetic relatedness and ecological similarity among species. *Ecol. Lett.* 11, 995–1003. doi: 10.1111/j.1461-0248.2008.01229.x

Meynard, C. N., Devictor, V., Mouillot, D., Thuiller, W., Jiguet, F., and Mouquet, N. (2011). Beyond taxonomic diversity patterns: how do α, β, and γ components of bird functional and phylogenetic diversity respond to environmental gradients across France? *Glob. Ecol. Biogeogr.* 20, 893–903. doi: 10.1111/j.1466-8238.2010.00647.x

Mikulyuk, A., Sharma, S., Van Egeren, S. J., and Hauxwell, J. (2011). The relative role of environmental, spatial, and land-use patterns in explaining aquatic macrophyte community composition. *Can. J. Fish. Aquat. Sci.* 68, 1778–1789. doi: 10.1139/f2011-095

Oksanen, J., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'hara, R., et al. (2016). *vegan: Community Ecology Package. R package version 2.4-1*. Available at: https://cran.r-project.org/web/packages/vegan/index.html

Pardo, I., Roquet, C., Lavergne, S., Olesen, J. M., Gómez, D., and García, M. B. (2017). Spatial congruence between taxonomic, phylogenetic and functional hotspots: true pattern or methodological artefact? *Divers. Distribut.* 23, 209–220. doi: 10.1111/ddi.12511

Peres-Neto, P., Legendre, P., Dray, S., and Borcard, D. (2006). Variation partitioning of species data matrices: estimation and comparison of fractions. *Ecology* 87, 2614–2625. doi: 10.1890/0012-9658(2006)87[2614:VPOSDM]2.0.CO;2

Pool, T. K., Grenouillet, G., and Villéger, S. (2014). Species contribute differently to the taxonomic, functional, and phylogenetic alpha and beta diversity of freshwater fish communities. *Divers. Distribut.* 20, 1235–1244. doi: 10.1111/ddi.12231

Santamaria, L. (2002). Why are most aquatic plants widely distributed? Dispersal, clonal growth and small-scale heterogeneity in a stressful environment. *Acta Oecol.* 23, 137–154. doi: 10.1016/S1146-609X(02)01146-3

Schweiger, O., Klotz, S., Durka, W., and Kühn, I. (2008). A comparative test of phylogenetic diversity indices. *Oecologia* 157, 485–495. doi: 10.1007/s00442-008-1082-2

Shmida, A., and Wilson, M. V. (1985). Biological determinants of species diversity. *J. Biogeogr.* 12, 1–20.

Shurin, J. B. (2007). How is diversity related to species turnover through time? *Oikos* 116, 957–965. doi: 10.1111/j.0030-1299.2007.15751.x

Strecker, A., Casselman, J., Fortin, M.-J., Jackson, D., Ridgway, M., Abrams, P., et al. (2011). A multi-scale comparison of trait linkages to environmental and spatial variables in fish communities across a large freshwater lake. *Oecologia* 166, 819–831. doi: 10.1007/s00442-011-1924-1

Tisseuil, C., Cornu, J. F., Beauchard, O., Brosse, S., Darwall, W., Holland, R., et al. (2013). Global diversity patterns and cross-taxa convergence in freshwater systems. *J. Anim. Ecol.* 82, 365–376. doi: 10.1111/1365-2656.12018

Tucker, C. M., and Cadotte, M. W. (2013). Unifying measures of biodiversity: understanding when richness and phylogenetic diversity should be congruent. *Divers. Distribut.* 19, 845–854. doi: 10.1111/ddi.12087

Viana, D. S., Santamaŕla, L., Schwenk, K., Manca, M., Hobæk, A., Mjelde, M., et al. (2014). Environment and biogeography drive aquatic plant and cladoceran species richness across Europe. *Freshw. Biol.* 59, 2096–2106. doi: 10.1111/fwb.12410

Veech, J. A., and Crist, T. O. (2007). Habitat and climate heterogeneity maintain beta-diversity of birds among landscapes within ecoregions. *Glob. Ecol. Biogeogr.* 16, 650–656. doi: 10.1111/j.1466-8238.2007.00315.x

Webb, C. O., Ackerly, D. D., Mcpeek, M., and Donoghue, M. J. (2002). Phylogenies and community ecology. *Annu. Rev. Ecol. Evol. Syst.* 33, 475–505. doi: 10.1146/annurev.ecolsys.33.010802.150448

Westlake, D. F. (1975). "Macrophytes," in *River ecology*, ed. B. Whitton (Berkeley: University of California Press).

Whittaker, R. J., Willis, K. J., and Field, R. (2001). Scale and species richness: towards a general, hierarchical theory of species diversity. *J. Biogeogr.* 28, 453–470. doi: 10.1046/j.1365-2699.2001.00563.x

Wiegleb, G., Gebler, D., De Weyer, K. V., and Birk, S. (2016). Comparative test of ecological assessment methods of lowland streams based on long-term monitoring data of macrophytes. *Sci. Total Environ.* 541, 1269–1281. doi: 10.1016/j.scitotenv.2015.10.005

Wiegleb, G., Herr, W., Zander, B., Bröring, U., Brux, H., and Van De Weyer, K. (2015). Natural variation of macrophyte vegetation of lowland streams at the regional level. *Limnologica* 51, 53–62. doi: 10.1016/j.limno.2014.12.005

Xu, J., Su, G., Xiong, Y., Akasaka, M., García Molinos, J., Matsuzaki, S.-I., et al. (2015). Complimentary analysis of metacommunity nestedness and diversity partitioning highlights the need for a holistic conservation strategy for highland lake fish assemblages. *Glob. Ecol. Conserv.* 3, 288–296. doi: 10.1016/j.gecco.2014.12.004

Zhang, M., García Molinos, J., Zhang, X., and Xu, J. (2018). Functional and taxonomic differentiation of macrophyte assemblages across the Yangtze River floodplain under human impacts. *Front. Plant Sci.* 9:387. doi: 10.3389/fpls.2018.00387

Zhang, P., Grutter, B. M. C., Van Leeuwen, C. H. A., Xu, J., Petruzzella, A., Van Den Berg, R. F., et al. (2019). Effects of rising temperature on the growth, stoichiometry, and palatability of aquatic plants. *Front. Plant Sci.* 9:1947. doi: 10.3389/fpls.2018.01947

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